

GE Healthcare

LightSpeed 2.X Installation Manual

Book 1 of 2: Mechanical Installation

This manual applies only to LightSpeed 2.X systems shipped in 2006 or later by GoldSeal.

OPERATIONS DOCUMENTATION



5266553-1EN
Revision 01

LEGAL NOTES

TRADEMARKS

Adobe, the Adobe logo, Acrobat, the Acrobat logo, Exchange, and PostScript are trademarks of Adobe Systems Incorporated or its subsidiaries and may be registered in certain jurisdictions.

Microsoft is a registered trademark and Windows is a trademark of Microsoft Corporation.

Sears Industrial Sales is an operation of Sears Commercial Sales. Sears is a trademark of the Sears, Roebuck and Co.

Snap-on is a registered trademark of Snap-on Incorporated.

All other products and their name brands are trademarks of their respective holders.

COPYRIGHTS

All Material, Copyright © 2000 by General Electric Company, Inc. All rights reserved.

This Page Intentionally Left Blank

IMPORTANT PRECAUTIONS

LANGUAGE

Предупреждение

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu áanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고
(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS
(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS
(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL
(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE
(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kilavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kilavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyariya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

Check for damage to property that may have occurred at the site during delivery, such as damage to floors, door frames or walls. If damage is found, notify the install specialist.

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact the local service coordinator for more information on this process.

Rev. Aug. 7, 2007

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE Healthcare uses its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE Healthcare equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, GE Healthcare Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, GE Healthcare Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION Risk of Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Healthcare Sales or Service representatives.

GE personnel, please use the GE Healthcare iTrak Process to report all omissions, errors, and defects in this publication.

This Page Intentionally Left Blank

Revision History

Revision	Date	Reason for change
1	03/28/08	Initial release of document meeting all current and regulatory conditions for systems shipped in 2006 or later by GoldSeal.

Table of Contents: Book 1

Preface

Publication Conventions 21

Section 1.0

Safety & Hazard Information 21

1.1 Text and Character Representation..... 21

1.2 Graphical Representation 22

Section 2.0

Publication Conventions 23

2.1 General Paragraph and Character Styles..... 23

2.2 Page Layout..... 23

2.3 Computer Screen Output/Input Character Styles 24

2.4 Buttons, Switches, and Keyboard Inputs (Hard & Soft Keys) 24

Chapter 1

Position Subsystems 25

Section 1.0

Installer/FE Notices 25

1.1 Shipping, Warehouse, and Transportation Warning..... 25

1.2 International Shipments 25

1.3 On Site Warning (Mechanical Installation Team) 25

Section 2.0

Install Table/Gantry Introduction 26

2.1 Skill Set Required 26

2.2 Floor and Room Preparation 27

2.2.1 Preparation 27

2.2.2 Flooring..... 27

2.3 Overview..... 27

2.4 Pre-Installation Template..... 27

2.5 Installation Alignment Tool Kit 28

2.6 Required Common Tools and Supplies 29

Section 3.0

Delivery and Inventory Procedure 31

3.1 Delivery Procedure 31

3.1.1 System Transportation - Temperature Extremes..... 31

3.1.2 Working with the Mover 31

3.1.3 Floor Protection 31

3.1.4 Equipment Delivery Route 31

3.1.5 Removing Gantry Dollies and Covers..... 31

3.1.6 Check for Shipping Damage..... 31

3.1.7 A1 Breaker..... 32

3.1.8 Installation Support Kits 32

Section 4.0

Install and Level Table/Gantry 33

4.1	Lay the Floor Template	33
4.1.1	Tools Required	33
4.1.2	Time & Personnel	33
4.1.3	Safety	33
4.2	Establish the Room Layout	33
4.3	Position the Gantry	35
4.4	Level the Gantry	38
4.5	Gantry Bearing Gap Inspection	40
4.5.1	Personnel Requirements	40
4.5.2	Tools and Test Equipment	40
4.5.3	Damage Indicators	40
4.5.4	Finalization - Mechanical Installers	42
4.5.5	FE Service Action Required	42
4.5.6	FE Inspection Completion	42
4.6	Position the Table	43
4.7	Install Alignment Bars	44
4.8	Level the Table	46
4.8.1	Before You Begin	46
4.8.2	Procedure	46
4.9	Table-Gantry Mechanical Alignment	48
4.9.1	Required Tools	48
4.9.2	Alignment Conditions	48
4.9.3	Alignment Specification	48
4.9.4	Procedure	48
4.9.4.1	Cradle Center Procedure	50
4.9.4.2	Measuring Cradle Center Procedure	52
4.10	Cradle/Table Parallel Check	53
4.10.0.1	Determining Table to Gantry Perpendicularity	56
4.11	Tighten the Lock Rings	56
4.12	Drill the Anchor Holes	57
4.12.1	Notes to Mechanical Installers	57
4.12.1.1	Note 1: Basic Anchoring Information	57
4.12.1.2	Note 2: Alternate Anchoring	57
4.12.1.3	Note 3: Non-Concrete Floors	57
4.12.1.4	Note 4: GE Notification	57
4.12.2	Requirements	57
4.12.2.1	Tools Required	57
4.12.2.2	Time and Personnel	58
4.12.3	Drilling Procedure	58
4.13	Gantry & Table Redundant Anchor Holes	62
4.14	Install the Anchors (102 mm (4 in.) Concrete Floors Only)	64
4.15	Inside Levelers	66

Section 5.0

Rear Entry Cable Box (Site Specific)	69
--	----

Section 6.0

Install Table Footswitch Assembly	70
---	----

Section 7.0

Remove Gantry Tilt Bracket	71
----------------------------------	----

Section 8.0	
Position the Power Distribution Unit.....	72
Section 9.0	
Install Operator Console.....	74
9.1 Unpack Console	74
9.2 Removing Shipping Dollies	74
9.3 Remove Console Covers	75
9.4 Adjust Table Top Height and Position Console	76
9.5 Attach Keyboard Table Top	77
9.6 Operator Drive Tray	77
9.7 Monitor Placement	77
Section 10.0	
Seismic Mounting.....	79
10.1 Console.....	79
10.2 Console (GOC1 and GOC2).....	79
10.3 Power Distribution Unit	80

Chapter 2

Power, Ground & Interconnect Cables 81

Section 1.0	
Introduction	82
1.1 System Component Identification	83
1.2 Cable Color Identifiers	83
Section 2.0System Interconnect Diagram.....	84
Section 3.0	
Contractor Connections	85
Section 4.0	
Console Connections	86
4.1 Accessing the Computer.....	86
4.2 SCIM, Keyboard, Trackball, and Mouse Installation	88
4.3 Connecting the Monitor.....	91
4.4 .Power Panel Connections	92
4.5 LAN Connections	92
4.6 Installing a DASM (Hardware Option).....	93
4.7 Bulkhead Connections	94
4.8 Installing the GOB Board	94
Section 5.0	
Install Options	95
5.1 Install Optional Bar Code Reader	95
5.2 Install Optional Remote Monitor	95
5.3 Install Cardiac Respiratory Gating Option	95
5.4 Install Respiratory Gating Option	95
5.5 Install Injector Option	95
5.6 Install The UVY Monitor and Stand	95
5.7 Customer Accessories (Head Holders and Extender)	95
5.8 Install UPS	95

5.9	Install Service Cabinet	96
5.10	Install UPS	96
Section 6.0		
Gantry Cable Connections.....		97
6.1	Installing the GOB Board	98
Section 7.0		
Option Cables		99
Section 8.0		
Table Connections.....		100
Section 9.0		
PDU Cable Connections & Configuration		101
9.1	A3 Panel - 380 - 480 VAC Mains "A1" Input Power Connection	103
9.2	A3 Panel - Circuit Breakers	104
9.3	A5 Panel - Transformer (480 VAC) Taps	104
9.4	A2 Panel - HVDC Connection	105
9.5	A3 Panel - 440 V Connection	105
9.6	A4 Panel - Gantry & Console Power Connections.....	106
	9.6.1 Console Power Cable Re-termination	106
	9.6.2 Gantry Power Cable Re-termination	107
9.7	A4 Panel – PDU Control Cable	107
9.8	A4 Panel – System Ground Connection	107
9.9	A4 Panel – X-ray Light and Door Interlock Connections.....	107
	9.9.1 X-ray Light Configuration and Connection	107
	9.9.2 Door Interlock Connections	109
Section 10.0		
System Ground Connections		110
 Chapter 3		
System Continuity & Ground Checks		113
Section 1.0		
System Continuity & Ground Checks (Mechanical Contractor).....		113
1.1	Tools Required.....	113
1.2	Procedure.....	113
Section 2.0		
Site Ground Continuity Check.....		116
Section 3.0		
Axial Head Holder Shim Installation		117
3.1	Overview	117
3.2	Requirements.....	117
	3.2.1 Personnel Requirements.....	117
	3.2.2 Tools and Test Equipments.....	117
	3.2.3 Repacement Parts.....	117
3.3	Required Conditions.....	117
	3.3.1 Introduction.....	117
	3.3.2 Selecting Shim Thickness	118

3.4	Procedure	118
3.4.1	Finalization.....	119

Section 4.0

Mechanical Installation Completion Section	120
---	------------

Appendix A

Removal & Installation of Covers.....	121
--	------------

Section 1.0

Gantry Side Covers 4	121
-----------------------------------	------------

1.1	Side Cover Removal	121
1.2	Side Cover Installation	122

Section 2.0

Gantry Top Covers	122
--------------------------------	------------

2.1	Top Cover Removal	122
2.2	Top Cover Installation	122

Section 3.0

Gantry Front Cover	123
---------------------------------	------------

3.1	Front Cover Dolly Setup	123
3.2	Removal.....	125
3.3	Installation.....	130

Section 4.0

Gantry Rear Cover.....	133
-------------------------------	------------

4.1	Removal.....	133
4.2	Installation.....	134

Section 5.0

Gantry Cover Alignment Adjustment Guide	135
--	------------

5.1	Overview	135
5.2	Tools Required	135
5.3	Adjustment Guide	135
4.0.1	Front/Rear Cover adjustment.....	135
4.0.2	Scan Window	136
4.0.3	Top Cover Adjustment	136
4.0.4	Side Covers	136
4.0.5	Finish	136

Section 6.0

Gantry Scan Window	137
---------------------------------	------------

6.1	Remove Scan Window	137
6.2	Install Scan Window	137

Section 7.0

Gantry Base Covers	138
---------------------------------	------------

Section 8.0

Foot-switch Covers	140
---------------------------------	------------

Appendix B

Pictorial Representation of Required Tools

141

Appendix C

Regulatory Clearance Quick Reference Guide.....

145

Section 1.0

Regulatory Code Description

145

Section 2.0

Terms and Definitions

146

Section 3.0

Regulatory Minimum Working Clearance by Major Subsystem.....

147

Section 4.0

Room Size

149

4.1

Suggested Room Size

149

4.2

Minimum Room Size

149

Section 5.0

System Specifications.....

150

5.1

Rooms With Less Than 711 mm (28 in) Egress Clearance at
Table Foot End

150

5.2

Cautions

150

Section 6.0

How to Measure

151

Section 7.0

Room Size and Requirement Layouts

152

Section 8.0

Recommended Room Size and Requirement Layouts

153

Preface

Publication Conventions

Become familiar with the conventions used within this publication before proceeding.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles are used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger, and warning identify important safety information. Text (Hazard) styles are applied to the paragraph contents applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information normally includes:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from the GEHC global document standard. These include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution indicates a hazard exists that can or could cause minor injury to self or others if the instructions are ignored. They include, for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER INDICATES A HAZARD EXISTS THAT CAN CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



WARNING
ROTATING
EQUIPMENT
BARE WIRES

WARNING INDICATES A HAZARD EXISTS THAT COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF THE INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to tag and lockout system power, which could allow for un-command motion.



NOTICE
Equipment
Damage
Possible

Notice indicates a hazard is present that can cause property damage, but has absolutely no personal injury risk. They can include:

- Disk drivecrash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

Do not ignore hazard statements in this document.

1.2 Graphical Representation

The exclamation point contained within a triangle  indicates Important information. In addition to text, several different graphical icons (symbols) are used to make the reader aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make the reader aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes highlight important non-safety related information. Paragraph prefixes (e.g., Purpose, Example, Comment, and Note) identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

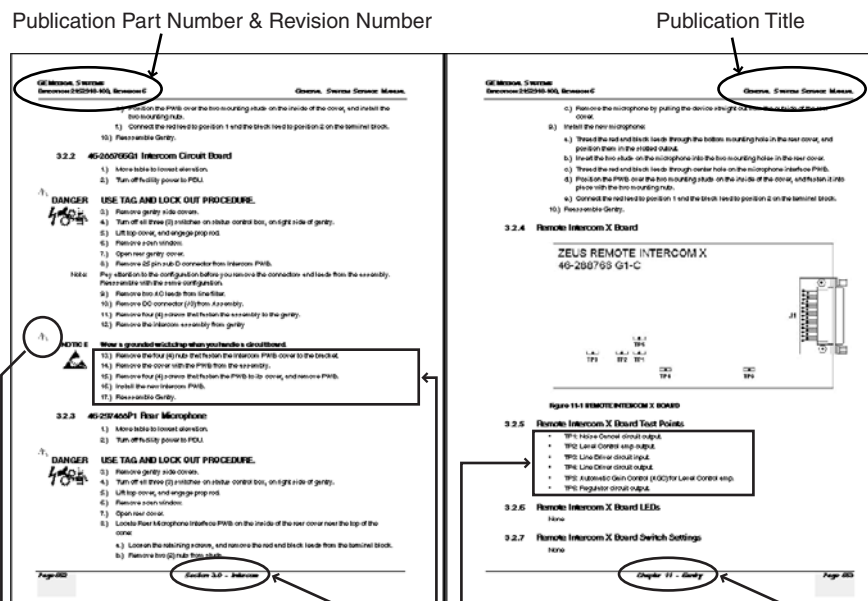
EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the chapter, section, or subsection. It conveys information that should be considered important to the reader.

Example: Makes the reader aware that the paragraphs that follow are examples of information stated previously.

Comment: Represents “additional” information that may or may not be relevant.

2.2 Page Layout



The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by Alphanumeric (e.g. numbers) characters is information that must be followed in a specific order.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by alphanumeric (e.g. numbers) characters is information that has no specific order.

Headers and footers in this publication allow the reader to identify the location within the document. The document part number and revision number appear in every header on every page. Odd numbered page footers indicate the current chapter, its title, and current page number. Even page footers show the current section and its title, and current page number.

2.3 Computer Screen Output/Input Character Styles

Different character styles within this publication indicate computer input and output text. Character styles (input, output, and variable) are applied to the text within a paragraph to indicate directions. Computer screen input and output also is formatted using mono-spaced (fixed width) fonts.

Example: Fixed Output This paragraph denotes computer screen fixed output. Output is considered fixed because it does not vary from application to application. This is the most commonly used style to indicate filenames, paths, and text.

Example: Variable Output *This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>*

Example: Fixed Input **This paragraph denotes fixed input. Fixed input is information that does not vary from application to application. The operator supplies fixed text as input.**

Example: Variable Input *This paragraph denotes computer input that can vary from application to application. The Operator is required to supply variable text as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped before input. Exceptions are noted in the text.*

2.4 Buttons, Switches, and Keyboard Inputs (Hard & Soft Keys)

Different character styles indicate actions requiring the reader to press either a hard or soft button, switch, or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer-generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style using both over and under-lined bold text. This is a hard key.

Example: Soft Keys The computer MENU button that requires the reader to click with the mouse or touch with the hand uses over and under-lined regular text. This is a soft key.

Chapter 1

Position Subsystems



NOTICE Potential for Data Loss and/or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from procedures completed in this chapter onto Form 4879, located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system. Any other revisions of this manual may not exactly match your system.



NOTICE If installing the CT system with a GRE Console, refer to the *GOC4 CT Console Upgrade and Service Info* manual, Direction# 5134894-800.

Section 1.0

Installer/FE Notices

1.1 Shipping, Warehouse, and Transportation Warning

- Move this gantry using the shipping dollies. Do not lift or move this gantry using a lift truck.
- Do not hoist gantry or table using dollies.

1.2 International Shipments

- Use dollies to remove the gantry from the shipping skid and transport it to the customer site.
- If lifting is required, refer to the Pre-Installation Manual for instructions.

1.3 On Site Warning (Mechanical Installation Team)

- This system requires a gantry bearing gap inspection *before* electrical calibration is started. See [Gantry Bearing Gap Inspection](#), on page 40.

Section 2.0

Install Table/Gantry Introduction

This chapter describes how to mount, position, and level the LightSpeed subsystems.

Note: Before you start the installation, ensure that the site preparation complies with conditions and instructions found in the pre-installation manual. Failure to comply, results in excessive installation delay and potential increased, unrecoverable installation costs. This product is designed to meet mechanical installation standards. Review these standards before installing this system.

2.1 Skill Set Required

NOTICE All Mechanical Installers shall meet all minimum requirements.

Only people experienced and trained in the installation process of a GE CT Scanner should perform the work described in this document. A person doing the installation should have:

- Completed three LightSpeed CT installations as a helper, before leading an installation. Installation leader must have completed project management, leadership training, and completed all required training before becoming a project leader.
- Reviewed the LightSpeed Installation Video, Safety Video, Installation Training CD, and successfully passed all related tests. (Refer to [Table 1-1](#).)

Media	Part Number
LightSpeed Installation CD	2289471-100
LightSpeed Installation Video	2289468-100
LightSpeed Installation Training CD	2289779-100
GE Safety CD	2290198-100
GE Safety Video	2290199-100

Table 1-1 Installation and Training Media

- A working understanding of the mechanical installation process before beginning.
- Completed environmental health and safety (EHS) training, understand and apply lockout/tagout processes, completed Hazcom training, completed Eye & Personal Protection Safety, and have taken Blood Borne Pathogens training.
- All mechanical vendor training and records must be verifiable and available to GE Healthcare.
- Mechanical vendors shall be responsible for training all employees involved in the installation of GE Healthcare CT products.
- It is the mechanical vendors responsibility to require and monitor that all of the above safety requirements are maintained and met on installation sites.
- Mechanical vendor installation teams must have basic measurement skills, system layout skills, and the ability to read and understand blue prints. Professional communication skills are required to communicate changes or additional requirements with customer before completing the installation project.

2.2 Floor and Room Preparation

2.2.1 Preparation

Consult your local GE Sales and service representative about your specific needs. It is the purchaser's (buyer's) responsibility to provide an approved support structure and an approved method of mounting. General Electric is not responsible for any failure of the support structure or method of anchoring.

The LightSpeedsystem has a total floor load of approximately 3021 kg (6660 lbs). About 2291 kg (5050 lbs.), including patient (204 kg (450 lbs)), is concentrated in the table-gantry assembly. Refer to the *LightSpeed 2.X Pre-installation* manual for more information.

2.2.2 Flooring

Do not place the scanner on any resilient flooring. Resilient tile or carpeting may slowly yield over a period of time and disturb the alignment of the table to the gantry. Refer to the floor template to determine locations where the resilient flooring material should be removed.

Limitations include:

- The support areas for the table and gantry must be higher than the floor surface within the table, gantry, or the two interface areas between the table and gantry.
- The floor structure must withstand the occupied weight of table and gantry, as well as the individual contact area loading of these components.
- The method and placement of anchors, or through bolts, must not reduce the structural strength of the floor.

2.3 Overview

Procedures in this chapter provide detailed instructions to position, level, and anchor the LightSpeedgantry and table securely for operation. The LightSpeedsystem uses adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads and six (6) auxiliary "leveler" pads located just inside the gantry base. There also are two leveling pads on each of the front and rear support frames. The auxiliary pads are sometimes referred to as "jig-feet" and/or "inside levelers." The table has five (5) pads used for leveling it.

Use the following process:

- 1.) Use the room-layout template to determine the general position of the gantry and table.
- 2.) Move the gantry into position.
- 3.) Level the gantry.
- 4.) Use the three alignment bars to position the table relative to the gantry.
- 5.) Level the table to the gantry, and anchor the system.

Use the template to position the system; however, use the gantry and table to locate and drill the anchor holes. Drill the anchor holes with the system in place. Refer to [Section 4.0, on page 33](#) and [Figure 1-19, on page 45](#), for an example of this procedure. This LightSpeedsystem installation procedure requires the items listed in [Section 2.5, on page 28](#) and [Section 2.6, on page 29](#).

2.4 Pre-Installation Template

Always use the room-layout template, P/N 2265473 (in two pieces), during installation. The gantry and table do not properly align using existing holes. The template shows the location of the gantry and table anchor holes.

This template is shipped with the system.

2.5 Installation Alignment Tool Kit

The installation personnel provide the installation tool kit, P/N 2110953. The tool kit contains the following special tools (See [Figure 1-1](#)). **This installation alignment toolkit requires yearly calibration and must have a current calibration sticker.**

- **Guide Pin (Bolt) Tool** included in the tool kit (not needed).
- **Long Alignment Bar** (in two pieces), P/N 2106369: use to position the table, relative to gantry
- **Two Short Alignment Bars**, P/N 2106368: use to position the table, relative to the gantry
- **Adjuster Tool**, P/N 2107863: use to turn the adjusters, which in turn level the system.
- **½ in. Drill Bushing**, P/N 2106205: use to center and guide the bit when drilling the anchor holes
- **Vacuum Attachment (funnel)**, P/N 2110944: use to vacuum debris out of the anchor holes
- **Spanner Wrench**, P/N 2110003: if necessary, use to turn the adjusters after anchoring system
- **Anchor Seating Tool**, P/N 2247965: used to properly seat anchors

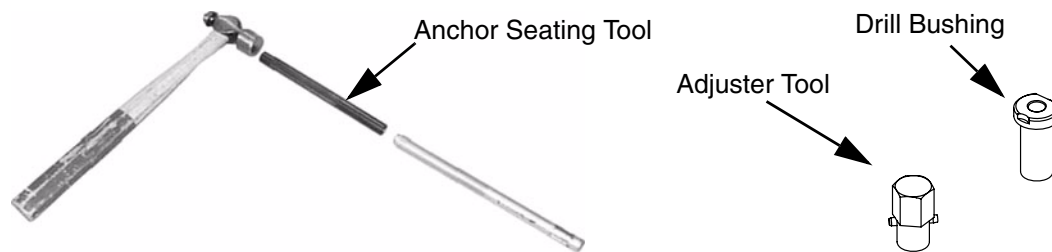


Figure 1-1 Installation Alignment Toolkit Accessories



Figure 1-2 Gantry/Table Alignment Toolkit

2.6 Required Common Tools and Supplies

The following tools and supplies are required for installation of the scanner. Please refer to [Section B](#), for pictorial descriptions of the tools and supplies.

WRENCHES

- Standard and Metric combination wrench sets
- Standard and Metric Hex Key (Allen wrench) sets
- ½ in. and 3/8 in. drive torque wrench: 0-100 N-m (0-100 ft.-lb.) Must be calibrated yearly.

SOCKETS AND EXTENSIONS

- 3/8 in. and ½ in. drive ratchet wrenches
- 1 in. 1-1/8 in., 1-¼ in. and 1-½ in. sockets for ½ in. drive
- ½ in. drive 3 in. and two 6 in. long extensions
- 3/8 in. drive universal joint
- 3/8 in. drive 12 in. long extension
- Metric hex bit set ¼ in. or 3/8 in. drive, including:
 - 14 mm hex bit 3/8 in. or ½ in. drive (14 mm ball hex helpful)
 - 10 mm hex bit 3/8 in. drive
- Standard and Metric 3/8 in. drive socket sets
- ¾ in. deep well socket 3/8 in.

SCREW DRIVERS

- Phillips screwdriver set (small, medium, and large)
- Straight blade screwdriver set (small, medium, and large)

DRILL BITS

- Complete set of standard (U.S.) drill bits
- Metric tap set
- ½ in. masonry bit, min. 8 in. long USA—18 in. optional (for rear table hole)
- 13 mm masonry bit, min. 203 mm long—457 mm optional (for rear table hole)
- 3 ft. (76 mm) hole saw with 1/4 in. (6 mm) masonry bit (to remove flooring)

POWER TOOLS

- 3/8 in. or ½ in. drill, cordless or electric
- Reciprocating Saw (Sawzall or equivalent) and assorted blades.
- Hammer Drill
- Sears 17740 Shop vacuum or equivalent, with “HEPA” or dry wall dust filter (Sears part number 17918) or equivalent
- 25 ft. Extension power cords

HAND TOOLS

- Ball-Peen Hammer (1 lb or 2 lb)
- Diagonal Cutting Pliers, Small
- Tongue and Grove Pliers (large)
- Large pry bar
- Diagonal Cutting Pliers, Large (to cut 1/0 ground)
- 4 ft., 2 ft. and 9 in. torpedo levels (see [Table 1-2: Recommended Levels](#))
- Framing Square (e.g., Empire 16 in. x 24 in. aluminum square)

	Johnson Magnetic Level, model 7500M
9 in.	Johnson Magnetic Level, model 4500 Stanley Magnetic Level
2 ft.	Johnson Professional Box Beam Level, model 9624 Empire Titan Professional Box Beam Level, model 900 series
4 ft. (nominal)	48 in. Johnson Professional Box Beam Level 42 in. Stanley Contractor Grade Level 48 in. Empire Titan Professional Box Beam Level, model 900 series 48 in. Stabila Aluminum Box Beam Level, Kit 24816

* Preferred levels

Table 1-2 Recommended Levels

ELECTRICAL TOOLS

- DVM
- Continuity tester
- Dale 600 Leakage Meter
- Temperature/humidity tool: Oregon Scientific Wireless Weather Station Model BAR608HGA (or equivalent)

PERSONAL SAFETY EQUIPMENT

- Safety shoes*
- Safety glasses*
- Gloves
- LOTO Kit*
- Hearing Protection*
- 6 ft. and 8 ft. Step ladder

* These PPE items are required for every installation job, with NO exceptions.

Note: A box labeled "Installation Support Kit" is shipped with each system. It contains paint, masking tape, towels, and other materials required for the CT scanner installation.

SYSTEM CLEANERS

Purchase locally:

- Windex glass cleaner or equivalent
- Scrubbing Bubbles bathroom cleaner or equivalent
- 409 grease cutting cleaner or equivalent
- WD-40

Section 3.0

Delivery and Inventory Procedure

3.1 Delivery Procedure

3.1.1 System Transportation - Temperature Extremes

When transporting the LightSpeed system, ensure it is not exposed to temperatures or humidity outside the following specifications:

Temperature: -18° to +49° C (0° to +120° F)

Humidity: 20% to 80%

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature before installation.



NOTICE Component Freezing occurs if the CT system is exposed to temperatures below -18° C (0° F) for a period longer than two days.

3.1.2 Working with the Mover

Follow the instructions provided by your Project Manager of Installation regarding working with equipment movers. Help direct movers on where to place equipment, and which items you need first.

Generally, movers should move all equipment into the customer room. Door removal and other site changes to move equipment should be done only as directed by the Project Manager of Installation.

For component sizes and weights, refer to the *LightSpeed Pre-Installation* manual.

3.1.3 Floor Protection

It is suggested that you use floor protection. Most equipment movers can provide floor protection during the equipment delivery. Installers should provide floor protection for the room.

3.1.4 Equipment Delivery Route

Before equipment delivery, review the delivery route with the movers. Refer to the Project Manager of Installation for any additional delivery instructions.

3.1.5 Removing Gantry Dollies and Covers

Do not remove gantry dollies or covers until the gantry is in place and you are told to remove the covers or dollies as part of the installation procedures.

3.1.6 Check for Shipping Damage

Check for the following damage, which may have occurred during shipping:

- Equipment damage: paint scrapes and cover damage.
- Damage to hospital property: floors, door frames, and walls.

If damage is found—or items are missing in shipment—notify the appropriate service personnel:

- For items missing in shipment or short shipped, contact the Project Manager of Installation to have the items shipped.
- Report damaged items to:
 - (800) 548-3366, Option 8
 - For further details, see [“DAMAGE IN TRANSPORTATION,”](#).

3.1.7 A1 Breaker

Note: The site **MUST** have a main disconnect with Lockout/Tagout capability.

Lockout and tagout the A1 breaker now.

A1 Breaker	UPS 2
USA 480 VAC: E4502AE	Phase Model: B7999PS
Europe and Asia 400 VAC: E4502AF	Phase Model: B7999PS

Table 1-3 A1 Breaker and UPS 2



Figure 1-3 Sample A1 Breaker

3.1.8 Installation Support Kits

An Installation Support Kit is shipped with every system. Locate this box now and open it. Use all included materials during the installation process. Leave these items on site for future service needs.

Cleaning supplies are not shipped with the system. Purchase cleaning supplies locally.

Section 4.0

Install and Level Table/Gantry

4.1 Lay the Floor Template

4.1.1 Tools Required

- Standard Install Tool Kit
- Install Support Kit
- GE Site Print
- Installation Manual
- Floor Template for your system
- 4 ft. Level
- Chalk line
- China marker

4.1.2 Time & Personnel

- 1.0 Hour labor on site
- 2 Engineers

4.1.3 Safety



CAUTION Potential for Injury

The LightSpeed gantry presents a variety of mechanical and electrical hazards. Use appropriate safety procedures when working on the CT system.

4.2 Establish the Room Layout

BEFORE YOU BEGIN

Place the main unit of the temperature and humidity tool in the control room and the remote unit in the scan room. Monitor environmental conditions at the installation site by recording the temperature and humidity of each room at the end of each day.

Use the GE site print developed for your site to establish the room layout, observing all operating and service clearances shown on the print. Likewise, before continuing, confirm that the site meets the following requirements:

- The site print is located at the facility and matches the actual dimensions of the room.
- The placement of equipment on-site matches the placement of equipment on the print.
- The site meets all regulatory requirements. Installation cannot proceed if the site does not comply with all regulatory requirements.

Note: If the site does not meet ALL of the above requirements, contact your management and your Project Manager of Installation.

PROCEDURE



NOTICE The floor must meet the minimum floor specification in order to begin this procedure. Refer to the *Pre-Installation Manual* for details.

- 1.) Use the supplied template to locate the anchor holes. Make sure they clear structural interferences in the floor.
- 2.) Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.
- 3.) Lay out the gantry template. Align the template per the GE print.

Note: If the floor has existing anchor penetration, the new anchors must be set at least 102 mm (4 in.) from the nearest penetration. The strength of the new anchors may be weakened if they are located too close to existing anchor points.

- 4.) Place the table template over the top of the gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.

Note:

- All regulatory and service clearances must be met to continue.
- If additional millwork is present that is not shown on the site print, confirm that there is a clear access route to the gantry right side service area.
- Check for grounded walls within the regulatory area.
- If you cannot meet all clearances, you must STOP the installation and escalate the issue to the Project Manager of Installation (PMI) until all clearances are met.

- 5.) Now, check the level of the floor (See [Figure 1-4](#)) across the templates.

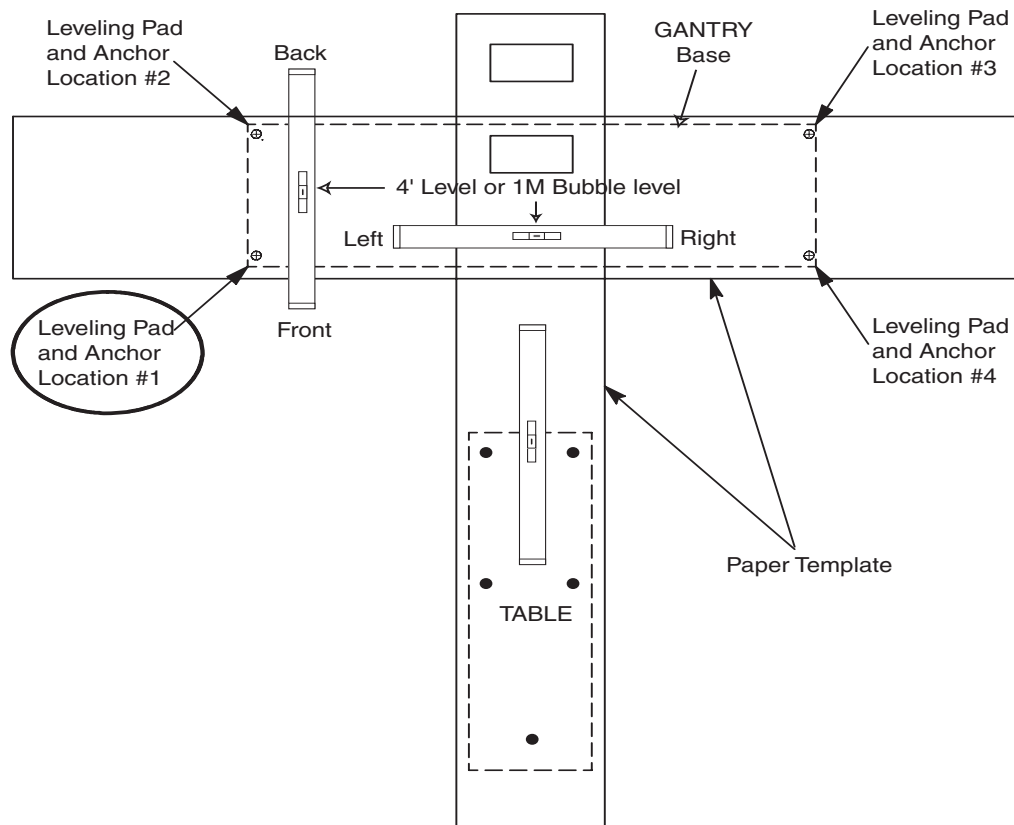


Figure 1-4 Gantry and Table Paper Templates

- 6.) Make a mark (e.g., use a center punch) at each of the gantry's mounting hole locations (there are four (4) of these). Also make marks for each of the gantry's six (6) leveler locations.
- 7.) Using a center punch, mark the three (3) table mounting holes and two (2) leveler locations.



NOTICE

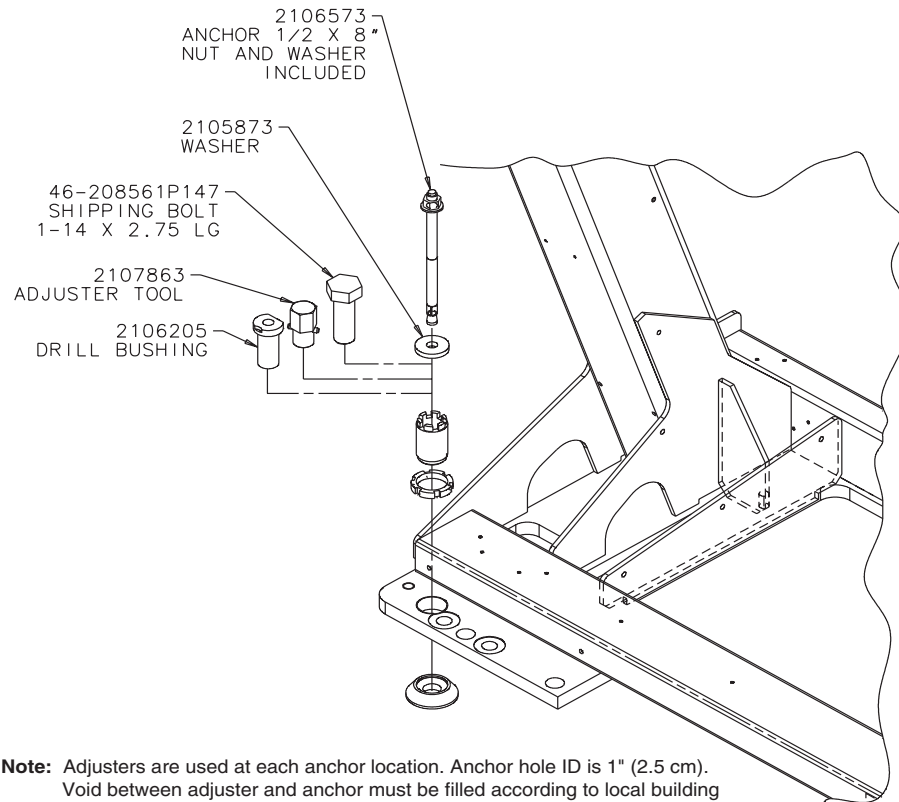
Positioning requires cutting 13 holes in the floor.

Before you drill or cut any flooring, make sure the appropriate hospital personnel have approved the location of the table/gantry.

- 8.) Cut tiles (or other resilient flooring) around all 15 holes punched in the template for the gantry and table. Use a utility knife with a heat gun, a 3 in. hole saw with a 1/4 in. masonry bit or other adequate tools to cut the flooring.
- 9.) Some sites may require sealing the floor penetrations after the flooring is removed.
 - Use RTV or other sealant to seal the floor covering, as necessary.
 - All documentation in this manual is based on mounting the table/gantry on a concrete floor only.

4.3 Position the Gantry

- 1.) Remove all the transportation packaging, except for dollies, from the gantry.



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm).
Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 1-5 Table Base Installation Hardware

- 2.) Position the gantry over the template appropriately.

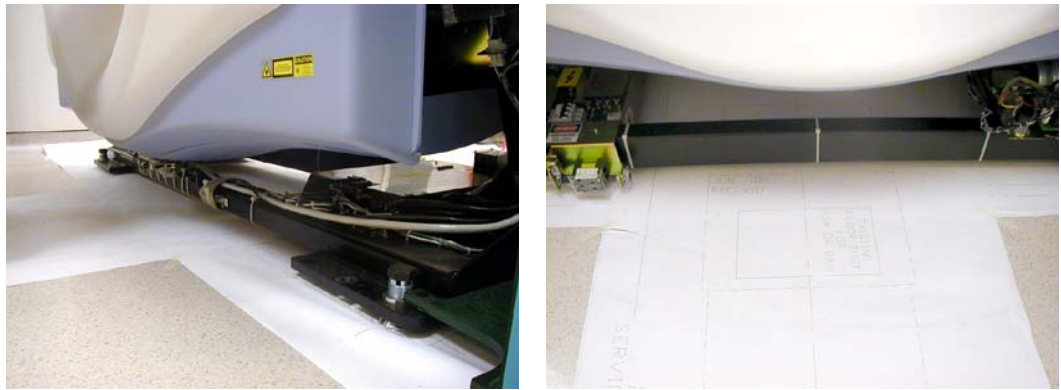


Figure 1-6 Gantry position on paper template

- a.) Locate the four (4) leveling pads, and position each of them beneath its associated adjuster.
- b.) Use the dollies to evenly lower the gantry, until it's just off of the floor (approximately 12.7 mm or ½ in.). Use a ½ in. ratchet to raise and lower the dollies.
- c.) Carefully rotate the gantry into the correct position over the template.

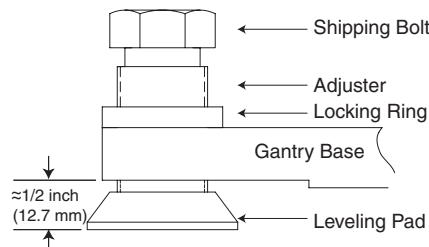


Figure 1-7 Gantry and Table Base Leveling Pads (Starting Positions)

- 3.) Remove the paper templates from the floor and discard of properly.
- 4.) Loosen the locking rings and shipping bolts so you can fine tune the leveling pads to compensate for slight variations in the floor surface.
- 5.) Position the gantry so that the adjusters are centered over their respective holes scribed earlier into the floor.

- Note: Make certain to route the gantry power cord under the two rear gantry rails, before removing the gantry shipping dollies.
- 6.) Using a ½ in. ratchet, gently lower the gantry until it rests on the floor, over the marked areas.

- 7.) Using a 14 mm hex socket, remove the dollies from the gantry by removing the three dolly bolts found at both ends of the gantry (Figure 1-8).



Figure 1-8 Gantry Dolly Bolts

- 8.) Remove the (4) gantry shipping bolts, using a 1½ in. socket.

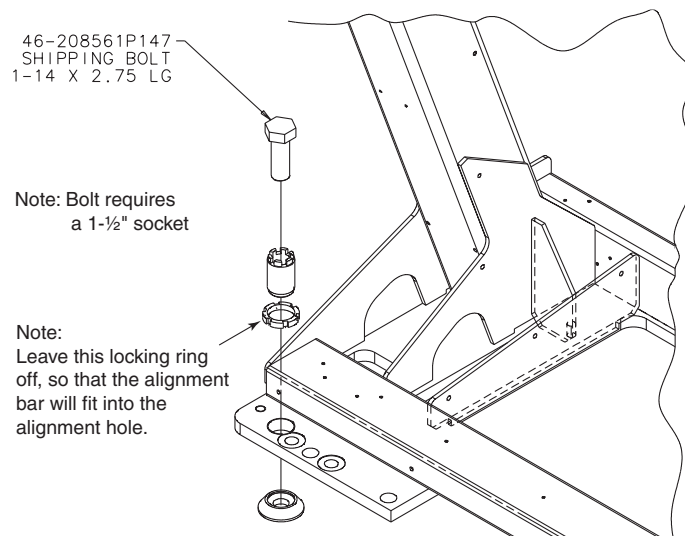


Figure 1-9 Gantry Shipping Bolts

4.4 Level the Gantry

The gantry uses two (2) bubble levels that are permanently mounted to machined surfaces on the stationary base to tell when it is level.

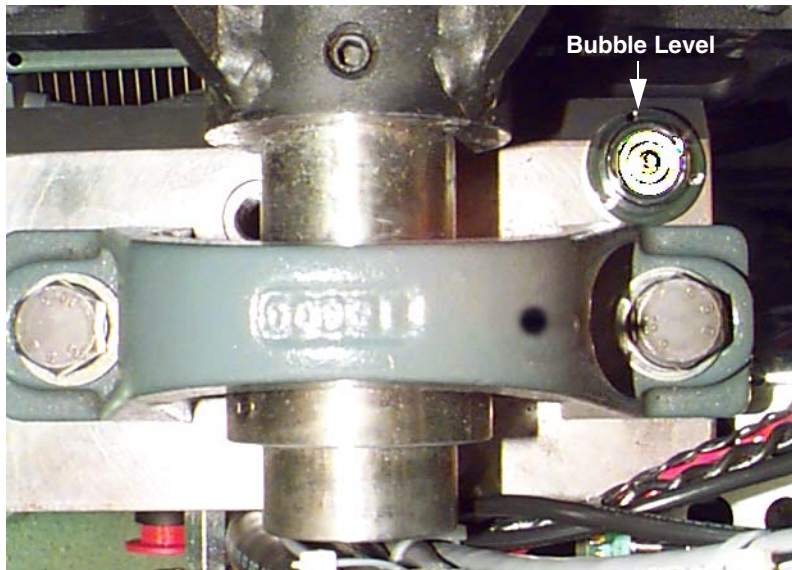


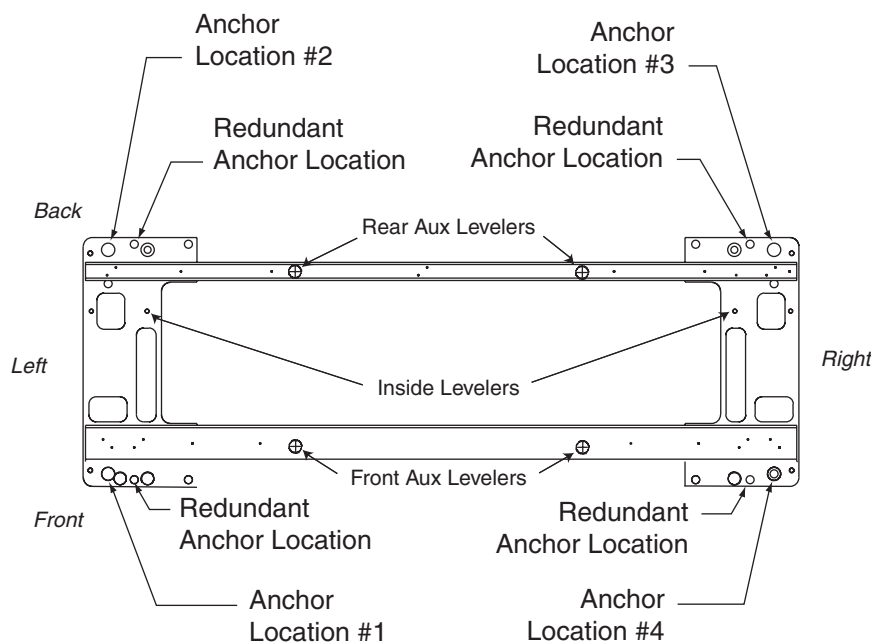
Figure 1-10 Gantry Bubble Level

The bubble levels are located on both ends of the gantry stationary base, near a point where the rotating structure pivots mount to the base structure. (See [Figure 1-10](#).) The gantry is properly leveled when the bubble is centered. (See [Figure 1-12, on page 39](#).)

- 1.) **Remove adjuster #1 lock ring** and only loosen all the other (#2, #3 and #4) adjuster lock rings. Do this using spanner wrench 2110003 (the spanner wrench can be found in the alignment toolkit). Check to make sure auxiliary levelers do not cause interference.
- 2.) Systematically turn each of the gantry's adjusters (locations 1, 2, 3, and 4, in [Figure 1-11](#)) until both bubble levels are centered left to right, and front to back.
 - Begin by turning each adjuster no more than 1 turn at a time.
 - Use the adjuster tool, 1 $\frac{1}{8}$ in. socket, and the $\frac{1}{2}$ in. drive ratchet to turn each adjuster. (Refer to [Figure 1-5, on page 35](#).)

Systematic Procedure for Leveling gantry follows:

- 1.) Level the left side from front to back by turning adjusters #1 and #2.
- 2.) Level the right side from front to back by turning adjusters #3 and #4.
- 3.) Level the side (right or left) that is higher, with respect to the other side. Turn both adjusters on a side equally until that side is level. The side should now also be level.



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm).
Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 1-11 Gantry Base "Adjuster" Locations —Top View

Note: "Levelers" are addressed later.

- 3.) When the bubble levels are centered ([Figure 1-12](#)), each of the four (4) leveling pads should be carrying a portion of the gantry weight. Distribution of the gantry weight prevents the base frame from rocking during normal operation. **DO NOT leave any adjuster un-loaded or floating.**

Correct level is 100% of bubble within small circle

Incorrect Level is less than 100% of bubble within small circle

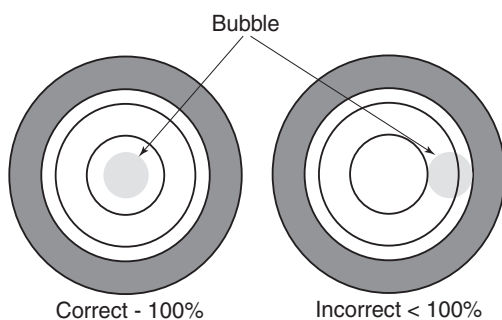


Figure 1-12 Bubble Level Centering

4.5 Gantry Bearing Gap Inspection

All CT systems require a Gantry Bearing Gap inspection before electrical calibration is started.

All international gantries are shipped in a wooden shipping crate that should not be removed until it arrives at the installation site. This shipping container is designed to reduce the risk of shipping damage. This inspection should be completed during system installation.

4.5.1 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
2 (mechanical suppliers or installation team)	15 min	15 min	5 min

4.5.2 Tools and Test Equipment

- Standard tool kit
- Inspection document
- 2.5 mm Allen wrench
- Rear cover dollies (2)
- Flashlight

4.5.3 Damage Indicators

On the inside edge of the black-colored bearing assembly, a mark similar to that shown in [Figure 1-13](#) is seen, if this is a Rotek bearing. I



Figure 1-13 Gantry Bearing - Rotek Label

The mark has a serial number in the same format as:
ROTEK 2TF4-44E1B-MA91960-8F-2372-REV13.

The gap to inspect is shown in [Figure 1-13](#) next to the serial number.

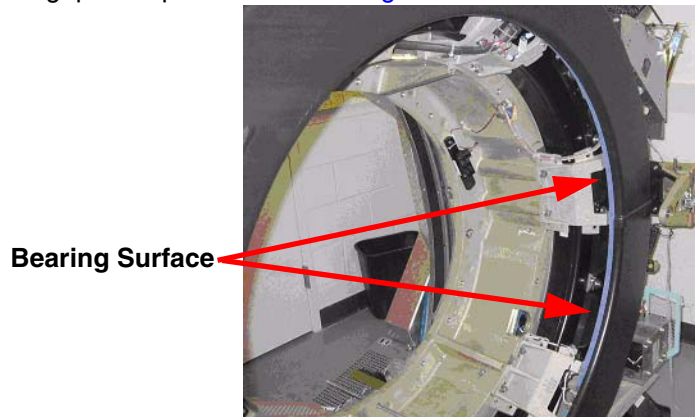


Figure 1-14 Gantry Bearing

On most systems, a change in the bearing gap does not cause the gantry to make unusual sounds, unless the gap is severe. If the gantry is badly damaged and the gap is severe, it can cause operation issues. Some systems are shipped with shock indicators that must be returned to Milwaukee.

A severe failure may be seen during installation as a problem rotating the gantry.

PROCEDURE

- 1.) Remove the scan window following the procedure in [Section A, Section 6.0](#).
- 2.) Remove the top and rear gantry covers, following the procedures in [Section A, Section 2.0](#) and [Section 4.0](#).
- 3.) Use a 2.5 mm hex wrench as a tool to measure the gap at the positions shown in [Figure 1-15](#). The location of the gantry components does not matter. Measure four (4) locations 90 degrees apart from each other.

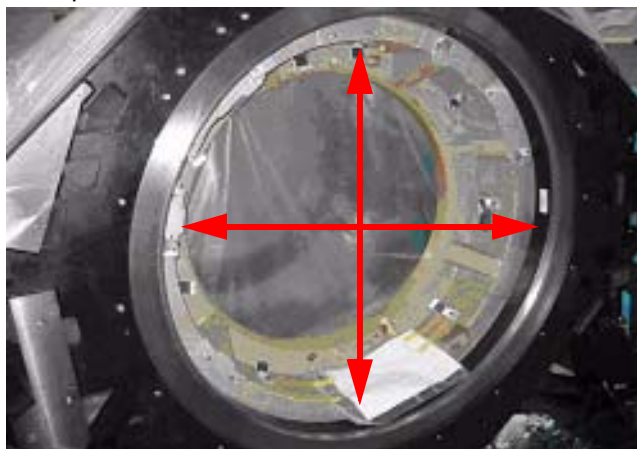


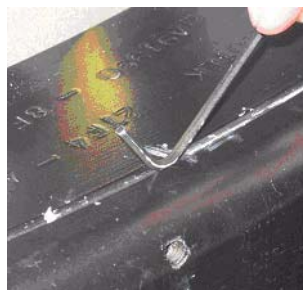
Figure 1-15 Inspection Locations

- 4.) If the 2.5 mm hex wrench easily fits without effort in the gap, the gap is out of spec. [Figure 1-16](#) shows a gap that is too large and one that is good. Notice that the hex wrench does not fit in the gap in the figure on the right, but does in the figure on the left

Note: Do not use force when putting the wrench in the gap. Either it slips in or it doesn't.



Figure 1-16 Gap is too large



Gap is good

- 5.) Replace the top and rear gantry covers, following the procedures in [Section A](#), [Section 2.0](#) and [Section 4.0](#).
- 6.) Replace the scan window, following the procedures in [Section A](#), [Section 6.0](#).

4.5.4 Finalization - Mechanical Installers

If the Bearing Gap Inspection passes, complete the signoff on the GE Form 4879 (located on the CD for this scanner's service manual), Installation Data verification form, that this inspection was completed.

If the Bearing Gap Inspection fails, contact your site FE.

4.5.5 FE Service Action Required

If the Bearing Gap Inspection fails, the mechanical installer notifies the site FE that the inspection failed.

The site FE should:

- 1.) Open a bearing inspection dispatch.
- 2.) Follow the inspection procedure described in this section.
- 3.) Record the bearing inspection results.

If no damage is found, close this dispatch and continue with the electrical calibration procedures.

If the system is damaged, go to the Equipment Delivery Quality web site and follow their instructions.

To enter a damaged in shipping claim, go to this web site:

<http://egems.med.ge.com/edq/home.jsp>

4.5.6 FE Inspection Completion

- 1.) After the Gantry Bearing Inspection passes, complete the opened service dispatch with the following information:
 - Gantry Serial Number
 - Gantry Type
 - System ID
 - Site Name
 - Installation date
 - Was the Gantry transported to the site in the shipping crate? (Yes/No)
 - Was the Gantry lifted or hoisted, were riggers used, or was the Gantry delivered via flatbed wrecker? (Yes/No)
 - Number of locations that fail the gap inspection if any: _____
- 2.) Close the service dispatch.

Should any follow-up be required after this inspection, the site engineer will be contacted directly by CT Engineering.

4.6 Position the Table

- 1.) Remove all the transportation packaging and boxes, except dollies, from the table.
- 2.) Wheel the table to its approximate position relative to the gantry, using the marks made earlier.
 - a.) Locate the table leveling pads and position them against the base of the table, using the adjusters with a 1½ in. socket and ½ in. ratchet.
 - b.) Use the dollies to evenly lower the table until it rests on the leveling pads using a ½ in. ratchet.
- 3.) **Loosen adjuster shipping bolts and locking rings “one turn” before you lower the table to the floor.** Loosen the locking rings and shipping bolts so you can **fine tune** the leveling pads to compensate for slight variations in the floor surface. Use a 1½ in. socket with 6 in. extension to loosen the shipping bolts.
- 4.) Remove the table left and right upper side covers (see [Figure 1-17](#)).

Note: The captive screws have a tendency to fall out occasionally.

Remove left and right side covers.

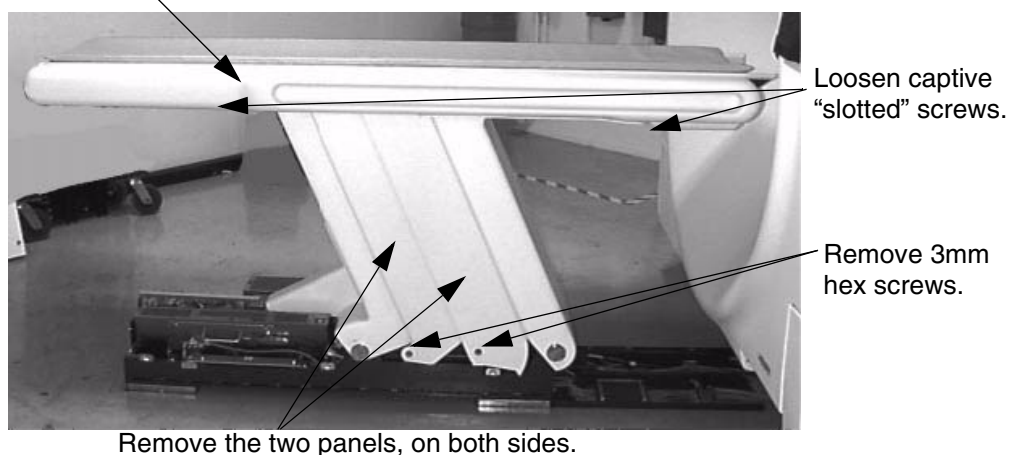


Figure 1-17 Table Cover Removal

- 5.) Remove the four side panels using a 5 mm hex key. Without disconnecting the ground straps at the z-channel, carefully lay the side panels on top of the padded cradle.



NOTICE
Potential for
Equipment
Damage

Take extreme care to prevent damaging the ETC and CPU printed wire assemblies. Use some packing material to protect these components.

- 6.) Remove the table base cover center support bar (see [Figure 1-18](#)).

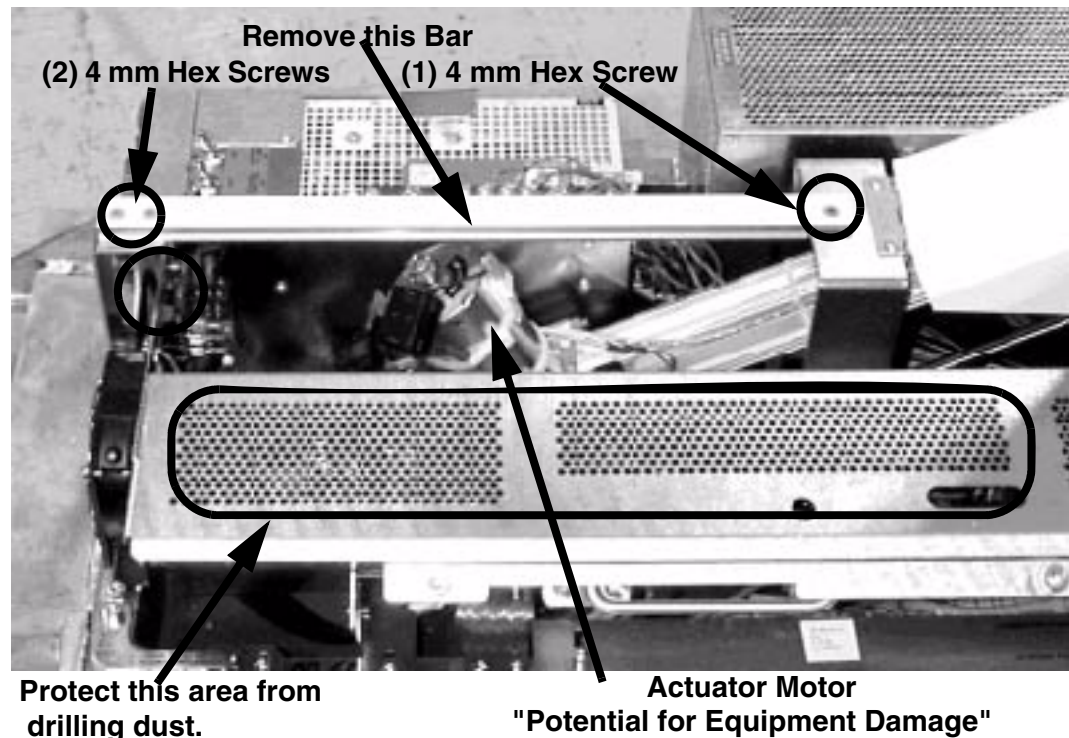


Figure 1-18 Center Support Bar

4.7 Install Alignment Bars

- 1.) Assemble the two (2) halves of the long gantry/table alignment bar (#3):
 - a.) Adjuster locking ring at location #1 (see [Figure 1-19](#)) must be removed to position the alignment bars properly. Also, observe the following:
 - * Bars should be at room temperature ($70^{\circ}\text{F} \pm 1\text{ degree}$ or $21^{\circ}\text{C} \pm 1\text{ degree}$).
 - * The alignment bar must be calibrated and have a valid calibration sticker to be accurate.
 - * Do not use or trust bars that have a missing locking pin or loose end cams.
 - b.) Insert the pin and rotate the handle to tighten the joint. If the joint remains loose, tighten the tip of the expanding pin to increase its diameter and try again.
 - c.) Check the serial numbers on both halves of the alignment bar; ensure that the serial numbers match.
- 2.) The short alignment bars (#1 and #2) are identical and interchangeable (see [Figure 1-19](#)):
 - a.) Install the short alignment bars (#1 and #2) first. The bars should fit snugly.
 - b.) Pivot the far end of the table until the long alignment bar (#3) fits into place.
 - c.) Adjust the position of the table until the three alignment bars fit into the $\frac{3}{4}$ in. alignment holes in the gantry base and the $\frac{1}{2}$ in. alignment holes in the table base.
 - d.) Fully seat all three bars on their mating surfaces.

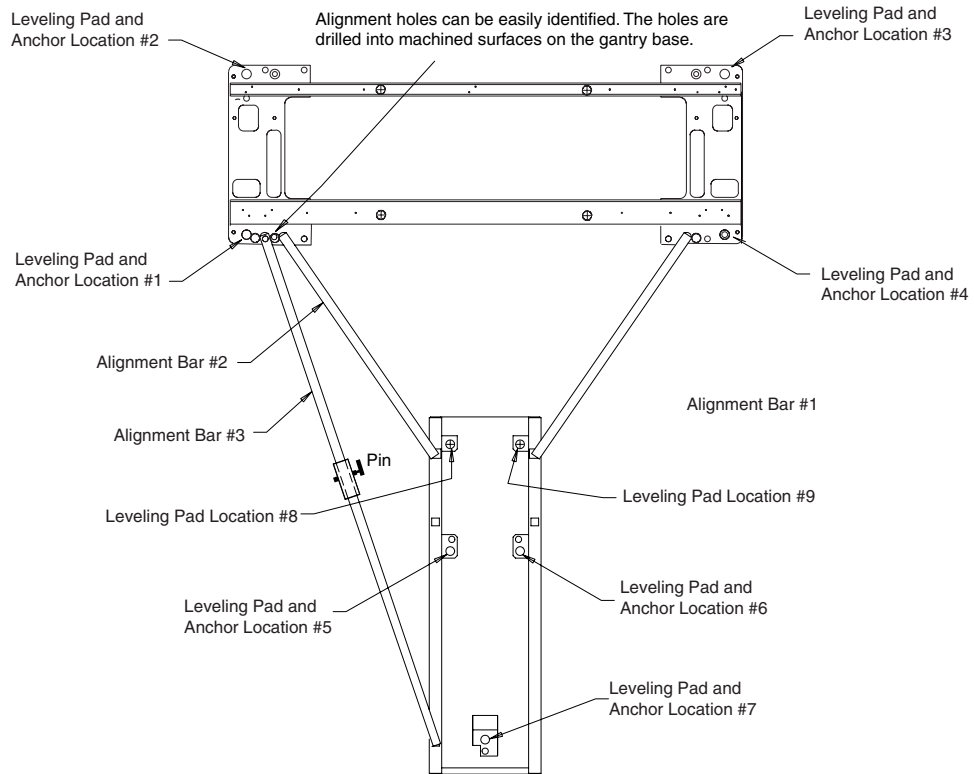


Figure 1-19 Table Base Leveling Pad and Anchor Locations

- 3.) After installing the alignment bars, lower the table to the floor and remove the transportation dollies (see [Figure 1-20](#)).



Figure 1-20 Table transportation dollies

- a.) Temporarily remove alignment bar #1 to remove the front table dolly.
- b.) Replace alignment bar #1 after you remove the front dolly.
- 4.) Remove the five adjuster shipping bolts and loosen the lock rings.
- 5.) Use the 1½ in. socket and 6 in. extension to remove the shipping bolts.

4.8 Level the Table

4.8.1 Before You Begin

- Only adjusters should be used to level the table.
- Use the adjuster tool, $1\frac{1}{8}$ in. socket, and $\frac{1}{2}$ in. drive ratchet to turn each adjuster.
- Place a mark on the adjuster to aid in counting adjuster rotation.

4.8.2 Procedure

- 1.) Loosen and unload adjusters #5 and #6 (see [Figure 1-21](#)). Do not remove the shipping bolts.
- 2.) Using a 4 ft. level, measure the levelness of the table at locations A, B, and C.
 - Place the level on the machined surfaces of the alignment bars.
 - Do NOT rest the level on the radiused surface on alignment bar #3 (see [Figure 1-19](#)).Because the leveling pads rest against the bottom of the base, raise the table from its lowest position to get it leveled.
- 3.) Map out the adjustments to level all three locations.
- 4.) Remove the shipping bolts from adjusters #7, #8, and #9 only.
- 5.) Systematically level locations A, B, and C (see [Figure 1-21](#)) by turning each adjuster. GE suggests a maximum of one (1) turn at a time.



NOTICE

Always use the adjuster tool to raise or lower the table.

- Use adjuster #8 to level Location A.
- Use adjuster #9 to level Location B.
- Use adjuster #7 to level Location C.
- Use adjusters #8 and #9 to level Location D.

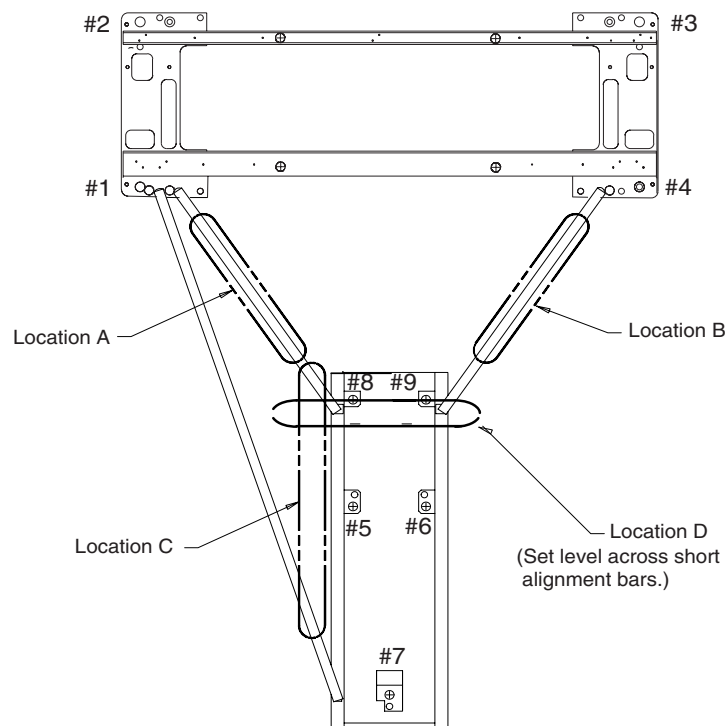


Figure 1-21 Table Base Gantry/Table Level Locations

- 6.) Continue to make adjustments until Locations A, B, and C are centered at the same time.
- 7.) Check location D. Fine tune only.
- 8.) Screw down the adjusters at table Locations #5 and #6, until each rests tightly against its leveling pad. Tighten each adjuster a little more than hand tight, but take care not to unload any of the other table adjusters. It is important that locations #5 through #9 (Figure 1-21) all bear part of the table load.
- 9.) There must be a minimum of 27 mm $\pm$ 3 mm between the bottom of the table base and the floor for the covers to fit correctly. Check the distance between the table base and the floor, to ensure that the covers fit properly (see Figure 1-22 and Figure 1-23).

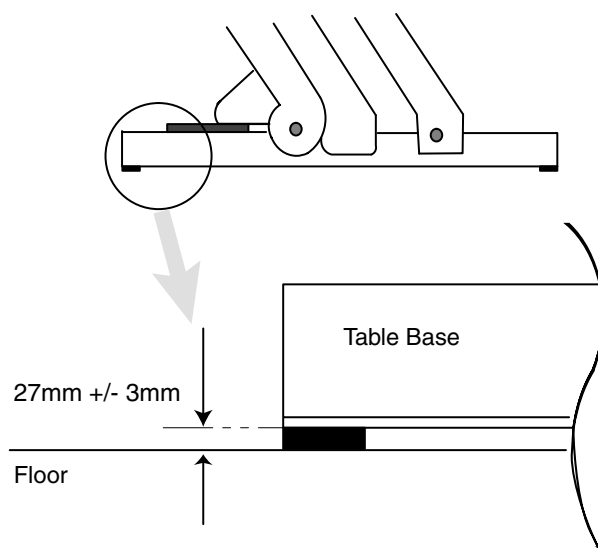


Figure 1-22 Typical Table Base Clearance

Comment: If the gap is excessive, you need to fill it.

If excessive clearance cannot be avoided, fill the gap. For example, use ½ in. round pipe insulation.



Figure 1-23 Excessive Table Base Cover Clearance

4.9 Table-Gantry Mechanical Alignment

4.9.1 Required Tools

- 2 ft. Carpenter Square
- 2 ft. Level
- Masking Tape
- Flat-blade Screwdriver
- 4 ft. Level
- 9 in. Level
- Metric Tape Measure
- 1 ft. Pry Bar

4.9.2 Alignment Conditions

- Complete all sections of this procedure before drilling any anchor holes.
- The alignment tool is used to align and level the table and gantry.
- The alignment tool **MUST** have a current calibration sticker.

4.9.3 Alignment Specification

- Mechanical alignment must be within ± 3 mm
- Table cradle travel must be parallel
- Table must be perpendicular to the gantry

4.9.4 Procedure



NOTICE Use caution while removing the gantry scan window.

- 1.) Remove the gantry scan window. (Refer to [Figure 1-24.](#)) Follow the procedures in [Chapter A, Section 6.0.](#)



Figure 1-24 Removing gantry scan (Mylar) window

- 2.) Do not remove the alignment bars unless told to do so in the procedure

Note: Do not place any object near or into the x-ray opening. This slot must not be nicked or marred (see [Figure 1-25](#)).

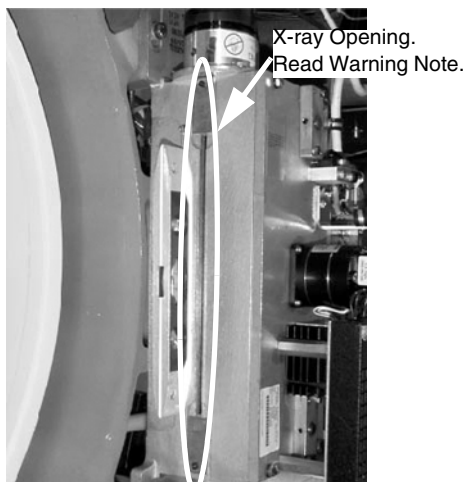


Figure 1-25 X-ray Opening

- 3.) Rotate the gantry by hand until the collimator face plate is at 3 o'clock (see [Figure 1-26](#)). Use a 9-in. level to verify that the plate is vertical. With the power off, the gantry movement is tight.

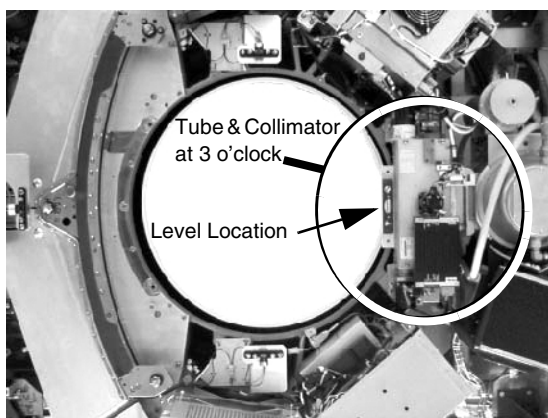


Figure 1-26 Three o'clock position

- 4.) Remove the cradle shipping bolt and the cradle right side cover. The cradle shipping bolt is located under the table/cradle assembly at the foot-end of the table. It has a RED TAG attached to it.
- 5.) Push the plunger on the right rear of the table to unlatch the cradle (see [Figure 1-27](#)).

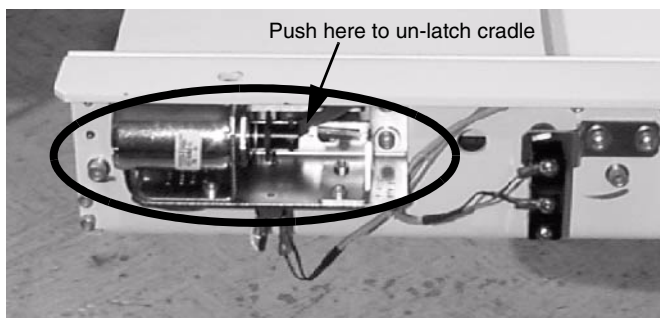


Figure 1-27 Unlatch Cradle

4.9.4.1 Cradle Center Procedure

- 1.) Center the cradle on the drive rollers assembly:
 - a.) Push the cradle into the gantry to its maximum position (1400 mm) and back to just before the latch position.
 - b.) Repeat six times. The cradle should be centered as shown in [Figure 1-28](#).



Figure 1-28 Centered Cradle

- 2.) Adjust the cradle as needed, using the six bolts located on the back of the cradle ([Figure 1-29](#)).

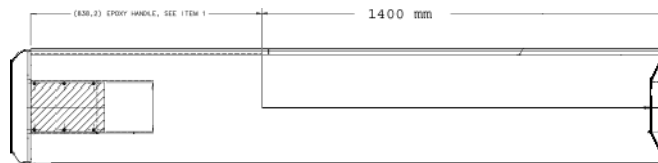


Figure 1-29 Use bolts to adjust cradle

- 3.) Unlatch and move the cradle toward the gantry a few inches until the front of the cradle is near the gantry opening.
- 4.) Use a 2 ft level to check the cradle level from side to side. Level the cradle using the table front level adjusters #8 and #9 (see [Figure 1-30](#)).

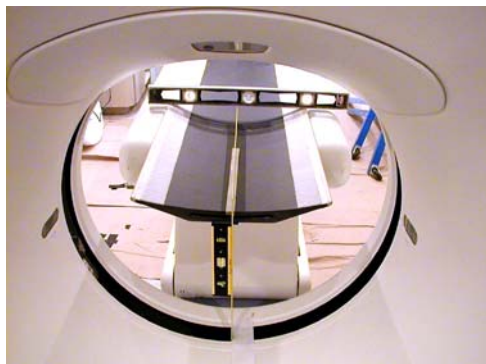


Figure 1-30 Place level across cradle

- 5.) Place the level in the center of the cradle and check that it is level (see [Figure 1-31](#)). The width and the length of the cradle must be level when finished. Re-tighten all adjuster lock rings.

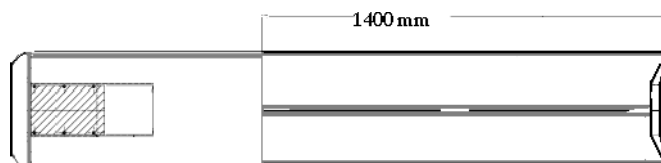


Figure 1-31 Place level along the length of the cradle

- 6.) Find the table center line marked on the front edge of the cradle. If your system has no center line mark, complete the following procedure to make a cradle center line.
- 7.) Using a metric tape measure or other suitable measuring device:
 - a.) Using supplied masking tape, place a 380 mm (15 in.) strip on the front of the cradle (see [Figure 1-32](#)).



Figure 1-32 Masking tape on the front of the cradle

- b.) Determine the cradle center-line by measuring the cradle width 421.1 mm (16.6 in.) (210.5 mm (8.3 in.) center). Use a square or straight edge to determine the center.

Note:

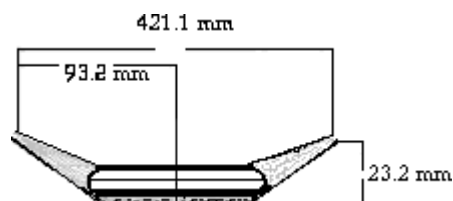


Figure 1-33 Front of Cradle

- c.) Find the center of the inside slot. Width of the inside slot is 186.4 mm (7.3 in.) (93.2 mm (3.7 in.) center). Use a square or straight edge to determine the center (see [Figure 1-34](#)).

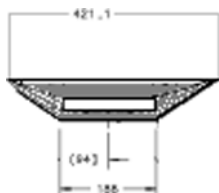


Figure 1-34 Determine the slot center

- d.) Use these two dimensions to determine the table center line by connecting the two marks. Measure back 300 mm (11.8 in.), then measure across the cradle and place a second cradle center mark on the tape. Use a square or straight edge to connect the two marks (see [Figure 1-35](#)).



Figure 1-35 Use a square to connect marks

4.9.4.2 Measuring Cradle Center Procedure

- 1.) Carefully measure from cradle center line to collimator faceplate. It may be helpful to use a square or straight edge to help determine the measured distance (see [Figure 1-36](#)).

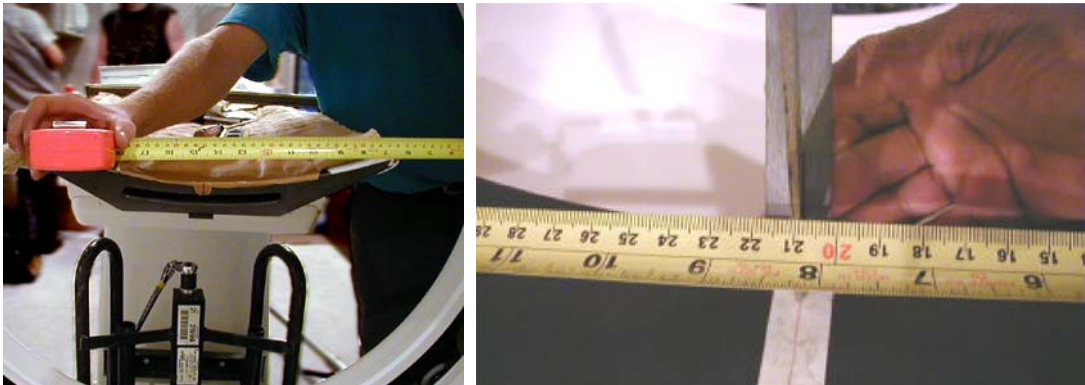


Figure 1-36 Measure Distance

Record this measurement as A on the worksheet ([Table 1-4](#)).

	FRONT		
	A	B	SPEC
	3 o'clock	9 o'clock	A-B ±1 mm
Start Position			
# 2			
# 3			

Table 1-4 Alignment Worksheet

- 2.) Carefully rotate the gantry by hand until the tube and collimator are in the nine o'clock position.
- 3.) Use a bubble level to position the collimator faceplate at the nine o'clock position.

- 4.) Carefully measure from the cradle center mark to the collimator faceplate. It may be helpful to use a square or straight edge to determine the measured distance (see [Figure 1-36](#)).
- 5.) Record this measurement as B on the worksheet.
- 6.) Perform the equation in the SPEC column to determine if the table should be moved. If so, move the table half the distance to correct the error.
- 7.) The table and gantry are in alignment when both measurements meet the specification.
- 8.) If A and B are not equal, move the table and repeat steps 1 through 6 until measurements meet the specification (± 1 mm).

4.10 Cradle/Table Parallel Check

Note: The table pictured in the photos in this section appears for demonstration purposes only. Your actual table may vary. Center line marks

- 1.) With the cradle in the home position, rotate the collimator to the 9 o'clock position. Confirm with a level placed on the collimator face plate.

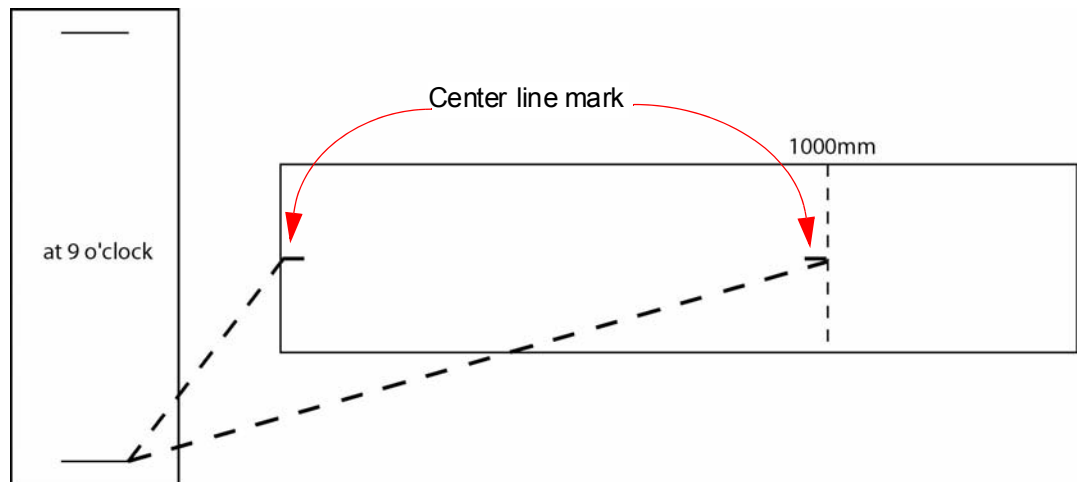


Figure 1-37 First Cradle/Table Parallel Check

- 2.) Measure the distance from the front top corner of the collimator face plate to the mark at the center front of the cradle (see [Figure 1-37](#), [Figure 1-38](#) and [Figure 1-39](#)). Note all measurements in [Table 1-4](#).



Figure 1-38 Collimator Face Plate to Front of Table—Overview

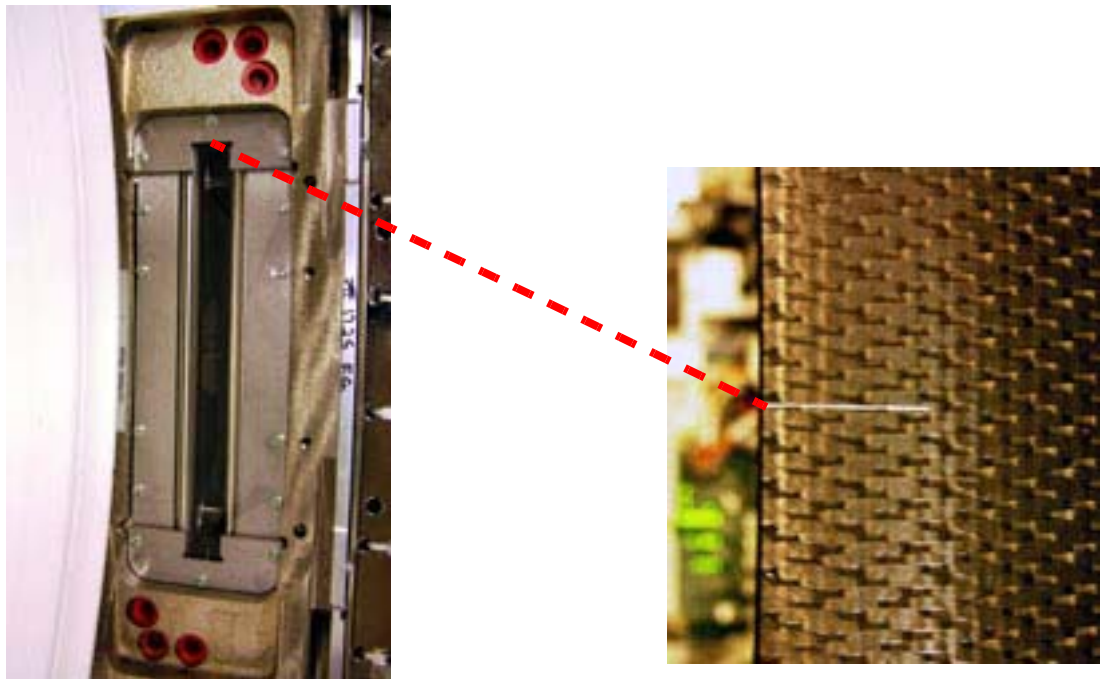


Figure 1-39 Collimator Face Plate to Front of Table—Detail

- 3.) Measure the distance from the same point of the collimator face plate to the center point on the cradle at the 1000 mm mark (see [Figure 1-37](#)).
- 4.) Rotate the collimator to the 3 o'clock position. Confirm with a level.
- 5.) Repeat [Step 2](#) and [Step 3](#) (see [Figure 1-40](#)).

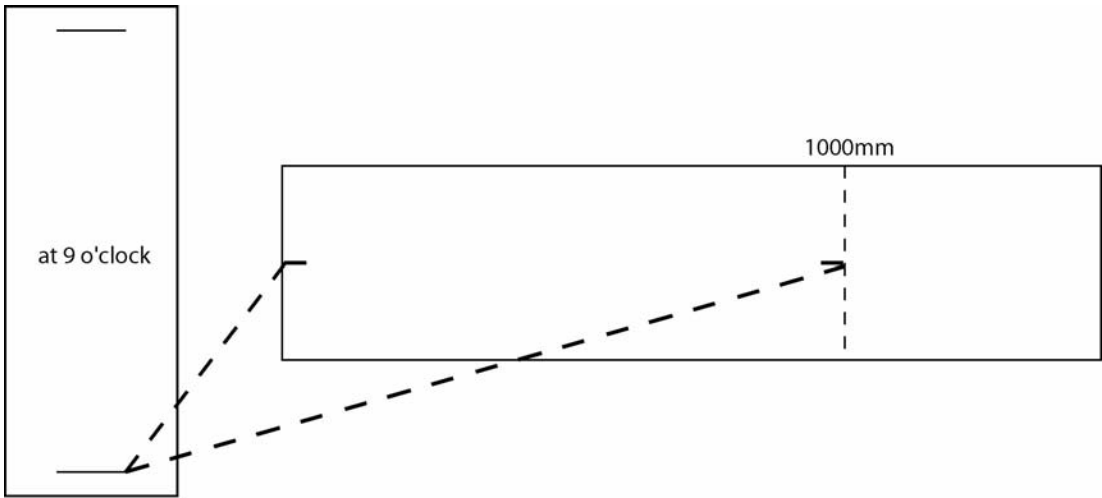


Figure 1-40 Second Cradle/Table Parallel Check

- 6.) If needed, move the table using an appropriate tool and re-measure each side until measurements are equal side to side ± 1 mm.

Note: This final adjustment may be slightly different than the placement obtained using the laser.

Table 1-5 Alignment Worksheet

MEASUREMENT	A: 3 O'CLOCK	B: 9 O'CLOCK	DIFFERENCE: A-B
Test #1			
To front of cradle			
To 1000 mm mark			
Test #2			
To front of cradle			
To 1000 mm mark			
Test #3			
To front of cradle			
To 1000 mm mark			

- 7.) Recheck the cradle level (front to back and across) and re-level as required.

4.10.0.1 Determining Table to Gantry Perpendicularity

Measure from each corner of the gantry base to the corner of the table as shown in [Figure 1-41](#). Each dimension should be equal when finished.

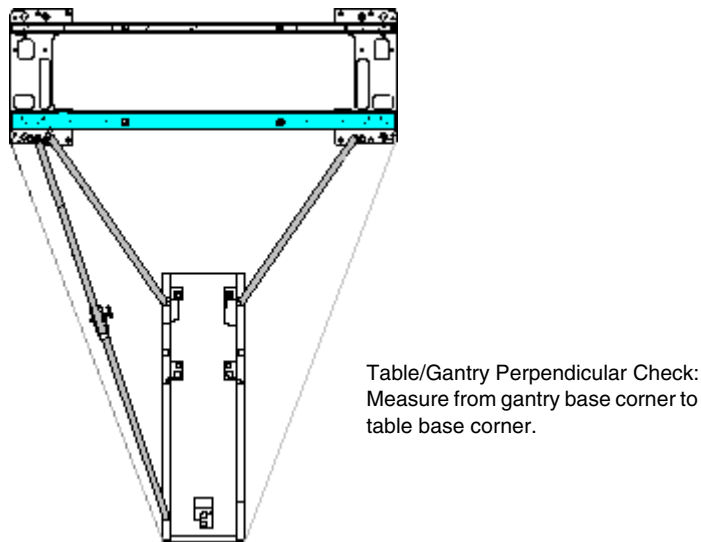


Figure 1-41 Table to gantry perpendicularity

4.11 Tighten the Lock Rings

- 1.) Re-check bubble levels and loading first. Refer to [Figure 1-21, on page 46](#).
 - a.) Re-check that bubbles are level at locations A through D on the table.
 - b.) Re-check that bubbles are level at both ends of the gantry.
 - c.) Re-check that adjusters at Locations #1, #2, #3, #4, #7, #8 and #9 carry a portion of the load.
 - d.) Re-check that the gantry “inside” levelers carry a portion of the load.



CAUTION



Eye protection is required when using a hammer and chisel.

- 2.) Tighten the lock rings at Locations #1 through #9 with the spanner where possible. Use a hammer and chisel to tighten the lock rings only where you can not use the spanner.

4.12 Drill the Anchor Holes



WARNING POTENTIAL FOR PATIENT INJURY.
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

4.12.1 Notes to Mechanical Installers

4.12.1.1 Note 1: Basic Anchoring Information

GE provided floor anchors are designed for use **ONLY** on concrete floors that meet the 102 mm (4 inch) concrete floor requirement. A trained contractor must install the supplied floor anchors. Set them to a minimum depth of 76.2 mm (3 in.) at each anchor point. Any anchors having more than 25.4 mm (1 in.) of thread showing above the nut when the torque is set to 75 N-m (55 ft. lb.), shall have a second anchor installed in the closest adjacent hole. The second anchor is required because the minimum anchor engagement length in the concrete was not met. Install the second anchor to the standard depth and torque specification. **Do not cut anchor bolts that extend longer than the 25.4 mm (1 in.) limit.**

4.12.1.2 Note 2: Alternate Anchoring

If at least four anchors cannot be set for the gantry, and at least three anchors set for the table, the installer must inform GE that the minimum anchoring cannot be met. Additionally, a structural engineering contractor must determine the anchoring method and certify that the anchoring meets the stated GE minimum load requirement and torque specification.

4.12.1.3 Note 3: Non-Concrete Floors

All other anchoring methods—on floor types other than the concrete minimum—must be determined, at customer expense, by a structural engineering contractor. The anchoring method must be certified to meet the stated GE minimum load requirement and torque specification.

4.12.1.4 Note 4: GE Notification

It is not the role of the mechanical contractor or installer to determine acceptable methods to install or anchor equipment on non-102 mm (4 in.) concrete floors. Notify your GE contact person that the facility floor type **DOES NOT MEET** the installation mounting requirement for the installation procedure (described in this Installation Manual), and therefore the table-gantry mounting process **CANNOT** continue.

4.12.2 Requirements

4.12.2.1 Tools Required

- Standard Install Tool Kit
- Hammer Drill
- ½ in. x 12 in. Drill Bit (Metric equivalent must not be used)
- ½ in. Drill Bushing (shipped in install support kit)
- Vacuum with HEPA or drywall dust filter
- Vacuum Hole Attachment – to clean debris from the holes
- PPE

4.12.2.2 Time and Personnel

- 0.5 Hours labor on site
- 2 Engineers

4.12.3 Drilling Procedure

- 1.) Ensure that the gantry scan window is in place, to prevent dust from entering the gantry.

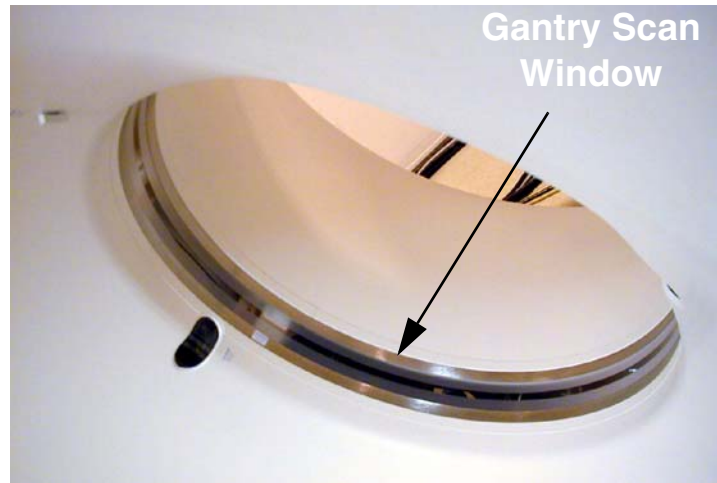


Figure 1-42 Gantry Scan Window

- 2.) Use a piece of tape to mark the drill bit depth of 190 mm ($7\frac{11}{16}$ in.) from the tip of the 13 mm ($\frac{1}{2}$ in.) masonry drill bit.



NOTICE
Potential for
Equipment
Damage from
Dust

You must cover all electronic assemblies in the table base before drilling to prevent damage due to the dust created during drilling.

- 3.) Use the 13 mm ($\frac{1}{2}$ in) bit to drill all seven (7) anchor holes (refer to [Figure 1-45](#) for locations).
 - Review [Figure 1-43](#), before drilling.

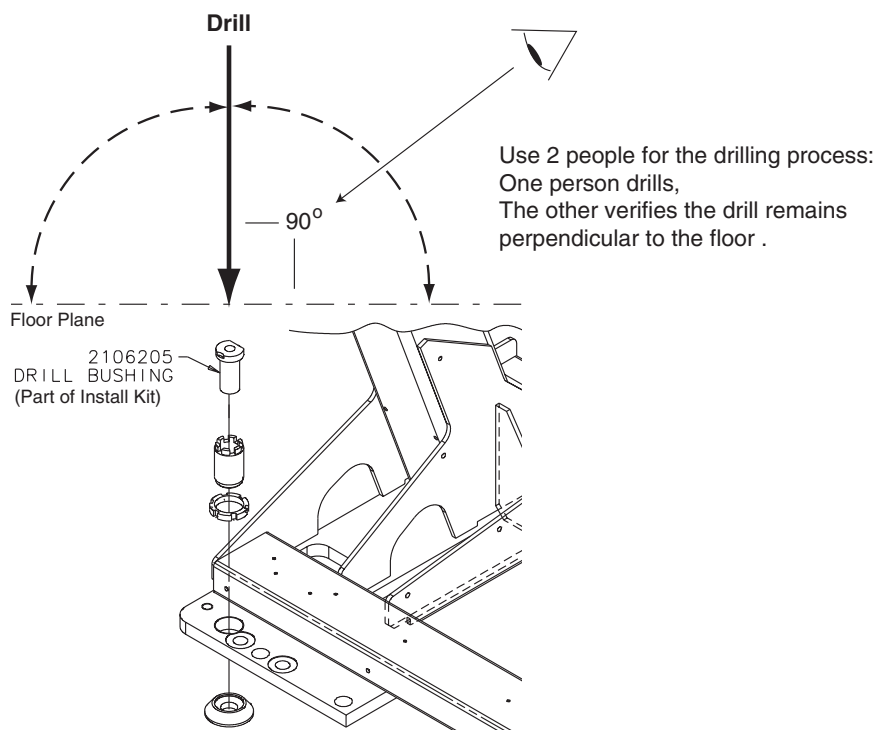


Figure 1-43 Drilling Position

- **Important** - Drill the holes perpendicular to the floor. While one person drills the holes, position a second person to watch the relationship between the drill bit and floor. Ensure that the bit remains perpendicular to the floor throughout the drilling operation.

Always use the mechanical guide supplied with the alignment kit.

- Stop drilling every 15 or 20 seconds and clear the hole of debris. This helps prevent binding of the drill bit.
- a.) Place appropriate protection to prevent damage and dust contamination to electronic assemblies. Refer to [Figure 1-18, on page 44](#), for placement of dust protection materials.
- b.) Drill each hole until the mark on the drill bit is even with the top of the drill bushing. (195 mm/ 7.7 in)
- c.) Place the drill bushing inside each adjuster, to keep the hole vertical, and centered within the adjuster.
 - * Use the drill bushing to center the anchor holes in adjuster Locations #5, #6, and #7. This provides maximum lateral alignment capacity when you center the cradle on isocenter during subsequent system testing.
 - * Take care not to injure yourself on the gantry cover brackets.
 - * Take care to avoid the actuator motor lead when you drill the Location #7 anchor hole. Remove the lead if necessary.

Note:

An 18 in. bit makes drilling anchor location #7 easy, and prevents component damage (see [Figure 1-44](#)).

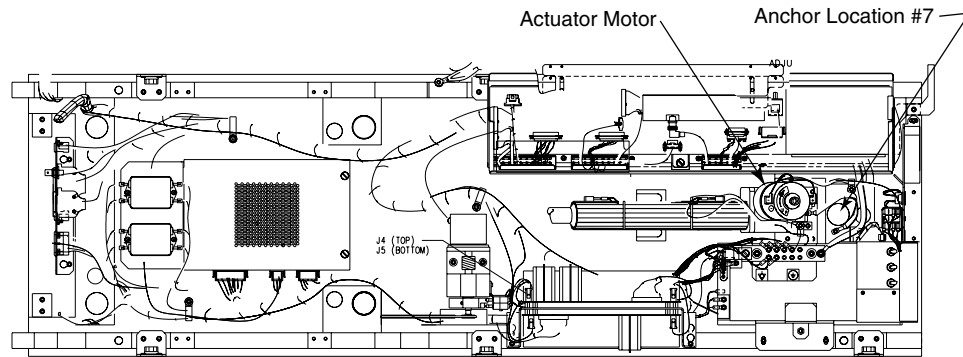


Figure 1-44 Table Assembly Showing Actuator Motor and Anchor Location #7

- d.) Stop the drill every 15 or 20 seconds and clear the hole of debris to prevent binding of the drill bit.

- 4.) Vacuum all debris from the inside of the hole:

Note:

A drywall dust filter should be used on the vacuum.

- a.) Place the funnel tip inside the hole; place the vacuum hose in the funnel.
 - * If you can, continue to vacuum while you drill to keep the gantry and table as free of dust contamination as possible.
 - * Stop drilling frequently to let the drill bit cool.
- b.) When you finish clearing the anchor hole, vacuum the debris from the surrounding area. All holes must be a minimum of 108 mm (4.25 in.) deep (see [Figure 1-48, on page 65](#)). Recheck all holes.

- 5.) Do not drill anchor holes at Locations #8 and #9. These locations help support the weight, but do not anchor the table into place.

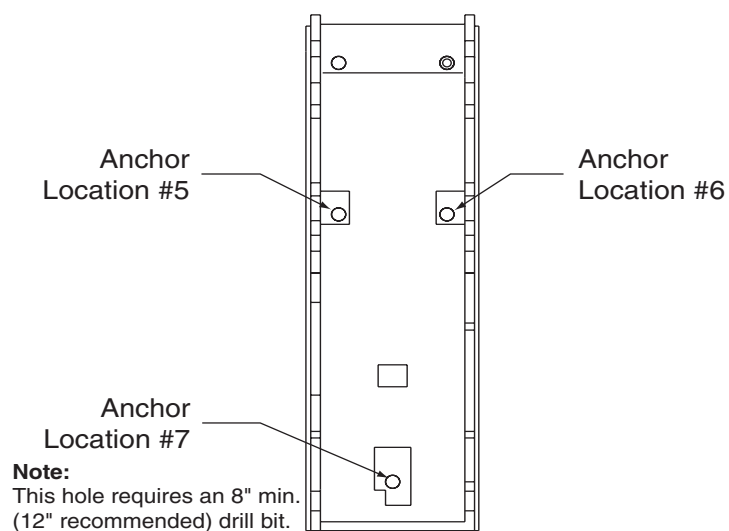
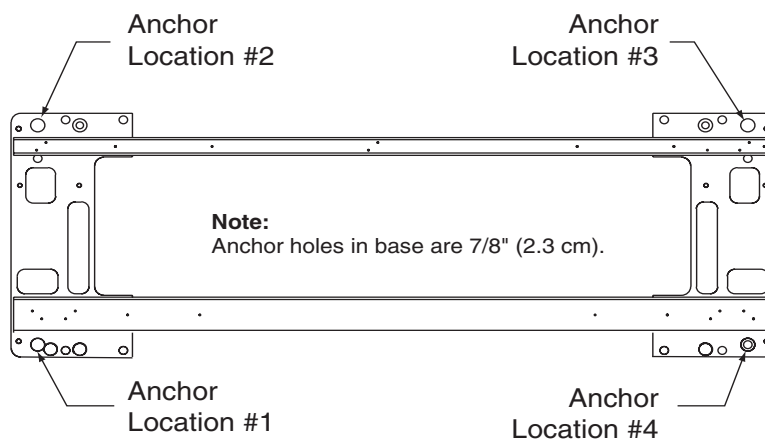


Figure 1-45 Anchor Locations (standard)

4.13 Gantry & Table Redundant Anchor Holes

If you cannot use one of the adjuster anchor holes due to structural interference, you must use one of the redundant anchor locations. Structural interference includes reinforcement bars in the concrete (see [Figure 1-46](#)).

- The gantry requires a minimum of four (4) anchors, one (1) in each corner.
- The table requires a minimum of three (3) anchors, one (1) at location #5, #6 and #7.

If you must use a redundant anchor hole in the gantry, you must remove the gantry covers. See [Section A](#), for gantry cover removal.



WARNING

POTENTIAL FOR PATIENT INJURY.

IMPROPERLY-SECURED TABLE MAY TIP, DISLODGING PATIENT.

PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements and/or through-bolting.

Note: GE is not responsible for anchoring methods other than those listed in the pre-installation manual. Provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 in.) concrete floor requirements.

MOUNTING REQUIREMENTS	GANTRY	TABLE
Minimum Floor Thickness:	101.6 mm (4 in)	101.6 mm (4 in)
Recommended Drilling Depth:	95.25 mm (3-¾ in)	95.25 mm (3-¾ in)
Average Anchor Embedment:	89 mm (3-½ in)	89 mm (3-½ in)
Minimum Anchor Embedment:	76.2 mm (3 in)	76.2 mm (3 in)
Available Alternate Anchor Locations:	Yes	Yes
Shipped Anchor Size:	203.2 mm (8 in)	203.2 mm (8 in)
Alternate Anchoring Methods:	Yes (see notes, above)	Yes (see notes, above)
Floor Levelness Requirement:	8 mm (0.3125 in) over 10 ft	8 mm (0.3125 in) over 10 ft

Table 1-6 Gantry and Table Mounting Requirements

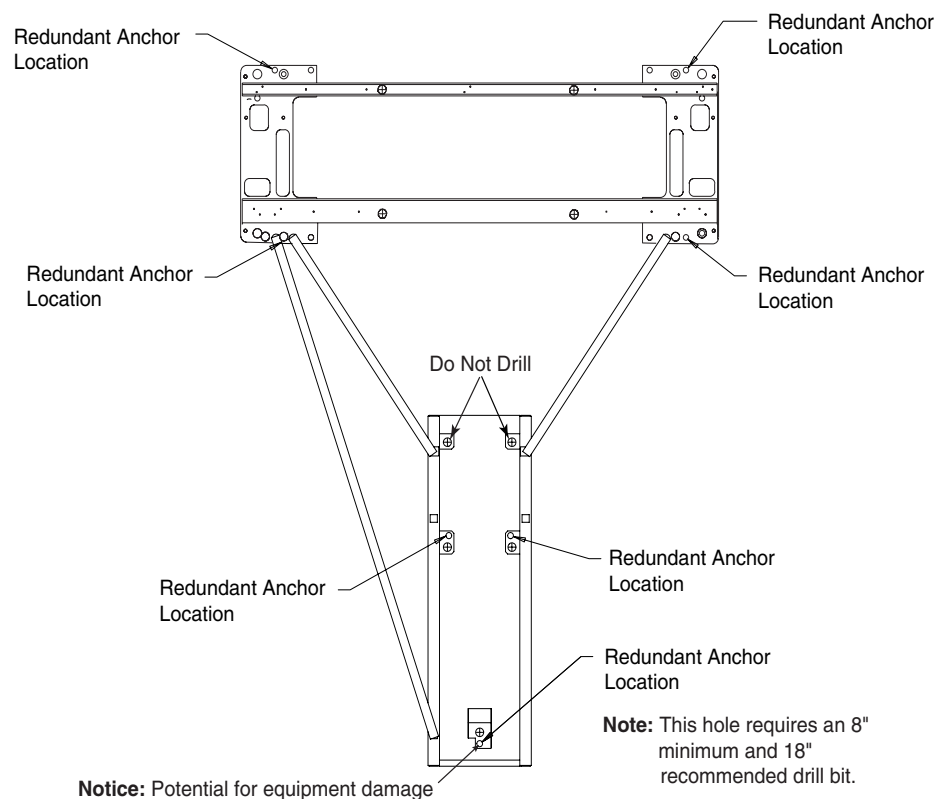


Figure 1-46 Table Base Redundant Anchor Locations

Mark the 13 mm ($\frac{1}{2}$ in.) drill bit with a piece of tape 170 mm ($6\frac{11}{16}$ in.) from the tip of the bit, and follow the instructions in [Section 4.12, on page 57](#).

4.14 Install the Anchors (102 mm (4 in.) Concrete Floors Only)



NOTICE These instructions apply to installing anchors in 102 mm (4 in.) (or greater) concrete floors **ONLY**.

(Refer to the notes in [Section 4.12.1, on page 57.](#))

Recommended - Use "Hilti Kwik-Bolt II" anchors for this procedure. P/N 2106573 (13 mm dia. x 203 mm) (½ in. dia. x 8 in long).

- 1.) Remove the three alignment bars and repack the install kit.
- 2.) Refer to [Figure 1-48](#). Assemble the anchors before you install them:
 - a.) Remove the nut and washer from the anchor.
 - b.) Add a 6.35 mm (¼ in.) thick washer (PN 2105873) under the regular anchor washer
 - c.) Reassemble the anchor washer and nut and position nut so top is flush with threads of anchor before hammering into hole (refer to [Figure 1-48](#)).



Figure 1-47 Measure Table Base Anchor Thread

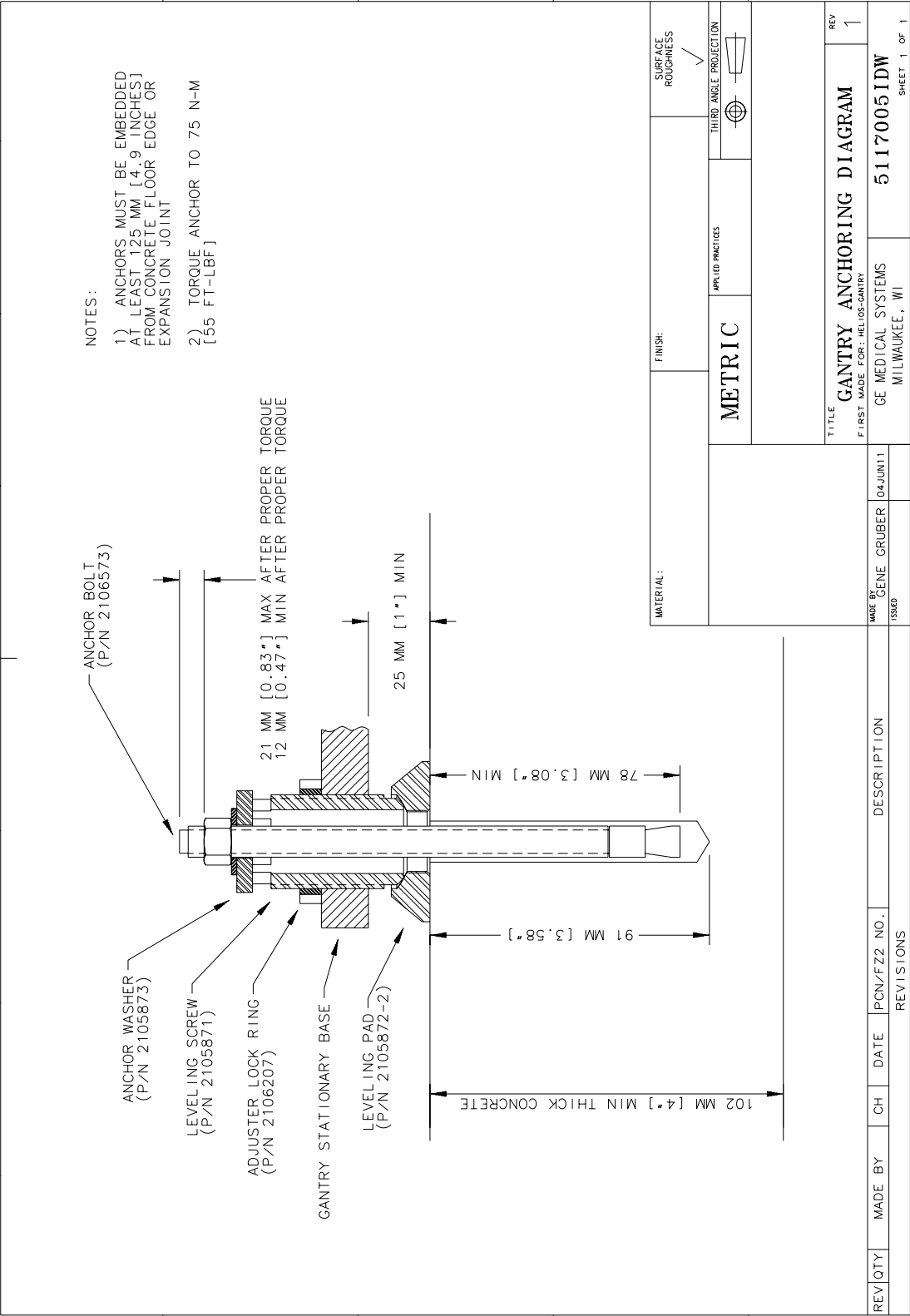


Figure 1-48 Table Base Anchor Assembly

- 3.) Fixed site Installations must be centered under mounting holes in table base, with 3 mm (0.12 in.) of space visible all of the way around the tapped bolt hole. [Figure 1-49](#).

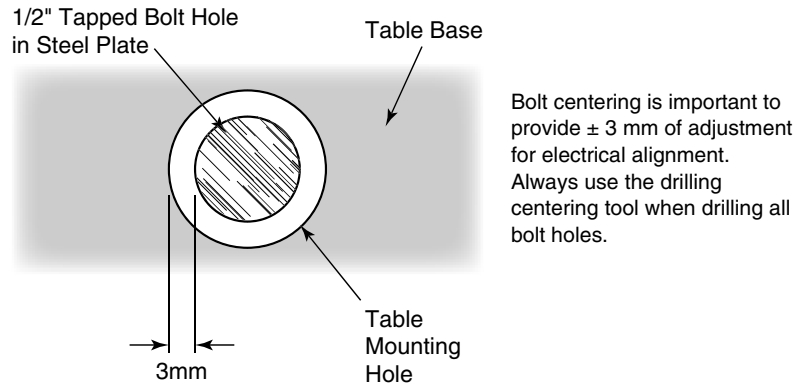


Figure 1-49 Center tapped holes under mounting holes in table base



CAUTION



Eye protection is required when using a hammer.

- 4.) Refer to [Figure 1-48](#) for instruction on inserting the anchors in each hole. Use the anchor seating tool to hammer the anchor downward to *insert the anchor completely in the hole*.
- 5.) Repeat [Section 4.9.4.2](#) and [Section 4.10.0.1](#).
This procedure is designed to mechanically center the table cradle within the gantry. All steps are important to achieve final electrical x-ray alignment of 1 mm (0.04 in). Alignment bars are required to level and position the gantry. Alignment bars must be calibrated yearly.
- 6.) Once alignment has been verified, torque all mounting bolts.
Tighten the Location #1 through #7 anchors and torque to 75 ± 6 N-m (55 ± 5 ft.-lb.).
- 7.) Replace all the table parts you removed during installation, including covers, ground strap and the center support bar. Check the motor lead.

Note: If you cannot replace the lower table cover because the floor interferes, adjust all of the table and gantry levelers. Use $\frac{1}{2}$ turn increments to raise the table/gantry until the lower table covers clear the floor. (Refer to [Figure 1-23, on page 47](#), for table base cover clearances.) To level the table, the gantry and tighten the locking rings, refer to [Section 4.4, on page 38](#), [Section 4.8, on page 46](#) and [Section 4.11, on page 56](#).

4.15 Inside Levelers

There are six (6) additional (auxiliary) levelers/stabilizers on this gantry. These levelers help to spread the gantry load across the base. The two “inside” levelers are located on the rear side of the gantry inside the base near adjusters #2 and #3. Two levelers/stabilizers each are located on both the front and rear cross bars. Ensure that they have remained at their lowest point (i.e., against the base) during shipment and do not interfere with positioning.

Note: **International Shipments**

The levelers are removed for international shipments and must be reinstalled.



Figure 1-50 Detail of Inside Leveler near Gantry base adjuster

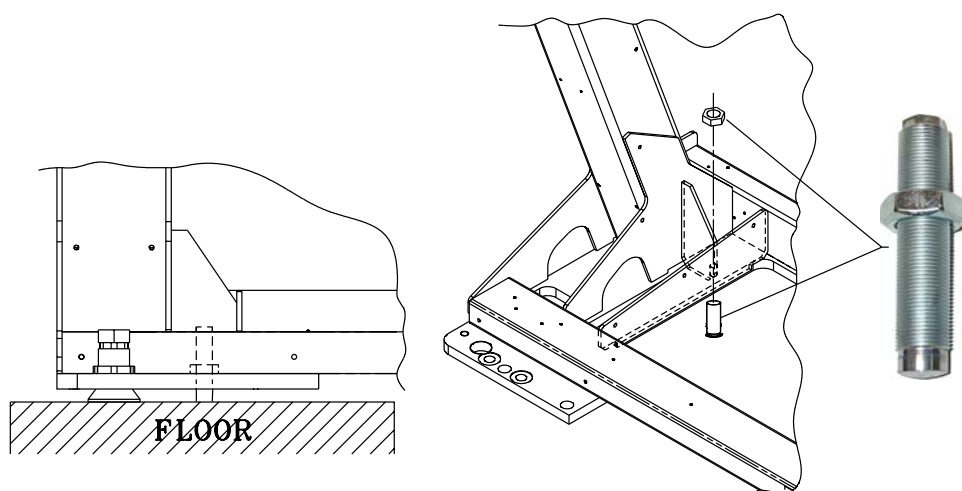


Figure 1-51 Inside Levelers

Adjust the six (6) auxiliary levelers to stiffen the gantry base. Torque the crossbar levelers, in the order shown in [Figure 1-52](#) (1, 2, 3 and 4), to 14 N-m (10 ft.-lbs.) Repeat at 27 N-m (20 ft.-lbs.) Then

torque inside levelers #5 and #6, to 27 N-m (20 ft-lbs.). Use the supplied nut to lockdown the levelers at all six locations.

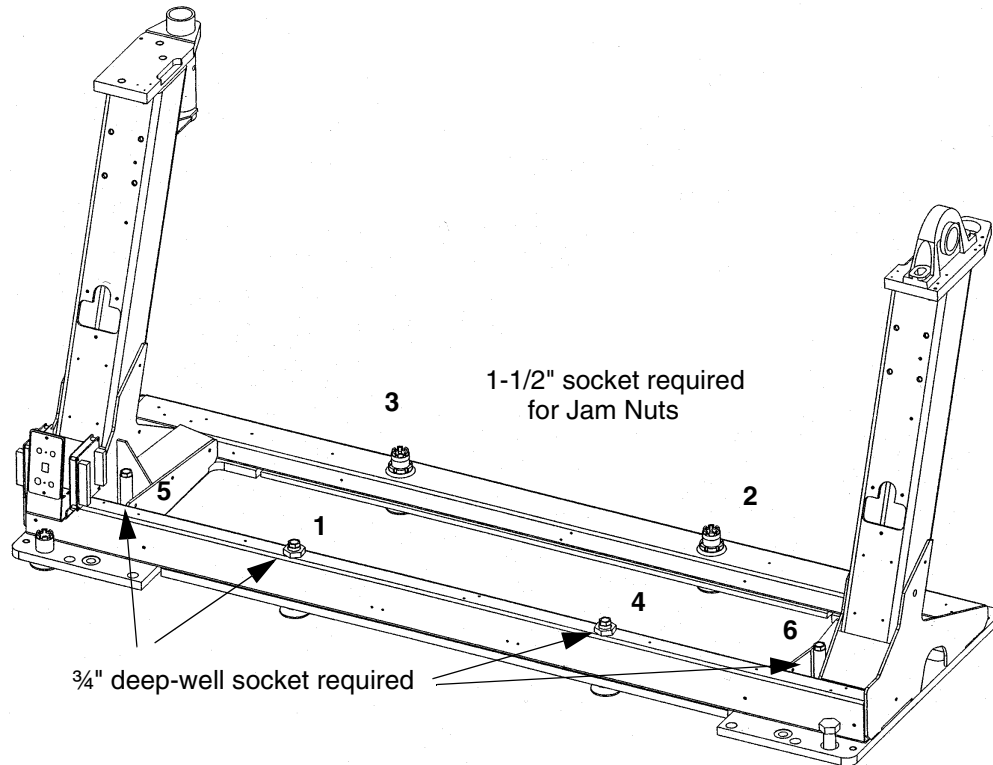


Figure 1-52 Auxiliary Leveler Layout, Rear View



Figure 1-53 Auxiliary Leveler, with Footpad

Section 5.0

Rear Entry Cable Box (Site Specific)

Use a rear entry cable box (B7850RC) when the cables to the gantry cannot be brought up inside the gantry base. The box is not supplied with the system and must be ordered separately.

The rear entry cable box frame attaches to the gantry base (see [Figure 1-54](#)).

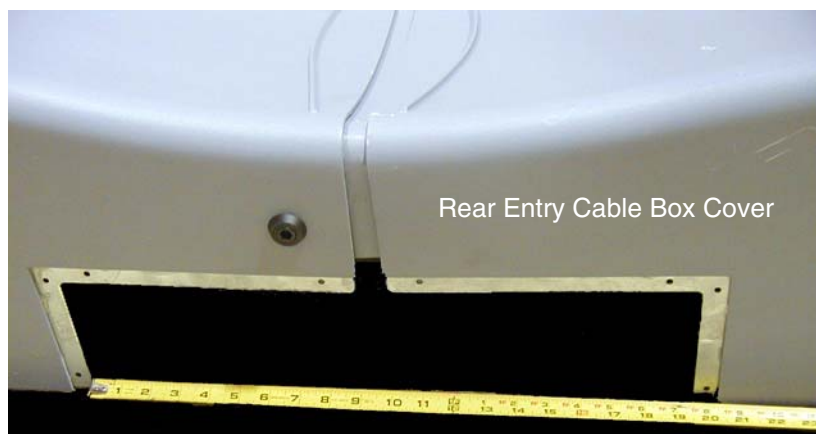
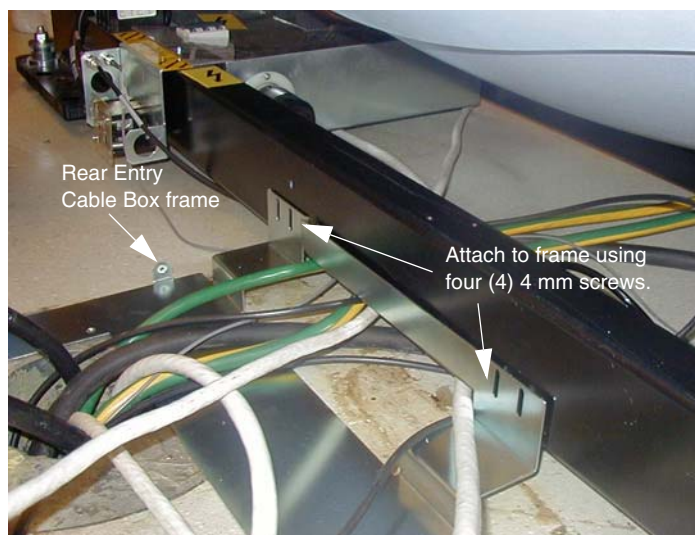


Figure 1-54 Rear Entry Cable Box

If necessary, cut the filler plate to fit the rear cable box into the cover opening. Use the supplied white molding around the filler plate.

Note: The ridge on the floor plate may need to be cut to accommodate the cables and conduit from the system.

Section 6.0

Install Table Footswitch Assembly

Install the table foot-switch assembly and filler strip as shown in [Figure 1-55](#).

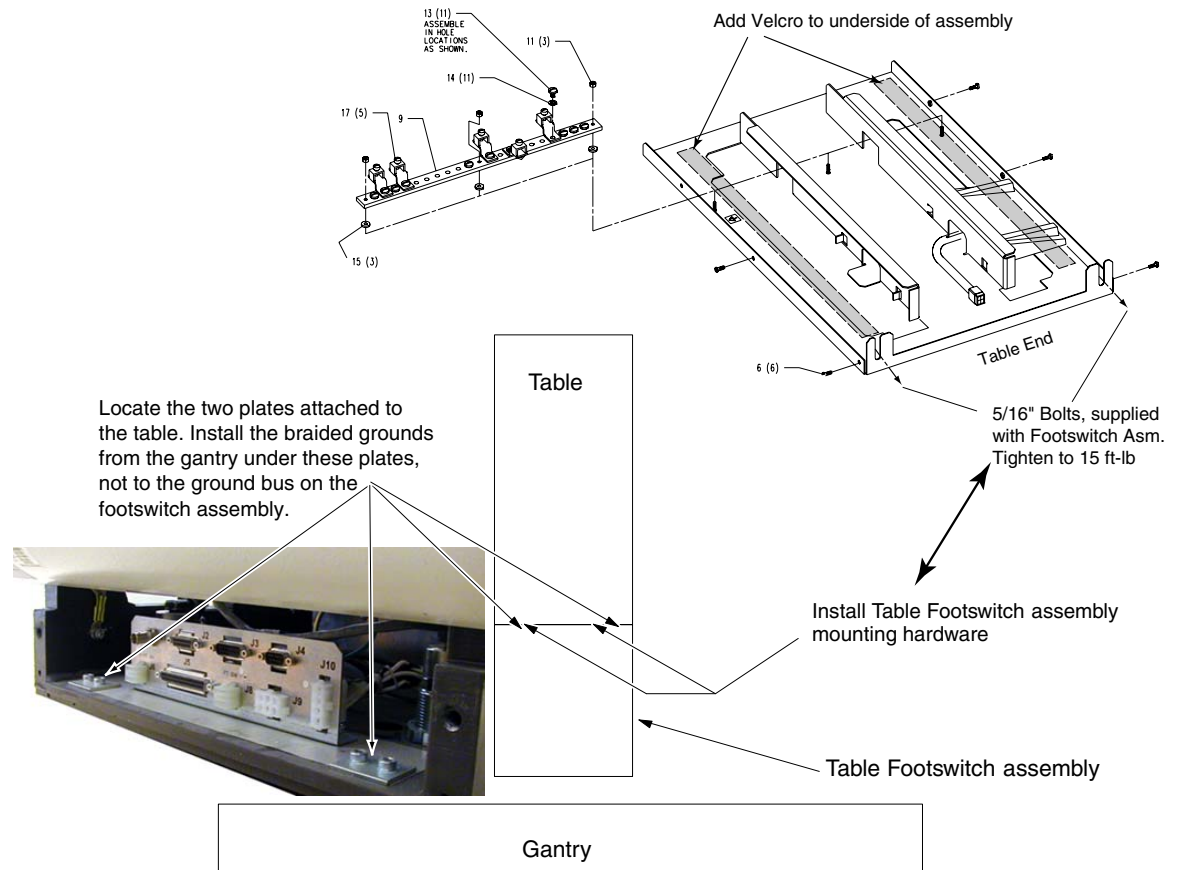


Figure 1-55 Install Table Foot-switch Assembly

- 1.) Remove the foot switch assembly and locate the hardware.
- 2.) Loosen six (6) hex head screws and remove the foot-switch assembly.
- 3.) Place the foot-switch cover in a secure place.
- 4.) Locate the 2 ft. velcro strip supplied in the install support kit, and apply two strips to the bottom of the foot-switch assembly.
- 5.) Using the two 5/16 in. bolts and blue Loctite (242), attach the foot-switch assembly.
- 6.) Remove the 2 in. braided strip and gantry cable that are secured to gantry base, and place them in the cable tray.

Section 7.0

Remove Gantry Tilt Bracket

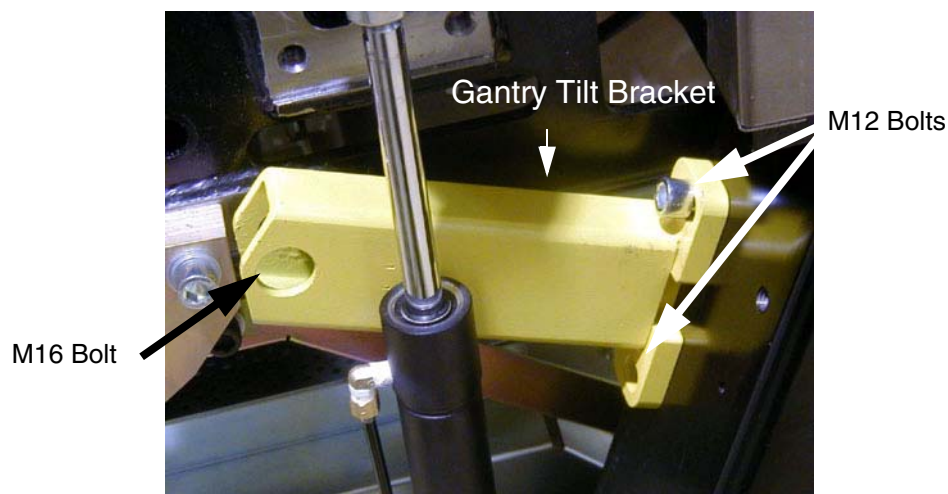


Figure 1-56 Gantry Tilt Bracket Removal

- 1.) Remove the M12 bolts using a 10 mm Hex wrench (see [Figure 1-56](#)).
- 2.) Loosen the M16 bolt 1-2 turns and check the gantry tilt bracket, it should be loose to the touch. If loose continue with step 4.



CAUTION

If the tilt bracket is not loose, stop and put the M12 bolts back in and tighten tilt bracket back in place. If there is a load on the tilt bracket, removal can suddenly cause the gantry to tilt all the way back. A lack of hydraulic pressure causes this movement.

- 3.) Check the hydraulic connections for leaks or lack of fluid. Wait until you can energize the system to use the tilt controls and relieve the load on the tilt bracket before removal. Do not use force to remove the bracket.
- 4.) If the bracket feels loose, remove the M16 bolt using a 14 mm Hex wrench.
- 5.) Remove the bracket.
- 6.) Close the gantry covers and reinstall the scan window.
- 7.) Store brackets in the service cabinet.

Section 8.0

Position the Power Distribution Unit



- WARNING**
- LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK.**
- 1.) Roll the PDU into position on its permanently-mounted casters. Leave at least 15.5 cm (6 in) between the PDU and back wall to allow cooling air to circulate.

Note: The customer’s electrical contractor connects the primary incoming power (steps 2 through 5, below).

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
A1	#1 #1 #1 #1/0	4A 4B 4C GND	A3 TB1 L1 A3 TB1 L2 A3 TB1 L3 A3 TB1 GND (Do NOT connect anything to neutral point.)	

Table 1-7 Contractor Connections

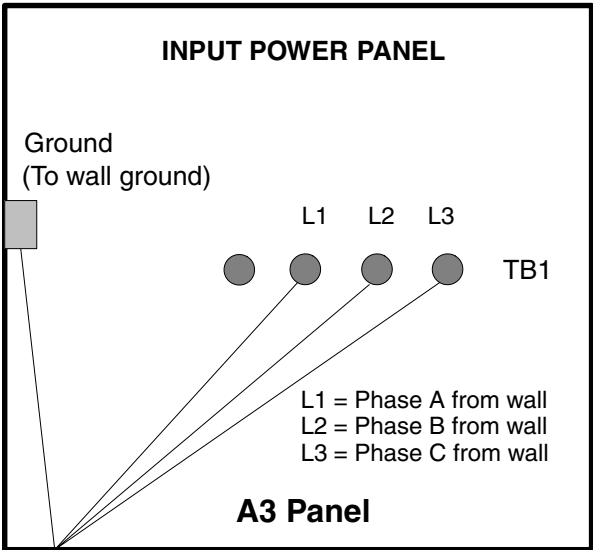


Figure 1-57 A3 Input Power Panel Connections

- 2.) Run the main input power conductors and ground through flexible metal conduit. Attach the conduit between the PDU chassis and room duct-work so you can move the PDU away from the wall during service.



Figure 1-58 Flexible Conduit for PDU Power

- 3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
- 4.) Cut the three phase and 1/0 ground wire to size.
- 5.) Place cable lugs on the wires and attach as shown in [Figure 1-57](#).

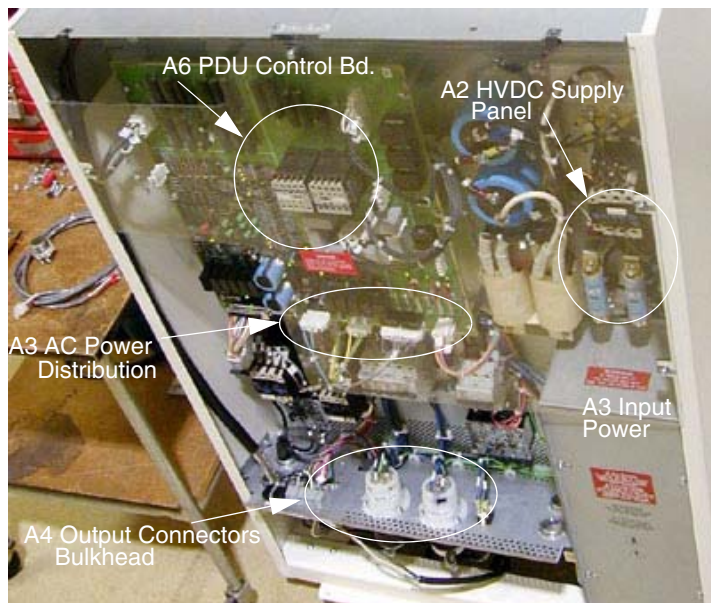


Figure 1-59 PDU Area Locations

Section 9.0

Install Operator Console

9.1 Unpack Console

- 1.) Remove all items from the console.
- 2.) Remove all packing materials and discard them.
- 3.) Move the console to the installation location.
- 4.) Attach the two pads to the console stabilizers before you remove the console shipping dollies (see [Chapter 1-60](#)).

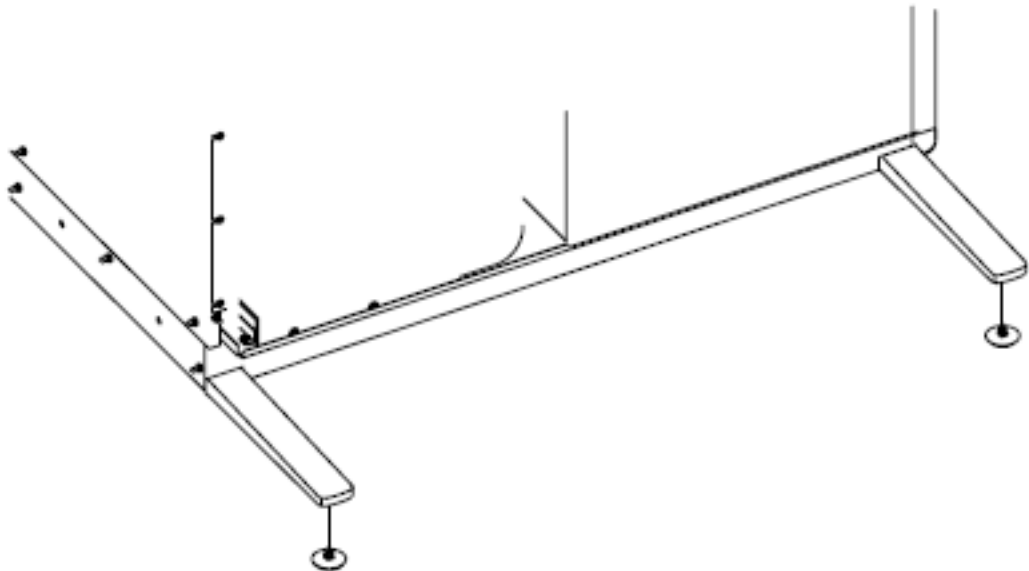


Figure 1-60 Console Stabilizer Pads

- 5.) Screw the pads all the way into the legs.

9.2 Removing Shipping Dollies

- 1.) Use a 1/2 in. electric drill with the 1/2 in. adaptor/drill adapter from the tool list **or** a 1/2 in. ratchet to loosen the bolts. Raise the console to its maximum height by raising all four bolts a little at a time to maintain console balance.
- 2.) When the console is at its maximum height, remove the shipping bolts and skid, and discard them.
- 3.) Lower the console by lowering all four bolts a little at a time to maintain console balance.
- 4.) After moving the console into position, lower the stabilizer pads until the console no longer rocks forward.
- 5.) Remove the console dollies.

9.3 Remove Console Covers

- 1.) Unlatch the two fasteners at the top corners of the front cover.
- 2.) Loosen the two quarter-turn screws located at the console bottom (by the D-rings).
- 3.) Loosen the bottom of the cover by pulling on the D-rings.
- 4.) Unlatch the front cover and slide the top edge forward 1/2 in.
- 5.) Lift the cover up and away from the console.

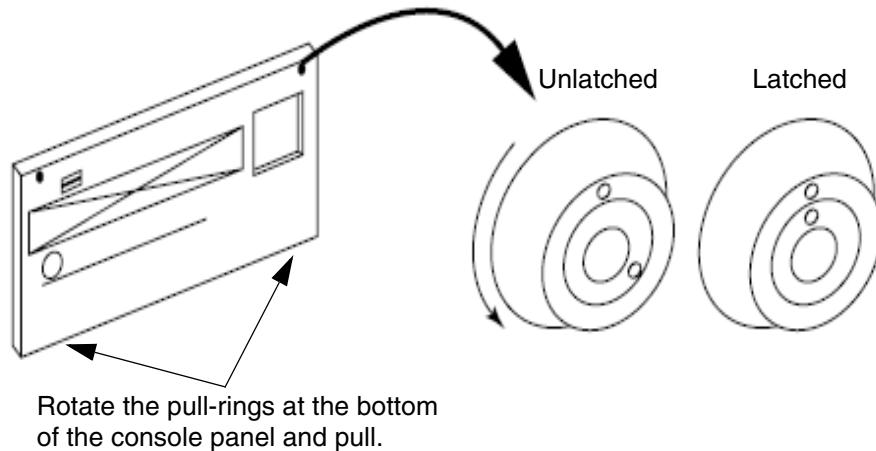


Figure 1-61 Remove Console Front Cover

- 6.) Remove the rear cable entrance cover by loosening four captive screws (see [Figure 1-62](#)).



Figure 1-62 Remove the console rear cable entrance cover

9.4 Adjust Table Top Height and Position Console

The console arrives with the bottom hole on the monitor top aligned with the fourth bolt from the bottom.

- 1.) For optimum operator comfort, align the bottom hole of the keyboard table top with the third hole from the bottom on the console.

Note: Your site may have different requirements and you may have to adjust the monitor top and/or keyboard table top up or down from this position.

Always select a table top height that permits the operator to see the patient on the table.

- 2.) Keep the one bolt-per-hole relationship between the monitor top and the keyboard table top.

Note: Use the extra washer between the arm and the console side to add extra space to the arm. This makes it easier to remove the front cover.

- 3.) Fasten the keyboard table top one hole lower than the console table top, as shown in [Figure 1-63](#).

- 4.) When you move the console into its final position, maintain a 15 cm (6 in.) minimum distance between the console and the wall.

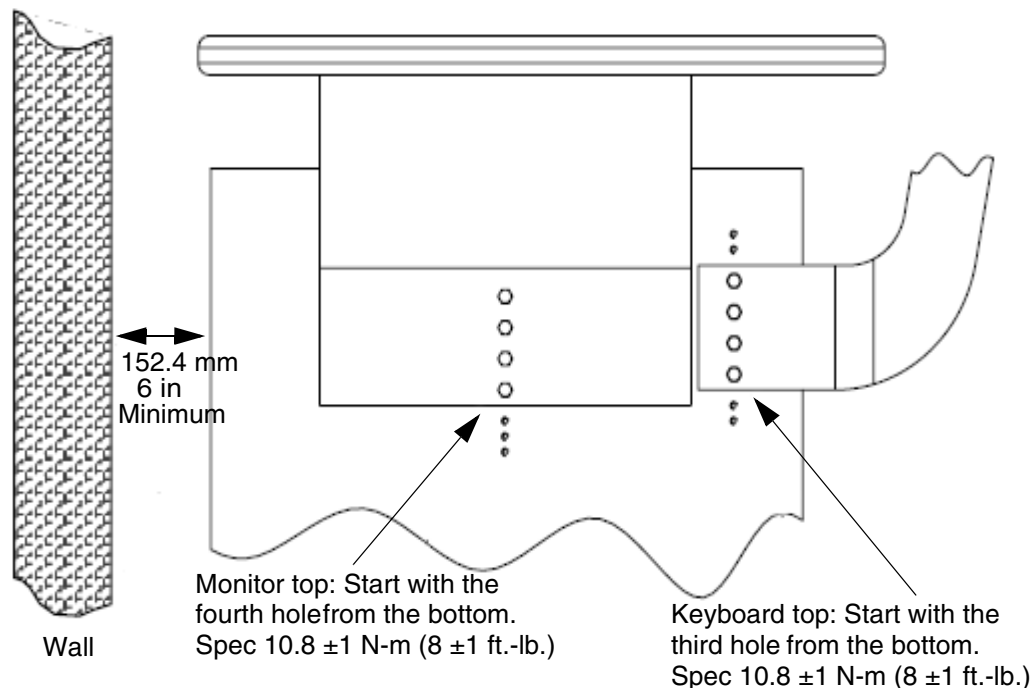


Figure 1-63 Console Position

- 5.) When finished, install the console side covers.

Note: There are left and right side covers.

9.5 Attach Keyboard Table Top

- 1.) Remove the bolts from the table top.
- 2.) Use the spring-loaded handles to attach the keyboard top to the support arms (see [Figure 1-64](#)).

Note: The keyboard top has a trackball/mouse cable raceway attached behind the front support.

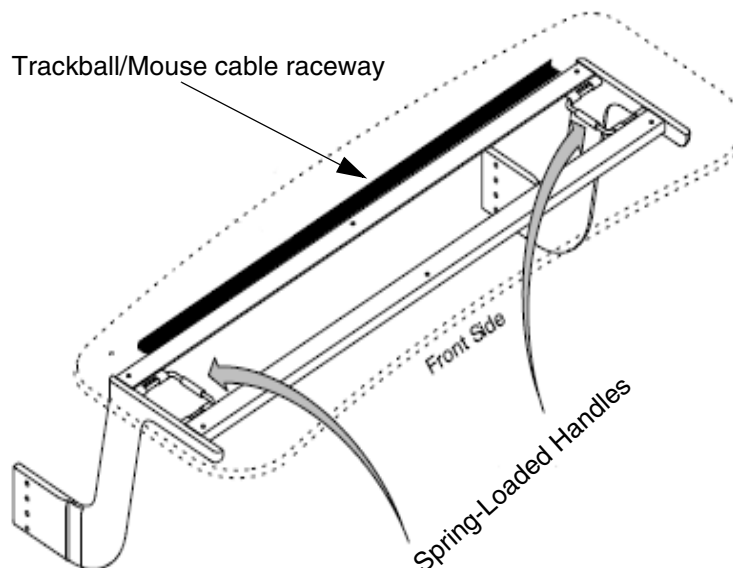


Figure 1-64 Attach the Keyboard Top

9.6 Operator Drive Tray

Remove the plastic cover film that protects the MOD and CD-ROM in the operator drive tray.



Figure 1-65 Operator drive assembly, covered with plastic film

9.7 Monitor Placement

- 1.) Locate and unpack the two monitors. Be careful as each monitor weighs approximately 36 kg (80 lbs.) each.
- 2.) Carefully place the monitors on the console desktop (see [Figure 1-66](#)).

Note: **This step requires two people.**

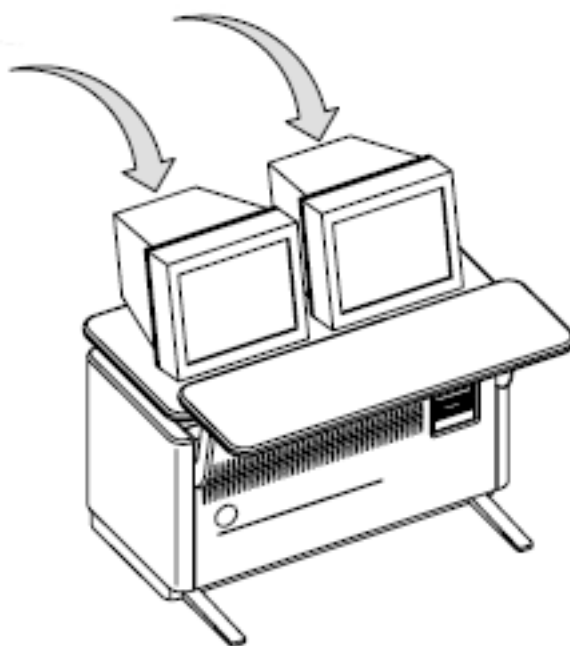


Figure 1-66 Console Monitor Locations

Section 10.0

Seismic Mounting

- Before proceeding with seismic mounting for any of the components in this section, allow sufficient space to unbolt and move the component from its mounted location for service.
- All four (4) mounting bolts may need to be removed.
- if removing the component requires lifting, use an appropriate-sized pry bar to lift each corner of the component
- Two installers may be required to complete the task safely.

10.1 Console

If site specifications require seismic mounting, use ½ in. bolts to mount the brackets to the floor. Refer to [Chapter 1-67](#) for hole placement. The console seismic brackets are shipped with the console. For the Global Console - Octane2, the brackets are shipped on the shipping skid. Remove and reuse these brackets as needed.

The Global Console is 2 in. narrower than the H3 console.

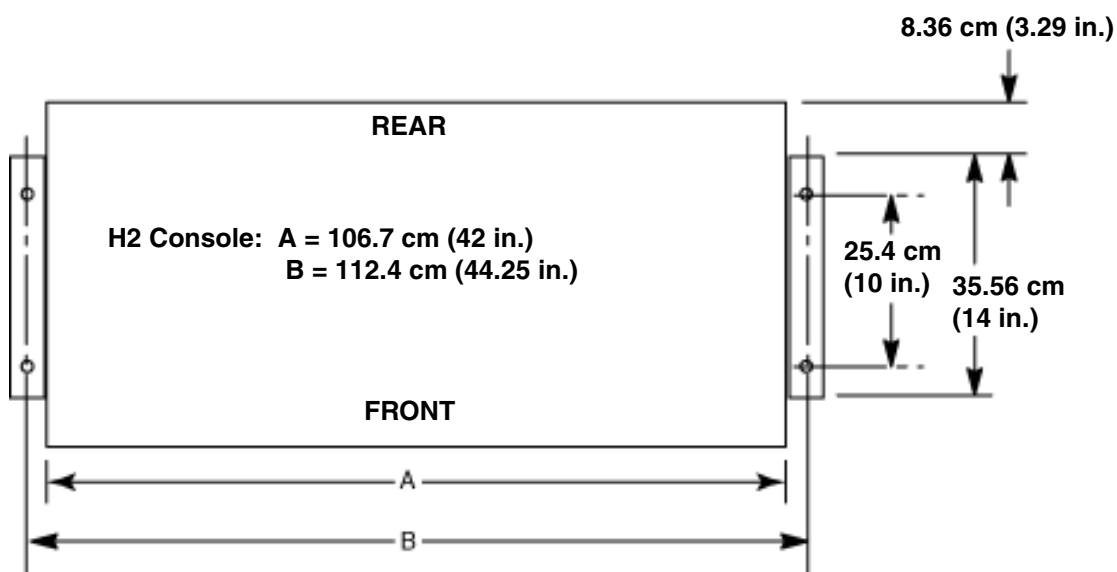


Figure 1-67 Seismic Console Mounting Hole Locations

10.2 Console (GOC1 and GOC2)

If site specifications require seismic mounting, use ½ in. bolts, or those supplied in the seismic mounting kit, to mount the brackets to the floor. Refer to [Figure 1-68](#) for mounting hole locations.

The console seismic brackets are shipped with the console. Console Seismic Kit (P/N 2336557) contains console mounting and SCSI tower mounting brackets.

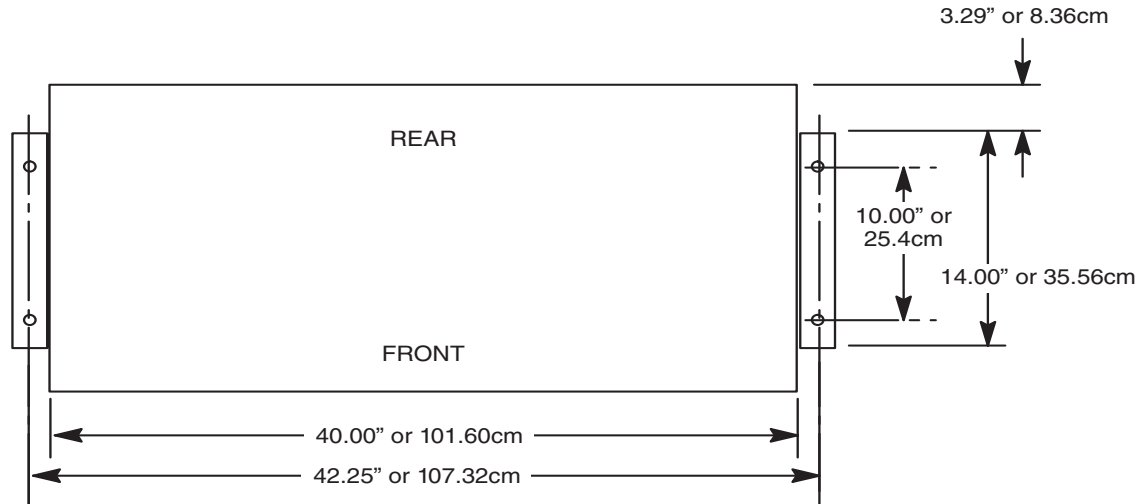


Figure 1-68 Seismic Console Mounting Hole Locations

10.3 Power Distribution Unit

If site specifications require seismic mounting, use the PDU seismic brackets that were shipped with the seismic kit. See [Figure 1-69](#) for hole placement.

Figure 1-69 Seismic PDU Mounting Hole Locations (CPU).

Chapter 2

Power, Ground & Interconnect Cables



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from procedures in this chapter on GE Form 4879 located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

Site use of conduit, floor duct, wall duct, or a raised computer floor, as well as the individual component layout, determines the system cable sequence. If your site has floor or wall ducts that interfere with placement of the table/gantry, have the movers unload the cable boxes (8 and 9) first. Run the cables while others unload the subsystems.

- Try to run the system cables after the contractor completes the contractor-supplied wiring.
- All ground wires and other contractor wiring should be complete to the point of equipment placement.



NOTICE
Potential for
Equipment
Damage

Do not store excess cable in the bottom of the PDU or gantry.

When possible, store excess cable length in a serpentine configuration, approximately one meter long. **(Do NOT coil excess cable.)**

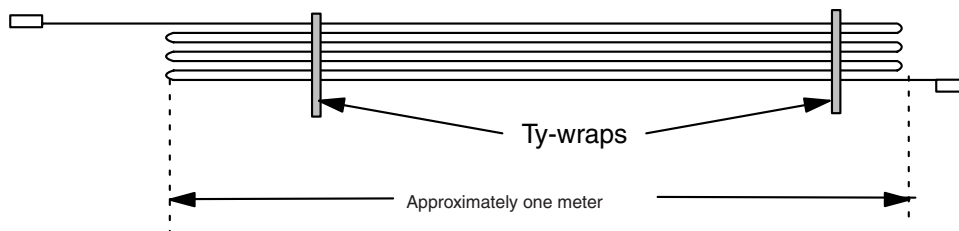


Figure 2-1 Excess Cable Storage Configuration

- Keep the signal and control cables away from power cables and power wiring. When you lay cables in a raceway, locate the signal cables in a separate section of the raceway, or a separate conduit.
- Check all connections for tightness.
 - Use suitable tools and judgment.
 - Check all visible connections, especially ground connections.
- Check for reasonable cable routing.
 - Take into consideration necessary take-up distances for equipment maintenance, etc.
 - Try to complete as neat a job as possible.

1.1 System Component Identification

Identify all system cables by the system component designators listed in [Table 2-1](#). Each end of a system cable has a label. There also may be a color near the connector (see [Table 2-2](#)) to indicate the component and the jack identifier of the component.

DESIGNATOR	SYSTEM COMPONENT
CT2	Gantry
CT1	Patient Table
PM	Power Distribution Unit
OC	Operator Console (Console Computer)
WL	X-ray ON Warning Light

Table 2-1 System Component Identifiers

1.2 Cable Color Identifiers

The ends of the cables may be marked with a piece of blue, yellow, red, or orange colored tape to help with the cable installation. [Table 2-2](#) lists the subcomponent and corresponding color.

SUBCOMPONENT	COLOR
Gantry	Blue
Table	Yellow
PDU	Red
Console Computer	Orange

Table 2-2 Cable Color Identifiers

QTY	RUN DESCRIPTIONS	PART NUMBER	
		LONG CABLE SET (B7850LL)	STD (SHORT) CABLE SET (B7850LS)
1 PC	PDU J5 TO CONSOLE POWER PAN	2335376*	
1 PC	GROUND - PDU A1 TO TABLE RACEWAY 1/0	2275844	2275844-3
1 PC	GANTRYPOWER PAN TO TS1 PDU HVDC	2267642	2267642-2
1 PC	PDU J4 TO GANTRY POWER PAN 120 VAC	2266880	2266880-2
1 PC	GANTRY POWER PAN TO PDU A3 HVAC	2267644	2267644-2
1 PC	GROUND - CONSOLE #2 TO TABLE RACEWAY	2275844-2	2275844-4
1 PC	GANTRY STC J8 TO PDU J2	2266014	2266014-2
1 PC	STC J7 TO CONSOLE SCAN INTL J20	2271060	2271060-2
1 PC	CONSOLE LAN TO GANTRY POWER PAN	2266887	2266887-3
1 PC	FIBER OPTIC CONSOLE J52 TO GANTRY PAN	2117848-2	2117848-6

Table 2-3 LightSpeed Cables, for use with CPDU

Section 3.0

Contractor Connections

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1 #1 #1 #1/0	4A 4B 4C GND	A3 TB1 L1 A3 TB1 L2 A3 TB1 L3 A3 TB1 GND (Do NOT connect anything to neutral point.)	
WL (Warning light) <i>See Figure 2-33.</i>	#12	WL-0A (Black) (from external low voltage (<30v) power source)	A4 WL Connection 1	
	#12	Neutral (White) (from external power source)	A4 WL Connection 2	
	#12	PM-A4 WL Connection 3	0A (Black) (to external warning light relay)	
	#12	PM-A4 WL Connection 4	Neutral (White) (to external warning light relay)	
	#12	Ground (from external power source)	Ground (to warning light ground)	
Door Interlock	—	Room Door Interlock Connections	A4 TS2 - 1, 2	

Table 2-4 Contractor Connections (Compact PDU)

Note: **IMPORTANT:** Add AWG #2 ground wire from the Table ETC frame to the table/gantry raceway ground bar (as shown in [Figure 2-37](#)).



WORK WITH THE ELECTRICAL CONTRACTOR TO ENSURE THAT THE EXTERNAL POWER SOURCE IS TURNED OFF.

Section 4.0

Console Connections

Note: The cables for the SCSIC tower, console, monitors, etc. are stored in the back of the console. Remove the back console cover to access the cables.

4.1 Accessing the Computer

 **NOTICE Potential for Equipment Damage**
To prevent overheating of the computer, remove the velcro strap before turning on the power.

- 1.) Remove the screws at locations A and B, and slide the drawer forward (see [Figure 2-3](#)).

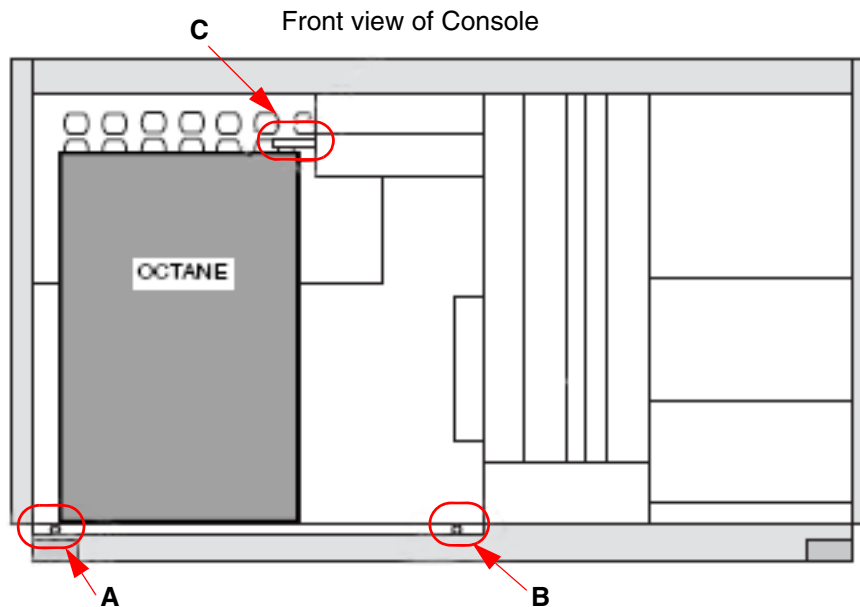


Figure 2-3 Octane Slide Drawer

- 2.) Remove the velcro strap before turning on the power. The strap blocks the cooling air vents. Store the strap for future use (see [Figure 2-4](#)).
- 3.) Remove and store the metal bracket (see [Figure 2-4](#)).

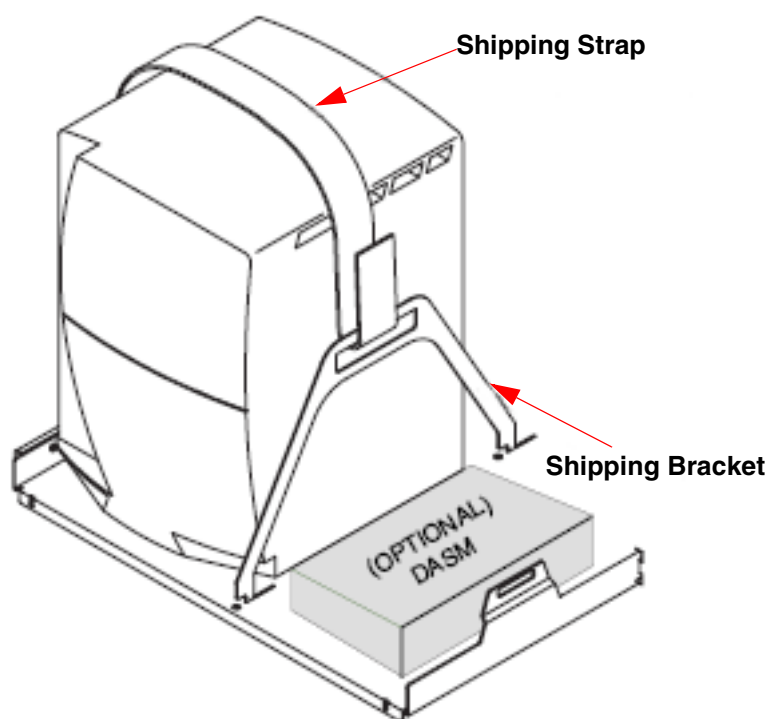


Figure 2-4 Shipping Strap Detail

- 4.) Complete the console cable installation, then return to Step 5.
- 5.) Slide the Octane tray back into the console.
 - Pay attention to the tight clearance points while sliding the tray into the console.
 - Ensure that the Octane computer clears this point without catching on the SCSI cable connector. (Refer to Location C on [Figure 2-3](#)).

4.2 SCIM, Keyboard, Trackball, and Mouse Installation

- 1.) Route the keyboard cable under the SCIM, as shown in [Figure 2-5](#).



Figure 2-5 SCIM control with keyboard cable routed through SCIM

- 2.) Route the mouse and keyboard cables through the console desktop raceway (see [Figure xxx](#)) and through the one-inch hole on the top of the console. Connect the mouse and keyboard cables directly to the back of the Octane computer (see [Figure 2-9](#)).

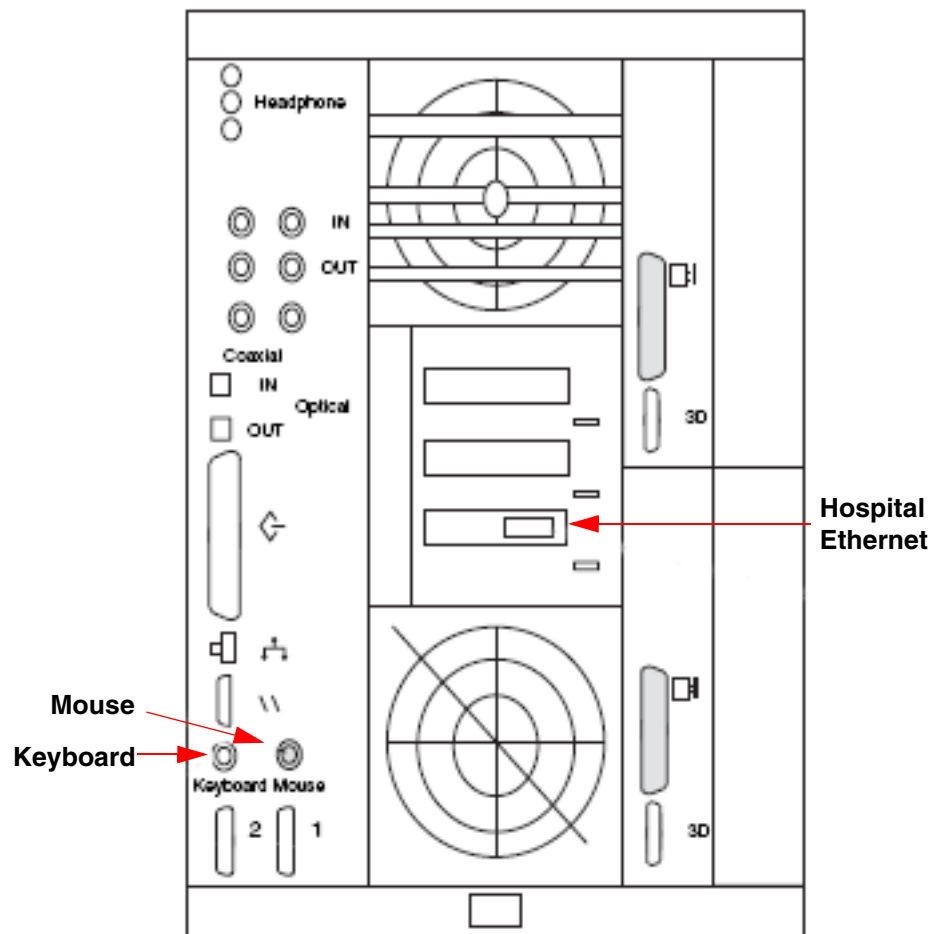


Figure 2-6 Rear view of Octane, showing mouse and keyboard ports

- 4.) Tywrap the keyboard and mouse cables to other cables to prevent them from being pulled out. Leave enough length so that the Octane can be moved in and out without pulling the cables

- 5.) Connect the SCIM cable to the SCIM as shown in [Chapter 2-7](#) (note the cable routing).

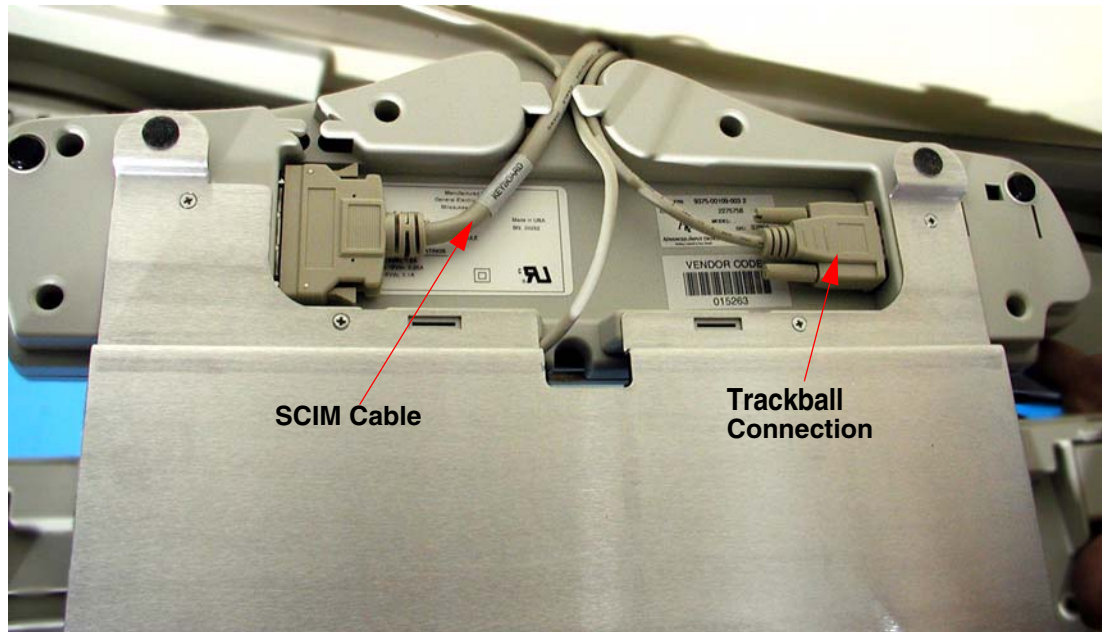


Figure 2-7 SCIM bottom, showing cables and keyboard mounting bracket

- 3.) Route the trackball cable through the console desktop raceway (see [Figure 2-8](#)), and connect it directly to the SCIM.
- The SCIM 50-pin cable (P/N 2262158-9) is run under the monitor table top and connected to the console back bulkhead (J19).

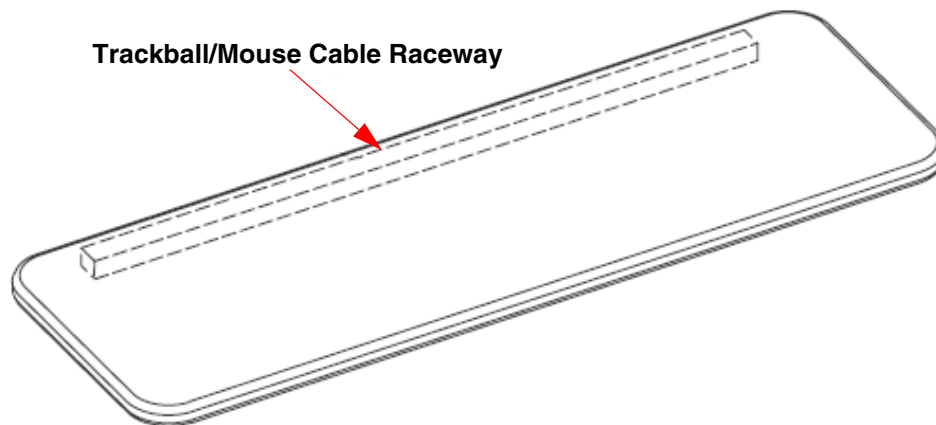


Figure 2-8 Trackball/Mouse Cable Raceway

- 4.) Select and install the proper overlay for your system (1) with tilt, and (2) without tilt.
- 5.) Verify that none of the buttons get caught or stuck under the overlay. Pay close attention to the prescribed tilt button on systems with the tilt feature.
- 6.) The keyboard attaches to the SCIM using the supplied velcro strip and fits snugly against the SCIM when finished (see [Figure 2-9](#)).



Figure 2-9 SCIM connected to the keyboard with the US English tilt overlay installed

- 7.) Before moving the computer tray in, check that NO cables are pinched and that a service loop is formed.

Note: Route the trackball and mouse cables through the cable raceway so that neither touches the floor when installation is finished.

4.3 Connecting the Monitor

- 1.) Turn off the monitor and computer.
- 2.) Connect the video signal cable to the monitor video input (see [Chapter 2-10](#)) and to the Octane video output (see [Chapter 2-10](#)).



NOTICE Equipment Damage Possible

Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, or the pins might bend.

- 3.) Connect the power cord to the monitor.

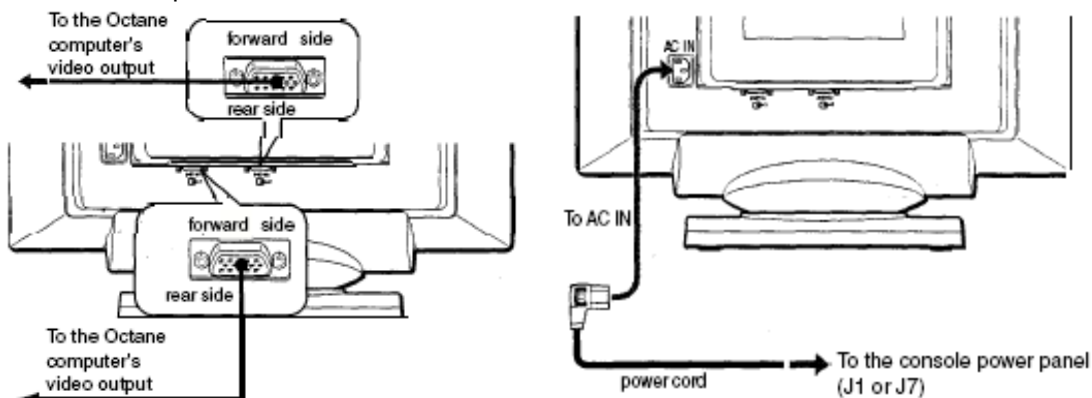


Figure 2-10 Monitor Connections

4.4 .Power Panel Connections

- 1.) Connect the console power cable to the console power panel.
- 2.) Connect console component power cords as listed in [Table 2-5](#). ("J numbers" increment from top to bottom, left to right)

J#	DEVICE	J#	DEVICE
J1	Rx Monitor	J9	Vieb (2)
J2	Octane	J10	DASM (2)
J3	Kieb (2)	J11	Image Printer (2)
J4	VCR (2)	J12	Modem
J5	Printer (2)	J13	Remote Monitor (2)
J6	Video Splitter (2)	J14	Fast Ethernet LAN Switch
J7	Display Monitor	J15	EMC Area
J8	Powered Transceiver	J16	Spare

Table 2-5 Power Panel Outlet Assignments

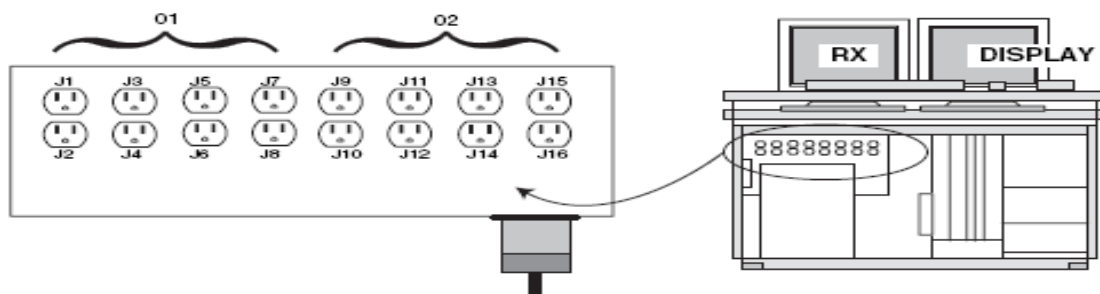


Figure 2-11 Console Power Panel

4.5 LAN Connections

Install the LAN cable onto the transceiver.



Figure 2-12 LAN Connection

4.6 Installing a DASM (Hardware Option)

If you are installing a DASM, perform the following procedure now. The DASM is placed on the Octane tray in the empty space provided.

Although DASMs are available in two types, digital and analog, setup is identical. Complete the following steps to connect a DASM to the computer.

- 1.) Turn the DASM over so you are looking at the bottom.
- 2.) Set the DASM SCSI ID to 1.
- 3.) Set the DASM termination to ON.

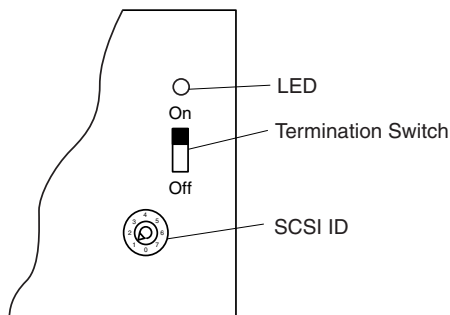


Figure 2-13 DASM Bottom

- 4.) If you have a digital DASM, set the RS232/RS422 switch to the RS232 position.
- 5.) Attach the SCSI cable supplied with the DASM kit to the back of the DASM.
- 6.) Attach the free end of the SCSI cable to the rear of the Host Computer (see [Figure 2-14](#)).

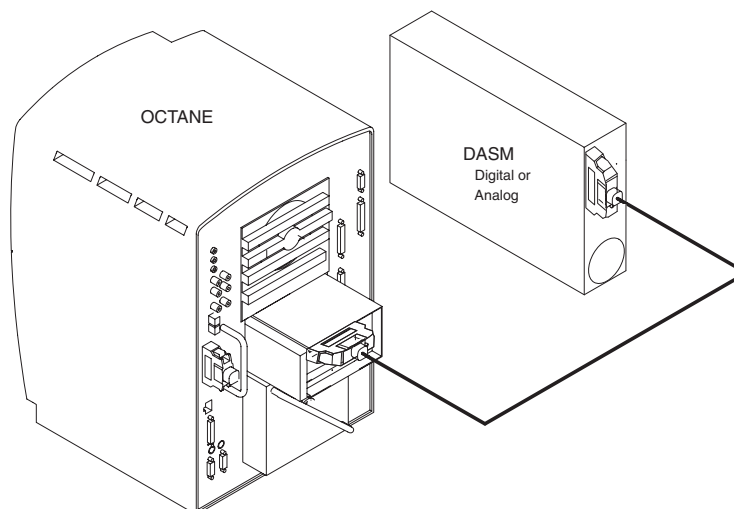


Figure 2-14 SCSI Cable Connection for DASM

- 7.) Plug the DASM power cord into J10.
- 8.) Attach the remaining cable(s) from the camera to the DASM and install the DASM to the right of the slide-out tray.

Note: See your *Software Installation Procedures* manual for instructions on configuring cameras.

4.7 Bulkhead Connections

J#	CABLE DESCRIPTION
J52	Fiber Optic Cable (2117848-2)
J20	Scan INTL Cable (2271060)
J19	SCIM (2262158-9)

Table 2-6 Cables Connections to Back Bulkhead

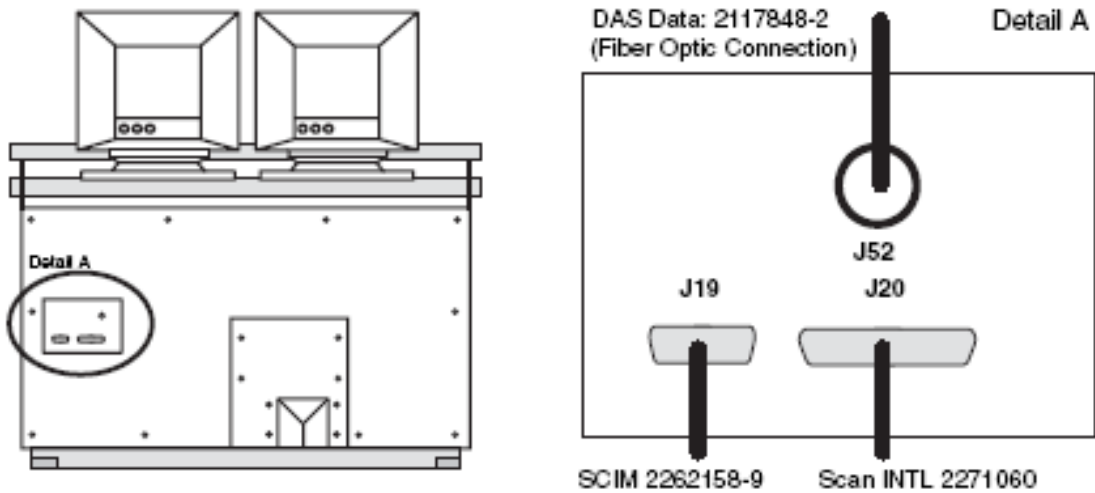


Figure 2-15 Back Bulkhead Detail

4.8 Installing the GOB Board

Connect the CAT5 LAN cable to the gantry base cardiac monitor base plate (GOB Board). Different plates are included with the cardiac option kit.

Note: The GOB connectors are not soldered to the board. DO NOT use the cable to pull on the board.

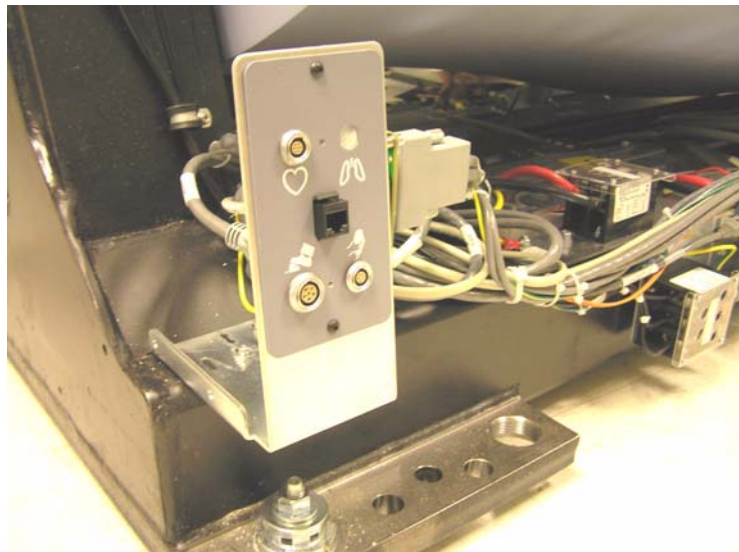


Figure 2-16 Cardiac Option Board

Section 5.0

Install Options

5.1 Install Optional Bar Code Reader

Follow the installation procedures in the Bar Code Reader box.
When finished, neatly dress all the cables.

5.2 Install Optional Remote Monitor

Follow the installation procedures in the Remote Monitor box.

5.3 Install Cardiac Respiratory Gating Option

Follow the instructions shipped with the option. The options board should be only mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

5.4 Install Respiratory Gating Option

Follow the instructions shipped with the option. The option is attached to the GOB, and should be only mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

5.5 Install Injector Option

Follow the instructions shipped with the option.

5.6 Install The UVY Monitor and Stand

Follow the instructions shipped with the Monitor and stand kit. Review the instructions carefully before assembling the stand and accessory basket to avoid repeated steps.

5.7 Customer Accessories (Head Holders and Extender)

Open the boxes and install the appropriate language warning labels.

The head holders ship with shims installed to assure proper fit. Check that shims are included and a pair is installed. The holder should fit snugly. Follow the procedure in [Chapter 3](#), [Chapter 3.0](#) to install Head Holders.

5.8 Install UPS

Follow the instructions shipped with the option.

5.9 Install Service Cabinet

The service cabinet you receive may ship disassembled. Assembly takes about 1.5 hours.

- 1.) Assemble the cabinet following the instructions located in the cabinet shipping box.
- 2.) When you complete assembly, place the cabinet in the location shown on the site print.
- 3.) Place all service materials shipped with the system in the service cabinet.

5.10 Install UPS

Follow the instructions shipped with the UPS option. The option ships with two sets of instructions, a GE set and a Powerware set.

The GE set instructs you to install the connections between the UPS and the PDU and between the UPS and the A1.

Note: A GE A1 Disconnect with UPS controls is required for this option.

The Powerware set instructs you to connect the batteries and do a power-up check. Refer to both manuals for additional guidance.

 **WARNING** LOCKOUT/TAGOUT IS REQUIRED WHEN WORKING IN THE A1 DISCONNECT

Section 6.0

Gantry Cable Connections

Please refer to for complete system interconnect details.

TO	FROM	CABLE DESCRIPTION
Gantry Power Pan	PDU	HVDC
Gantry Power Pan	PDU	440 VAC
Gantry Power Pan	PDU	120 VAC
Gantry Power Pan	Console	Fiber - <i>Take extreme care when you install the fiber optic DAS data cable. Do not step on, kink, or sharply bend this fragile DAS cable.</i>
Gantry Power Pan	Console	LAN
STC Backplane (J7)	Console	Control
STC Backplane (J8)	PDU	Control

Table 2-7 Gantry Cable Connections

- 1.) If using a rear cable entry box (B7850RC), install it now, before routing cables to the gantry.
- 2.) Install the cables to the gantry power pan. The power pan is located on the rear of the gantry at its base. (See "Detail B" [Figure 2-17](#) of the system interconnect drawing for connections.)

Note:

The gantry 120 VAC cable may not fit under the gantry frame. Install this cable before gantry placement—or remove the power plug—to route it under the gantry.



NOTICE
Potential for
equipment
damage.

Observe the correct polarity when connecting the high voltage DC power. Reversing these leads results in serious equipment damage. The HVDC positive conductors have red insulation and are labeled "ONE." The HVDC negative conductors have black insulation and are labeled "TWO." Lead "ONE" must be connected to lead "ONE," and lead "TWO" must be connected to lead "TWO."

Observe the correct phase rotation when connecting the axial motor power. Phases one, two and three should be connected left to right, as shown in [Figure 2-17](#).

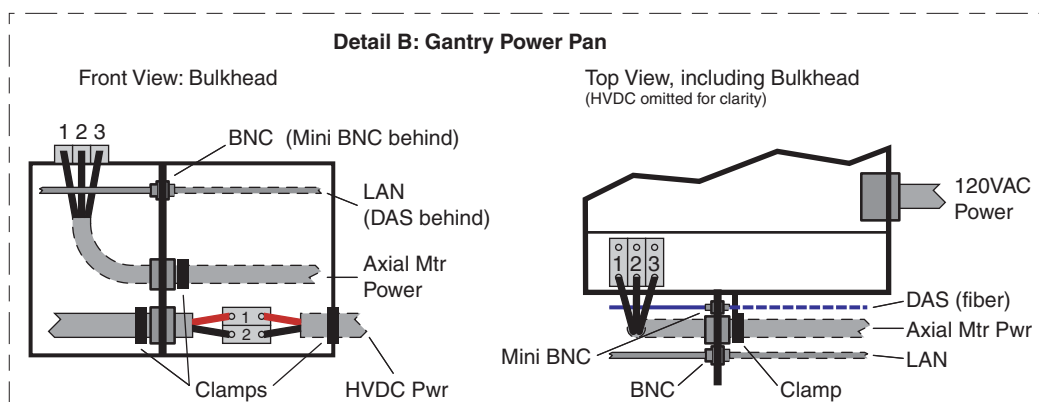


Figure 2-17 Power Pan Connections

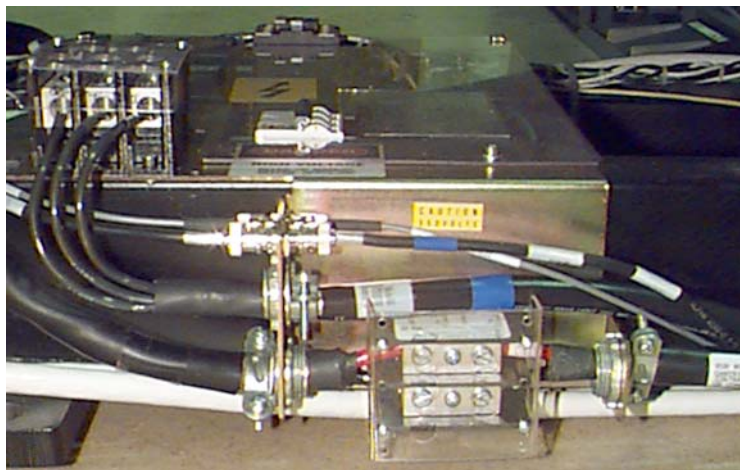


Figure 2-18 Gantry Power Pan

- 3.) Install the cables to the STC backplane.

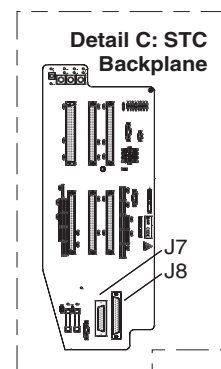
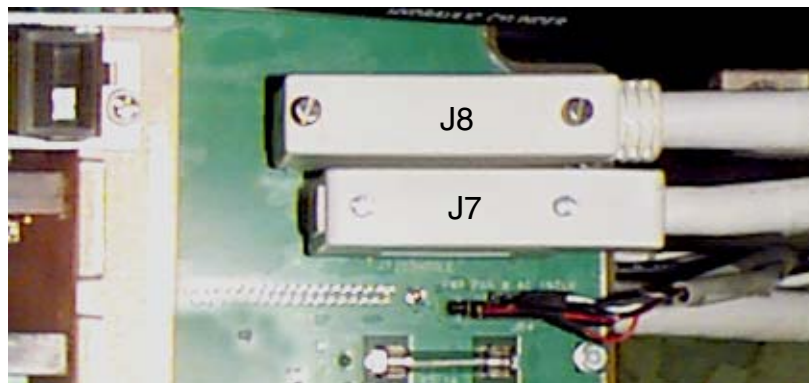


Figure 2-19 A1 STC Connections

6.1 Installing the GOB Board

Connect the CAT5 LAN cable to the gantry base cardiac monitor base plate (GOB board). Different plates are included with the cardiac option kit.

Note: The GOB connectors are not soldered to the board. DO NOT use the cable to pull on the board.



Figure 2-20 Cardiac Option Board

Section 7.0

Option Cables

Run all option cables that connect to the table, gantry, or console.

Section 8.0

Table Connections

Pull and connect the following cables:

J#	CABLE DESCRIPTION
J1	LAN (2265881-2)
J2	Tilt Limit Switch (2268222)
J3	CAN BUS (2263663)
J4, J9, J10	120 VAC bundle (2265683)
J5	Table IF (2271059)
J6, J8	Raceway Foot switches
ETC frame	AWG #2 ground to table/gantry raceway

Table 2-8 Cables Connected to Table

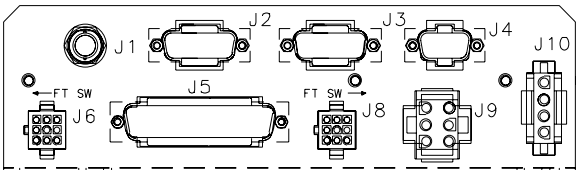


Figure 2-21 Table Bulkhead Connections

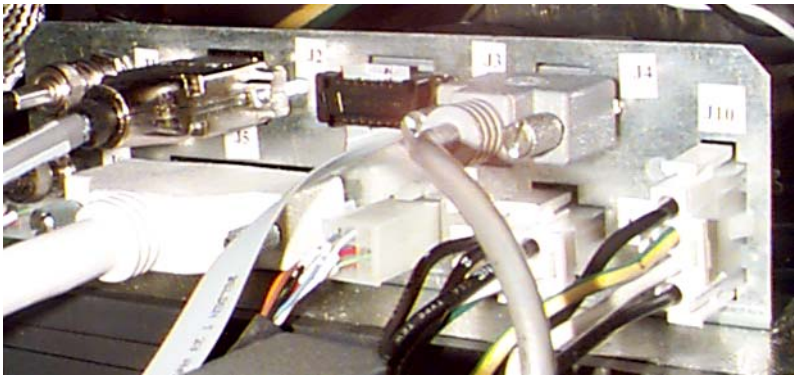


Figure 2-22 Table Bulkhead

Section 9.0

PDU Cable Connections & Configuration



CAUTION



Do not work on an energized PDU. When working on the PDU, follow this simple rule: Always tag and lockout power to the PDU at the “main” disconnect. Failure to do so can result in electrocution or death.

Do apply power to the PDU until all work is completed and all PDU covers are in their proper place.

Some of the cables must be connected to the Power Distribution Unit before it operates properly. In addition: taps, jumpers, and switches may have to be moved from their default positions. In this section, the installer is connecting cables to the PDU and verifying and/or changing settings to match the site-specific needs.

Some cables must be installed throughout the PDU (see [Figure 2-23](#)). Specific details on each connection are found in the sub-sections that follow. Use [Figure 2-23](#) for reference. The PDU was designed to have cables routed into the PDU from behind and/or beneath it.

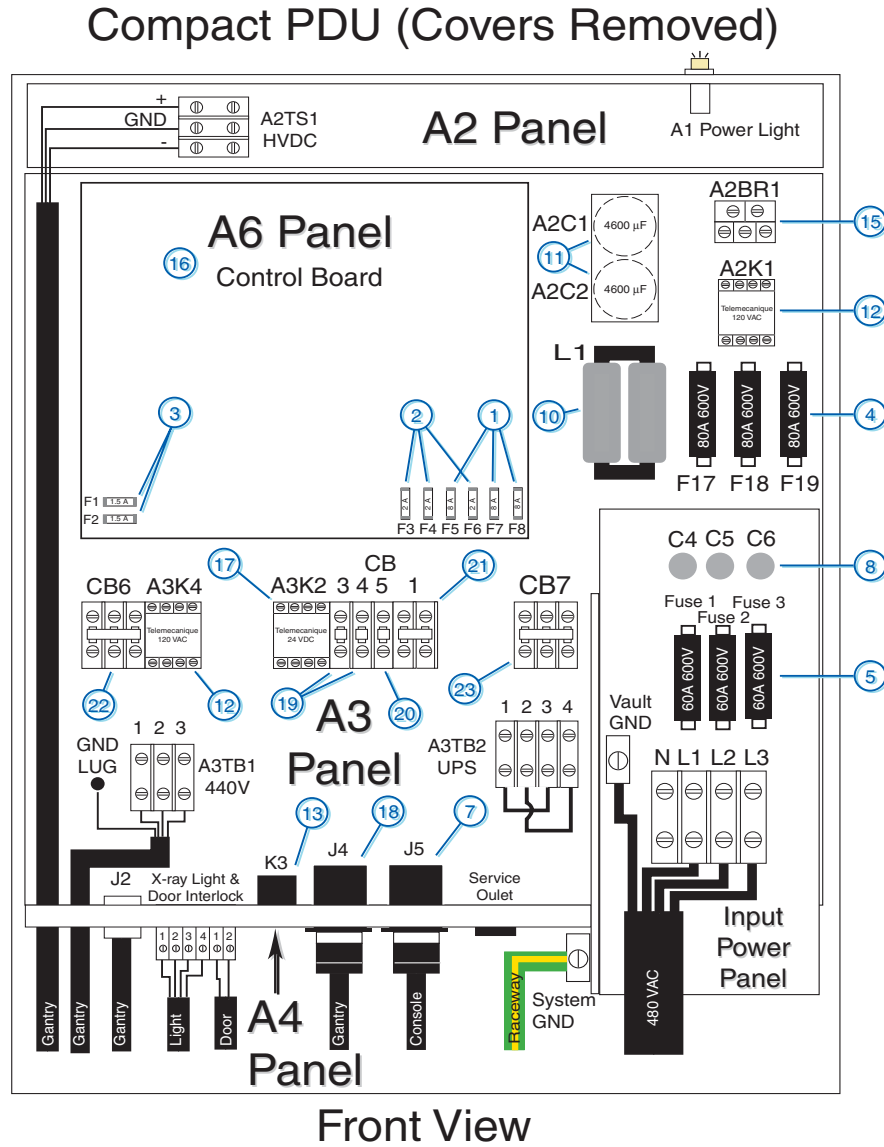


Figure 2-23 PDU Cable Connections - Front

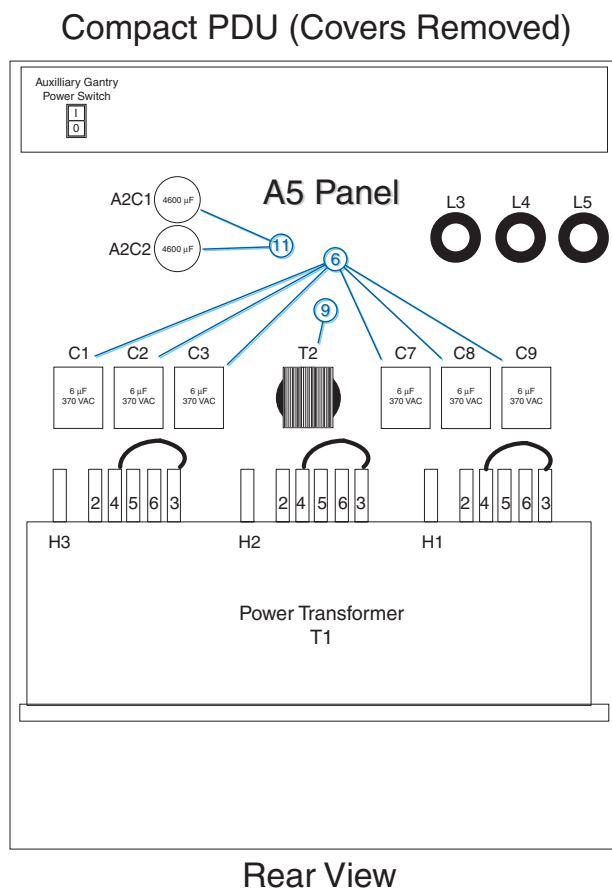


Figure 2-24 PDU Cable Connections - rear

9.1 A3 Panel - 380 - 480 VAC Mains “A1” Input Power Connection

- 1.) Remove the A3 panel front cover.
- 2.) Strip the wires to fit securely on the power block.
- 3.) Observe incoming phases (L1, L2, and L3), and insert the bare leads into the power block.
- 4.) Insert the “vault” ground into the PDU “vault” ground lug.
- 5.) Tighten all fasteners securely and replace the A3 front panel.

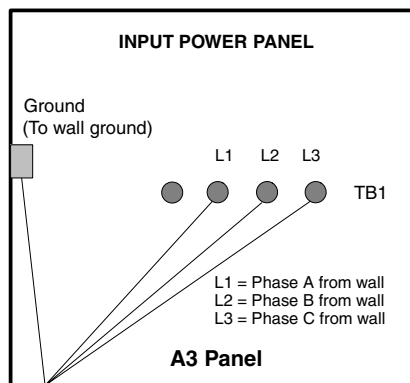


Figure 2-25 A3 Input Power Panel Connections

When Mains power is available to the PDU, the “A1” power light is illuminated (see [Figure 2-23](#)).

9.2 A3 Panel - Circuit Breakers

Place the circuit breakers in the “off/down” position during installation, even with Mains incoming power tagged and locked out. After completing work on the PDU, return the circuit breakers to the “ON” positions.

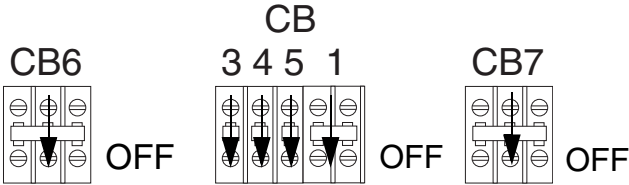


Figure 2-26 A3 Circuit Breaker Panel

By design, when CB7 is in the “OFF” position, circuit breakers 3, 4, 5, and 1 are switched “OFF.” CB7 is essentially in series with these breakers.

CIRCUIT BREAKER	DESCRIPTION
CB6	Axial Drive Power, 440 VAC
CB3	Service Outlets (PDU, Table, and Gantry), 120 VAC
CB4	24 Hour Power (STC and Table), 120 VAC
CB5	Gantry Rotating Power (Cover fans and axial drive)
CB1	Console Power, 208 VAC
CB7	Master Power (CB3, 4, 5, and 1), 120 and 208 VAC

Table 2-9 A3 Panel Circuit Breaker Descriptions

9.3 A5 Panel - Transformer (480 VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory. The taps should be set as shown in [Figure 2-27](#).

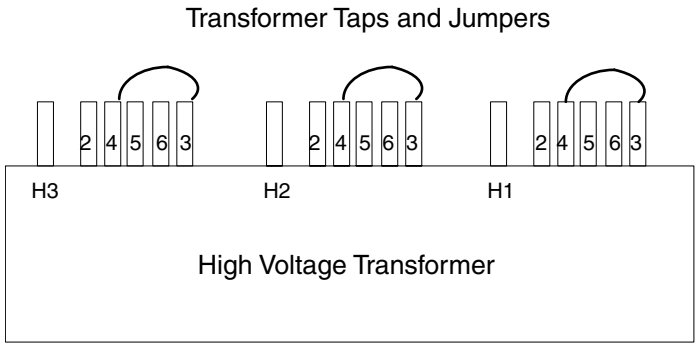


Figure 2-27 PDU Tap Positions for 480 volt operation

Adjustment of the 480 VAC and these “taps” is done in [Chapter 5, Section 3.0](#).

9.4 A2 Panel - HVDC Connection

Connect the internally shielded HVDC cable to A2TS1 on the A2 panel. (See [Figure 2-23](#) for the location of the connector and [Figure 2-28](#) for details.) Observe polarities and grounds.

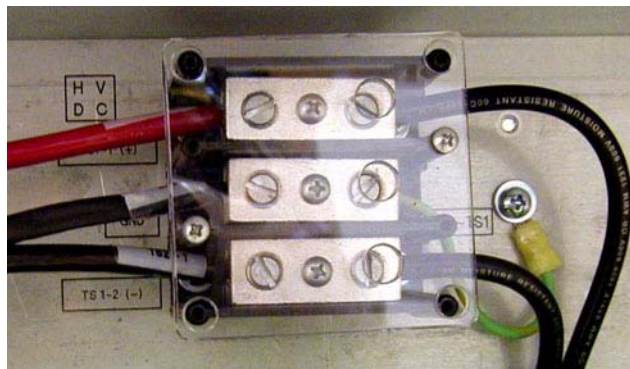
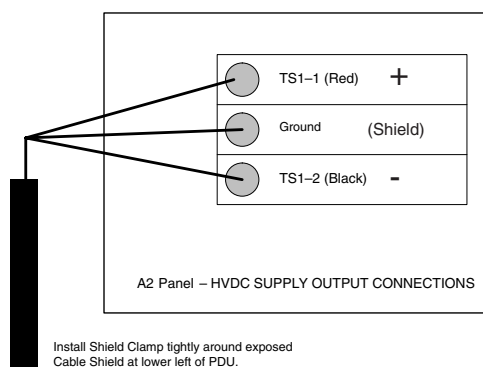


Figure 2-28 HVDC Connection

The HVDC cable can be re-terminated to eliminate excessive cable length, if necessary. Observe cable colors and polarities during re-connection. The cable shield acts as the ground.

9.5 A3 Panel - 440 V Connection

Connect the internally shielded 440 V cable from the gantry to A3TB1 on the A3 panel. (See [Figure 2-23](#) for the location of the connector, and [Figure 2-28](#) for details.) Observe the labels on the cable leads for proper identification and orientation.

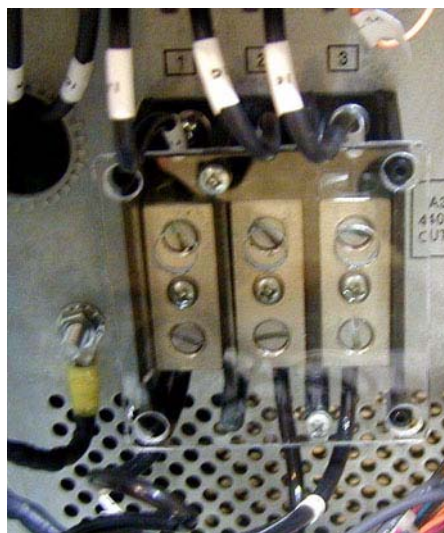
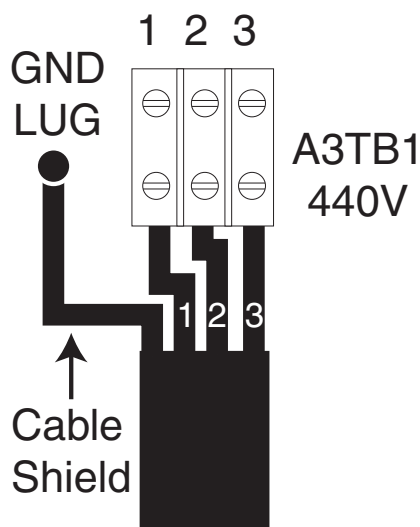


Figure 2-29 440 VAC Connection

The 440 V cable can be re-terminated to eliminate excessive cable length, if necessary.

9.6 A4 Panel - Gantry & Console Power Connections

Both gantry and console power cables come pre-terminated. Plug the gantry power cable into J4 and the console power cable into J5, as shown in [Figure 2-30](#).



Figure 2-30 Gantry & Console Power Connections

If excessive cable length prevents the proper storage, both the gantry and console power cables can be re-terminated at the PDU end.

9.6.1 Console Power Cable Re-termination

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-10](#). The terminals are labeled X, Y, W, and G on the plug.

TERMINAL	X	Y	W	G
Description	Hot	Hot	Neutral	Ground
Color				

Table 2-10 Console Power Cable Termination

- 2.) Cut the cable to the desired length and dress the ends.
- 3.) Re-install the power plug, according to the orientations recorded in [Table 2-10](#).
- 4.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU PLUG	TO CONSOLE PLUG END	
CB1 -11 (Black) (J5 Phase X (Brown))	Phase X (Brown)	☐ Check box when complete
CB1-12 (Black) (J5 Phase Y (Black))	Phase Y (Black)	☐ Check box when complete
A3 Neutral Buss Bar (Blue) (J5 -13 Phase W)	Phase W (Blue) Neutral	☐ Check box when complete
A3 Ground Buss Bar (J5 -22 Ground Green)	Ground (Green) Ground Green screw	☐ Check box when complete

Table 2-11 Resistance Verification Points

9.6.2 Gantry Power Cable Re-termination

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-12](#). The terminals are labeled X, Y, Z, W, and G on the plug.

TERMINAL	X	Y	Z	W	G
Description	Hot	Hot	Hot	Neutral	Ground
Color					

Table 2-12 Gantry Power Cable Termination

- 2.) Cut the cable to the desired length and dress the ends
- 3.) Re-install the power plug, according to the orientations recorded.
- 4.) Resistance checks are verified later.

9.7 A4 Panel – PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Plug the cable into J2 on the A4 panel. Secure it using the fasteners integrated into the cable connector shell.

9.8 A4 Panel – System Ground Connection

Connect the ground wire (green with a yellow strip) from the table/gantry raceway ground bus to the system ground lug in the PDU (see [Figure 2-23](#), and [Figure 2-31](#)).



Figure 2-31 PDU System Ground Connection

9.9 A4 Panel – X-ray Light and Door Interlock Connections

9.9.1 X-ray Light Configuration and Connection

- 1.) Configure the PDU control board for the correct X-ray light operation. Determine whether the X-ray light should come on when high voltage is present, or when X-ray is present. The board is shipped with the X-ray light turning “ON” when X-ray is present only. To change this, go to the PDU Control Board (see [Figure 2-23](#)), and move the jumper from position A to position B (see [Figure 2-32](#)). In the B position, the X-ray light illuminates whenever high voltage is present.

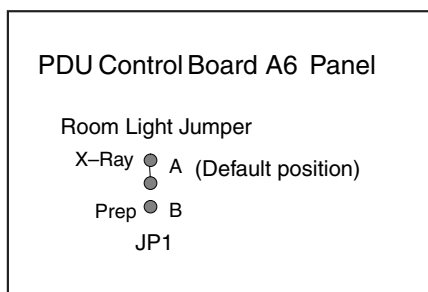


Figure 2-32 X-ray Light Configuration

- 2.) This step is site-specific. By default, the PDU is configured for “no” external X-ray light connection. If you have an external X-ray light, see [Figure 2-33](#) for proper connection.

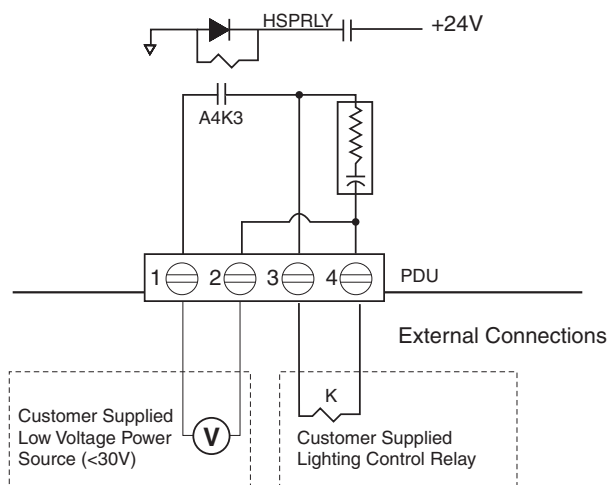


Figure 2-33 External X-ray Light Connection

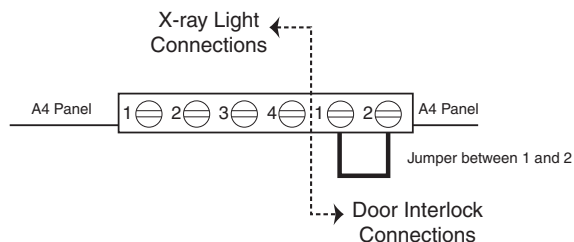
It is recommended to use the four (4) wire method of adding an X-ray warning light to a room, as shown in [Figure 2-33](#). When using this method, you:

- Minimize the EMC interference.
- Increase contact life of the relay used in the PDU.

9.9.2 Door Interlock Connections

Door interlocks prevent X-rays from generating when the scan room door is open.

The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate an X-ray until disengaged. A short must exist between pins 1 and 2 to generate an X-ray. Use a small piece of wire to short pins 1 and 2 together (see [Figure 2-34](#)).



If jumper is not in place, exposures will not be made. Check this jumper if you get scan interlock errors.

Figure 2-34 Without a Door Interlock

To use the system with a door interlock, wire a normally open switch between pins 1 and 2 that is attached to the interlock.

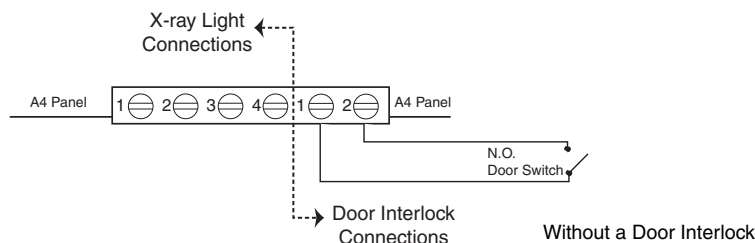


Figure 2-35 With a Door Interlock

Section 10.0

System Ground Connections

As seen in [Figure 2-36](#), the table/gantry raceway ground bus centralizes all system grounding. The system ground is tied to the vault ground at the PDU through its chassis.

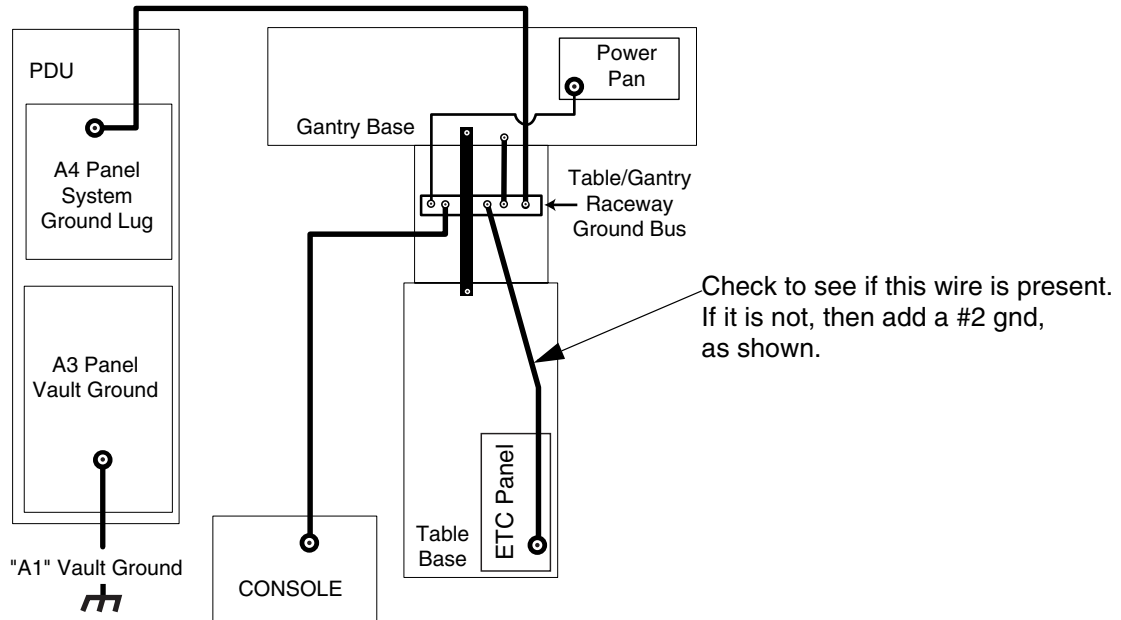


Figure 2-36 CT System Ground Connections (CPDU)

The gantry is tied to system ground at several points. It is important that all of these ground connections are securely made (see [Figure 2-37](#)).

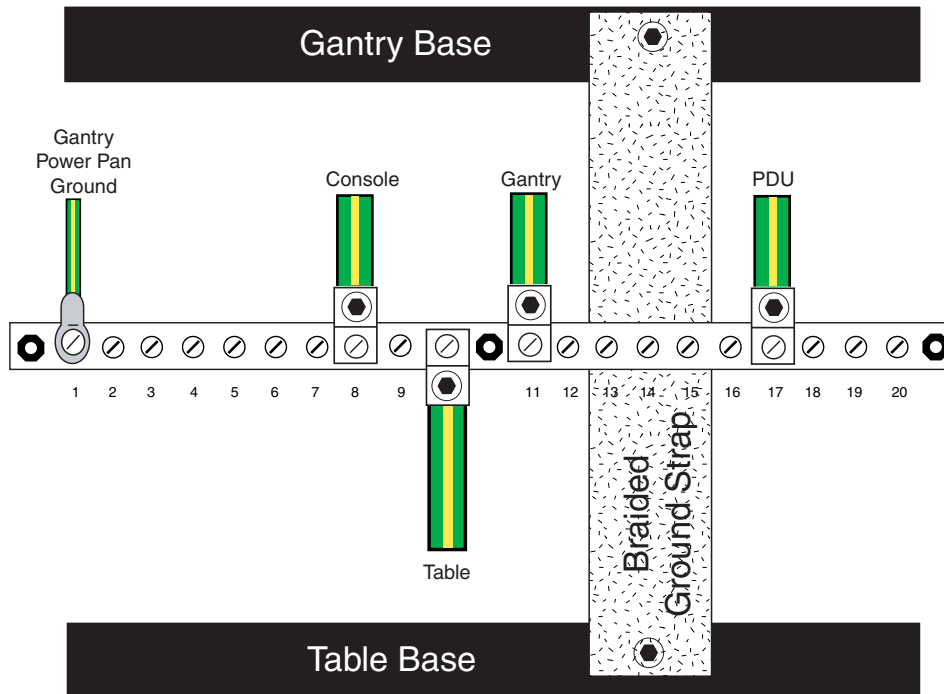


Figure 2-37 Table/Gantry Raceway Bus - Grounds

Various types and sizes of wire ground the system. Please use the type and sizes specified in [Table 2-13](#).

AWG #	CONNECTION TO	CONNECTION TO
1/0	PDU	Power Main
#10	Gantry (Power Pan)	Raceway
#2	Console	Raceway
#1/0	Gantry	Raceway
#2	Table (ETC Frame)	Raceway
#1/0	PDU	Raceway
GND Braid (2)	Table Frame	Gantry Frame

Table 2-13 System Ground Connections

Chapter 3

System Continuity & Ground Checks



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- When directed, record data collected from procedures in this chapter into GE Form 4879 located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

System Continuity & Ground Checks (Mechanical Contractor)

Use this section to check cable and ground connections.

1.1 Tools Required

- Digital VOM
- 30 ft of #18 wire
- 600 VAC meter leads

1.2 Procedure

Reference [Figure 3-1](#) (*Front View of PDU, with Covers Removed*) and [Figure 3-2](#) (*Gantry Power Pan*).



WARNING



USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.

- 1.) Remove all system power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
- 2.) Put the UPS in the service position.
- 3.) Remove the PDU A3 input power panel cover.
- 4.) With a voltmeter, verify that mains power is disconnected.
- 5.) Verify that less than 1 ohm of resistance exists between the following ground connections:

FROM A1 MAIN DISCONNECT PANEL	TO PDU A3 INPUT POWER PANEL	
A1 Breaker, wall ground connection	System gnd	☐ Check box when complete
L1, L2, L3	L1, L2, L3	☐ Check box when complete

Table 3-1 Mains Connections to PDU

6.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU	TO GANTRY POWER PAN	
A2 TS1-1 (+HVDC)	TS1-1 (RED) (+HVDC connection)	☐ Check box when complete
A2 TS1-2 (HVDC RTN)	TS1-2 (BLACK) (HVDC RTN connection)	☐ Check box when complete
A4 HVDC Cable Shield Braid Clamp	HVDC Cable Shield Braid Clamp	☐ Check box when complete
A3K4-2 T1 (Black) (A3 TB1-1)	TS2-1	☐ Check box when complete
A3K4-4 T2 (Black) (A3 TB1-2)	TS2-2	☐ Check box when complete
A3K4-6 T3 (Black) (A3 TB1-3)	TS2-3	☐ Check box when complete
HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	☐ Check box when complete
A3 System Ground	Table Raceway (Green/Yellow)	☐ Check box when complete

Table 3-2 Resistance Verification Points

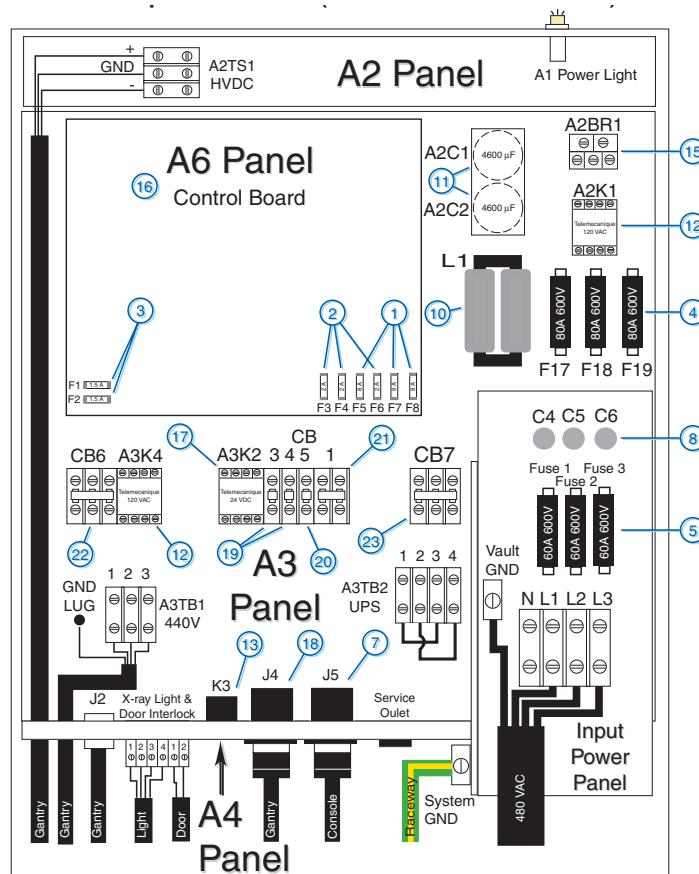


Figure 3-1 Front View of PDU, with Covers Removed

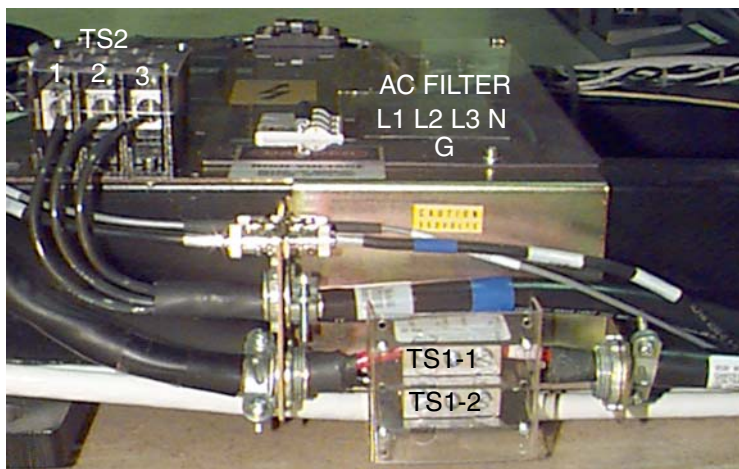


Figure 3-2 Gantry Power Pan



WARNING



TURN OFF ALL PDU CIRCUIT BREAKERS.

- 7.) Set an ohmmeter to the lowest scale. Check between the following points for shorts-to-ground. Verify that no continuity exists between the following points:

FROM PDU	TO A1 BREAKER BOX	
A2 TS1-1 (+HVDC)	vault ground	☐ Check box when complete
A2 TS1-2 (-HVDC)	vault ground	☐ Check box when complete

Table 3-3 No Continuity Verification Points

- 8.) Leave the metal cover off the PDU A3 input power panel until you complete the checks in [Section 2.0](#) of this chapter.

Section 2.0

Site Ground Continuity Check

Use an ohmmeter to verify the presence of **less than 1.0 ohm of resistance** between each of the following points:

FROM	TO	
PDU A3 System Ground	Vault Ground A1 Breaker Box	☐ Check box when complete
PDU A3 System Ground	Table/Gantry Raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry	☐ Check box when complete
Table/Gantry raceway ground point	Table	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console	☐ Check box when complete
All Display or Computing Options	Operator Console	☐ Check box when complete
Any room facility patient ground points	PDU A4 Ground Bus	☐ Check box when complete

Table 3-4 Resistance Verification - Site Ground (Compact PDU)

Section 3.0

Axial Head Holder Shim Installation

3.1 Overview

This procedure applies to the following table types:

GT 2000 All head holders
GT 1700 All head holders
HP 1600 All head holders

3.2 Requirements

3.2.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	10 min.	15 min.	5 min.

3.2.2 Tools and Test Equipments

Standard FE Tool Kit

3.2.3 Repacement Parts

Shim Kit



NOTICE Understand and follow all general table safety procedures.

3.3 Required Conditions

Check the head holder for a tight fit. If the head holder fit is loose, follow this procedure:

3.3.1 Introduction

Some Axial Head Holders have a large free-play in the horizontal directions, which can potentially lead to motion, resulting in image artifacts.

Installation of the 2327335 rubber shim kit can minimize this motion.

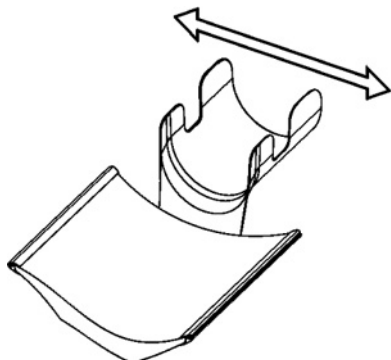


Figure 3-3 Head holder horizontal motion

3.3.2 Selecting Shim Thickness

While selecting the best shim size, do not attach the rubber shim with the adhesive on the back. Use a piece of tape to hold on the shim in order to see if the size is correct.

Selecting a shim size that is too thick may result in:

- Difficulty attaching the head holder properly. (The head holder must latch so that a patient is not injured.)
- Damage to the plastic latch or to the plastic screws that secure it.

3.4 Procedure

- 1.) Place the two 4.0 shims (thickest size—the thickness is printed on the shim) onto both edges of the head holder as shown in [Figure 3-4](#). (Use a piece of tape to secure them temporarily.)

Note: Place the shim with the tab facing out.

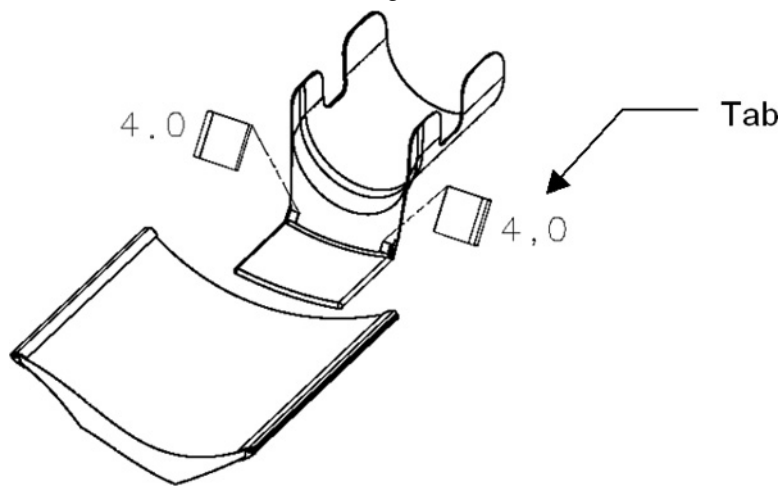


Figure 3-4 Attaching the shims

- 2.) Insert the head holder into the cradle.
- 3.) Latch the head holder onto the first step of the plastic latch mechanism (see [Figure 3-5](#) and [Figure 3-6](#)). It is not necessary to latch the head holder onto the second step.

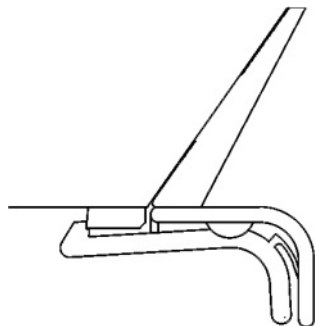


Figure 3-5 Head holder latched

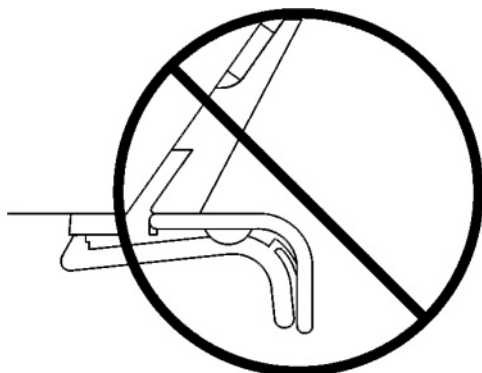


Figure 3-6 Head holder is NOT latched

- 4.) Check if the head holder has a small free-play in the horizontal direction.

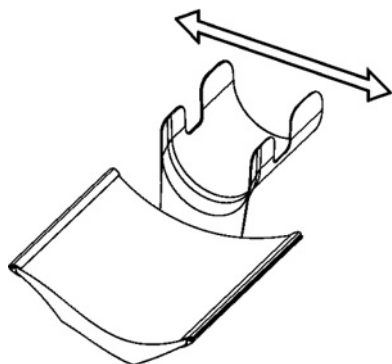


Figure 3-7 Head holder with small, horizontal free-play

- 5.) If the rubber is too thick, repeat steps 1 through 4 using a thinner shim (3.5, 3.0 ...0.5 mm). Continue until the head holder is latched (without excessive force) and fits in the cradle securely.
- 6.) If the thinnest shim (0.5 mm) is too tight, cut the tab off to reduce the thickness (see [Figure 3-8](#)).

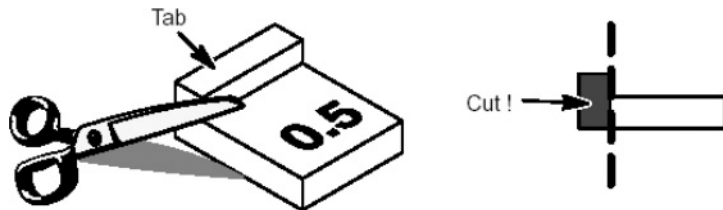


Figure 3-8 Cut the shimtab

- 7.) Use alcohol to clean off the surfaces where the shim mounts.
- 8.) Peel off the paper from the back of the selected shims and attach them with the tabs facing out. Hold each shim with your fingers for a few seconds to attach it to the head holder.

3.4.1 Finalization

Review latching of the head holder with the customer after installation.

Section 4.0

Mechanical Installation Completion Section

Align and check all covers.

Complete the Installation Completion checklist in [Chapter 15](#) and return it to GE.

Appendix A

Removal & Installation of Covers

Section 1.0 Gantry Side Covers 4

1.1 Side Cover Removal

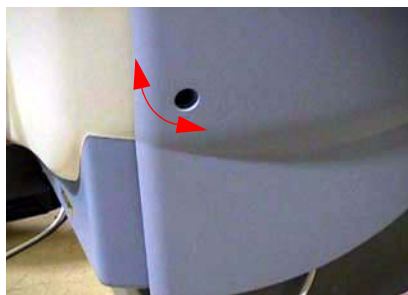


CAUTION

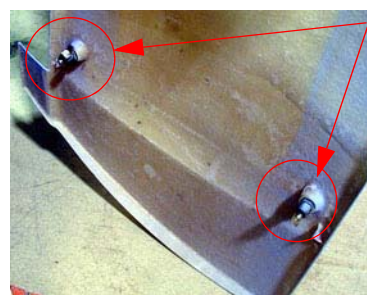
1.) Lower the table to the home (lowest) position.

Potential for injury if covers removed and power is left "ON"
Always remove the right side cover first, and turn OFF power at the STC..

2.) Use an 8 mm Hex wrench to unlatch the side cover from the front cover (see [Figure A-1](#)).



Turn the latch $\frac{1}{4}$ turn. The side cover latches and unlatches from the front cover.



Inside view of latches

Figure A-1 Side Cover Latches

3.) Remove the right side cover by lifting it upward to release the two (2) latches located on the top edge of the cover. Once removed, the STC backplane is exposed.

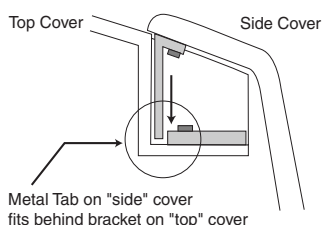


Figure A-2 Side and Top Cover Clasp

4.) Turn off all three (3) power switches on the STC backplane.

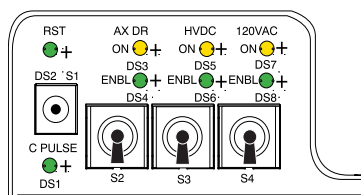


Figure A-3 STC Power Switches

5.) Repeat steps 1-3 for the left side cover.

1.2 Side Cover Installation

- 1.) Place the side cover over the top cover. Slide the side cover down until the two (2) side cover latches slide behind the metal tabs located on the top cover (see [Figure A-2](#)).
- 2.) Use a Hex wrench to secure the side cover to the front cover by turning the bolts a quarter turn (see [Figure A-1](#)).

Section 2.0 Gantry Top Covers



CAUTION Before you remove the top covers, always ensure that the three (3) power switches have been turned off (see [Figure A-3](#)).

2.1 Top Cover Removal

- 1.) Remove the associated side cover if you have not already removed it.
- 2.) Take the end of the top cover nearest to the side cover and tilt it upwards.
- 3.) Slide the cover down 75 mm (3 in.). This disengages the cover tab from the mounting bracket (see [Figure A-4](#)).
- 4.) Lift the cover clear.
- 5.) Repeat steps 1 through 4 for the other cover.

2.2 Top Cover Installation

The top cover consists of two (2) pieces. Install the front and rear gantry covers, if not already installed. Refer to [Section 3.0](#), and [Section 4.0](#) in this chapter.

- 1.) Take one of the top covers and align the tabs on the cover with its associated bracket. Lift and slide the cover into place. Position the cover to fully engage the fan interlock switch.

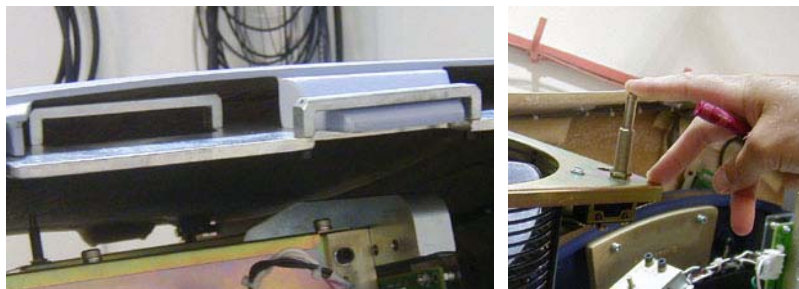


Figure A-4 Top cover tabs and bracket, plus fan interlock switch.

- 2.) Take the other top cover and align the tabs on the cover with its associated bracket. Lift and slide the cover into place, while ensuring that to engage the fan interlock switch.

Section 3.0

Gantry Front Cover

3.1 Front Cover Dolly Setup

The front cover dollies are a folding design for easy storage. This design is stable and assembly/disassembly can be done in 30 seconds.



DANGER



DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED FOR USE ON FLAT, LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.



WARNING

Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms reduce the stability of the dollies when supporting the front cover. Do not lubricate.

Note: For clarity, rotating arms are shown in the inverted, or upside down, position.

- 1.) Arrange the dolly sections for assembly. The base and stand can be assembled only one way (see [Figure A-5](#)).
 - The stand has a large stud and three (3) smaller studs that engage the base assembly.
 - The large stud provides stability.
 - The three small studs engage the base arms, preventing them from folding under if accidentally tipped.
 - The locking pin engages the fourth base arm and prevents accidental separation when assembled.

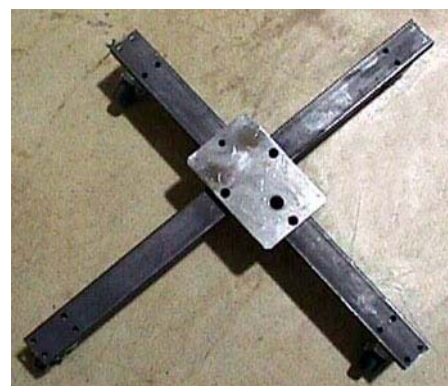


Figure A-5 Front Cover Dolly base and stand disassembled

- 2.) Unfold the base (as shown in [Figure A-5](#)) and place it on a flat surface.
- 3.) Install the stand in the base, insert the base locking pin, unfold the stand top, and secure it with a palm screw. Ensure that the additional safety bracket and washer are installed (see [Figure A-6](#)).

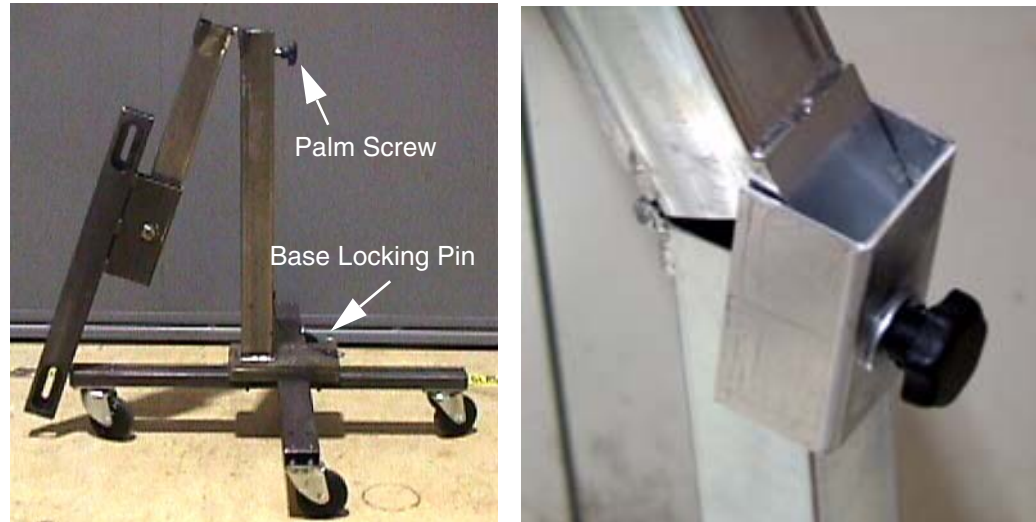


Figure A-6 Front Cover Dolly Stand and Base with additional Safety Bracket

- 4.) Assemble the second dolly. Both look similar (as in [Figure A-7](#)). However, the stand portion of the dollies are side-specific. The “right” dolly has the locking pin for cover rotation. The “left” dolly does not have a cover rotation locking pin.
- Both dolly stands should have an “R” or an “L” indicating right or left.
 - The base assemblies are interchangeable.



Figure A-7 Front Cover left and right Fully Assembled

3.2 Removal

- 1.) Position the table at its lowest position.
- 2.) Remove the gantry side and top covers, if you have not already done so. Refer to [Section 1.0](#) in this chapter. Ensure that the three (3) power switches have been turned off (see [Figure A-3](#)).
- 3.) Assemble the front cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the gantry securely. This makes cover installation easier (see [Figure A-8](#)).

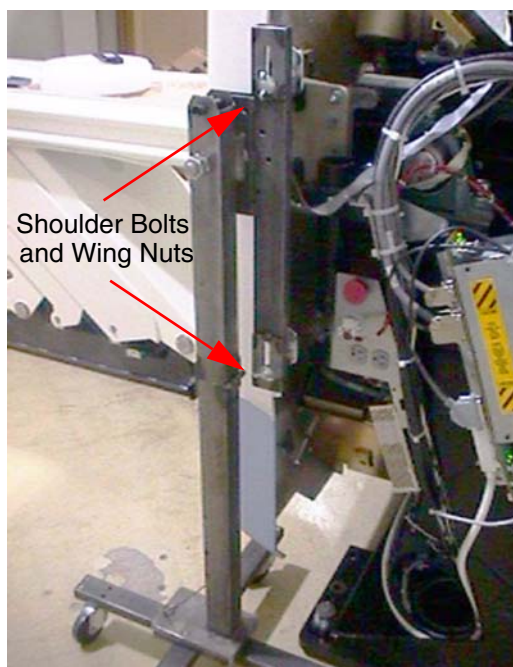


Figure A-8 Front Side Dolly

- b.) Attach the side dolly to the shoulder bolts and secure the assembly with two (2) wing-nuts.
 - c.) Repeat steps a and b to assemble the other side dolly.
- 4.) Detach front cover J3, J2, and BKHD J1 cables.
- 5.) Remove the front cover.
 - a.) Disengage the upper and lower cantrell brackets on both sides of the cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers (see [Figure A-9](#)).

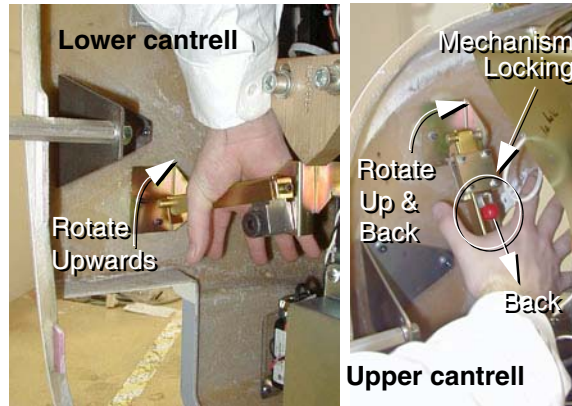


Figure A-9 Releasing cover brackets

- 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This releases the locking mechanism and allows the cantrell to rotate upward with steady and firm pressure.
- b.) Disengage the rubber retaining straps on both sides. It may be helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.
- c.) Also lift and rotate the cover locking arm to the unlocked position.

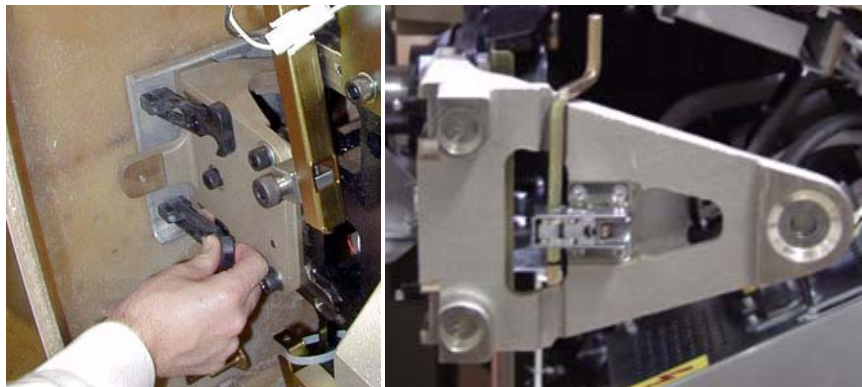
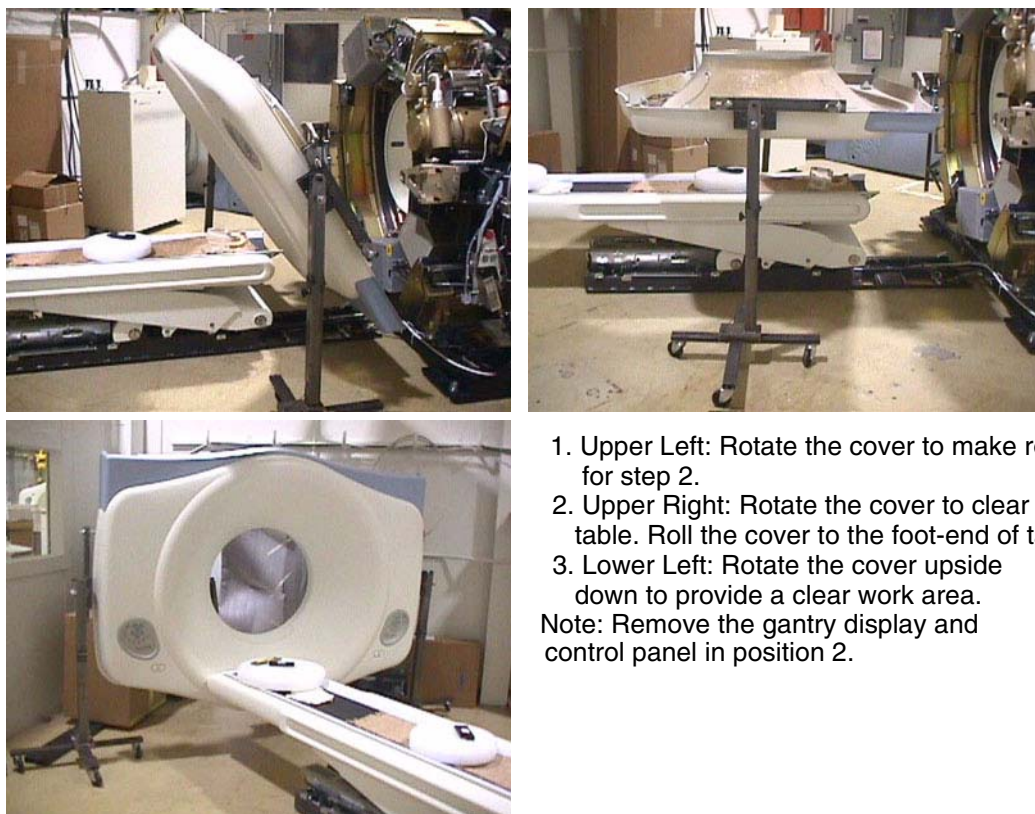


Figure A-10 Rubber retaining straps and Cover Locking mechanism

- 6.) Rotate the front cover away from the gantry.
 - a.) Move the front cover away from the gantry, giving enough space (about 1.5 m/5 ft.) between the front cover and the gantry.
 - b.) Pull the locking pin and rotate the front cover away from the gantry. Place the locking pin in one of the side dolly perforations (see [Figure A-11](#)).



Figure A-11 releasing Front Cover Dolly Hinge



1. Upper Left: Rotate the cover to make room for step 2.
 2. Upper Right: Rotate the cover to clear the table. Roll the cover to the foot-end of table.
 3. Lower Left: Rotate the cover upside down to provide a clear work area.
- Note: Remove the gantry display and control panel in position 2.

Figure A-12 Front Cover Removal Sequence

- 7.) Rotate the cover horizontally and move it back and over the table to a safe location. Once in a safe location, over-rotate the cover to full vertical, but upside down.
- 8.) Remove the gantry display and one (1) of the cover control assemblies, and place them into the service positions.
 - a.) Remove the gantry display and place it into its service position.
 - * The gantry display is held in place with five (5) thumb screws. Use a flat-blade screwdriver to remove the display (see [Figure A-13](#)).
 - * Mount the display on the right gantry fan.
 - * There are two (2) mounting methods, both use the cables connected to the REAR GANTRY COVER.
 - * Disconnect the cabling at the right rear gantry cover. Only one (1) cable connects to the gantry display.
 - * Position A: use one of the display T-hooks to hang in the "T" slot on the side of the right gantry fan assembly (see [Figure A-14](#)).
 - * Position B: place the display in the cradle across the top of the right gantry fan assembly. Use the thumb-screw to secure the display on the right side (see [Figure A-15](#)).



Figure A-13 Gantry Display Removal

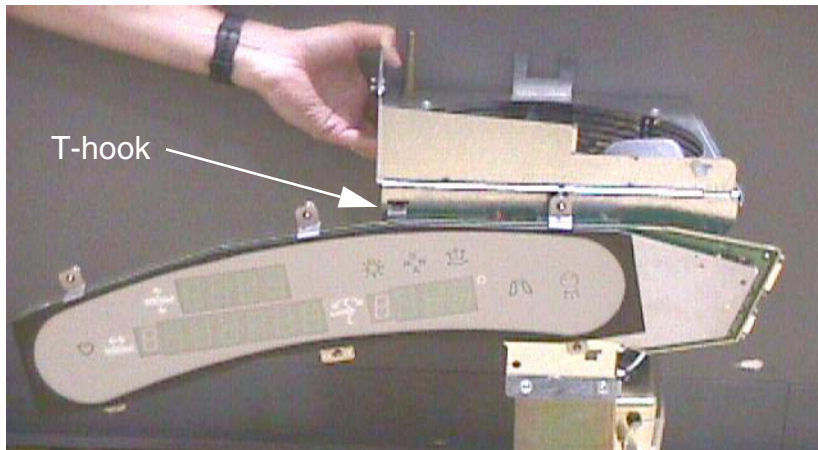


Figure A-14 Gantry Display Service Mounting Location "A"

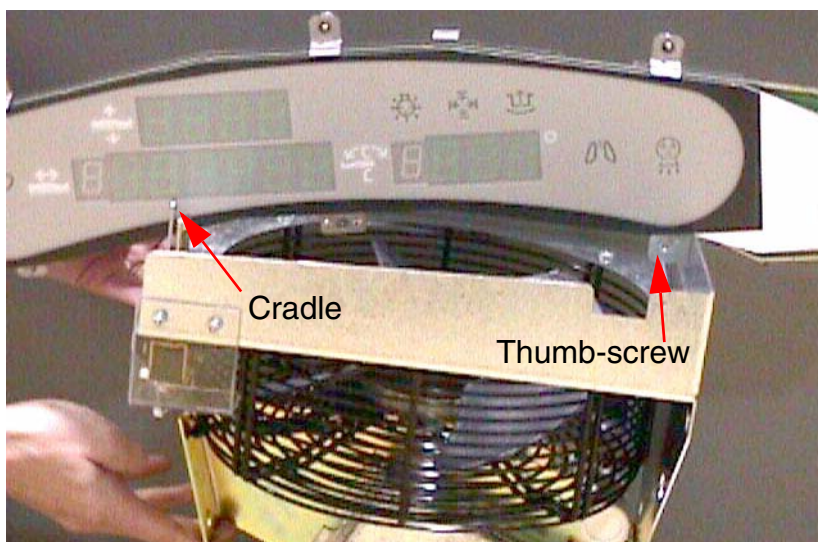


Figure A-15 Gantry Display Service Mounting Location "B"

- b.) Remove a gantry control and place it into its service position.
 - 1.) Press on each ball stud until the panel is released. Keep one hand on the control panel at all times to prevent it from dropping to the floor.



Figure A-16 Gantry control panel removal

- 2.) Align the ball studs with their associated receivers and snap the panel into place.

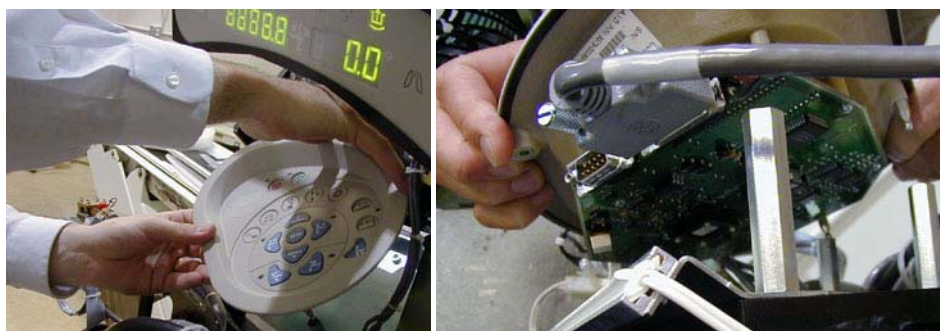


Figure A-17 Control panel service position

Note:

- 3.) Connect the cable to the terminator located on the cantrell arm (see [Figure A-18](#)).
There are three (3) cables, each of which is unique. The ribbon cable is not used in the service configuration. The other two (2) cables only fit in either the terminator or the control panel, not both.



Figure A-18 Gantry Service Mode Cable Terminator

3.3 Installation

- 1.) Remove the gantry display and control assembly from their service positions and re-attach them to the gantry cover.
 - a.) Disconnect the cables from the display and gantry control panels.
 - b.) Install the gantry display in the front cover. Secure the five thumbscrews. With a flat-blade screw driver, gently tighten the screws past finger tight.
 - c.) Install the gantry control panel making sure that the ball studs are secure within the receivers.
 - d.) Re-attach the cables.
- 2.) Rotate the gantry back to its vertical position.



NOTICE Potential for front cover damage.

When you rotate the gantry back to its vertical position, ensure that not to scratch the front cover with the edge of the table cradle.

- 3.) Attach the front cover.
 - a.) Align the studs on both sides of the front cover with each associated receiver. The receiver is located on the gantry frame.



Figure A-19 Cover stud and mounting bracket receiver

- b.) Insert the stud on one side into its associated receiver.
- c.) Attach the rubber retaining straps.
- d.) Insert the stud on the other side into its associated receiver.
- e.) Attach the rubber retaining straps.

It may be helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

- 4.) Re-attach the upper and lower cantrell brackets on both sides.
 - a.) Remove the upper cantrell brackets from the service position and rotate them into position over their associated retaining pins (see [Figure A-20](#)).

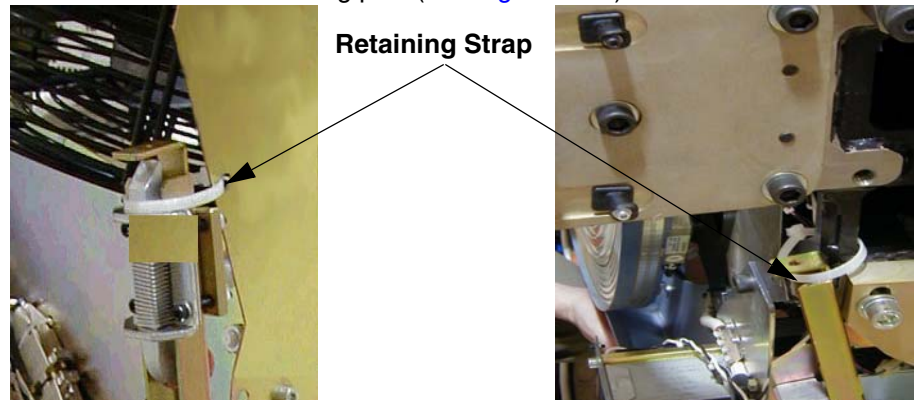


Figure A-20 Service position of upper and lower cantrell brackets.

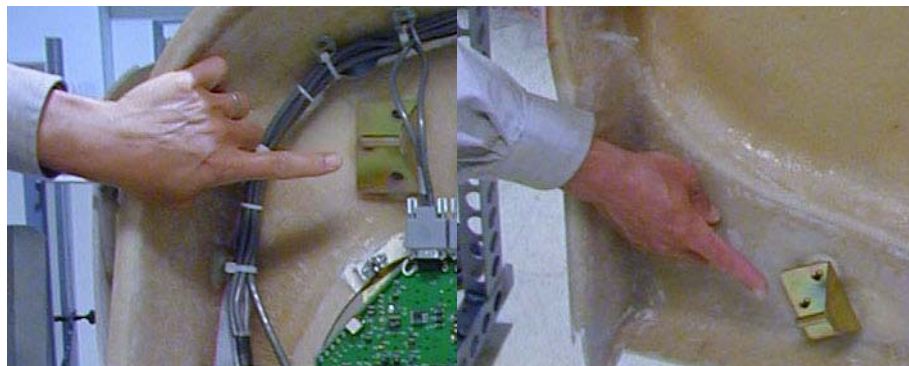


Figure A-21 Cover retaining pins (top and bottom)

- b.) Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides (see [Figure A-22](#)).



Figure A-22 Locking the cover brackets into place.

- c.) Remove the lower cantrell brackets from the service position (see [Figure A-20](#)), and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place (see [Figure A-22](#)).

Note: Mis-adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not require adjustment during normal use.

- 5.) Remove, disassemble, and store the dolly safely away for later use.
- 6.) Re-attach the cables to the cover.

Section 4.0

Gantry Rear Cover

4.1 Removal

- 1.) Assemble the rear cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the rear cover.

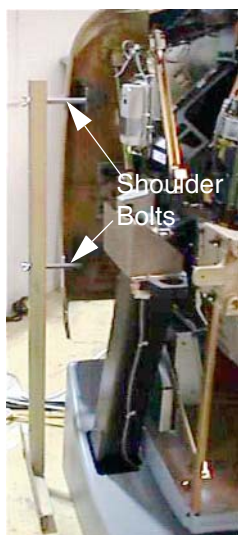


Figure A-23 One side of the Rear cover dolly

- b.) Fit the side dolly through the shoulder bolts and secure the assembly with two (2) wing-nuts (see [Figure A-23](#)).
 - c.) Repeat steps a and b for the other side dolly.



CAUTION Potential for injury if covers are removed and power is left "ON".

- 2.) Disconnect the cables on the right side of the rear cover.
- 3.) Remove the rear cover.
 - a.) Disengage the upper and lower cantrell brackets on both sides of the rear cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers (see [Figure A-9](#)).
 - 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This releases the locking mechanism and allows the cantrell to rotate upward with steady and firm pressure.
 - b.) Disengage the rubber retaining straps on both sides.

4.2 Installation

- 1.) Position the cover in back of the gantry.
 - 2.) Attach the rear cover.
 - a.) Align the studs on both sides of the rear cover with the receivers located on the gantry frame.
 - b.) Insert the stud on one side into its associated receiver.
 - c.) Attach the rubber retaining straps.
 - d.) Insert the stud on the other side into its associated receiver.
 - e.) Attach the rubber retaining straps.
- Note: It may be helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.
- 3.) Reattach the upper and lower cantrell brackets on both sides.
 - a.) Remove the upper cantrell brackets from the service position and rotate them into position over their associated retaining pins.
 - b.) Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides.
 - c.) Remove the lower cantrell brackets from the service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place.
- Note: Adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not require adjustment during normal use.
- 4.) Remove, disassemble, and store the dolly safely away.
 - 5.) Re-attach the cables to the cover.
 - 6.) Re-install the Mylar (scan) window. Carefully bend the scan window and place it into the channel (groove) provided in the covers.



Figure A-24 Installing the Mylar window

Section 5.0

Gantry Cover Alignment Adjustment Guide

5.1 Overview

This section explains the adjustment capability available for the Gantry covers and guidelines for performing adjustments.

5.2 Tools Required

3 mm, 5 mm, 8 mm, Hex wrenches

5.3 Adjustment Guide

4.0.1 Front/Rear Cover adjustment

The front and rear covers are held in a fixed vertical position by the cover locking mechanisms. These are located on each side of the cover at the vertical middle. The only adjustment capability is found on each of the cover cantrells (see [Figure A-23](#)), one near each corner of the front and rear covers. Adjustments are best done at zero tilt.

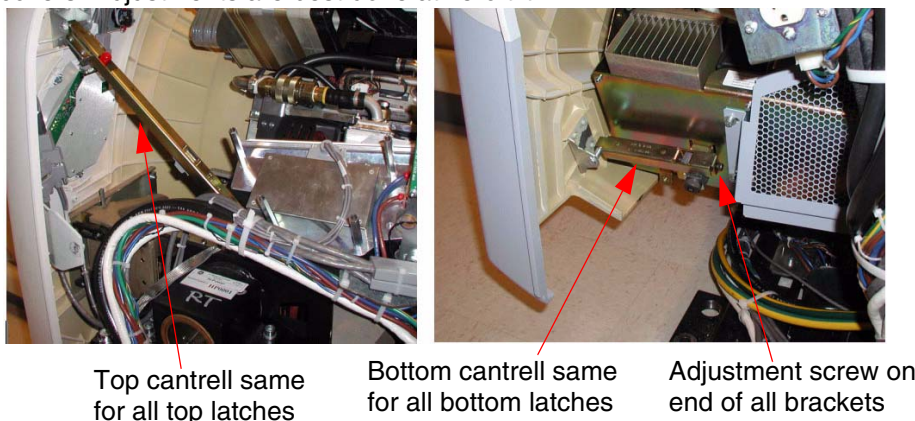


Figure A-25 Cantrells

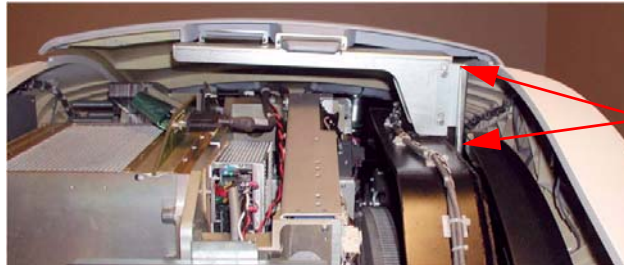
- 1.) Remove the Gantry side covers using a 8 mm hex wrench to release the bottom latches. Refer to [Section 1.0](#) of this chapter for further details.
- 2.) Disable Gantry rotation, 120 VAC, and HVDC via the service switch panel.
- 3.) Disconnect the fan power via the white Molex connector (one for each top cover). Remove the top covers by lifting up and pulling to the side of the gantry.
- 4.) Remove the scan window to avoid creases or kinks during cover movements.
- 5.) Use a 5 mm hex wrench to adjust the bolt on the gantry-end of the cover cantrells (see [Figure A-23](#)). Lengthen or shorten the cantrell arm that pushes or pulls the cover corner. The distance between the inside edges of the front and rear covers is typically 27 in. + 1/8 in. (689 mm). Hang the covers level using a level on the inside vertical edge. Maintain the distance along the inside edges from top to bottom.

4.0.2 Scan Window

Position the scan window in the opening to verify that it seats all the way around with no gaps. The scan window should seat easily if the front and rear covers have been adjusted properly. If not, go back and readjust such that the scan window seats properly. There is no easy solution other than adjusting and looking for fit.

4.0.3 Top Cover Adjustment

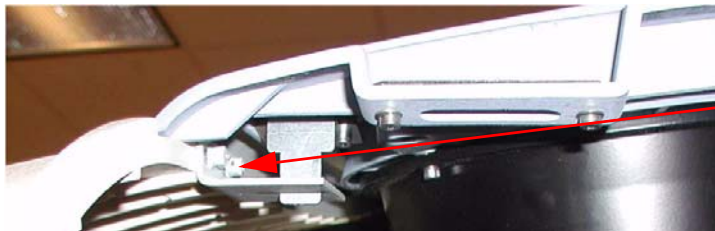
The top covers rest in a top bracket that is mounted to the stationary gantry frame, and then just rests on top of the front/rear covers. There also are also two alignment fingers on the outer edge of the top covers that must sit in brackets on the front/rear covers.



Screws on both sides
allow up/down only

Figure A-26 Top Cover Bracket

- 1.) Set the top covers back on the gantry and look for proper fit. The top covers should rest on the front/rear covers all along the edges. If they do then skip the rest of this section.
- 2.) If the top covers do not rest on the gantry at the top, lift them off. Use a 5 mm hex wrench to loosen the bolts holding the top cover bracket.
- 3.) Use a 3 mm hex wrench to adjust the brackets on the front/rear covers for the fingers on the outer edge of the covers. Slide the bracket until it lines up with the top cover finger (see [Figure A-25](#)).



Adjustment bolts for
aligning bracket to
top cover finger

Figure A-27 Top Cover Fingers

- 4.) Verify that the top cover tapered edges fit over the lip along the top edge of the front/rear covers. If they do not, go back to [Section 4.0.1](#) and readjust the cantrells as necessary.
- 5.) When done adjusting the top covers, reconnect the fan power Molex connections.

4.0.4 Side Covers

The side covers just hang from the top covers by two fingers that are not adjustable. When set into the top covers, the side covers should hang straight down. The tapered edges fit over the top of the lip of the front/rear covers.

4.0.5 Finish

- 1.) Re-enable gantry rotation via the Axial Enable switch on the service switch panel.
- 2.) Replace the side covers and latch using an 8 mm hex wrench on the side cover bottom latches.
- 3.) Run an axial scan to ensure proper system operation before turning the system over to the customer.

Section 6.0

Gantry Scan Window

6.1 Remove Scan Window

- 1.) Grab the window at the top and pull firmly downward.
- 2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.
- 3.) Hold the top and bottom portions of the scan window together.
- 4.) Grasp both sides of the scan window and move them together.
- 5.) Lightly pull upward until you can free the window from between the front and rear covers.



Figure A-28 Scan Window Removal

6.2 Install Scan Window

- 1.) Install the front and rear covers.
- 2.) Deform the scan window (as shown in [Figure A-29](#)), and nest it at the bottom of the opening between the front and rear covers (see [Figure A-30](#)). The rivets are in the six o'clock position. Remember, the rivets must be in the 12 o'clock position when the Mylar window is fully installed.
- 3.) After you complete the initial seating of the scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.
- 4.) Ensure that you position the window with the rivets at the 12 o'clock position. Place the Mylar window slit at either the 3 or 9 o'clock position.

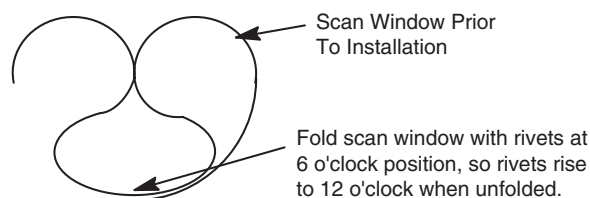


Figure A-29 Install Scan Window

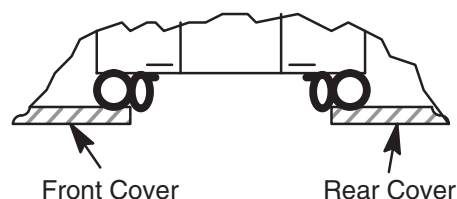


Figure A-30 Scan Window Nested Between Front and Rear Cover

Section 7.0

Gantry Base Covers

Refer to [Figure A-31](#) for the following assembly sequence.

Note: Tighten means torque to 2.3 N-m

- 1.) Assemble two (2) brackets (item 11) to the front cover (item 5), using four (4) hardware items 1, 2, and 3.
- 2.) Align the leg of the brackets parallel to front edge of the cover and tighten.
- 3.) Position the cover on the gantry base with the bracket slots aligned to the gantry holes.
- 4.) Center the cover left to right, and attach with four (4) hardware items 1, 2, and 3 (as shown), and tighten.
- 5.) Assemble two (2) brackets (item 12), and two (2) brackets (item 13), to the gantry base, using eight (8) hardware items 1, 2, and 3.
- 6.) Finger-tighten the hardware with the brackets moved outward to the end of the slots.
- 7.) Install the side covers (items 6 and 7) on the base, pushing the brackets (items 12 and 13) inward until properly aligned with front cover.
- 8.) Remove the side covers and tighten the fasteners.
- 9.) Replace the side covers using two (2) hardware items 1, 2, and 3, and one item 4 on each cover, and tighten.
- 10.) Assemble the last bracket (item 13) loosely to the gantry base with two (2) hardware items 1, 2, and 3.
- 11.) Assemble the latch (item 10) to the rear cover (item 8), oriented as shown.
- 12.) Install the rear cover (item 8) to the base, properly aligned to the side cover (item 6).
- 13.) Attach the rear cover to the bracket with hardware items 1, 2, 3, and 4, tightening all fasteners. Lock the latch (item 10).
- 14.) Assemble two (2) latches (item 10) to the rear cover (item 9), oriented as shown. Place the cover on the gantry base, aligned to covers 7 and 8. Lock both latches.

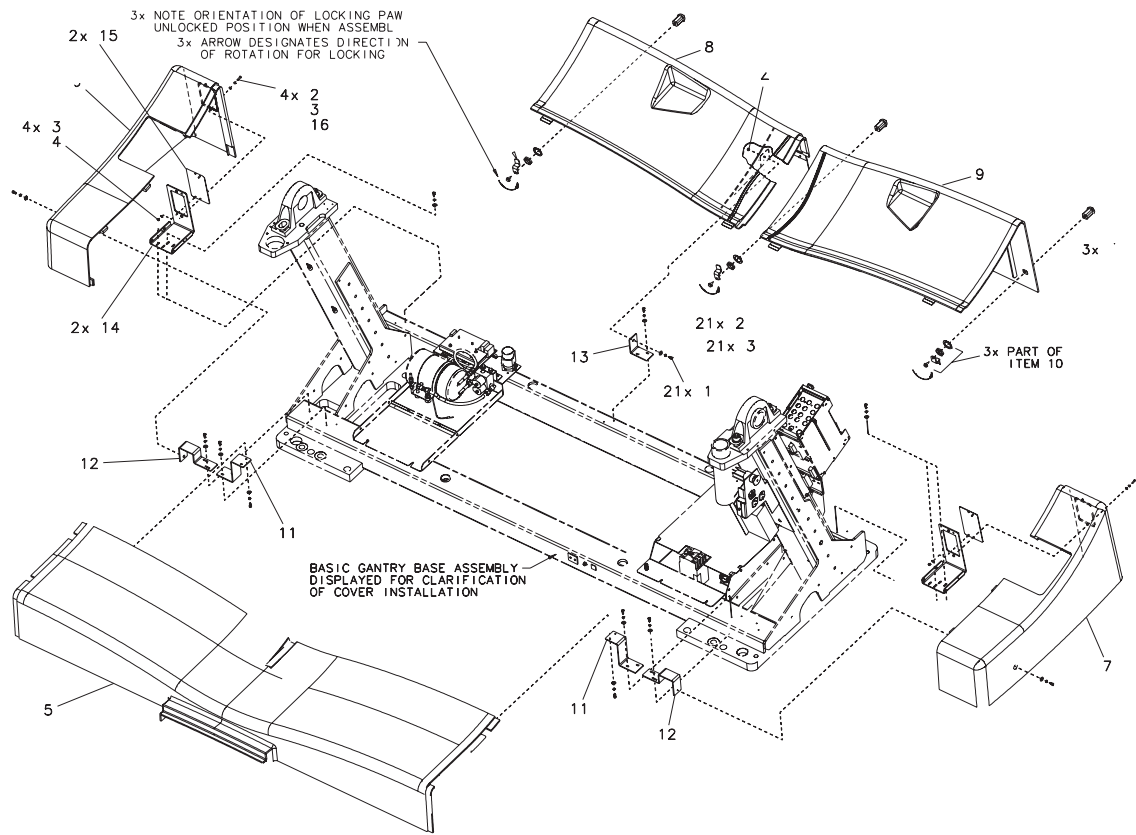


Figure A-31 Gantry Base Covers

Section 8.0

Foot-switch Covers

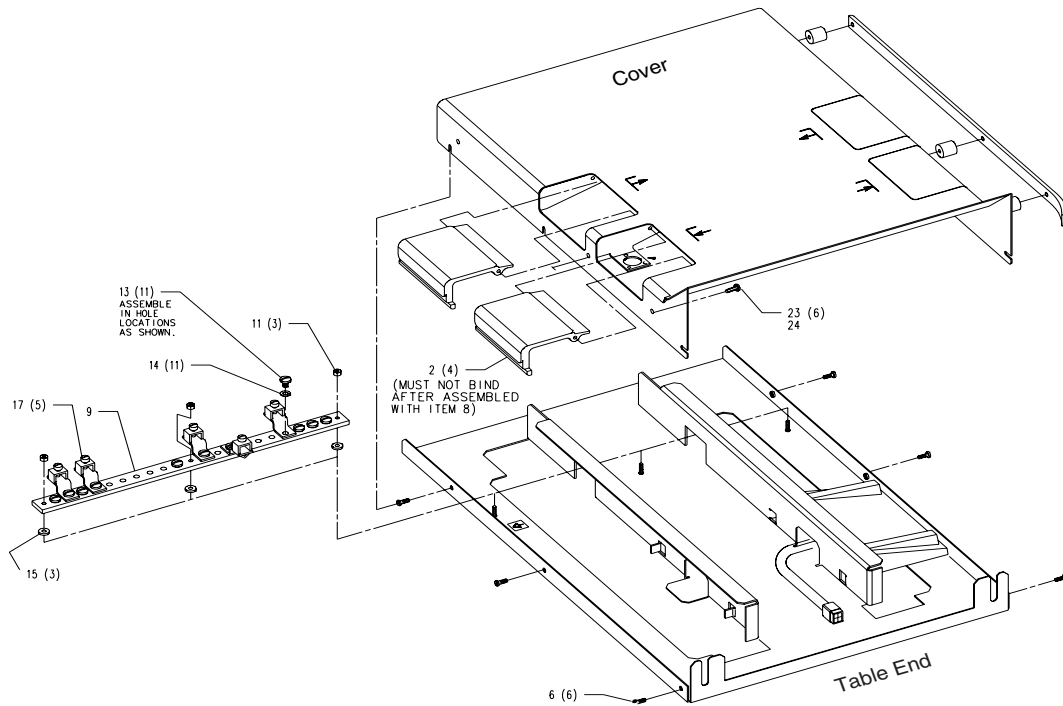


Figure A-32 Table Foot-switch Assembly

Appendix B

Pictorial Representation of Required Tools

Use the following guide as a reference, if you are unsure of a tool listed in [Section 2.6, on page 29](#).

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Adapter		Sears Industrial: $\frac{3}{8}$ " to $\frac{1}{2}$ " (9-4258)
Ball-Peen Hammer		Sears Industrial: 1lb/2 lb (9-38465)
Canned Air		Miller Stephenson: Aero Duster (MS-222N)
Clamp on Amp Meter		Sears Industrial: 9-WTAD105
Combination Wrench Set		Sears Industrial: U.S. Standard & Metric (9-44048)
Cordless Screwdriver		Sears Industrial: 9-MU65401
Deep Well Socket		Sears Industrial: $\frac{3}{4}$ " X $\frac{3}{8}$ " (included with 9-34496)
Dental Pick		
Diagonal Cutting Pliers		Sears Industrial: Small (9-45077)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Drill		Sears Industrial: $\frac{3}{8}$ " or $\frac{1}{2}$ " (9-27859)
Drill Adapter		Sears Industrial: 3" X $\frac{3}{8}$ " (9-APSZ24)
Drill Bit Set		Sears Industrial: U.S. Standard (9-66084)
DVM		Sears Industrial: 9-82028 Sears Industrial: 9-FL873
Extension for Ratchet Wrench		Sears Industrial: 3" X $\frac{1}{2}$ " (9-44133)
Gloves		Sears Industrial: Large (9-40502)
Hammer Drill		Sears Industrial: $\frac{1}{2}$ " (9-27205)
Hex Bit Set		Sears Industrial: $\frac{1}{4}$ " (9-SK45508)
Hex Key (Allen Wrench) Set		Sears Industrial: U.S. Standard (9-46284)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Level		Sears Industrial: 4' (9-39856)
Masonry Bit		
Open-End Wrench (Thin or Standard Tappet)		Snap-on: 10 mm (SRSM10) & 21 mm (LTAM2124)
Ratchet Wrench		Sears Industrial: 3/8" (9-43175)
Reciprocating Saw with Blades		Sears Industrial: 9-MU650921
Safety Glasses		Sears Industrial: 9-18650
Safety Shoes		
Screwdriver Set		Sears Industrial: Phillips & Straight (9-41505)
Socket Set		Sears Industrial: Standard 3/8" (9-34496)
Sockets		Sears Industrial: 1 1/8" X 1 1/2" (9-47516)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Step Ladder		Sears Industrial: 6' (9-WN6006)
Tongue & Groove Pliers		Sears Industrial: Large (9-CL440)
Torpedo Level		Sears Industrial: 9" (9-39829)
Torque Wrench		Sears Industrial: $\frac{3}{8}$ " (9-WR3470)
Universal Joint		Sears Industrial: $\frac{3}{8}$ " (9-4435)
Vacuum Cleaner		Sears Industrial: 8 Gal (9-17780)
* Part Numbers given for reference only. GEMs does not endorse any tool brand name.		

Table B-1 Required Tools (Continued)

Appendix C

Regulatory Clearance Quick Reference Guide

Section 1.0 Regulatory Code Description

Egress: 29 CFR 1910 Subpart E (OSHA) and NFPA 101 (Life Safety Code) define the minimum requirements for means of egress. The requirement most applicable to equipment installation and room layout is minimum width of exit access. Under OSHA 1910.37(f)(6), the minimum width of exit access shall in no case be less than 28 in. from any potentially occupied point in the room.

Under NFPA 101 (2006 edition) 7.3.4.1, the minimum width of any means of egress is 36 in. However, NFPA allows this to be reduced to 28 in. around furniture or equipment, if a 36 in. clearance would otherwise be available without moving permanent walls.

Electrical Clearance: 29 CFR 1910 Subpart S (OSHA) and NFPA 70E (Standard for Electrical Safety in the Workplace) define minimum clearance requirements for the workspace around electrical equipment. Under both OSHA 1910.303(g)(1) and NFPA 70E (2004 edition) 400.15, electrical equipment with parts operating at 600 volts or less, which may require examination or service while energized, require a minimum clear space of 36 in. depth (with minimum 30 in. width and 78 in. height) in front of the equipment.

This safety clearance requirement applies to all GEHC equipment. Where parts operate above 150 volts but less than 600 volts, standards require an increased minimum clearance under the following circumstances:

- If the wall or surface directly facing the electrical equipment is grounded (e.g. brick, concrete, or tile), or includes grounded protrusions, a 42 in. clearance depth is required. Protrusions include:
 - medical gas ports
 - metal door or window frames
 - water sources and metallic sink structures
 - metallic cabinets,
 - electrical disconnects or emergency off panels
 - air conditioners or vents
- If exposed and unguarded live parts on both sides of the workspace could exist, a 48 in. clearance depth is required. For example, if a power distribution unit is positioned on the wall directly facing the GEHC equipment.

Section 2.0

Terms and Definitions

Egress: The path of exit from within any room. U.S. regulations require a minimum of 28 in. (711.2 mm) of continuous and unobstructed space, including trip hazards along the path of exit.

Workspace: The dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. The U.S. regulation minimum is 36 in. (914.4 mm), but additional conditions can increase the minimum dimension requirement. GEHC defines this as the envelope of the component superstructure with the external covers in place.

Service Access Width: The width of the workspace in front of the equipment. A minimum of 30 in. (762 mm), or the width of the equipment, whichever is greater.

Head Clearance: The height dimension of the workspace. The height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). 78 in. (1981.2 mm), or the height of the equipment, whichever is greater.

Grounded Wall: Any wall that can be electrically conductive to earth ground. Masonry, concrete, and tile are considered conductive. Additional commonly found aspects of a wall should also be considered grounded. This is not an all-inclusive list:

- Medical gas ports and plates
- Metal doors and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinets
- A1 main disconnect panel
- Equipment Emergency Off panels
- Industrial equipment (such as air conditioners and vents)
- Expansion joints
- Surface raceway
- Exposed wall conduits
- Floor outlets boxes

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks
- Ceiling tile grids

Section 3.0

Regulatory Minimum Working Clearance by Major Subsystem

Requirements apply to equipment operating at 600 V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.

Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.

Required regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance.

Workspace Requirement	Minimum Clear Space	Additional Conditions
Direction of service access: Front of console	36 in. (914 mm)	No exposed live part hazards with cover in place. If the console is placed under a counter, the front edge of the console must be even with the vertical edge of the console workspace. Note: This component is typically serviced from the front with access to the rear.
Service access width: Left-Right of workspace	30 in. (762 mm)	This is the width of the workspace in front of the equipment. 30 in. (762 mm) minimum or the width of the equipment, whichever is greater.
Head clearance	78 in. (1981.2 mm)	The height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). 78 in. (1981.2 mm) or height of equipment, which ever is greater.

Note: Distances are measured to the finished covers.

Table 3-5 Console Subsystem

Workspace Requirement	Minimum Clear Space	Additional Conditions
Direction of service access: Front of PDU	36 in. (914.4 mm)*	No exposed live part hazards with cover in place. This component is typically serviced from the front with access to the rear. *48 in. (1219.2 mm), if exposed live parts of 151 V - 600 V are present on both sides of workspace, with the operator between. *42 in. (1066.8 mm), if the opposite wall is grounded and exposed live parts of 151 V - 600 V are present.
Service access width: Left-Right of workspace	30 in.(762 mm)	This is the width of the workspace in front of the equipment. 30 in. (762 mm) minimum or the width of the equipment, whichever is greater.
Head clearance	78 in.(1981.2 mm)	The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s). 78 in. (1981.2 mm) or the height of equipment, which ever is greater

Note: Distances are measured to the finished covers.

Table 3-6 PDU Subsystem

Workspace Requirement	Minimum Clear Space	Additional Conditions
Direction of service access: All sides	36 in. (914.4 mm)*	No exposed live part hazards with cover in place. This component is typically serviced from all four sides. *48 in. (1219.2 mm), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. *42 in. (1066.8 mm), if the opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service access width: Left-Right of workspace	30 in. (762 mm)	This is the width of the workspace on each side of the equipment. 30 in. (762 mm) minimum or the width of the equipment, whichever is greater.
Note: Distances are measured to the finished covers.		

Table 3-7 Gantry Subsystem

Workspace Requirement	Minimum Clear Space	Additional Conditions
Direction of service access: Table head or foot	36 in. (914.4 mm)*	No exposed live part hazards with cover in place. This component is typically serviced from all four sides This is the width of the workspace on each side of the equipment. 30 in. (762 mm) min or the width of the equipment, whichever is greater.
Direction of service access: Table sides	36 in. (914.4 mm)*	
Direction of service access: Table foot [CT ONLY]	28 in. (711.2 mm)*	*18 in. (457.2 mm) minimum for Front Gantry Cover removal only if unobstructed egress space of 28 in. (711.2 mm) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service access width: Left-Right of workspace	36 in. (914 mm)	This is the width of the workspace in front of the equipment. 36 in. (914 mm) minimum or the width of the equipment, whichever is greater.
Note: Distances are measured to the finished covers.		

Table 3-8 Table Subsystem

Section 4.0 Room Size

4.1 Suggested Room Size

The suggested room size configuration 4267 mm x 6096 mm (13 ft. 6 in. x 22 ft) offers the most flexibility for future upgrades. It has sufficient workspace and space to add mill work, while meeting all regulatory requirements. This room is compatible with most two-step future installations.

4.2 Minimum Room Size

The minimum room size configuration 4267 mm x 6706 mm (13 ft. 6 in. x 19 ft., 3 in.) allows for no future upgrades. It has limited workspace and no in-room mill work, but meets all regulatory requirements. This room is not compatible with two-step future installations.

Section 5.0

System Specifications

System	Suggested Room Size	Minimum Room Size
LightSpeed 2.X	4267 mm x 6096 mm (13 ft. 6 in. x 22 ft.) ¹	4267 mm x 6706 mm (13 ft., 6 in. x 19 ft., 3 in.) ¹

¹ Same Regulatory requirements apply.

Table 3-9

5.1 Rooms With Less Than 711 mm (28 in) Egress Clearance at Table Foot End

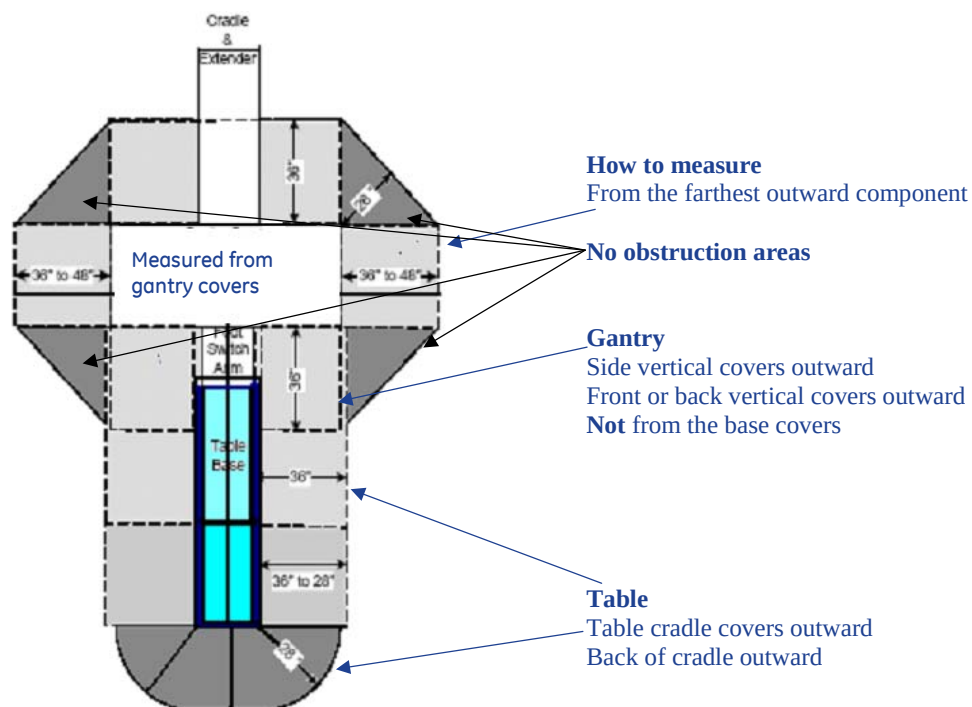
Egress requires a clear unobstructed route out of the room, either around the back of the gantry or around the foot-end of the table. If your egress route is not around the foot-end of the table, maintain a clearance of 457 mm (18 in) between the foot-end of the table and the wall. Additionally, there must be a continuous width of 3200 mm (126 in)—1600 mm (63 in) on each side of the table center-line—to any obstruction so that the front cover can be removed. This may limit placement of mill work within the room. Refer to the Pre-Installation manual for more details on service clearances.

5.2 Cautions

Rooms smaller than 4267 mm x 6706 mm (13 ft., 6 in. x 19 ft., 3 in.) may require construction to meet the minimum requirements. The design center or your Project Manager of Installation may have additional recommendations for your room size.

Section 6.0

How to Measure

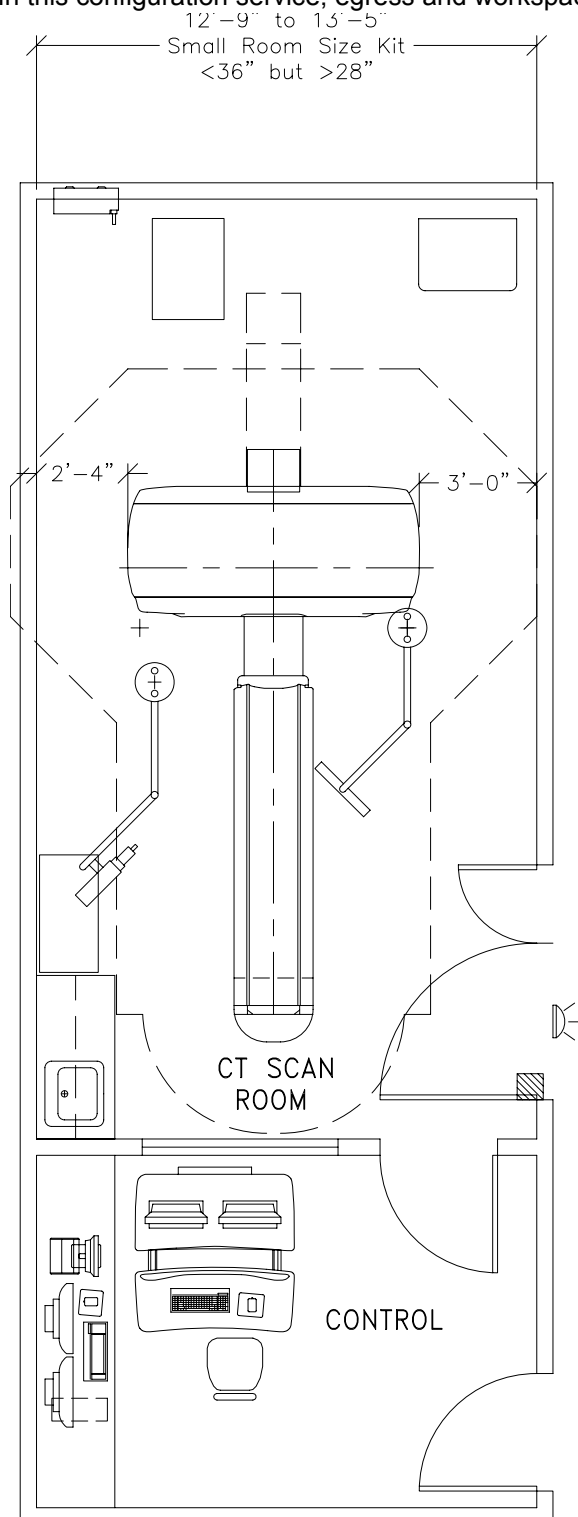


Section 7.0

Room Size and Requirement Layouts

Less than 36 in. (914 mm) but greater than 28 in. (711 mm) measured from the covers to the left sidewall.

In this configuration service, egress and workspace are acceptable around the Gantry.

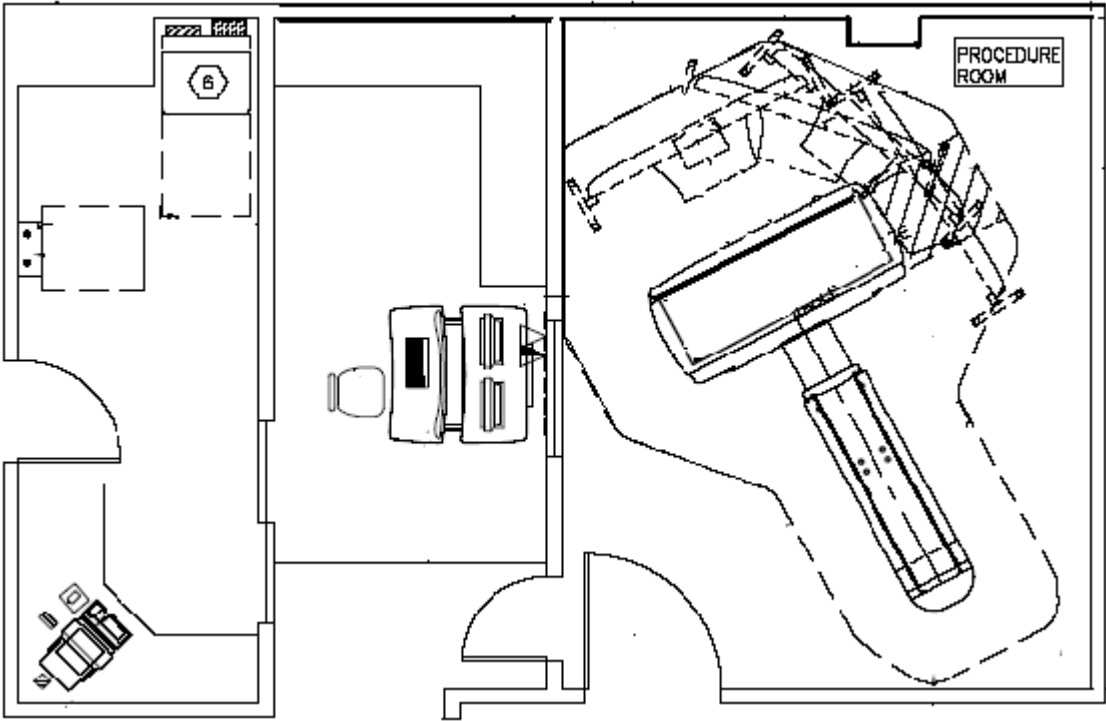


English to Metric Conversions:

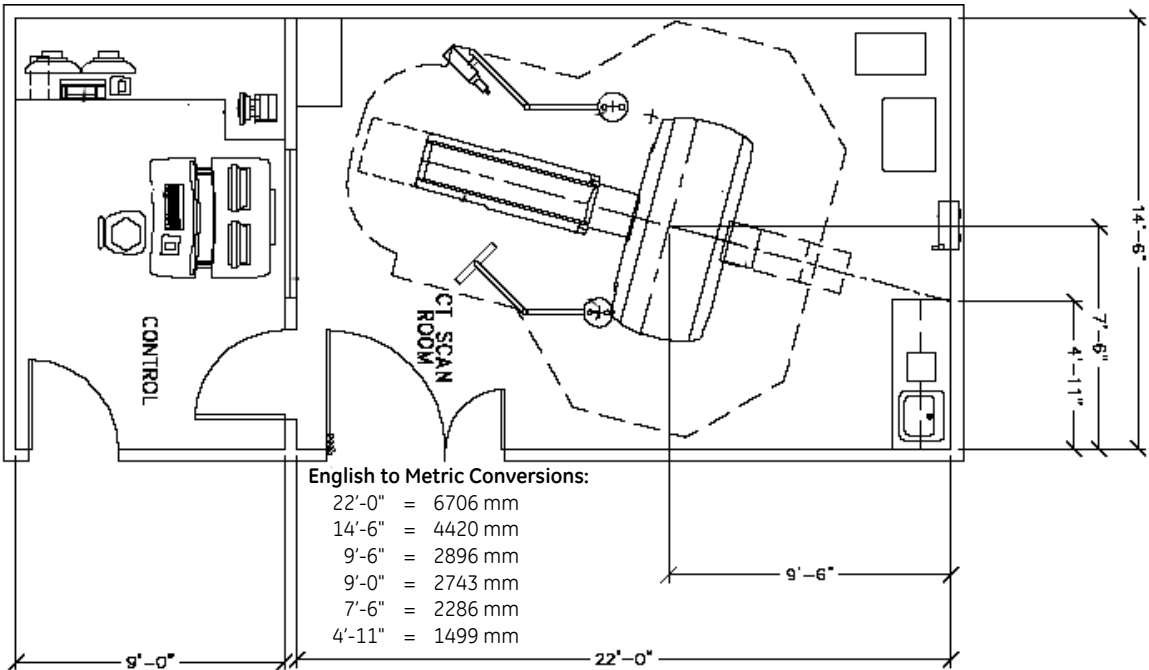
13'-5"	=	4089 mm
12'-9"	=	3886 mm
3'-0"	=	914 mm
36"	=	914 mm
2'-4"	=	711 mm
28"	=	711 mm

Section 8.0

Recommended Room Size and Requirement Layouts



Note: Your room layout may meet the recommended or suggested room requirements but look different from the room shown above. Contact your salesperson to have a detail room layout completed for your site.



This Page Intentionally Left Blank

(c) 2000, General Electric Company.

GE Medical Systems, a General Electric Company, going to market as GE Healthcare.

3200 N. Grandview Boulevard

Waukesha, Wisconsin 53149

USA

www.gehealthcare.com



GE Healthcare

LightSpeed 2.X Installation Manual

Book 2 of 2

Electrical Installation

OPERATIONS DOCUMENTATIONi



5266553-1EN
Revision 01

Effectivity

This manual applies to all LightSpeed 2.X systems released after 2006.

Table of Contents: Book 2

Chapter 4

Electrical Introduction.....	167
Section 1.0	
Introduction	167
Section 2.0	
Review Mechanical Hand Off Material.....	167
Section 3.0	
Required FE Common Tools and Supplies.....	168
3.1 FE Calibration and Service Tool List	168
3.2 Electrical Tools	168
3.3 Image Quality Calibration Tools.....	168
3.4 Detector Service Tools	168
3.5 Additional Calibration Tools	169
3.6 Safety Materials	169
3.7 Cleanliness	169
Section 4.0	
Requirements/Assumptions.....	170

Chapter 5

Electrical Power ON and Ground Checks	171
Section 1.0	
Compact PDU - Electrical Configuration.....	171
1.1 Required Tools	171
1.2 Initial PDU Configuration	172
1.2.1 Circuit Breakers	172
1.2.2 Relay Board	172
1.2.3 Power Switches	172
1.2.4 Hardware and Connection Check.....	172
1.2.5 Covers	172
Section 2.0	
Suite Emergency Off Checks	173
Section 3.0	
Line Transformer Settings.....	174
3.1 Requirements	174
3.2 Line Input Conditions	174
3.2.1 Compact PDU.....	174
Section 4.0	
System Power Up.....	176
Section 5.0	
Test HVDC Enable Switch	177

Section 6.0	
Emergency Stop Check.....	179
 Chapter 6	
Computer Integration.....	181
Section 1.0	
Introduction.....	181
Section 2.0	
System Configuration Data Sheets.....	182
2.1 Requirement.....	182
2.2 Manual Film Composer Options.....	183
2.3 System Network Configuration.....	183
2.4 Host Ethernet Address	183
2.5 Network Application (Image Transfer) Configuration	184
2.6 Camera Application Configuration	184
Section 3.0	
Configuration	186
3.1 Restore System State	186
3.2 Install Customer Options.....	187
3.2.1 Software	187
3.2.2 Camera.....	189
3.3 Monitor Setup.....	190
3.3.1 connecting the Sony CRT Monitor	190
3.3.1.1 Color Adjustments.....	190
3.3.1.2 Using Menu Buttons.....	191
3.3.1.3 Adjusting Brightness and Contrast.....	192
3.3.2 Connecting the LCD Monitor	192
3.4 Shut Down Application	193
3.5 Check/Set Date and Time	193
3.6 Reconfigure the Operator Console	193
3.6.1 Overview	193
3.6.2 Procedure.....	194
3.7 Applications Startup	199
 Chapter 8	
Table and Gantry Integration	207
Section 1.0	
Introduction.....	207
Section 2.0	
Autovoice/Intercom Check	208
2.1 Gantry Speaker	208
2.2 Console Speaker.....	209
2.3 Autovoice Volume	209
Section 3.0	
Table/Gantry Characterizations.....	210

3.1	Tilt Characterization	210
3.2	Elevation Characterization	211
3.3	Cradle Characterization	212
Section 4.0		
Interference Test		213
Section 5.0		
Tilt Speed Adjustment Procedure.....		214
Section 6.0		
Collimator Characterization and Flash Download		215
Section 7.0		
System Checks		216
7.1	Alignment Lights	216
7.2	Check Alignment Light Position	216
7.2.1	Internal Axial Lights	216
7.2.2	External Axial to Internal Axial Distance	217
7.2.3	Coronal Lights.....	217
7.3	Check X-ray Lights	217
7.4	Check Warning Labels.....	218
7.4.1	Console.....	218
7.4.2	Gantry	218
7.4.3	Table	218
7.4.4	Accessories	218
7.4.5	Laser	218
7.5	Process Product Locator Cards	219
Section 8.0		
Install the Service Cabinet.....		219
 Chapter 7		
DAS Integration.....		201
Section 1.0		
Introduction		201
1.1	Failure Protocol.....	201
1.2	Special Handling Precautions	201
1.3	Anti-static Handling	201
1.4	Cleanliness	202
1.5	Required Tools	202
1.6	Process Overview	202
Section 2.0		
DAS Tools		203
Section 3.0		
Verify X-Ray/Signal to Noise Test.....		205

Chapter 11

System Alignments 259

Section 1.0

Introduction 259

- 1.1 Required Tools 260
- 1.2 Gantry Geometry Definitions 260
- 1.3 Tube Heat Definitions 260

Section 2.0

Plane of Rotation (POR) 261

- 2.1 Overview 261
- 2.2 Tools Required 262
- 2.3 Procedure 262
 - 2.3.1 Start the POR Software 262
 - 2.3.2 Measure Tube Alignment 263

Section 3.0

Beam on Window Alignment (BOW) 267

- 3.1 Overview 267
- 3.2 Cool Tube 268
- 3.3 Access the Software 269
- 3.4 Adjustment Procedure 269

Section 4.0

CBF/SAG Alignment Process 271

- 4.1 Cool Tube 271
- 4.2 Access the Software 272
- 4.3 Adjustment Procedure 273

Section 5.0

ISO Alignment 274

- 5.1 Overview 274
- 5.2 Cool Tube 274
- 5.3 Access the Software 274
- 5.4 ISO Adjustment Procedure 275

Section 6.0

HOT ISO Alignment 276

- 6.1 Accessing the Software 276
- 6.2 Adjustment Procedure 276

Chapter 12

Generator Calibration 277

Section 1.0

Introduction 277

- 1.1 Process Overview 277
- 1.2 Required Tool 277

Section 2.0

mA Meter Check 278

Section 3.0	
Auto mA Cal/X-ray On Light Check	280
3.1 X-ray Lights.....	281

Chapter 10 System Tests..... 229

Section 1.0	
Introduction	229

Section 2.0	
Calibration Process.....	230

2.1 Reference Procedures	230
2.1.1 Scanning with Service Protocols.....	230
2.1.2 Center Phantom.....	230
2.2 Prepare the QA Phantom	230
2.3 Calibration Process Introduction.....	230
2.4 Fast Cal	231
2.5 Table/Gantry Alignment Procedure	231
2.5.1 Conditions	231
2.5.2 Procedure	231
2.6 Auto CT N Number Adjustment	232
2.7 Tomographic Plane Indication	232
2.8 System Scanning Test.....	233
2.9 Collimator Aperture Test.....	234
2.9.1 Small Aperture Test	234
2.9.2 Large Aperture Test.....	234
2.10 Generator Calibration	235
2.11 Process Overview	235
2.12 Required Tool	235
2.13 mA Meter Check	235
2.14 Auto mA Cal/X-ray On Light Check	237
2.15 X-ray Lights.....	238

Section 3.0	
Image Series	239

3.1 Scan Protocol	239
3.2 Data Recording: Means and Standard Deviation	239
3.3 Term Definitions & Screens	239
3.4 48cm Phantom Image Series Image Performance Verification	241
3.4.1 Acquiring the 48cm Phantom Image Series.....	241
3.4.2 Image Performance Verification.....	242
3.4.2.1 Brightness Uniformity and Noise	242
3.4.2.2 Failure Recovery.....	244
3.5 20cm QA Phantom Image Series Image Performance Verification	245
3.5.1 Acquiring the 20cm QA Phantom Image Series	245
3.5.1.1 High Contrast Spatial Resolution.....	245
3.5.1.2 Failure Recovery.....	248
3.5.2 Low Contrast Detectability	250
3.5.2.1 Image Performance Verification	250
3.5.2.2 Failure Recovery.....	252

3.5.3	QA#3 Phantom Brightness Uniformity and CT#.....	252
3.5.3.1	Performance Verification.....	252
3.5.3.2	Failure Recovery	255
3.5.4	QA#3 Phantom Noise.....	255
3.5.4.1	Performance Verification.....	255
3.5.4.2	Failure Recovery	257
Section 4.0		
Save System State.....		258

Chapter 13

Hospital/System Network Integration..... 283

Section 1.0		
Overview		283

Section 2.0		
Before You Start		284

2.1	Network Physical Requirements	284
2.2	LightSpeed Network Identity Information	284
2.3	LightSpeed to DICOM Remote Hosts Network Information	284
2.4	LightSpeed to DICOM HIS/RIS Interface Network Information	285
2.5	LightSpeed to DICOM Printers Network Information	285

Section 3.0		
Declaring the System on the Hospital Network		286

3.1	Enter Configuration Routine.....	286
3.2	Configure Network Settings	287
3.3	Initiate System Reconfiguration	289

Section 4.0		
Declaring Remote Hosts on the System.....		290

4.1	Enter Remote Host Configuration Screen.....	290
4.2	Declaring Advantage NET Remote Hosts on the LightSpeed.....	290
4.3	Declaring DICOM Remote Hosts on the LightSpeed	291

Section 5.0		
Declaring the System on Remote Hosts.....		294

5.1	Declaring LightSpeed on Advantage NET Protocol Devices/Systems.....	294
5.2	Declaring LightSpeed on DICOM Protocol Devices/Systems	294

Section 6.0		
Declaring DICOM HIS/RIS Interfaces on the System.....		295

6.1	Prerequisites	295
6.2	Loading ConnectPRO Software Option on the System	295

Section 7.0		
Declaring DICOM Filming Devices on the System		297

7.1	Prerequisites	297
7.2	Declaring DICOM Filming Devices on the System.....	297
7.3	Troubleshooting Tips.....	308

Section 8.0		
Teleradiology (Framegrabber Type) Systems and Auxiliary Monitors.....		310

8.1	Introduction	310
8.2	Auxiliary Monitors	311
8.3	GEHC Analog Filming Interface Specs (Video & Serial)	312
 Chapter 10		
GE Forms.....		151
Section 1.0		151
1.1	Customer Site Data	151
 Chapter 15		
Installation Completion Checklist.....		327
 Appendix A		
Additional Characterization Procedures		163
Section 1.0		
Tilt Characterization		163
Section 2.0		
Elevation Characterization		164
2.1	Prerequisite.....	164
2.2	Procedure	164
Section 3.0		
Cradle Characterization		165

Chapter 4

Electrical Introduction



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- When directed, record data collected from procedures in this chapter into GE Form 4879 located on the CD for this scanner's service manual.
- Only use the installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

Use the continuity and ground checks to:

- verify that the system power connections have not shorted to ground
- the ground and neutral connections are intact.

Section 2.0

Review Mechanical Hand Off Material

Complete the Mechanical Hand Off checklist:

- ☐ All options were installed. If not, contact your install specialist.
- ☐ Check for short ships.
- ☐ Review cable connections with mechanical team.
- ☐ Complete mechanical vendor evaluation.

Section 3.0

Required FE Common Tools and Supplies

3.1 FE Calibration and Service Tool List

Special CT Tools used for mechanical alignments (found in the install support kit):

- 5 mm hex bit for 3/8 in. drive 6 in. long (Snap-On FAML5E or equivalent)
- 10 mm hex bit for ratchet wrench (ball end) (Snap-On FABM10E or equivalent)
- 2.5 mm Allen hex bit for 1/4 in.
- Standard FE Tool Kit (not included in install support kit)
- 10 mm open-end thin wrench (Snap-On SRSM10 or equivalent)
- 21 mm open-end thin wrench (Snap-On LTAM2124 or equivalent)
- 14 mm ball hex socket bit for 3/8 in. drive
- 14 mm hex socket bit for 1/2 in. drive

3.2 Electrical Tools

These tools must be calibrated yearly.

- Fluke 87 DVM or equivalent
- Clamp on amp meter
- Dale Meter

3.3 Image Quality Calibration Tools

- GE Performance Phantom
- QA Phantom (2206352)
 - 20 cc (2144721)
 - 40 cc (2144721-2)
 - Phantom Holder

3.4 Detector Service Tools

These tools can be found in the install support kit.

- 2.5 mm hex bit
- Antistatic bags
- Electric Screw Driver
- Canned Air (Grounded) with metal nozzle (Miller Stephenson Aero Duster MS-222N or equivalent)
- Ground Wrist Strap
- Grounded Mat
- Dental Pick

3.5 Additional Calibration Tools

Needed for warranty recalibration:

- GE HV Bleeder
- Scope with 10X probes
- Beckman CT231 clamp-on Amp probe: 46-194427P228 (Fits up to #2 size wire)
- Beckman CT 232 clamp-on Amp probe: 46-194427P270 (Fits #1 size wire and larger)
- Scope Probe-to-bleeder Cable 46-219921G1 (Belden #8422 cable, 30 ft.)
- Polaroid type 52 film and developer
- Radial dial indicator (mm or in) and mounting bracket
- Caliper Dial indicators (mm or in)
- Ground Rod Tester Clamp-on Amp Probe
- Torque Wrench kit

3.6 Safety Materials

- Lockout/Tagout kit, or equivalent
- Safety Glasses
- Nitrile Rubber Gloves (part number 2207303-6 [large], 2207303-7 [extra large])

3.7 Cleanliness

Any dirt on the surface increases leakage current on the filter or converter cards. This causes the DAS to fail the drift spec. Wear Nitrile gloves when you handle the DAS. Fingerprints on the board cause problems during humid conditions. Use only clean, new Nitrile gloves. Do not use hospital grade gloves.



NOTICE **Never use an eraser to clean any part of the DAS.**
Dust covers are required for installations on construction sites.

Section 4.0

Requirements/Assumptions

- All stations in a suite must have the same Suite ID.
- You need the name of all hosts in the suite and their corresponding Internet/Ethernet numbers.
- You need the Internet Protocol (IP) addresses the first time you execute a reconfig on the system.
- When you connect the system to a network, contact the system administrator to obtain the IP addresses for all the computers in the suite.
- You also need an IP address for each gateway (second) Ethernet board in any OC or IC.
- You can use the default internet number on stand-alone systems (stand-alone = not connected to any network).
- For more detailed information and instructions regarding Network Integration, see [Chapter 13 Hospital/System Network Integration](#).

Chapter 5

Electrical Power ON and Ground Checks



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- When directed, record data collected from procedures in this chapter into GE Form 4879 located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Compact PDU - Electrical Configuration



WARNING



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

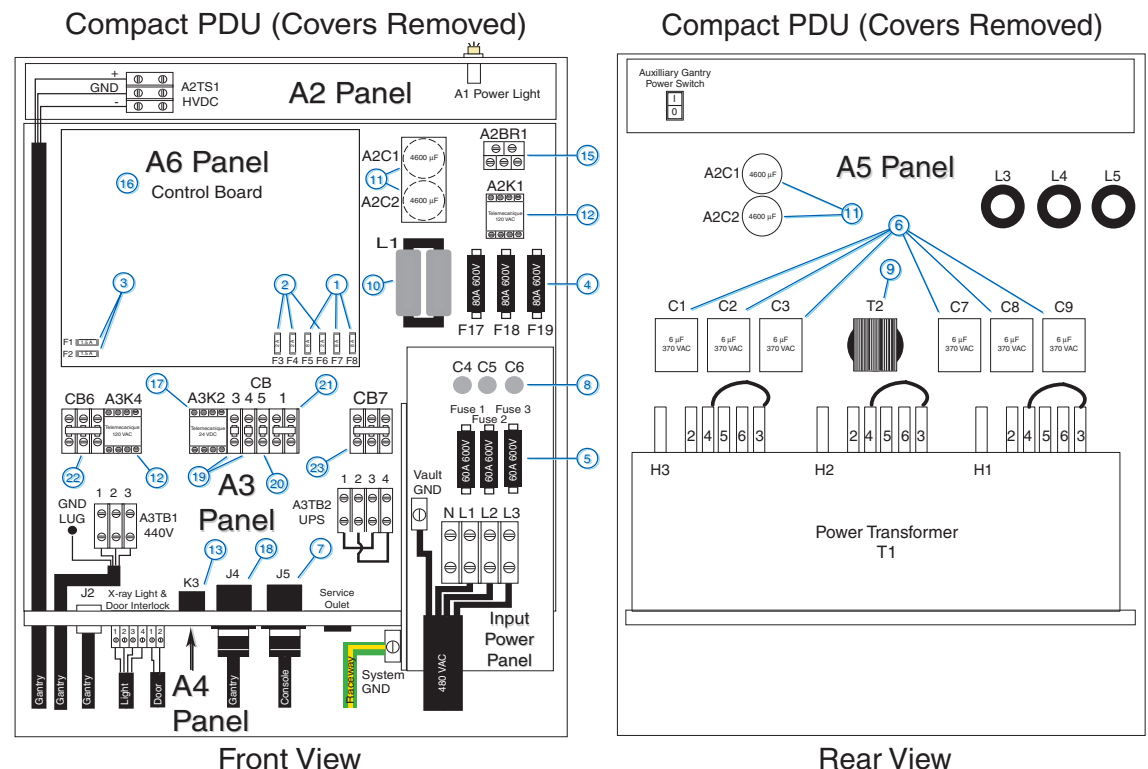
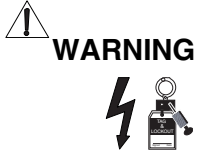


Figure 5-1 Compact PDU

1.1 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

1.2 Initial PDU Configuration



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

1.2.1 Circuit Breakers

Set all circuit breakers to OFF.

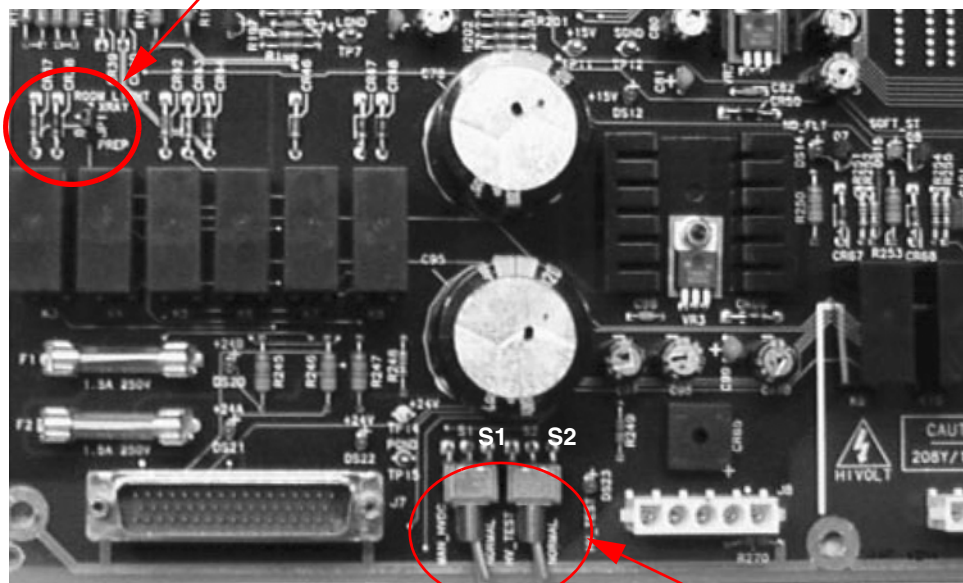
1.2.2 Relay Board

- 1.) Set SW1 and SW2 to the normal position.
- 2.) Set the Room Light jumper (JP1) to meet hospital requirements (see [Figure 5-2](#)).
 - Position A enables the room warning light during X-ray only.
 - Position B enables the room warning light as soon as the HVDC turns on.

Note:

Use position B unless the hospital requires otherwise.

Verify Room Light jumper is in the prep/ b position.



Verify both switches are in the normal position.

Figure 5-2 PDU Control Board (A6 Panel)

1.2.3 Power Switches

Turn OFF all power switches on all subsystems (gantry status display box, STC, DAS, table, console, computer, and SCU).

1.2.4 Hardware and Connection Check

Check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

1.2.5 Covers

Install, or verify the presence of, all the Lexan safety covers.

Section 2.0

Suite Emergency Off Checks

Use this procedure for systems with 2133533 (PDU).



WARNING



VERIFY THAT ALL PERSONNEL HAVE CLEARED THE SYSTEM BEFORE YOU TURN ON THE WALL POWER.

- 1.) Turn the wall power ON to the PDU.
- 2.) **Press the suite emergency off button and verify it turns off wall power to the PDU.**
[Typically, this red palm button is located on the wall close to the console, within the scan suite.]
- 3.) Verify that all "Emergency Off" buttons are working properly.
- 4.) Leave power "OFF."

Section 3.0

Line Transformer Settings

3.1 Requirements

- 1.) The PDU is shipped configured for 480 VAC.
- 2.) Complete only if your site uses a voltage other than 480 VAC.
- 3.) If the PDU is configured for 480 VAC, go to [Section 4.0](#). Otherwise, proceed to [Section 3.2](#).



WARNING

TURN OFF, TAG AND LOCKOUT THE MAIN WALL POWER BEFORE CHANGING TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION. TURN OFF THE WALL POWER TO CONNECT OR MOVE METER LEADS, OR TO REMOVE OR INSTALL COVERS.

3.2 Line Input Conditions

3.2.1 Compact PDU

- 1.) Remove the metal cover over the right A3 panel.
- 2.) Monitor the No Load Line-to-Line Voltage at A3 L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 3.) Remove the PDU’s back cover. (Refer to [Figure 5-1](#) for a back view of the PDU with the cover off.)
- 4.) After you determine the nearest nominal line, verify the tap connections match. Refer to [Table 5-1](#) and [Figure 5-3](#) for tap locations.

Transformer Taps and Jumpers

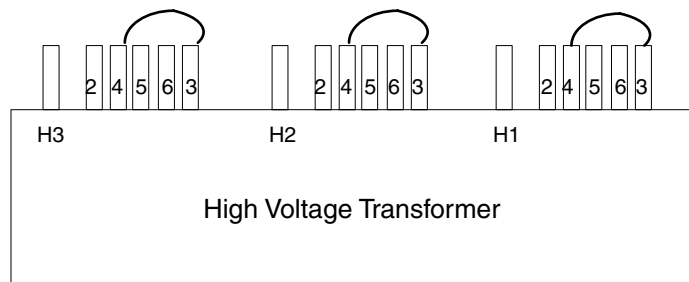


Figure 5-3 PDU Tap Positions (Rear)

Note:

Taps should be shipped as shown for 480 VAC only. For all others, move the taps. The mechanical installer completes the tap check.

- 5.) Verify that the No Load Line-to-Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 5-1](#).
- 6.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at A3 L1, L2, and L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



WARNING



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

7.) Replace the metal cover over the right A3 panel.

NO LOAD		TAP CONNECTIONS		
Line to Line Voltages		(All 3 phases must have same the configuration)		
Nominal	Maximum Range (8%)	Phase A Connection	Phase B Connection	Phase C Connection
480 V*	442 to 518*	3-4*	3-4*	3-4*
460 V	423 to 497	3-5	3-5	3-5
440 V	405 to 475	3-6	3-6	3-6
420 V	386 to 454	2-4	2-4	2-4
400 V	368 to 432	2-5	2-5	2-5
380 V	350 to 410	2-6	2-6	2-6
* Factory Default				

Table 5-1 PDU 2269902 Line Tap Connections

Section 4.0

System Power Up

- 1.) Turn OFF the wall power.
- 2.) Turn ON all PDU A3 circuit breakers.



CAUTION Verify that all personnel have cleared the system before you turn on the wall power.

- 3.) Turn ON the wall power.
- 4.) Check the A1 Emergency Off push button.

SUB-SYSTEM POWER-UP

- 1.) Turn ON switch S3 in the table (120vac 24hr power).
- 2.) Turn ON the gantry 120VAC enable switch on the STC. Tube fans turn on.
- 3.) Leave AXIAL DRIVE ENABLE OFF.
- 4.) Leave HV DC ENABLE OFF.
- 5.) Turn ON the DAS power switch located on the outside of the DAS power supply assembly.
- 6.) Turn ON Table switches S1 and S2.
- 7.) Turn ON the console power.
- 8.) Ensure that the switch on the back of the PDU is ON.
- 9.) Turn the HEMRC Power ON.
- 10.) Verify that the gantry 210 VAC is ON at the Gantry Power Pan.

Section 5.0

Test HVDC Enable Switch

Note: Console must be booted and operational.

- 1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.



- 2.) Click the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select TABLE/GANTRY INTEGRATION.
- 5.) Select HV DC ENABLE SWITCH.
- 6.) Select START.
- 7.) Select the following parameters in the Rotor Functional window:

PARAMETER	MX200
Test Selection:	HEMRC
Rotor Speed:	LOW
Rotor Loop:	OPEN
Rail Diff Voltage:	Normal
Test Duration:	10 seconds
OBC Service Input:	Disabled

Table 5-2 Rotor Functional Window Parameters

- 8.) Verify that all personnel have cleared the system *before* you enable the HVDC.
- 9.) Verify the following:
 - a.) HVDC enable switch on the STC is ON
 - b.) "HVDC ENABLE ON" light on the STC is ON



Figure 5-4 STC Service Switches

- 10.) Select RUN. If the system asks for a diagnostic download at this time, answer YES, and wait 90 seconds.
- 11.) Verify the "HVDC BACKUP CONTACTOR CLOSED LIGHT" on the STC status display turns ON.
- 12.) When the test finishes, verify that the system displays the following message along the bottom of the screen: Diagnostics have completed with zero errors reported.

VERIFY/PERFORM

- 1.) Turn OFF the HVDC enable switch on the gantry service switch box.
- 2.) Select RUN to run the test again.
- 3.) Verify the test aborts.
- 4.) Verify the message log contains the following error: *Gentry I/O backup contactor error.*
- 5.) Select FILE → EXIT.
- 6.) Select DISMISS.
- 7.) Select CLEANUP.
- 8.) Wait 90 seconds for the following message to appear on the screen: "Scanning Hardware Reset Successful"

AXIAL ENABLE SWITCH TEST

- Note: For the initial condition, do not leave the tube at the 2:30 position.
- 1.) Turn OFF the axial drive enable switch S2 on the STC backplane.
 - 2.) Clear the gantry area for rotation.
 - 3.) Press the alignment light push button.
 - 4.) Verify that the gantry did not rotate.
 - 5.) Turn ON axial drive enable switch S2 on the STC backplane.



WARNING

MAKE SURE THAT THERE ARE NO OBSTRUCTIONS AROUND THE GANTRY. PRESSING THE ALIGNMENT LIGHT PUSHBUTTON CAUSES THE GANTRY TO ROTATE.

- 6.) Press the alignment light push button.
- 7.) Verify that the gantry rotates.

ROTATION SAFETY CHECKLIST

- 1.) Turn OFF enable switch S2 on the gantry.
- 2.) Turn OFF HV enable switch S3.
- 3.) Press the red E-STOP button.
- 4.) Manually rotate the gantry 360 degrees. (Keep one finger at the console E-STOP button.)
 - Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - Listen for any loose parts.
(Tighten parts as needed.)
- 5.) Turn ON the enable switch S2 on the gantry.
- 6.) Turn ON the drive power switch S1.
- 7.) Leave the large gantry bay door half open. (This makes it easier to hear any loose or interfering parts.)
- 8.) Using DDC, prescribe a four-second X-ray OFF scan. (Keep one finger at the console E-STOP button.)
 - Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - Listen for any loose parts.
(Tighten parts as needed.)
- 9.) After completing the four-second scan, repeat Step 8 with 2.0, 1.0, 0.7, and 0.5 second scans

Section 6.0

Emergency Stop Check

- 1.) Use the gantry push buttons to advance the cradle about 0.5 m (2 ft.) from the home position.
 - 2.) Press one of the E-STOP buttons on the gantry.
- Note: Emergency Stop buttons are located on the front and rear of the gantry (eight in all). They are also located on both sides of the table base (four in all) and one on the SCIM (see [Figure 5-6](#)).
- 3.) Ensure that the TABLE POWER shuts off, and the green LED flashes.
 - 4.) Depress one of the table elevation buttons to verify the emergency stop disabled table elevation.
 - 5.) Depress one of the cradle drive buttons to verify the emergency stop disabled the cradle drive.
 - 6.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER.
 - 7.) Press the other E-STOP button on the gantry.
 - a.) Ensure that the table power shuts off.
 - b.) Manually move the cradle to the home position to ensure that the cradle clutch released.
 - c.) Ensure that the cradle latches securely in the home position.
 - 8.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER.
 - 9.) Press one of the four table tape switches to ensure that the X-RAY DRIVES POWER shuts off.
 - a.) Turn on TABLE POWER.
 - b.) Repeat with the three remaining table tape switches.
 - 10.) Press the console emergency stop switch; ensure that the Table Power shuts off.
 - 11.) Turn on STC 120 power.
 - 12.) Exit.
 - 13.) Return to Service Desktop home page.

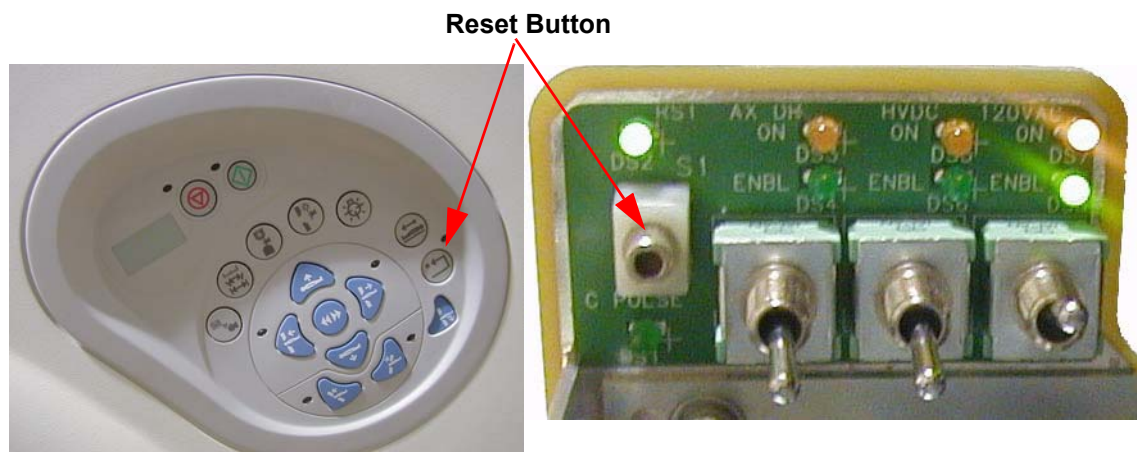


Figure 5-5 Reset buttons on the Gantry and STC Service Switch bank

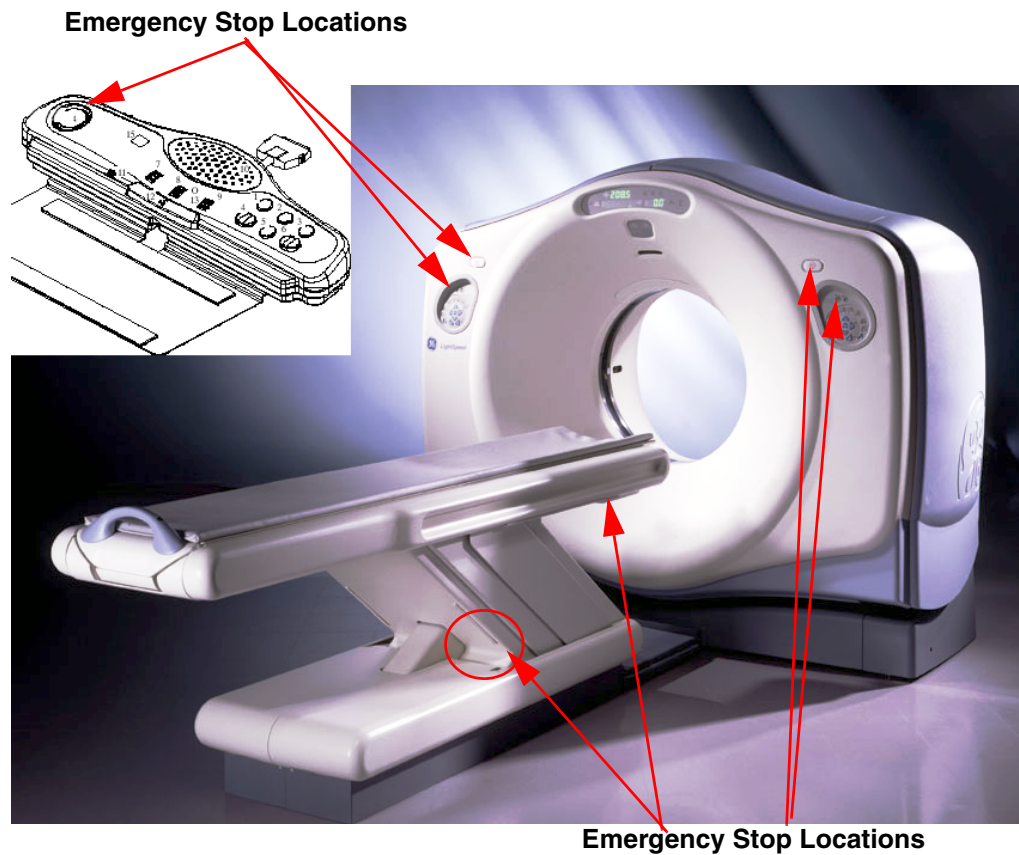


Figure 5-6 Emergency Stop button locations

Chapter 6

Computer Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- When directed, record data collected from procedures in this chapter into GE Form 4879 located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

This chapter describes the reconfiguration, system state restore, options, and monitor adjustment procedures.

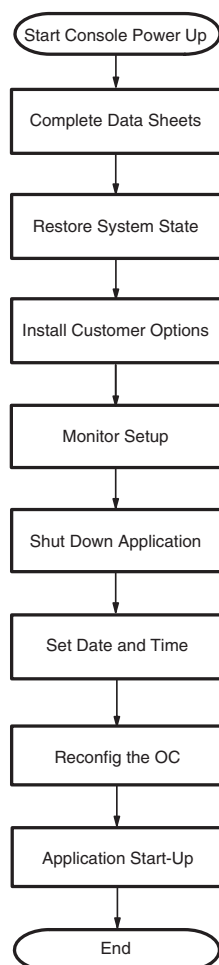


Figure 6-1 Computer Integration Process Overview

Section 2.0

System Configuration Data Sheets

2.1 Requirement

Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their LightSpeed scanner, and their network and filming expectation reduces the time required to reconfigure the system.

- [Table 6-1 Manual Film Composer Options](#)
- [Table 6-2 System Network Configuration](#)
- [Table 6-3 Networking Application \(Image transfer\) Configuration](#)
- [Table 6-4 DASM Laser Camera Configuration](#)
- [Table 6-5 DICOM Print Camera Configuration](#)
- [Table 6-6 DICOM Print Camera Advanced Configuration](#)

2.2 Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS

Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table 6-1 Manual Film Composer Options

2.3 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* ask customer or checklog	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table 6-2 System Network Configuration

2.4 Host Ethernet Address

_____ : _____ : _____ : _____ : _____

2.5 Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

Networking Application (Image transfer) Configuration				
AE Title or Host Name	Network Address	Network Protocol	Port Number	Comments

Table 6-3 Networking Application (Image transfer) Configuration

2.6 Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM Laser Camera Configuration	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table 6-4 DASM Laser Camera Configuration

DICOM Print Camera Configuration	
Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table 6-5 DICOM Print Camera Configuration

DICOM Print Camera Advance Configuration	
Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table 6-6 DICOM Print Camera Advanced Configuration

Section 3.0 Configuration

Note: Type the text shown in **boldface**, and press the ENTER key on the keyboard.

3.1 Restore System State

Tools Required: None

Your system should have a system state MOD, located in the software collector box.
The system state MOD contains:

- Collimator Characterization
- Phantom Calibrations

The installation process uses all the system state files. At this time, use the system state DVD or MOD to restore all of the files.

If you cannot locate an existing system state MOD or DVD, manually characterize the system and perform calibrations.

1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.

2.) Click the PM icon.



3.) Select SYSTEM STATE.

4.) Insert the MOD in the MOD drive.

5.) Select CHARACT.

6.) Select CALS.

7.) Select RESTORE to restore the system characterization and phantom calibration files to the system.

Note: **Restore State can take as long as 10 minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.**

If any error occurs during the restore process, see the Software Load Procedure manual for information regarding possible error messages and their recovery.

8.) Click NO for Reset Scan Hardware pop-up message.

9.) Select DISMISS.

3.2 Install Customer Options

Note: Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Refer to:

- GE Prints and schematics for mechanical (physical) location of each option.
- FDO shipment for identification of items.
- Project Manager of Installation for installation instructions, if they differ from the printed version.

CT OPTIONS

- ☐ Injector functional tests completed.
- ☐ AW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
 - * Direct 3D
 - * SmartScore Pro
 - * CT Perfusion
- ☐ Teleradiology Connections (Refer to [Chapter 13, Section 8.0, Teleradiology \(Framegrabber Type\) Systems and Auxiliary Monitors.](#))
- ☐ Auxiliary Monitor
 - * Flat Panel
 - * CRT

3.2.1 Software

Tools Required: None

Note: **Your system may have a MOD that contains customer purchased options. If your system has an options MOD, install it at this time.**

Ensure that the options MOD is NOT write-protected at this time. The MOD must be write-enabled for initial installation, but you can do subsequent installations with the MOD write-protected.

- 1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.
- 2.) Click the UTILITIES icon.
- 3.) Select INSTALL.

- 4.) Select INSTALL OPTIONS.

An Options Window is displayed (Figure 6-2):

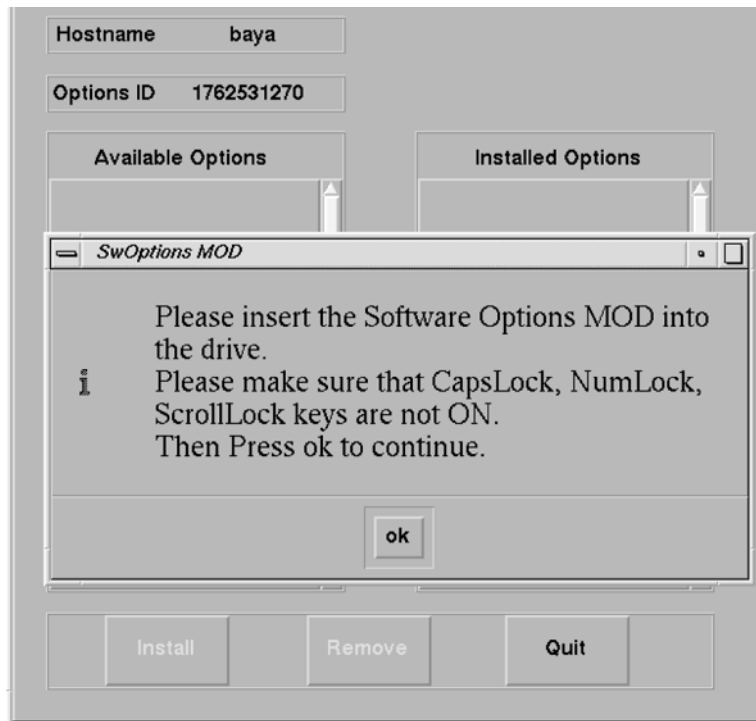


Figure 6-2 Options Window when first selected

- 5.) Insert the options MOD into the MOD drive and click OK. (If you do not have an options MOD, click OK anyway, wait for the abort pop-up, and abort the process.)

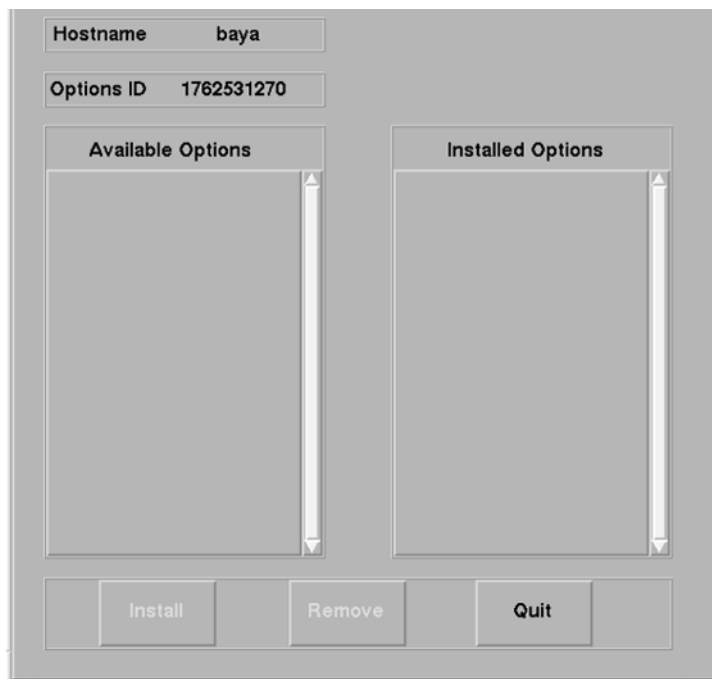


Figure 6-3 Example: Options Window

- 6.) Select all of the options in the left-hand column to install the corresponding software.

- 7.) Select INSTALL. A box may appear while the options are loading. When an option is displayed in the *Installed Options* list, then installation of that option is complete.

Note: Some options take a fraction of a second to install, while options like 3D may take a half minute (due to the fact that they are installing software).

- 8.) After the options are installed, select QUIT.
- 9.) Select OK.
- 10.) Remove the MOD and write-protect the side with options.
- 11.) When the system prompts to Reboot, click YES, and reboot the system to complete the installation.

3.2.2 Camera

Tools Required: Small flat-blade screw driver

Note: DASM hardware must be installed before proceeding.

For details on camera configuration, refer to the Software Load Procedures manual.

For details on troubleshooting the camera, refer to the System Service manual.

- 1.) Click on the SERVICE DESKTOP icon.
- 2.) Select UTILITIES.
- 3.) Select INSTALL CAMERA.
- 4.) Select ADD.
- 5.) Select DASM or DICOM.
- 6.) Follow procedures on the screen.
- 7.) Return to the Home Page.

3.3 Monitor Setup

Note: If using a Sony CRT monitor, complete the setup in [Section 3.3.1, connecting the Sony CRT Monitor](#).

If you are using an LCD monitor, use the setup procedures in [Section 3.3.2, Connecting the LCD Monitor](#).

3.3.1 connecting the Sony CRT Monitor

Tools Required: Sony Service Manual (located in monitor box)

3.3.1.1 Color Adjustments

- 1.) Adjust the ambient room light to the level the operator uses during image review.
- 2.) Select the COLOR menu to adjust the picture color temperature. (You can use this to match the monitor colors to a printed picture colors.)

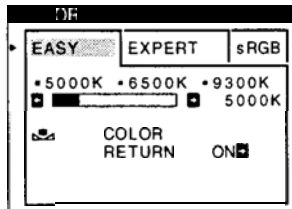


Figure 6-4 Color Menu

- 3.) Set the monitor color temperature to 6500 K. (Refer to the system service manual for this procedure, if necessary.)
- 4.) Use one of the following methods:



- a.) Click the IMAGEWORKS icon and proceed to step 3.

OR

- b.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.



- c.) Click the IMAGE QUALITY icon.

- d.) Select INSTALL SMPTE IMAGE and proceed to step 3. (Approximately 3-4 minutes.)

- 5.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.
- 6.) Increase the Contrast to maximum.

Note: Adjust the monitor contrast until the operator sees the anatomical structure (window width) at the expected window levels.

- 7.) Increase the Brightness to maximum.

Note: Adjust the monitor brightness until the operator sees the anatomical structure (window width) at the expected window levels.

- 8.) Decrease the Brightness until the raster fades into, and matches, the monitor screen background. The 5% and 95% patches should be just visible.

If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the Brightness control.

If the CRT image exhibits any tearing or smearing of the alphanumeric characters, reduce the contrast setting slightly until the tearing/smearing is eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.

You should always finish by displaying and filming images of anatomy (typical heads and bodies). Ask the technologist to compare the CRT image to the film image.

- 9.) To view SMPTE on the text monitor, swap the video cable.

3.3.1.2 Using Menu Buttons

- 1.) Press the Main Menu button to display the main menu on your screen.



Figure 6-5 Main Menu Button

- 2.) Press the / buttons to highlight the menu you want to adjust. Press the Main Menu button to select the menu item.



Figure 6-6 Select Menu Item

- 3.) Press the / buttons to select the desired adjustment item. Press the / buttons to make the adjustment.



Figure 6-7 Adjust the Menu

- 4.) Press the Main Menu button once to return to the main menu, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.



Figure 6-8 Main Menu Button

- 5.) Press the reset button to reset the adjustments back to the factory settings.



Figure 6-9 Reset Button

3.3.1.3 Adjusting Brightness and Contrast

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu. These settings are stored in memory for the signals from the currently selected input connector.

- 1.) Press either one of the Brightness or Contrast buttons. The BRIGHTNESS/CONTRAST menu appears on the screen.

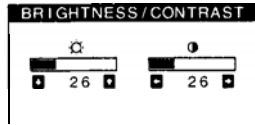


Figure 6-10 Brightness/Contrast Menu

- 2.) Press the / buttons to adjust the brightness and the / buttons to adjust the contrast.

3.3.2 Connecting the LCD Monitor

With the monitor and computer switched off, connect the video signal cable to the monitor video input.



NOTICE Equipment Damage Possible

Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, otherwise the pins might bend.

Connect the following cables:

SCIM:

- SCSI cable
- Template
- Keyboard

Scan Monitor:

- Video cable
- Power cable
- Cable keeper

Display Monitor:

- Video cable
- Power cable
- Cable keeper

Note: A DARC video cable may be shipped with the system. Connect it only when troubleshooting. Do not connect the cable during normal installation.

3.4 Shut Down Application

Tools Required: None

- 1.) If Applications is currently running, you must shutdown system applications to the CT user desktop.
 - a.) Click the SERVICE DESKTOP button.
 - b.) On the desktop toolbar select UTILITIES.
 - c.) Select APPLICATIONS SHUTDOWN (to bring down applications only).
- 2.) Open a UNIX Shell window.
 - a.) Type: `su -` ENTER at the prompt.
 - b.) Enter the root (super user) password.

3.5 Check/Set Date and Time

Tools Required: None

Check the date and time.

- 1.) Type/enter `setdate` ENTER.
- 2.) Follow the instructions to set the correct time and date.

3.6 Reconfigure the Operator Console

Note: The document collector box that arrived with your system contains the LightSpeed *Software Load Procedure* manual, which documents the reconfiguration procedure in more detail.

3.6.1 Overview

On the following screens, you should make the changes necessary. Press the corresponding button at the top of the screen to move from screen to screen. When finished, press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

```
/var/adm/install.log.YYYYMMDDWWHHMMSS
```

Where

`YYYYMMDDWWHHMMSS` is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens are incorrect, DO NOT try to stop the reconfiguration. Wait for the reconfiguration to complete, and rerun reconfig, entering the correct parameters.

3.6.2 Procedure

- 1.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the prompt.
- 2.) Launch the LightSpeed Install utility:
Type: `reconfig` ENTER at the prompt.
The OC displays the LightSpeed Install Utility Window as shown in [Figure 6-11](#).

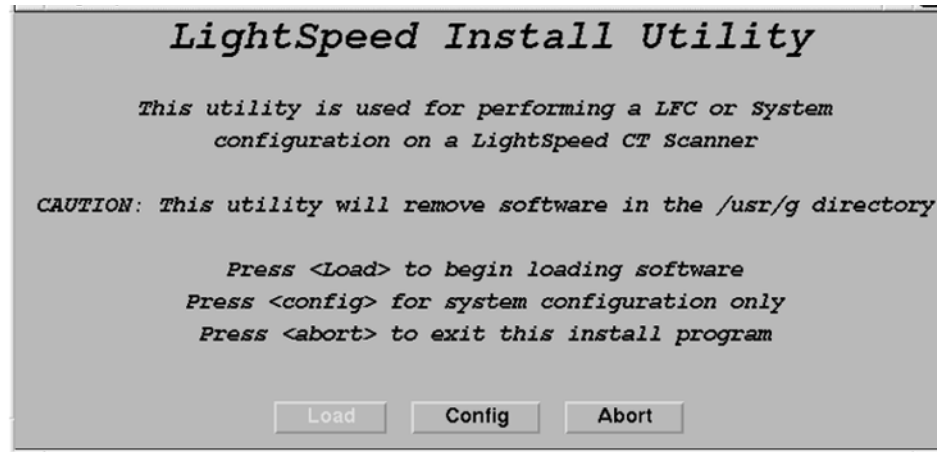


Figure 6-11 Install Utility Window

- 3.) Using the mouse, click the CONFIG button.
The OC displays the LightSpeed System Configuration - System Settings Screen as shown in [Figure 6-12](#).

Comment: The following pages show examples of the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens vary depending on the current configuration of your system.

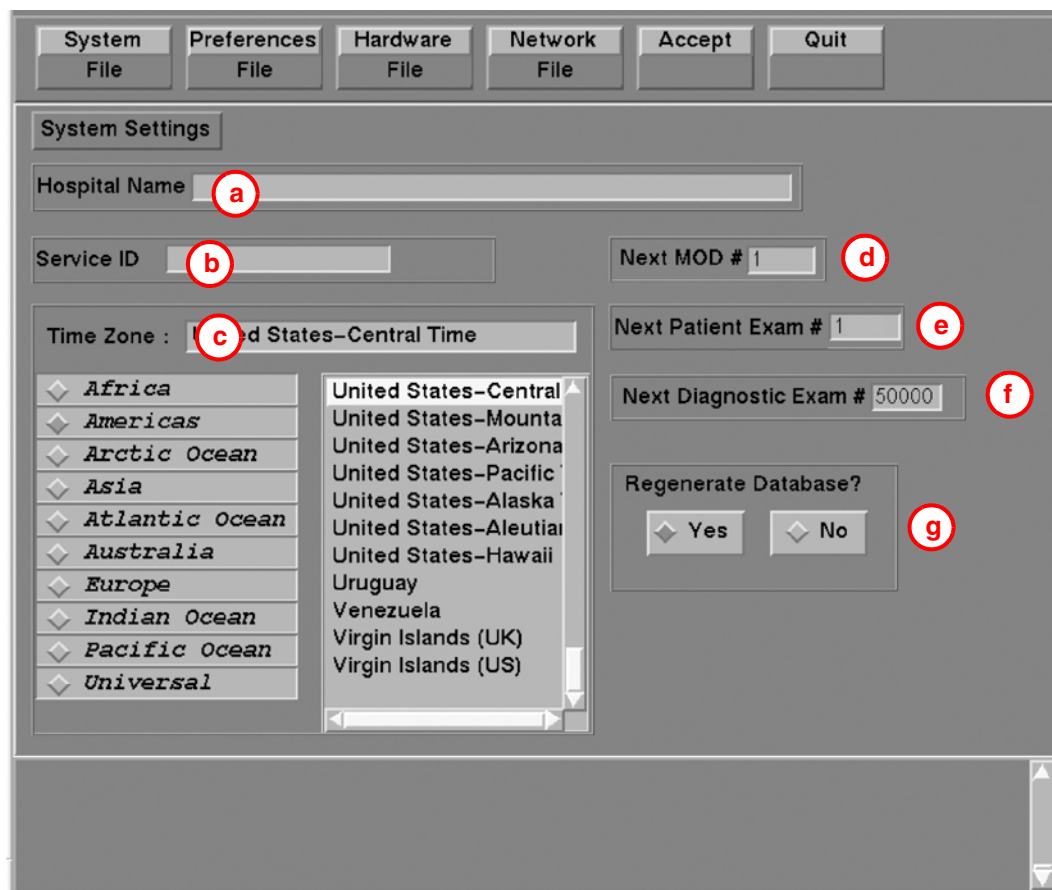


Figure 6-12 System Settings Screen

4.) Configure System Settings (refer to [Figure 6-12](#))

- a.) Hospital Name: configures the name that displays on images produced by this scanner.
Example: ST MARYS HOSPITAL
- b.) Service ID: issued by the Service organization.
Example: 262785CT2 (no spaces)
- c.) Time Zone: select the time zone for the site.

Note:

Use the scroll bar at the bottom of the time-zone selection list to view the entire description of the time-zone you are want to use. This ensures that you select the correct time-zone for your location.

If the time-zone of your location is not in the list, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time do not take effect. See the Load from Cold manual if you require more information regarding time-zone setting and selection.

- d.) Next MOD #: *Customer selected.*
- e.) Next Patient Exam #: configures the next Exam number the scan user interface uses.
At initial system installation, type: 1
- f.) Next Diagnostic Exam #: *Customer Selected.*
- g.) Regenerate Database: determines if the Scan Database is regenerated during reconfiguration.

Note: If YES is selected, all scan data is deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Ensure that all images are reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See the Load from Cold manual if you require more information regarding Scan Database Regeneration.

- 5.) Select the PREFERENCES button to display the Preference Settings Screen as shown in Figure 6-13.

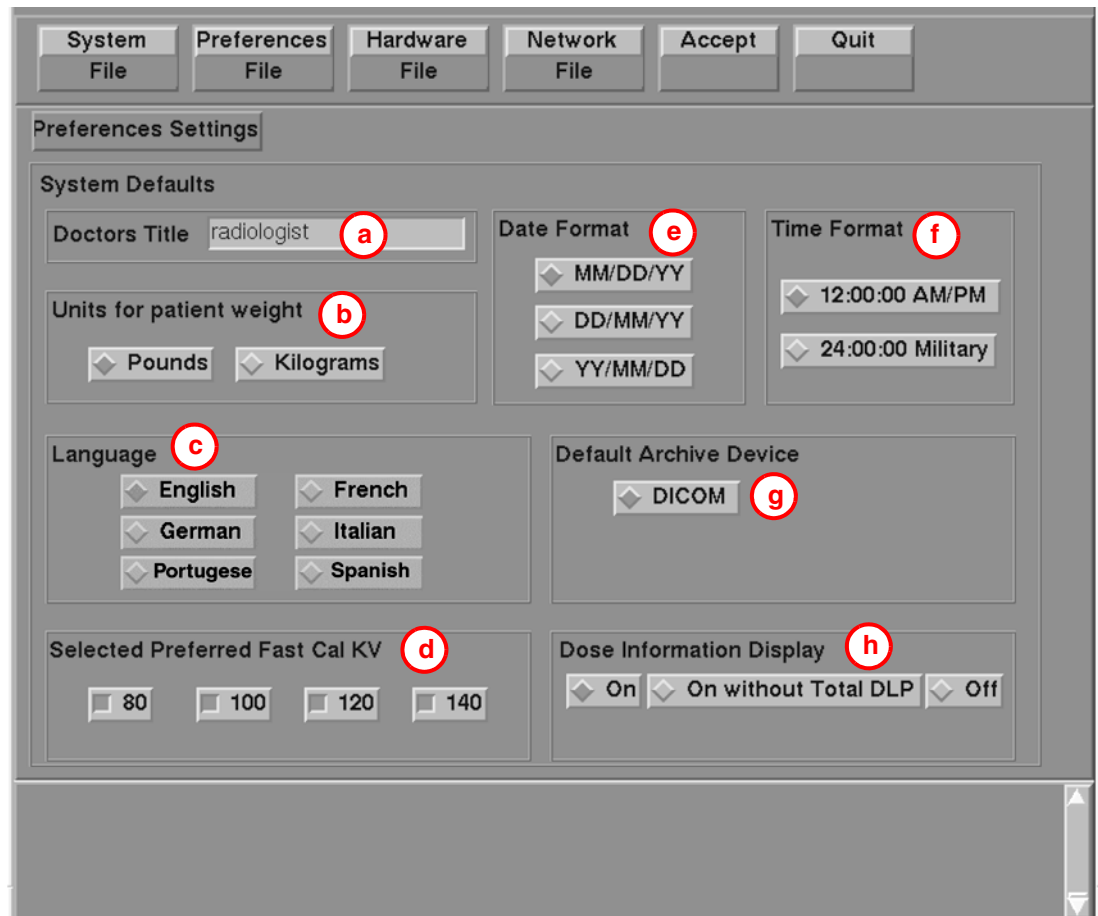


Figure 6-13 Preferences Setup Screen

- 6.) Configure Preferences Settings (refer to Figure 3-14)
 - a.) Doctors Title: enter the title for the doctor (i.e. radiologist).
 - b.) Units for Patient Weight: tells the software whether pounds or kilograms are being used.
 - c.) Language: configures the language to display on the Application screens (ENGLISH, FRENCH, GERMAN, ITALIAN, PORTUGUESE, SPANISH).
 - d.) Selected Preferred Fast Cal KV: configures the preferred kV that the Fast Cal Routine calibrates (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.

- Comment:* These kVs should include all kVs the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs
- e.) Date Format: configure the format for the date displayed on the images.
 - f.) Time Format: configure the format for the time displayed on the images.
 - g.) Default Archive Device: the device is DICOM.
 - h.) Dose Information Display: select the site preferred option for the site to use in monitoring calculated Patient Dose:
 - * Select ON (full CTDIw Display)
 - * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - * Select OFF (no CTDIw Display)
 - 7.) Select the HARDWARE button (see [Figure 6-13](#)) to display the Hardware Settings Screen (see [Figure 6-14](#)).

Figure 6-14 Hardware Settings Screen

- 8.) Configure Hardware Settings (refer to Figure 3-15)
 - a.) DAS Type: configures which DAS the software should expect.
 - b.) Tube Type: configures which X-ray Tube the software should expect.
 - c.) PDU Type: configures which PDU the software should expect.
 - d.) Collimator/Filter Type: configures which Collimator/Filter the software should expect. Presently, LightSpeedsystems only support HELIOS G1.
 - e.) Network Image Printer: use only if the site plans to use a network-connected postscript printer (such as Codonics printers) with this system. When selected:
 - * Enter the printer Hostname in the Name field.
 - * Enter the printer IP Address in the IP Address field.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e., a computer with a printer attached); a Network Image Printer (which is supported) is a printer attached directly to the network.

- f.) Camera setup:

Comment: Camera set up is not performed using Reconfig, but rather is performed from the Service Desktop once Applications are started. See ["Declaring DICOM Filming Devices on the System" on page 297](#).
- g.) Is O2 present: selects when O2 is present.
- h.) Gantry Type: selects the type of gantry connected to this console.

- 9.) Select the NETWORK button to display the Network Settings Screen as shown in [Figure 6-15](#).

Figure 6-15 Network Settings Screen

- 10.) Configure Network Settings

Comment:

This screen provides the ability to declare the LightSpeedsystem on a hospital network. Key information such as Host Name, IP Address, Net Mask (for LightSpeed systems on a subnet) must be obtained from the hospital network administrator.

Refer to [Chapter 13](#) for more information and complete details of setting the Hospital/System Network Configuration.

- a.) Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface is <Suite Name>, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

- b.) Host Name.

The hospital-provided Host Name identifies the network hostname and AE Title of the LightSpeed system to the hospital network. The Host Name:

- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _.

Comment:

The Host Name is typically stmary or ct01.

- c.) IP Address: enter the hospital-provided address for the system.
d.) Net Mask(if the LightSpeedsystem is on a subnet): enter the hospital-provided Net Mask.
e.) Broadcast Address.

Comment: The Broadcast Address should be the same as the IP Address, except for the bits of the host ID portion (last digit group). These set to 1' or 0 depending on the configuration of the network. The standard default is 1, but older Sun OS machines used 0.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

- f.) Default Gateway (if applicable): enter the hospital-provided Default Gateway IP Address. If the site network does not use a default gateway, leave the field blank.
- g.) Internet Option:
- h.) Advanced Option:
- i.) Use NIS? Select NIS (a.k.a. Yellow Pages database) only if requested by the hospital network administrator, as follows:
 - * Select ADVANCED OPTIONS button on the Network Settings Screen.
 - * Select USE NIS button.
 - * Enter the hospital-provided Domain Name (from the hospital network administrator).

11.) Select ACCEPT on the LightSpeedSystem Configuration Screen.

Comment: The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window that closes when the loading process is complete. All the window output is logged to a file named:

- 12.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click YES (see Figure 6-16).



Figure 6-16 Reboot Screen

- 13.) The system automatically logs in as ctuser after the reboot. Select OK on the Autostart Disabled pop-up message.

3.7 Applications Startup

Open the Console shell window, and type: **st ENTER**. The applications desktop appears on the monitor.

This Page Intentionally Left Blank

Chapter 7

DAS Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- When directed, record data collected from procedures in this chapter into GE Form 4879, located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.
- Refer to DAS Cleaning in the GE Healthcare *LightSpeedSystem Service Manual for your scanner*.

Section 1.0 Introduction

1.1 Failure Protocol

Follow the standard troubleshooting protocol to verify all failures.

- If you are moving boards, ensure that you follow special handling procedures (e.g., use ESD protection).
- Record all board moves on the DAS board tracking sheet.
- When you verify a failure, identify the defective part and remove it from the system.

Before running any verification tests, turn on the DAS and let it warm up. It takes two (2) minutes of warm-up time for every minute that the DAS was turned off, up to two (2) hours.

If you encounter failures in any section, restart the procedure at the DASTools.

1.2 Special Handling Precautions



NOTICE



Potential for Static Electricity Damage

DAS circuit cards are more sensitive to both STATIC DAMAGE and DIRT than any other circuit cards in the LightSpeed system. Read Sections 1.3 and 1.5, and observe all precautions.

1.3 Anti-static Handling

You **MUST** wear a grounding wrist strap anytime you remove the boards from the DAS or a conductive bag. **NEVER** place the DAS boards on non-conductive surfaces, such as white bench tops, plastic covered books, or non-conductive plastic bags.

When removing the covers from the DAS, put on a wrist strap, and push all the DAS boards firmly into their slots on the DAS backplane. Do **NOT** remove and re-seat any boards at this time.

1.4 Cleanliness

Any dirt on the surface increases leakage current on the converter cards. This causes the DAS to fail the drift spec. Dirt also causes microphonic failures, which cause image artifacts. Wear anti-static finger cots (part number 2221235 [small] and 2221235-2 [large]) when handling the DAS. Fingerprints on the board can cause problems during humid conditions. Use only clean, new anti-static finger cots.

Follow cleaning and de-ionizing procedures in the System Manual.



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.

1.5 Required Tools

TOOLS REQUIRED

- Ground Wrist Strap
- Aero Duster, Part #MS-222N (if cleaning is needed) with static-free metal nozzle inserted
- Lint-free towels, Part #46-307237P1
- Amax Contact and Circuit Cleaner, Part #2274104
- DAS/DET Interface Kit 2227089 (if cleaning is needed)

1.6 Process Overview

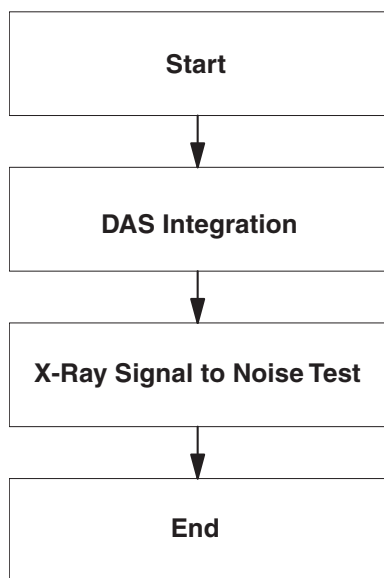


Figure 7-1 DAS Integration Flowchart

Section 2.0

DAS Tools

- 1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.



- 2.) Click the UTILITIES icon.



- 3.) Select INSTALL.

- 4.) Select DAS INTEGRATION.

- 5.) Select DAS TOOLS to open the corresponding menu window.

- 6.) Select ACCEPT.

DAS Tests		Test Name	Number of Iterations	Remaining Iterations	Total Successes	Total Failures
Auto Test		DC CAL Absolute	1	1	0	0
Manual Test		Offset Drift	1	1	0	0
Interconnect Test		Pop Noise & microphonics	10	10	0	0
Pop Noise Microphonics Test		DC Noise & DC Offsets	1	1	0	0
Auxillary Channel Test						
X-ray Verification Test						

Options

DAS Viewers		Remaining		
DAS Architecture				
View Log				
View Specification				
Dismiss				

Number of repetitions for Group	1	1	Accept
Delay between groups (msecs.)	200		
Total Failure before stopping	128	128	Stop Tests

Figure 7-2 DAS Tools Screen (Auto Test Selected)

- 7.) Select AUTO TEST to run all available DAS Tools.
- Use the default iteration values.
 - The auto test displays **comfort dots** to show activity, and displays iteration pass/fail information on the screen.

-or-

Manually select the following tests:

- DCCal: 1 iteration
- Offset Drift: 1 iteration
- Pop Noise: 10 iterations
- Microphonics: 10 iterations (see [Figure 7-3](#))
- Offset Means: n1 iteration
- Offset Noise: 1 iteration

Auto Test takes approximately 35 minutes to complete one pass.

- 8.) After DAS Tools successfully completes one pass of the auto test, or the designated number of manual test iterations, select DISMISS to exit the test. If the Total Failures column lists any failures, troubleshoot before proceeding.

DAS Tests		Test Name	Number of Iterations	Remaining Iterations	Total Successes	Total Failures	
Auto Test		Pop Noise & microphonics	10	10	0	0	Options
Manual Test							
Interconnect Test							
Pop Noise Microphonics Test							
Auxiliary Channel Test							
X-ray Verification Test							
DAS Viewers							
DAS Architecture							
View Log							
View Specification							
Dismiss							

Pop Noise & microphonics Options			
☒ Rotor On	☒ Rotating		
☐ Rotor Off	☐ Non-Rotating		

	Remaining	
Number of repetitions for Group	1	Accept
Delay between groups (msecs.)	200	
Total Failure before stopping	128	Stop Tests

Figure 7-3 DAS Microphonics Test screen

Section 3.0

Verify X-Ray/Signal to Noise Test

This section uses X-ray to test the DAS for opens, lows, and shorts.

System Configuration:

- All covers ON
- Nothing in detector FOV

1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon, .

2.) Click the UTILITIES icon, .

3.) Select INSTALL.

4.) Select DAS INTEGRATION.

5.) Select START.

6.) Select X-RAY VERIFICATION.

7.) Select X-RAY VERIFICATION (DDC).

8.) Follow on-screen instructions to acquire each set of X-ray verification scans, for a total of four sets.

9.) Select ACCEPT RX.

10.) Press **START SCAN** when it flashes.

11.) Select DISMISS to close the X-ray verification window.

12.) Select X-RAY VERIFICATION ANALYSIS.

13.) If X-ray verification scans do not pass, troubleshoot with DAS Tools and System Service Manual before continuing.

14.) Complete the Tube Signal/Output check on GE Form 4879, located on the CD for this scanner's service manual.

This Page Intentionally Left Blank

Chapter 8

Table and Gantry Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- When directed, record data collected from procedures in this chapter into GE Form 4879 located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

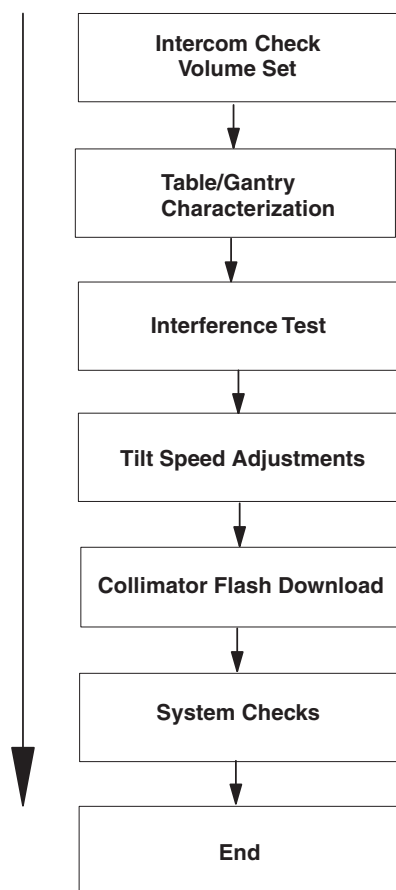


Figure 8-1 Table/Gantry Integration Flow Chart

REQUIRED TOOLS

- Multimeter
- Scope (or Multimeter with Frequency Measurement Capability, such as the FLUKE 87.)

Section 2.0

Autovoice/Intercom Check

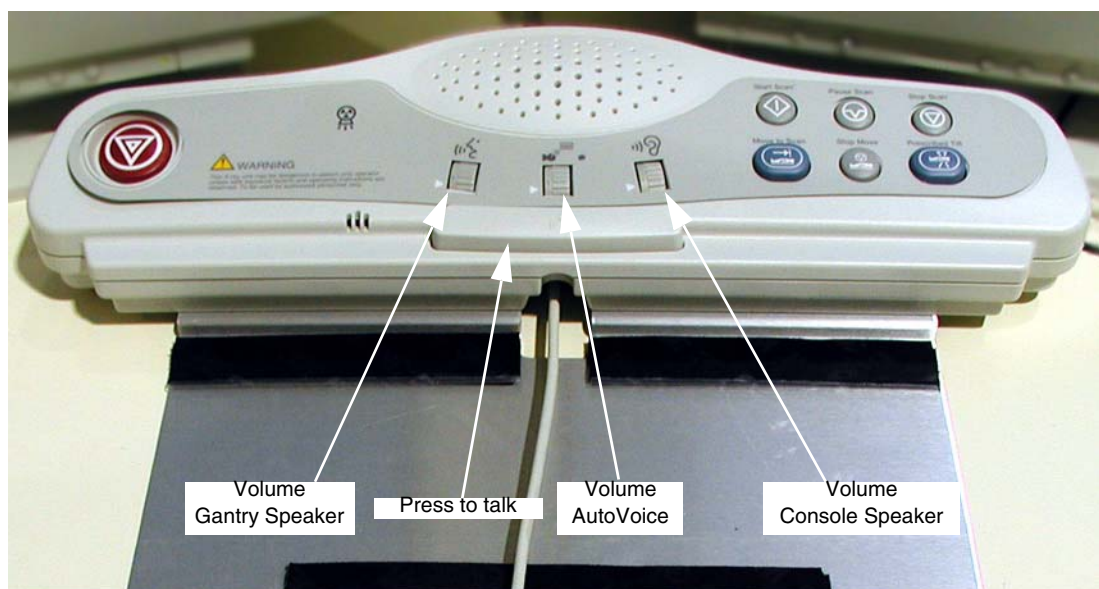


Figure 8-2 SCIM Volume Controls

2.1 Gantry Speaker

- 1.) To adjust the gantry speaker volume, speak into the console microphone (press the bar on the SCIM) and adjust the left volume thumb wheel.
- 2.) Click on the Autovoice volume control from the tool chest.
- 3.) Click on the Audio Panel icon to open it.
- 4.) Leave all audio panel sliders at their default positions, as shown in [Figure 8-3](#).

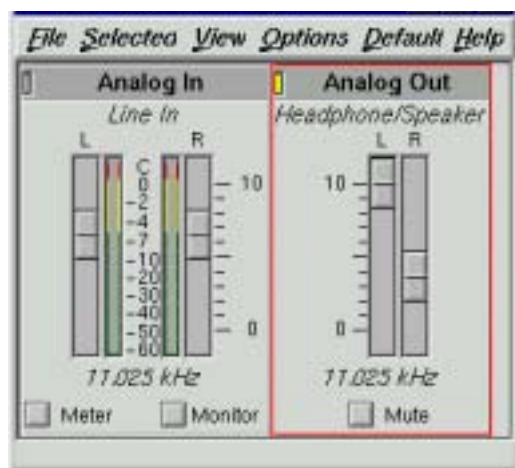


Figure 8-3 Audio Panel

- 5.) When finished, select **FILE-SAVE** to retain your settings.

2.2 Console Speaker

To adjust the console speaker volume, adjust the right volume thumb wheel until a person speaking near the gantry is clearly heard.

2.3 Autovoice Volume

- 1.) To set volume for the gantry speaker;
 - a.) Adjust the left volume thumb wheel.
 - b.) With Autovoice playing, adjust the center volume thumb wheel, while Autovoice is playing,.
- 2.) To set the volume from the gantry microphone to the speaker on the SCIM, adjust the right volume thumb wheel.

Section 3.0

Table/Gantry Characterizations

3.1 Tilt Characterization

Tools Required

- Hex (metric) set
- Flat blade screw driver

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select TABLE GANTRY INTEGRATION.
- 5.) Select CHARACTERIZATION.
- 6.) Select TILT.
- 7.) Follow the Tilt Characterization instructions on the screen.

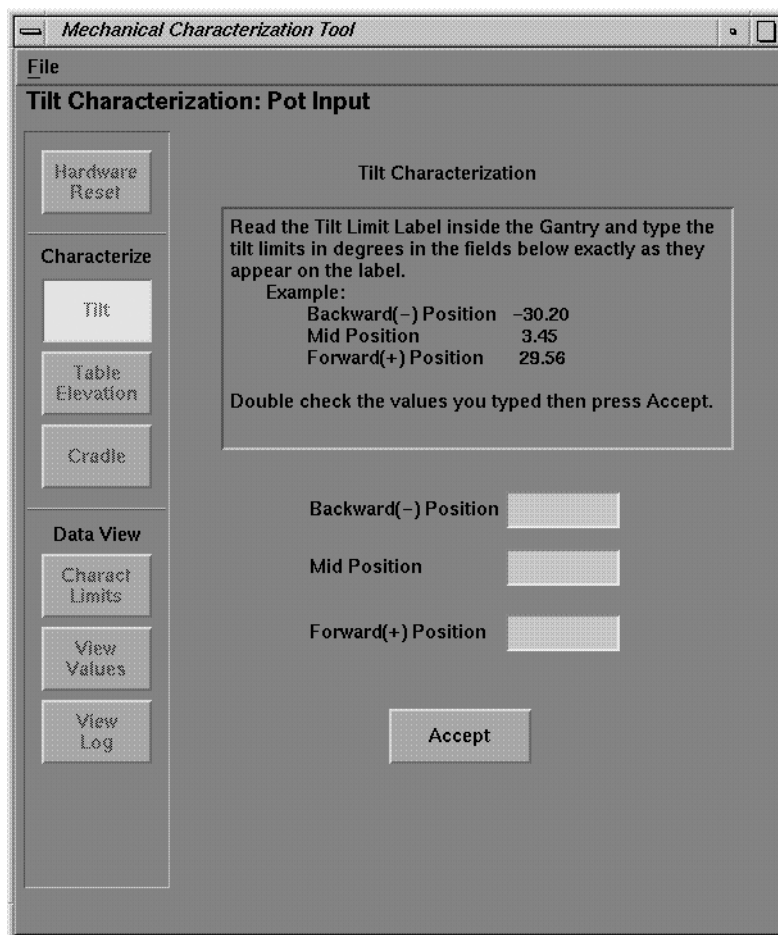


Figure 8-4 Gantry Tilt Characterization Screen

3.2 Elevation Characterization

Tools Required:

- Metric tape measure
- Flat blade screwdriver
- Metric hex key set
- 9-inch level

Note: The table to gantry MUST be level before performing this procedure.

- 1.) Remove the upper-right and lower-right table covers.
- 2.) Remove the small, L-shaped measurement block from its storage position (stored on top of the bigger measurement block).
- 3.) Install it in its characterization portion, at the right rear of the table.
- 4.) Move the measurement plate on the right side of the table to the **out** position.
- 5.) Tighten the mounting screws for both plates.
- 6.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 7.) Click on the UTILITIES icon.
- 8.) Select INSTALL.
- 9.) Select MECHANICAL CHARACTERIZATION.
- 10.) Select TABLE ELEVATION.
- 11.) Follow the elevation characterization instructions displayed on the screen
 - a.) Measure the elevation distance as vertically as possible.
 - b.) Measure between the bottom surface of each measurement block and the center of the rear leg upper pivot pin (see [Figure 8-5](#)).

Note: Measure the exact elevation distances. Incorrect measurements result in Elevation Characterization failure.

Table Low = 374.1 mm Table High = 849.1 mm

- 12.) Return the measurement plates to the storage position.

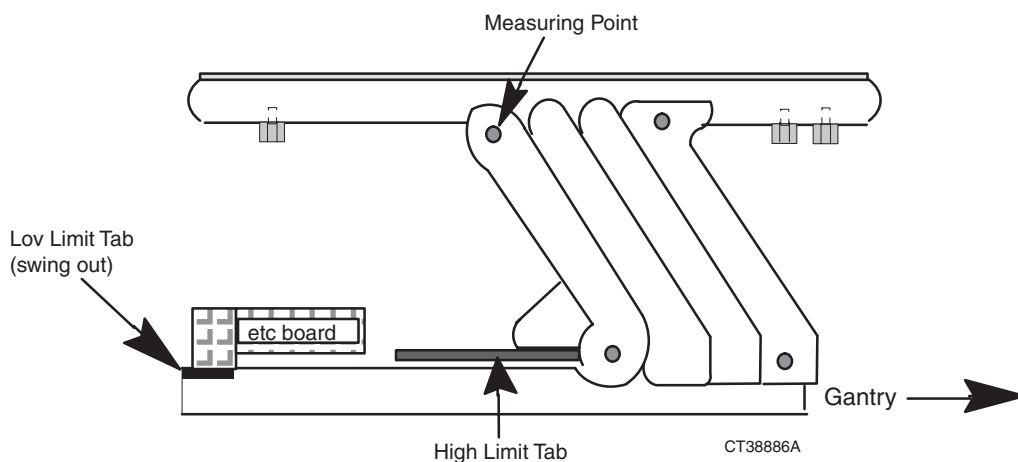


Figure 8-5 Measure Table Elevation Distance

3.3 Cradle Characterization

Tools Required:

- Hex (metric) set
- Metric tape measure

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select MECHANICAL CHARACTERIZATION.
- 5.) Select CRADLE.
- 6.) Follow the cradle characterization instructions displayed on the screen.

Note: If the display flashes “888” a table/gantry reset is required before proceeding.

To reset the table: Lower the table to its lowest position, then raise the table to its maximum position.

- a.) In the Alignment Light procedure (screens 3 and 4) use the Black Dot on the cradle right-side bumper.
Elevate the table to at least 40 mm from isocenter, so the black dot reaches the external scan plane alignment light before the cradle reaches its home position.
- b.) In the End-of-cradle Position procedure (screen 5), use the same Black Dot on the cradle right-side bumper.

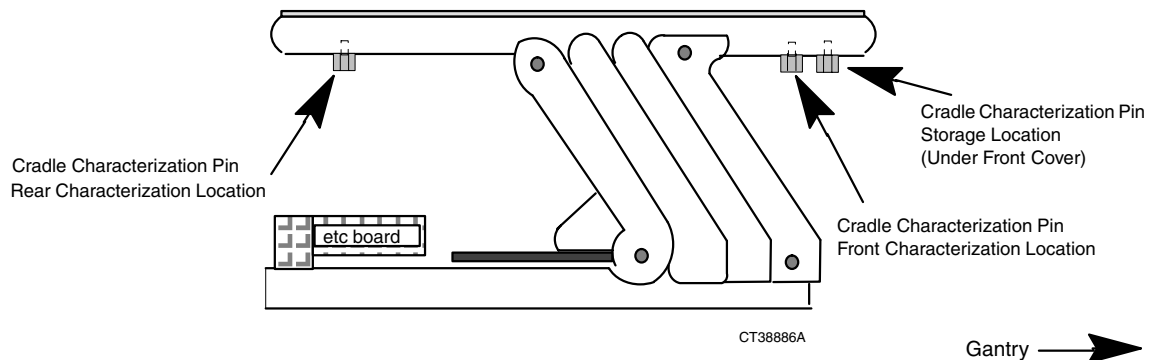


Figure 8-6 Cradle Characterization Pin Locations

Section 4.0

Interference Test

Reset the hardware to download the new characterization values to the ETC, ICE, and STC before proceeding to the table/gantry interference test on this page.

If the following elevation and/or tilt display readings fail to meet specifications, DO NOT adjust the limit switches. Instead, re-characterize elevation and/or tilt.

1.) Select HARDWARE RESET.

Press only the **fast** push-button to display the elevation limits for the present elevation/tilt combination.

- Move the cradle to the home position.
- Use the DOWN pushbutton to lower the table to the minimum height.
- Make sure the elevation display reads 500.0 ± 3 mm.
- Use the UP pushbutton to raise the table to the maximum height.
- Make sure the elevation display reads 25.0 ± 3 mm.

Note: If this step fails, check table/gantry level, and run table elevation characterization.

“I” tilt means the top of the gantry tilts away from the table. “S” tilt means the top of the gantry tilts toward the table.

2.) Advance the cradle about 914.4 mm (36 in) from the home position.

- Tilt the gantry to ± 23 degrees.
- Use the DOWN pushbutton to lower the table until it automatically stops.
- Make sure the elevation display reads 115.5 ± 3 mm.
- Make sure the table clears the gantry covers by at least $\frac{1}{2}$ in.

3.) Tilt the gantry to ± 20 degrees.

- Use the DOWN pushbutton to lower the table until it automatically stops.
- Make sure the elevation display reads 210.0 ± 3 mm.
- Make sure the table clears the gantry covers by at least $\frac{1}{2}$ in.

4.) Use the UP push-button to raise the table until the elevation display reads between 115.5 mm and 210 mm.

- Tilt the gantry to ± 10 degrees.
- Tilt the gantry away from the table; make sure it stops at $\pm 22.0 \pm 0.5$ degrees.
- Tilt the gantry toward the table; make sure it stops at $\pm 18.5 \pm 0.5$ degrees.

5.) Return the cradle to home position.

- Make sure you can tilt the gantry to 30.0 ± 0 degrees.
- Leave the gantry tilted to ± 30 degrees for now.

6.) Raise the table until you can move the cradle.

- Advance the cradle about 305 mm (12 in.) from the home position.
- Use the HOME Button to return the cradle to the home position, tilt the gantry to 0 degrees, and lower the table to minimum height.

Make sure you cannot move the cradle at table heights below 210.0.

- Verify that the *cradle lock* indicator on the gantry display is lit.

7.) Select FILE —> EXIT from the pull-down menu.

8.) Select DISMISS and then CLEAN UP.

Section 5.0

Tilt Speed Adjustment Procedure

- 1.) Remove rear gantry base covers.
- 2.) Using a stop watch or suitable time piece, adjust for one (1) degree per second motion.



NOTICE Adjustments are very sensitive. Improper speed adjustments result in overspeed error generation.

Note: Tilt speed varies based upon hydraulic fluid temperature. Make adjustments at normal scan room temperature settings.

- 3.) From zero (0) degrees tilt, press and hold the tilt forward button,  until the gantry display reads S25. Observe the tilt speed using a time piece.
- 4.) Press and hold backward tilt button,  until the gantry display reads I25. Observe the tilt speed using a time piece.
- 5.) Adjust the forward speed control valve for one (1) degree per second motion (see [Figure 8-7](#)).
- 6.) Adjust the backward speed control valve for one (1) degree per second motion (see [Figure 8-7](#)).
- 7.) Repeat steps 3 through 6 until the tilt speed for both directions is correct.
- 8.) Now tilt the gantry forward to S30. Observe the S25 to S30 speed. Do this several times. If a noticeable difference is observed, then check the hydraulic fluid levels.

Note: This is a self-bleeding system. Trapped air can cause slowed or limited tilt range. Exercising the full range of motion several times should purge any trapped air from the hydraulic system.

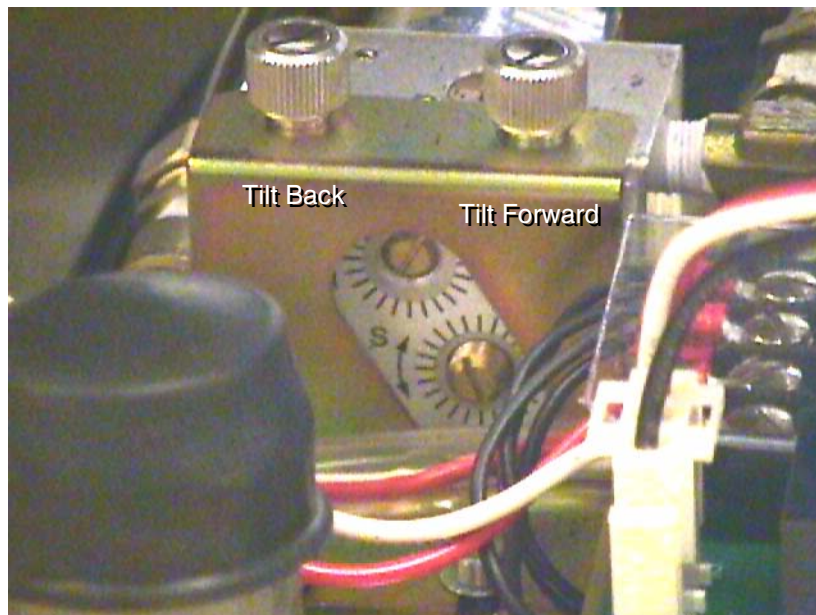


Figure 8-7 Tilt Speed Adjustment Screws

Section 6.0

Collimator Characterization and Flash Download

The Collimator Flash Download diagnostic compares the values in flash memory on the collimator control board with the values stored on the system, and updates them if necessary. If the memory values don't match, the diagnostic prompts you to type/enter the collimator serial number to complete the update.

To prevent invalid test results, remove any obstructions from the gantry opening.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select FLASH DOWNLOAD TOOL.
- 5.) Select UPDATE.
- 6.) Look for "Download Successful" message.
- 7.) Select DISMISS when finished.

Note: New installations upload values from FLASH.

Section 7.0 System Checks

7.1 Alignment Lights

- 1.) Press the Alignment Lights button
- 2.) Verify that the alignment light at the back and front of the gantry are on.

7.2 Check Alignment Light Position



CAUTION Verify that all personnel have cleared the system. The gantry rotates to 180 degrees during this check.

- 1.) Press the alignment light button on the gantry control panel to position the gantry.
- 2.) Press the alignment light button on the gantry control panel, again, to turn the lights off



**WARNING
LASER EYE
INJURY** NEVER STARE DIRECTLY INTO THE LASER BEAMS WHEN YOU OPERATE THE ALIGNMENT LIGHTS. STARING INTO THE BEAMS CAN CAUSE PERMANENT EYE DAMAGE.

7.2.1 Internal Axial Lights

- 1.) Place a sheet of plain white paper over the output port of each light.
- 2.) Verify that the two laser lines coincide and appear as a single line.

Note: **GE designed the internal axial lasers on the current CT system to shine *down* on the collimator. Do NOT adjust the internal alignment lights at this time. The tomographic plane tests use the QA phantom to check the internal axial laser's alignment to the collimator.**

7.2.2 External Axial to Internal Axial Distance

- 1.) Raise the table to its highest elevation.
- 2.) Extend the cradle until you see both the internal and external laser lights shining on the cradle.
- 3.) Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line. Verify this distance equals $240.0 \text{ mm} \pm 1.0 \text{ mm}$.
- 4.) Place the rule on the left edge of the cradle and measure again.
- 5.) Leave the cradle in its current position and lower the table to the minimum elevation.
- 6.) Measure the distance between the internal and external lights on both edges of the cradle, as above. Verify the distance remains equal to $240.0 \text{ mm} \pm 1.0 \text{ mm}$.

7.2.3 Coronal Lights

- 1.) Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light. Verify the two coronal lines coincide.
- 2.) Move the paper to the right side of the patient opening. Verify the two coronal lines coincide.
- 3.) Place the paper in the center of the gantry opening. Use a level to verify that the coronal lines are horizontal.

7.3 Check X-ray Lights

- 1.) Perform several scans following the steps below.
- 2.) Verify that the X-ray ON lights are ON during the scans.
- 3.) When finished, check the boxes in .
- 4.) Make sure the axial drive enable and HVDC enable switches are ON.
- 5.) If you are not on the Service Desktop, click on the Service Desktop icon.
- 6.) Select diagnostics
- 7.) Select Diagnostic Data Collection.
- 8.) Set the scan time to 2.00
- 9.) Set the kV to 80.
- 10.) Set the mA to 40.
- 11.) Press the accept Rx button.
- 12.) Press the Start Scan button when it flashes.
- 13.) Record the information on Form 4879 located on the CD for this scanner's service manual.

Light On?	Warning Light Locations
	SCIM
	Gantry Front
	Gantry Back *use a mirror to view)
	Room Light (outside of the room)

Table 8-1 X-ray Light Chart

7.4 Check Warning Labels

All labels are installed in English and present on the PDU, console, table, gantry, and accessories. Replace the labels listed below with the appropriate language labels for the country in which this scanner is installed. Additionally, if any other warning labels are present, apply them to the appropriate places on the equipment. Obtain and install replacements for any missing labels.

Record all information on GE Form 4879, verifying that the labels are installed. GE Form 4879 is located on the CD for this scanner's service manual.

7.4.1 Console

Make sure the following labels are in the correct location and in the appropriate language:

- X-ray warning label on the SCIM
- Overlay on the SCIM
- Keyboard warning label
- Function key overlay
- Keyboard back labels

7.4.2 Gantry

Make sure the following labels are in the correct location and in the appropriate language:

- Laser warning labels on the gantry near the laser opening
- Warning labels on the lower right side of the gantry front cover
- Scan window laser labels
- Laser window warning labels
- Front cover laser warning labels
- Front cover information labels
- Gantry system GIB (rating) labels

7.4.3 Table

Make sure the Pinch Hazard warning labels are in the correct location and in the appropriate language on:

- Each side of the front cover
- Each side of the rear cover
- Each side of the back cradle pan

7.4.4 Accessories

Make sure the following warning labels are in the correct location and in the appropriate language:

- Table foot extender label
- Coronal head holder label
- Sagittal head holder label

7.4.5 Laser

Make sure all laser warning labels are in the correct location on the outside of the gantry in the appropriate language.



Figure 8-8 Laser Warnings and Precautions

7.5 Process Product Locator Cards

- 1.) Collect the product locator cards shipped with the system. The average system has approximately 28 product locator cards.
- 2.) Update the online product locator website with the required hospital information.
- 3.) Confirm that the serial numbers on the cards shipped with the system match those found on the website for that GON number. Update as required.
- 4.) Place the cards in a plastic bag; then place them in the service cabinet.

Section 8.0 Install the Service Cabinet

Note: Some service cabinets may require assembly. Cabinet assembly takes about 1.5 hours to complete. Assemble the cabinet following the instructions located in the cabinet's shipping box.

- 1.) Place the cabinet in the location shown on the site print.
- 2.) Place all service materials shipped with the system in the service cabinet.

This Page Intentionally Left Blank

Chapter 9

Customer Options Installation & Verification

Note: Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 CT Options

- ☐ Injector functional tests completed.
- ☐ AW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
 - * Direct 3D
 - * SmartScore Pro
 - * CT Perfusion
- ☐ Teleradiology Connections (Refer to [Chapter 13, Section 8.0, Teleradiology \(Framegrabber Type\) Systems and Auxiliary Monitors.](#))
- ☐ Auxiliary Monitor
 - * Flat Panel
 - * CRT

Section 2.0

Install Options

Refer to:

- GE Prints and schematics for mechanical (physical) location of option
- FDO shipment for identification of items
- Project Manager of Installation for installation instructions if they differ from print

2.1 Software

TOOLS REQUIRED:

None

Note: **Your system may have a MOD that contains customer purchased options. If your system has an options MOD, install it at this time.**

Ensure that the options MOD is NOT write-protected at this time. The initial install requires a write-enabled MOD; subsequent installs can be done with the MOD write-protected.

- 1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.
- 2.) Click the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select INSTALL OPTIONS.

An Options Window is displayed (Figure 9-1):

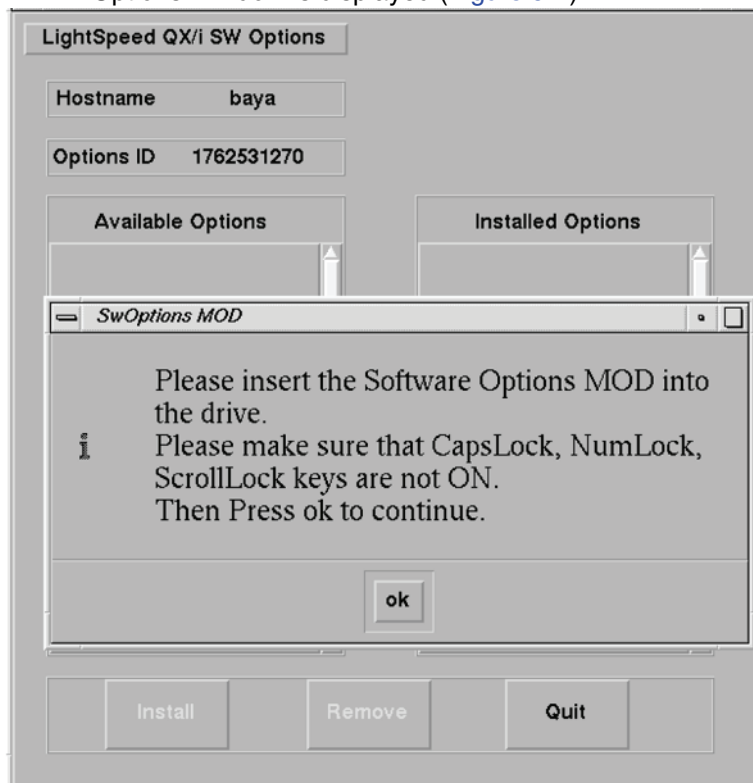


Figure 9-1 Options Window when first selected

- 5.) Insert the options MOD into the MOD drive and click OK.
If you do not have an options MOD:
 - a.) Click OK.
 - b.) Wait for the abort pop-up.
 - c.) Abort the process.

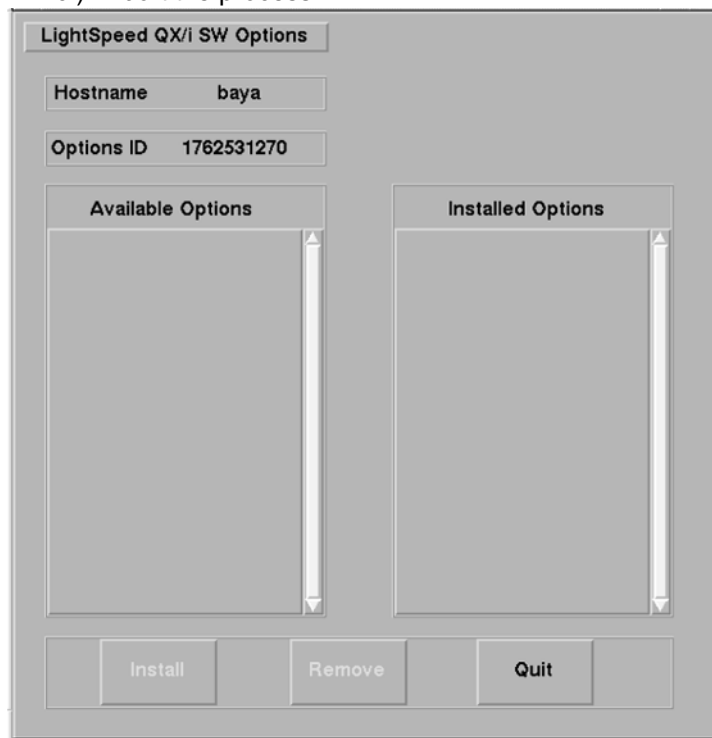


Figure 9-2 Example Options Window

- 6.) Select all of the options in the left-hand column to install the corresponding software.
 - 7.) Select INSTALL. A box may appear while the options are loading. When an option is displayed in the *Installed Options* list, then installation of that option is complete.
- Note: Some options take a fraction of a second to install. Options like 3D may take a half-minute because they are installing software.
- 8.) After the options are installed, click QUIT.
 - 9.) Click OK.
 - 10.) Remove the MOD and write-protect the side with the options.
 - 11.) When the system prompts to Reboot, click YES, and reboot the system to complete the installation.

2.2 Camera (Filming Device)

Refer to documentation shipped with the camera

- Chapter 6, [Section 2.2](#) of this manual for configuration checklists.
- DICOM 2210573 GE Document
- DICOM Print 2152913

2.3 Monitor Setup

Note: If you are using a Sony CRT monitor, you must do the following setup.
If you are using an LCD monitor, use the setup procedures in [Section 2.4](#) of this chapter.

Tools Required: Sony Service Manual (located in monitor box)

2.3.1 Overview

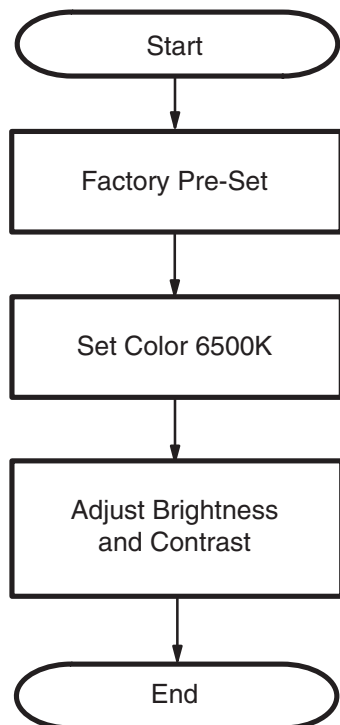


Figure 9-3 Monitor Setup Flowchart

2.3.2 Color Adjustment

- 1.) Adjust the ambient room light to the level used by the operator during image review.
- 2.) Select the COLOR menu to adjust the picture color temperature. (You can use this to match the monitor colors to a printed picture's colors.)

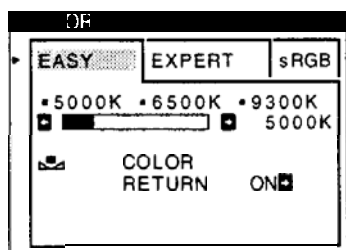


Figure 9-4 Color Menu

- 3.) Set monitor color temperature to 6500 K. Refer to the system service manual for this procedure, if necessary.

- 4.) Use one of the following methods:
 - a.) Click the IMAGEWORKS icon and proceed to step 3.
 - OR**
 - b.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.
 - c.) Click the IMAGE QUALITY icon.
 - d.) Select INSTALL SMPTE IMAGE and proceed to step 3. (Approximately 3-4 minutes.)
- 5.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.
- 6.) Increase the Contrast to maximum.

Note: Adjust monitor contrast until the operator sees the anatomical structure (window width) at the expected window levels.

- 7.) Increase the Brightness to maximum.

Note: Adjust the monitor brightness until the operator sees the anatomical structure (window width) at the expected window levels.

- 8.) Decrease the Brightness until the raster just fades into, and matches, the monitor screen background. The 5% and 95% patches should be visible.

If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.

If the CRT image exhibits any tearing or smearing of the alphanumeric characters, reduce the contrast setting slightly until the tearing/smearing is eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.

Always finish by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the CRT image to the film image.

- 9.) To view SMPTE on text monitor, swap the video cable.

2.3.3 Using Menu Buttons

- 1.) Press the Main Menu button to display the main menu on your screen.



Figure 9-5 Main Menu Button

- 2.) Press the / buttons to highlight the menu you want to adjust. Press the Main Menu button to select the menu item.

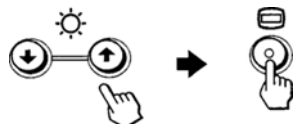


Figure 9-6 Select Menu Item

- 3.) Press the / buttons to select the desired adjustment item. Press the / buttons to make the adjustment.



Figure 9-7 Adjust the Menu

- 4.) Press the Main Menu button (Figure 9-5) once to return to the main menu, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.
- 5.) Press the reset button to reset the adjustments back to the factory settings.



Figure 9-8 Reset Button

2.3.4 Adjusting Brightness and Contrast

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu. These settings are stored in memory for the signals from the currently selected input connector. Press either one of the brightness or contrast buttons. The BRIGHTNESS/CONTRAST menu appears on the screen.

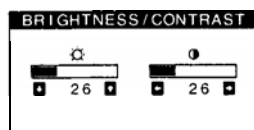


Figure 9-9 Brightness Contrast Menu

Press the ☀/🌑 buttons to adjust the brightness and the 🔵/🔴 buttons to adjust the contrast.

2.4 Connecting the LCD Monitor

Note: If you are using an LCD monitor, do the following setup.
With the monitor and computer switched off, connect the video signal cable to the monitor video input.



NOTICE Equipment Damage Possible

Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, otherwise the pins might bend.

Connect the following cables:

SCIM:

- SCSI cable
- Template
- Keyboard

Scan Monitor:

- Video cable
- Power cable
- Cable keeper

Display Monitor:

- Video cable
- Power cable
- Cable keeper

Note: A DARC video cable may be shipped with the system. Only connect the cable when troubleshooting. Do not connect it during normal installation.

2.5 Injector

Refer to documentation shipped with the injector.

Complete a Functional test of the injector.

- The position of the ceiling mount assembly must allow the ceiling stand movement and gantry to tilt, typically 20 degrees, without interference.

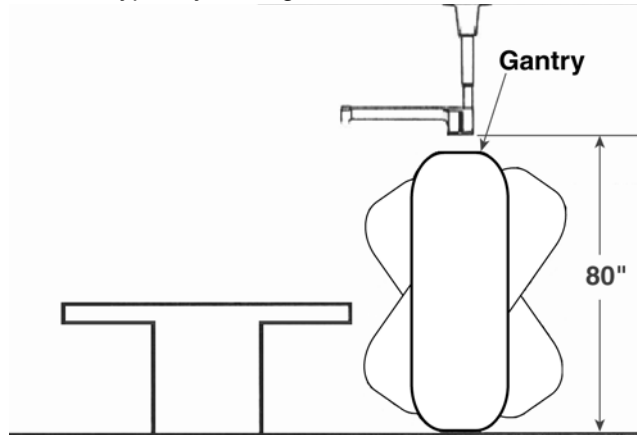


Figure 9-10 Ceiling Mount Position

2.6 Advantage Windows Workstation (AWW)

Refer to the directions provided:

- Pre-install 2111833
- Service 2111831

2.7 Uninterruptible Power System (UPS)

Refer to directions shipped with uninterruptible power system (UPS):

- Powerware B7999PS
- Liebert B7999PT
- Installation support phone numbers found in shipped documentation or in *LightSpeed 2.X Pre-Installation Manual*.

2.8 Other Options

Refer to CT-2211224-100 *LightSpeed Options* for all other options.

This Page Intentionally Left Blank

Chapter 10

System Tests

 **NOTICE**
Potential for
Data Loss and/
or Equipment
Damage

- To prevent potential data loss, please do the following:
- When directed, record data collected from procedures in this chapter into GE Form 4879 located on the CD for this scanner’s service manual.
 - Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

This chapter contains descriptions of the procedures to test system operation after installation, alignment and calibration, but prior to customer turnover and acceptance. The Image Series sections instruct you to scan system phantoms, record the means and standard deviations, and evaluate the image quality of the resulting images.

Use the system tests in the following sections to exercise all aspects of the system, to assure system integrity, before turnover to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

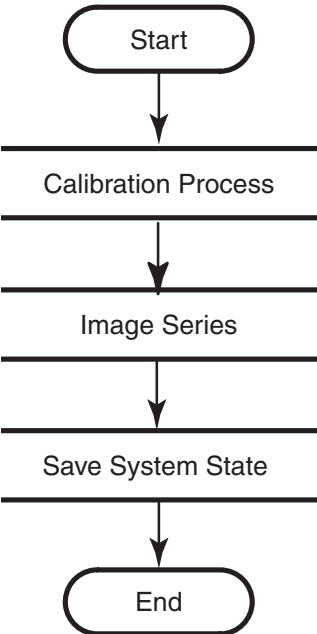


Figure 10-1 System Test Overview

Section 2.0

Calibration Process

2.1 Reference Procedures

Do NOT perform these procedures now. They are placed here as reference for later procedures. You will be directed to follow these instructions at several points throughout this chapter.

2.1.1 Scanning with Service Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

Note: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

- 1.) Select the NEW PATIENT icon.
 - 2.) Enter a Patient ID (e.g., getest).
 - 3.) Click on the infant anatomical figure icon.
 - 4.) Click on the area below the infant's feet (inside the circle):
 - 5.) Select a service protocol from the list to display the corresponding view edit screen.
- Optional Method: Enter the protocol number into the Protocol Number Field on the Exam Rx Screen.

2.1.2 Center Phantom

- 1.) Select SCANNER UTILITIES.
- 2.) Select CENTER PHANTOM.
- 3.) Follow the on-screen procedures.
- 4.) The phantom center spec is ± 1 mm .
- 5.) Select QUIT when the phantom is within spec.

2.2 Prepare the QA Phantom

Note: **The QA phantom is shipped water-filled.**

- 1.) Locate the multi-language sticker packet in the QA phantom shipping box.
- 2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

2.3 Calibration Process Introduction

If your system has a factory-supplied state MOD, you used it to load the system calibration files during the Restore System State ([Section 3.1, "Restore System State"](#) in this manual).

2.4 Fast Cal



- 1.) Select Daily Prep to warm up the tube.
- 2.) Select FAST CALIBRATION from the Daily Prep menu.

Note: **Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)**

- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, click on QUIT.

2.5 Table/Gantry Alignment Procedure

2.5.1 Conditions

This procedure may require the removal of the table bottom, side, and rail covers.

This procedure requires that mechanical final alignment was completed and passed.

2.5.2 Procedure

- 1.) If this is a Non-Fixed Site, check the van level. Correct it, if necessary.
- 2.) Check the table level. Correct it, if necessary.
- 3.) Drive the table to its highest elevation (phantom holder removed).
- 4.) Drive the table cradle in and out five times to seat the rollers.
- 5.) Turn on the alignment lights. Check the scan window for proper installation.
- 6.) Advance the end of the cradle (black dot on cradle) to the internal light.
- 7.) Complete [Section 2.1.1, "Scanning with Service Protocols"](#) (in this chapter) now.
- 8.) Select the service protocol, PERPENDICULAR ALIGNMENT.
- 9.) Add two scout scans to this procedure. Use 120 kV 40 mA tube at zero, 120 kV 40 mA tube at ninety.
- 10.) Set the Internal Landmark. The red boxes disappear from the screen.
- 11.) Click on CONFIRM, then press the START SCAN button when lighted to perform the scan.
- 12.) Use over/under multi-image display to view both images at the same time:
 - a.) Click on IMAGE WORKS
 - b.) Highlight the scans just performed and click on VIEWER.
 - c.) Click on FORMAT, and select the "over/under" view (two boxes aligned vertically).
- 13.) Use the upper image (cradle front) to verify that the table is centered:
 - a.) Click on the "grid" icon.
 - b.) Click on MEASURE, and select the straight (angled) line.
 - c.) Using the mouse, position one end of the line on the vertical axis.
 - d.) Position the other end at one outside edge of the cradle, forming a perfectly horizontal line (angle = 90). Read the distance displayed on the screen.
 - e.) Repeat step d to measure the distance from the vertical axis to the other edge.
- 14.) Repeat step 11 for the lower image (cradle at 1000 mm).

- 15.) If the edges are off by greater than 1 mm (front-to-back), perform the following:
 - a.) Loosen the table anchors at locations #5, #6, and #7.
 - b.) Move the table half the measured distance.
 - c.) Return to Step 8. Repeat the procedure until the table edges line up (± 1 mm). If needed, complete a scout scan to check table horizontal presentation image.
 - i.) Check the two scout scans.
 - ii.) Check the tube at zero scan for table/gantry perpendicularity. Make table adjustments as needed
 - iii.) Check the tube at 90 scan for cradle level and visual irregularity with the white side molding. Molding strips should be parallel within ± 3 mm. Cradle should be level within ± 3 mm.
 - d.) Tighten the anchors and torque to 75 N-m (55 ft.-lb.) for fixed sites only, 102 N-m (75 ft.-lb.) for non-fixed systems.
 - e.) Reinstall all table parts removed to access the anchors, including covers, ground straps, elevation motor leads, and the center support bar.
- 16.) Install all table and gantry covers.
- 17.) Use ERASE ALL to remove lines and grids from the images.
- 18.) Click on END EXAM to exit the program.

2.6 Auto CT N Number Adjustment

Note: **If it was more than 30 minutes since the calibration scans, rerun FASTCAL (approx. 15 min.). N Number Adjustment needs the tube warm, or CT numbers change.**

Run the small and large phantom CT N number adjustment before you acquire the image series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Place the QA Phantom in the holder.
- 2.) From the application desktop, select SCANNER UTILITIES.
- 3.) Select AUTO CT NUMBER ADJUSTMENT.
- 4.) Follow directions on the screen.
 - a.) When centering the phantom:
 - * Centering may be easier to perform from the gantry right side.
 - * Center on the water portion of the phantom (not over the internal block).
 - b.) The program automatically makes adjustments. If the program is unable to make adjustments, it stops at the technique it is unable to adjust. Troubleshoot with manual CT # Adjust before continuing.

2.7 Tomographic Plane Indication

- 1.) Place the QA phantom on the phantom holder.
- 2.) Center the phantom. (Refer to [Section 2.1.2](#) in this chapter.)
- 3.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening until the line on the phantom lines up with the internal laser lights.
- 4.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check the table/gantry, cradle, and/or laser alignment.
- 5.) Center the phantom in the scan plane with the calibration program. (See [Section 2.1.2](#), for details on the phantom centering procedure.)

- 6.) Click on NEW PATIENT.
- 7.) Enter a Patient ID. (Enter any numbers for the purpose of testing.)
- 8.) Select MANUFACTURING.
- 9.) Select the service protocol, TOMO PLANE INDICATION. (See [Section 2.1.1](#) in this chapter for details on scanning with service protocols.)
- 10.) Display the image series and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.
- 11.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 12.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0$ mm. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0$ mm requirement.
- 13.) Repeat the Tomographic Plane Indication test with the external alignment lights.
 - a.) Use the external alignment light, and press the external landmark.
 - b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
 - c.) The scan plane indication must fall within the $S0.0 \pm 1.0$ mm specification.
- 14.) Check the box on **Form 4879**.

2.8 System Scanning Test

Use the System Scanning Test to verify hardware functionality. Review images for visible artifacts, and review the message log for unacceptable errors.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

- 1.) Place the QA phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the line on the phantom with the internal laser lights.

Never scan above 50 mA without first placing a phantom in the field of view. Levels in excess of 50 mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation-damaged detector, the cals may cause artifacts in subsequent image series scans.

- 2.) Click on NEW PATIENT.
- 3.) Enter a Patient ID. (Enter any numbers for the purpose of testing.)
- 4.) Select MANUFACTURING.
- 5.) Select the service protocol SYSTEM SCAN/CUST QUAL RELI.

Note: **Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.**

- 6.) Complete the scans.

2.9 Collimator Aperture Test

2.9.1 Small Aperture Test

- 1.) Clear all obstructions out of the gantry SFOV.

- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon,



- 3.) Click on the UTILITIES icon.



- 4.) Select INSTALL.

- 5.) Select COLLIMATOR APERTURE TEST.

- 6.) Select SMALL APERTURE SCAN to display a scan protocol window and start gantry activity (refer to [Figure 10-2](#)).

- 7.) Press the START SCAN key when it flashes to initiate the first scan.

- 8.) When the test completes, the screen updates to display the Small Aperture values.

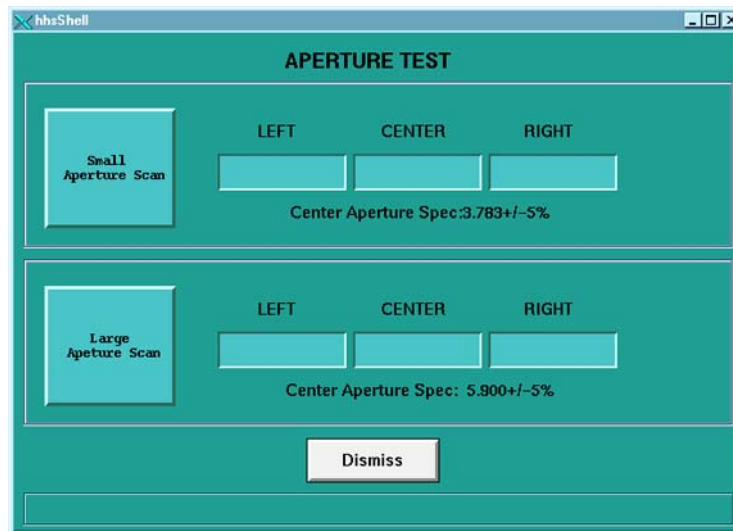


Figure 10-2 Collimator Aperture Test Screen

2.9.2 Large Aperture Test

Note: The center channel is the only channel with a spec. The left and right channels may or may not pass. Small aperture center spec is $3.783 \pm 5\%$. Large aperture center spec is $5.900 \pm 5\%$.

- 1.) Select LARGE APERTURE to display a scan protocol window, and start gantry activity (see [Figure 10-2](#)).
- 2.) Press the START SCAN key when it flashes to initiate the first scan.
- 3.) When the test completes, the screen updates to display the large aperture values.
- 4.) Record the large and small aperture values on GE Form 4879 (see).

2.10 Generator Calibration

2.11 Process Overview

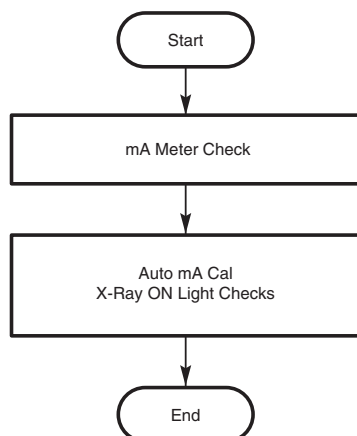


Figure 10-3 Chapter Flowchart

IMPORTANT:

- You must record the data collected in this chapter on GE Form 4879 (located on the CD for this scanner's service manual).
- Locate the HHS data, sent from the factory, in the software collector box.
- Retain this data with the HHS data collected at the site.
- A complete generator RECAL is not required.

Note: If any one of the calibration procedures fail, data collected is different, or you replace any component in the High Voltage section, YOU MUST RECALIBRATE THE SYSTEM. To re-calibrate the system, refer to procedures in the *LightSpeedSystem Service Manual for your scanner*.

2.12 Required Tool

DVM with meter leads

2.13 mA Meter Check

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select GENERATOR CALIBRATION.
- 5.) Switch OFF the HVDC (550) ENABLE inside the gantry.
- 6.) Switch OFF the AXIAL DRIVE inside the gantry.
- 7.) Rotate the anode tank to the two o'clock position.
- 8.) Pin the gantry.
- 9.) Remove the OBC cover (if necessary).
- 10.) Use a DVM as a DC mA meter. Set the meter range to one decimal place.
- 11.) Connect the DVM black lead to TP8 (ACAL1) on the mA board.

- 12.) Connect the DVM red lead to TP11 (ACAL2) on the mA board.

Collimator
HEMRC
MA Control
KV Control
Gentry I/O
LSCOM
Heurikon (CPU)

Figure 10-4 OBC Board Slot Assignment

- 13.) Select MA METER CHECK.
- 14.) Select ACCEPT to begin the test after the time delay expires.
- 15.) Once the test begins, the software enables the meter circuit for four seconds.
- 16.) Record the displayed cathode and anode mA values on GE Form 4879 in the Circuit ON section of the kV Meter Verification table.
- 17.) The Circuit ON values of the cathode and anode sides should read approximately 15 to 25 mA 140 to 200 mA.

Note: The cathode side reads approximately 19 mA during Circuit ON. The anode should read 140 to 200 mA.

- 18.) Repeat the test for the cathode side:
- Use a DVM as a DC mA meter; set the meter range to one decimal place.
 - Connect the DVM red lead to TP9 (CCAL1) on the mA board.
 - Connect the DVM black lead to TP14 (CCAL2) on the mA board.

- 19.) Select ACCEPT.
- 20.) Record the displayed and measured cathode mA values in the Circuit OFF and Circuit ON table in space provided below. Acceptable Circuit ON values: 140 to 200mA.

Note: **The anode side reads approximately 19 mA during Circuit ON. The cathode should read 140 to 200mA, as stated above.**

- 21.) Disconnect the test equipment.
- 22.) Switch ON the HVDC (550) enable.
- 23.) Select DISMISS.

Circuit ON		
	A/D	Range
Anode mA		15 - 25
Cathode mA		15 - 25
DVM - A/D = Delta Compare Delta to Limits		

Table 10-1 mA Meter Verification (Table 1)

2.14 Auto mA Cal/X-ray On Light Check

Use this procedure to verify and complete Auto mA Cal on the system. If you have just run a Heat Soak and Seasoning, wait 30 minutes before you run the Auto mA Cal.

Note: **The system automatically warms up the tube.**

- 1.) Select AUTO MA CAL.
- 2.) Press START SCAN when it flashes. The program automatically runs Ductility and Auto mA Cal.
- 3.) During this test, verify that the X-ray on indicators, located on the gantry display and the rear gantry shroud, illuminate during X-ray on.

If the system cannot complete the Auto mA Cal because HV aborts, run the "Tube Heat Soak and Seasoning" scan protocol (see the *LightSpeedSystem Service Manual* for your scanner for details). Place a 48 cm phantom in the X-ray beam to protect the detector from radiation damage during this procedure. Do not worry about the image quality of the scans reconstructed during the Tube Heat Soak and Seasoning.

- 4.) Record the data on Form 4879 (located in).

The MX200 tube uses a metal/ceramic frame, which means some electrons hit the target and actually bounce off, and are then captured by the frame. This causes a frame current, which results in an anode mA measurement of slightly less than the cathode mA, typically 5 to 10% less. Because the cathode mA is the true indicator of radiation output, the CT QX/i system controls cathode mA rather than anode mA.

2.15 X-ray Lights

- 1.) Perform several scans (see below).
- 2.) Verify that the X-ray ON lights are on during the scans. Check the boxes in [Table10-2](#)
- 3.) Record data on **GE FORM 4879** (see).

PERFORMING A SCAN

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Select DIAGNOSTICS.
- 3.) Select DIAGNOSTIC DATA COLLECTION
- 4.) Set the scan time to 2.00
- 5.) Set the kV to 80
- 6.) Set the mA to 40.

Light On?	Light Locations
☐	SCIM
☐	Gantry Front
☐	Gantry Back (use a mirror to view)
☐	Room Light

Table 10-2 X-ray Light Chart

Section 3.0 Image Series

Note: Run ALL Image Series tests in Auto Mode.

3.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images, and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements, and keeping a record of failures that occur during the image series.

Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512 x 512 matrix size

Note: **Consider any image series scan that does not meet specifications as failing.**

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least 10 additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

3.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a 10 additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

3.3 Term Definitions & Screens

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

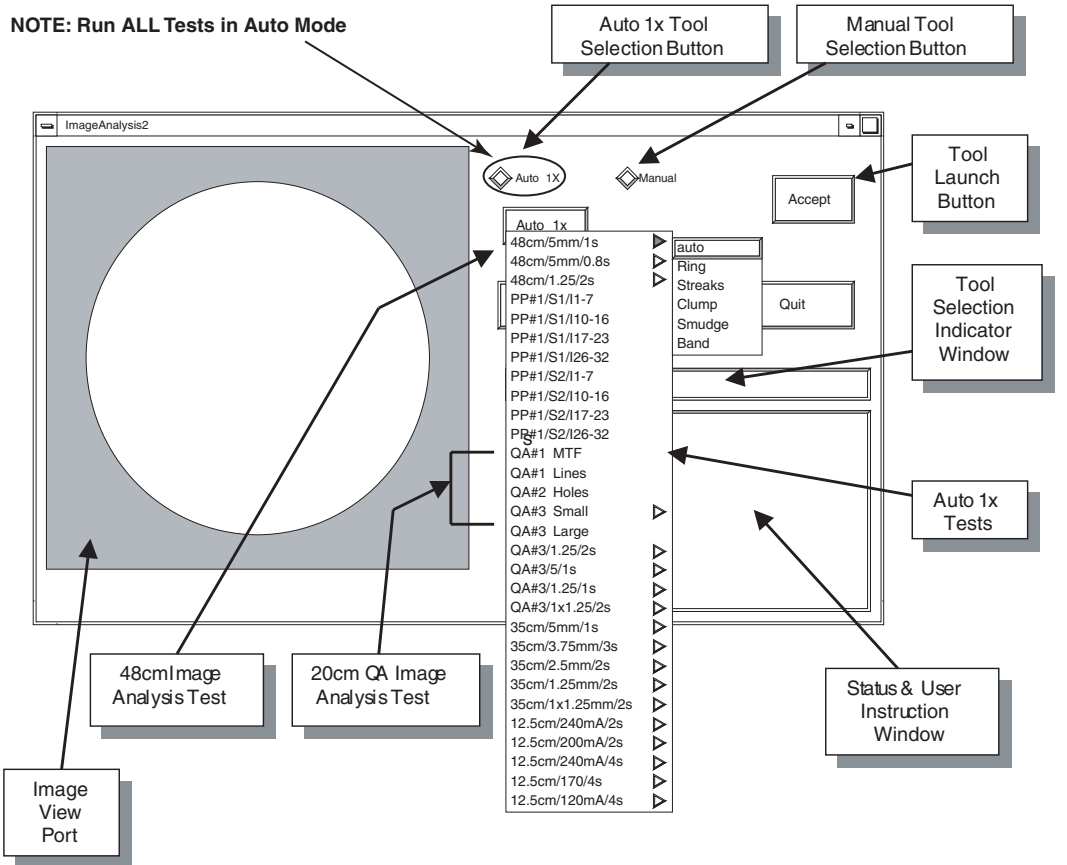
AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

Figure 10-5 AvSDo - Average image noise (standard deviation) for the outside of a phantom:

Average of all outside coordinates of all the slices in an exam. **Image Analysis Tool User**

NOTE: Run ALL Tests in Auto Mode



Interface – Auto 1x Test Pull Down Menu

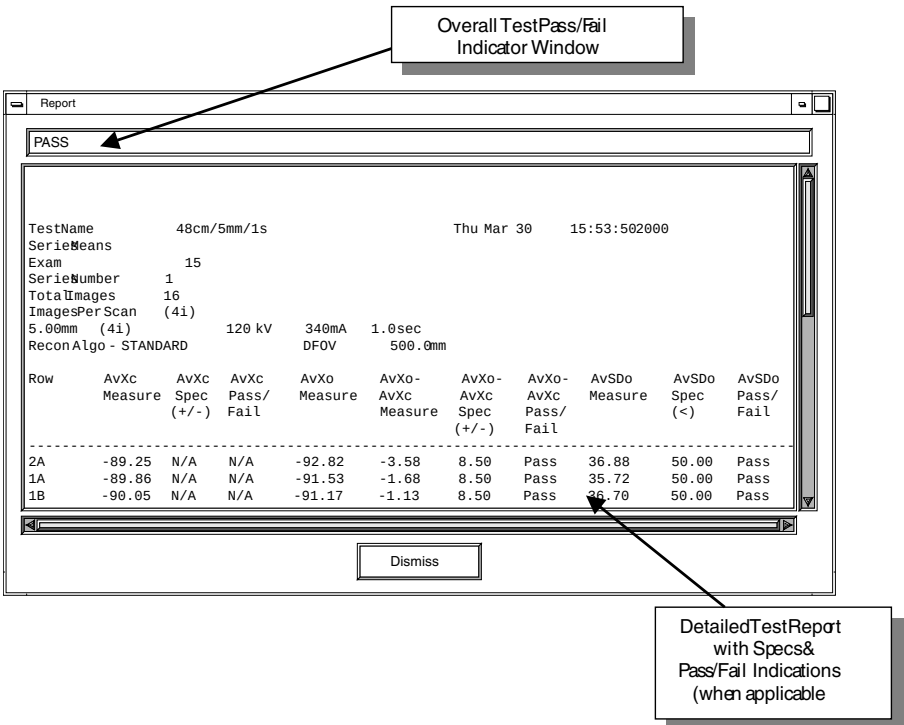


Figure 10-6 Analysis Tool User Interface – Test Results Report Window

3.4 48cm Phantom Image Series Image Performance Verification

3.4.1 Acquiring the 48cm Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 48 cm phantom on the Phantom Holder.
- 3.) Align, level, and center the 48 cm phantom.
 - Align the phantom using the internal laser lights.
 - Level the phantom using a bubble level and the Z Axis knob on the Phantom Holder.
 - Center the phantom using the CENTER PHANTOM utility in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan a four-scan, 16-image, 48 cm phantom image series.
 - a.) On the Exam Rx desktop, select NEW PATIENT.
 - b.) Type the following entries in the two listed Patient Information fields:
Patient ID: Service
Name: 48cm Phantom Image Series
 - c.) On the Service tab, select MANUFACTURING,
 - d.) On the Service Category protocol List, click on the IMAGE SERIES 48 CM button .
 - e.) Select the first series (Series Description filed should display 48cm Lg Series 1)
- 5.) Set the internal landmark.
- 6.) Acquire a four-scan, 16-image, 48 cm phantom image series.

3.4.2 Image Performance Verification

3.4.2.1 Brightness Uniformity and Noise

- 1.) Select the 48 cm phantom image series exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS desktop.
 - b.) From the Image Works desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 45 x 45 pixel reference ROI box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square 45 mm x 45 mm (2025 mm²) box. Tolerance: 45 mm $\pm$ 5 mm (1600 mm² to 2500 mm²).
If required, magnify the image to adjust it to the proper dimensions. (See [Figure 10-7](#) for additional ROI size and placement information.)
- 3.) Collect Mean and Standard deviation values for five reference ROI box positions on the sixteen 48 cm phantom images.
If required, magnify the image to adjust to proper dimensions. (See [Figure 10-7](#) for additional ROI placement information.)
 - a.) Position the reference ROI Box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
 - b.) Type **prop a** in the Image Works Accelerator Line and press ENTER to propagate Box #1 ROI (size and position) on the remaining images in the series.
 - c.) Click on the MEASURE button and select the box ROI icon to display Box #2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box #1.
 - d.) Reposition Box #2 to the left center portion on the first image (Box #2 position in [Figure 10-7](#)).
 - e.) Type **prop a** in the Image Works Accelerator Line and press ENTER to propagate Box #2 ROI (size and position) on the remaining images.
 - f.) Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in [Figure 10-7](#) (Box #3 through #5).
 - g.) Record the Mean and Standard Deviation values of the 16 images in the series for each of the five box positions in [Table 10-3](#).
Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

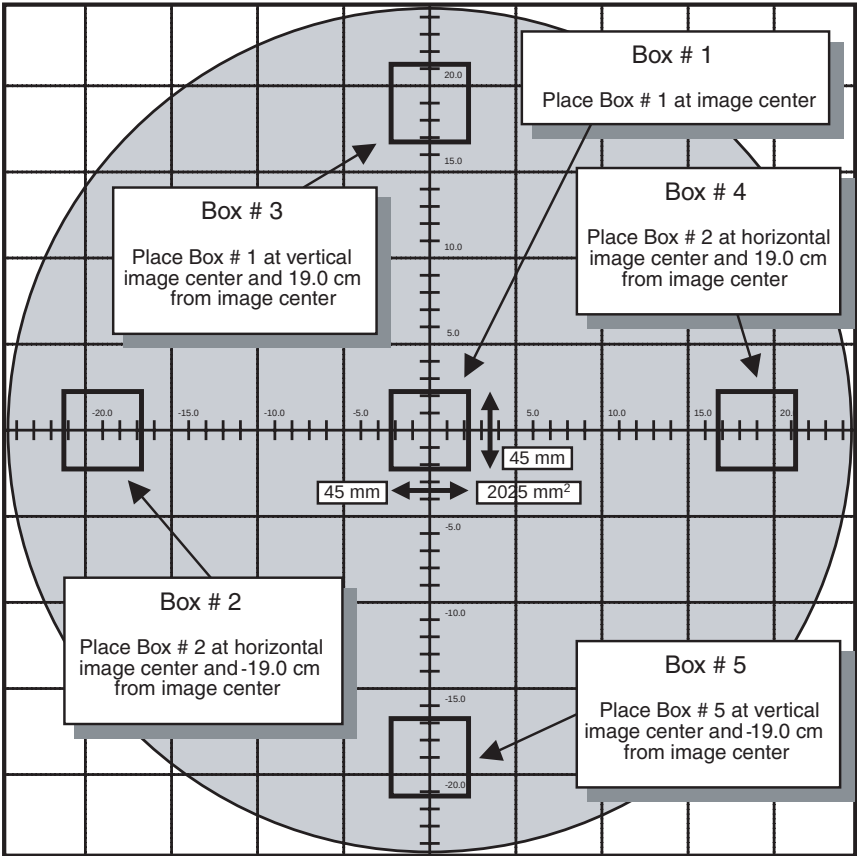


Figure 10-7 48cm Phantom Brightness Uniformity & Noise Measurement – Building Placing Reference ROI Boxes

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											

Table 10-3 48cm Phantom CT# Brightness Uniformity & Noise Image Performance Worksheet

Box Size = 1600 mm² to 2500 mm²
45 mm (± 5 mm) x 45 mm (± 5 mm)
45 (± 4 pixels) x 45 (± 4 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
Box 2 = 0 mm x -190 mm
Box 3 = 190 mm x 0 mm
Box 4 = 0 mm x 190 mm
Box 5 = -190 mm x 0 mm

Row	Images	Center Box Means (AvXc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo - AvXc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
2A	1, 5, 9, 13					
1A	2, 6, 10, 14					
1B	3, 7, 11, 15					
2B	4, 8, 12, 16					
Specifications		n/a	n/a	± 8.5	< 70	

Table 10-4 48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet

- 4.) Calculate the Brightness Uniformity and Noise values for each image in the series; record the values in [Table 10-4](#), and compare the to the specifications.
 - a.) Calculate and record the average means and average standard deviation values for the four outside boxes (boxes #2 through #5) for each of the images, and record them in [Table 10-3](#).
 - b.) Calculate and record the center box (Box #1) average means values (AvXc) for each row in [Table 10-4](#).
 - c.) Record the outside boxes (Box #2 through #5) average means values (AvXo) and average standard deviation values (AvSDo) for each row in [Table 10-4](#).
 - d.) Calculate the Brightness Uniformity (AvXo - AvXc) value for each row and record it in [Table 10-4](#).
 - e.) Verify that the Brightness Uniformity (AvXo - AvXc) value and Noise value (AvSDo) for each row meets specifications listed in [Table 10-4](#).
 - f.) Record the 48 cm phantom Brightness Uniformity (AvXo - AvXc) value and Noise value (AvSDo) for each row in the HHS Record Tables.

3.4.2.2 Failure Recovery

Specifications

Each Row (2A, 1A, 1B, and 2B) of the series *must* pass 48 cm Brightness Uniformity and Noise (for the first series scan parameters) specifications:

- AvXo – AvXc: < ± 8.5
- AvSDo: < 70.0

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify 48 cm Phantom Image Performance.

3.5 20cm QA Phantom Image Series Image Performance Verification

3.5.1 Acquiring the 20cm QA Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 20 cm QA phantom on the Phantom Holder.
- 3.) Align, level, and center the 20 cm QA phantom.
 - Align the etched line (QA#1 position) on the phantom using the internal laser lights.
 - Level the phantom using a bubble level and the Z Axis knob on the Phantom Holder.
 - Center the phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection, and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan the QA#1, QA#2, and QA#3 positions on the 20 cm QA phantom.
 - a.) On the Exam Rx desktop, select NEW PATIENT.
 - b.) Type the following entries in the listed Patient Information following fields:
 - * Patient ID: **Service**
 - * Name: **20cm QA#3 Phantom Image Series**
 - c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
 - d.) On the Service Category Selection window, click on MANUFACTURING.
 - e.) Click on the IMAGE SERIES QA button.
 - f.) When the scan prescription appears on the left monitor, select the first series.
- 5.) Set an internal landmark.
- 6.) Acquire the QA#1, QA#2 and QA#3 positions of the 20 cm QA phantom.

3.5.1.1 High Contrast Spatial Resolution

- 1.) Select the first 20cm QA phantom image series (QA#1) from the exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS desktop.
 - b.) From the Image Works desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and the first series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 17 x 17 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square of approximately 8 mm x 8 mm (64 mm²) box. If required, magnify the image to adjust to the proper dimensions. (See Figure 10-8 for additional ROI size and placement information.)
- 3.) Collect Mean and Standard deviation values for three reference ROI box positions on the four 20 cm QA#1 phantom images.

If required, magnify the image to adjust to proper dimensions. (See Figure 10-8 for additional ROI placement information.)

 - a.) Position the reference ROI box built in step 2 directly over the 1.6 mm Line Pair Pattern on the QA#1 phantom image. Fine adjust the size of Box #1 ROI to ensure that the box does not exceed the line pair boundaries (see Figure 10-8).
 - b.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box #1 ROI (size and position) on the remaining images in the series.

- c.) Click on the MEASURE button and select the box ROI icon to display Box #2. The system places an ROI box labeled #2 at the center of the image with the exact same dimensions as Box #1.
- d.) Reposition Box #2 over the plexiglas portion of the QA#1 phantom image (Box #2 position in [Figure 10-8](#)).
- e.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box #2 ROI (size and position) on the remaining images.
- f.) Click on the MEASURE button and select the box ROI icon to display Box #3. The system places an ROI box labeled #3 at the center of the image with the exact same dimensions as Boxes #1 and #2.
- g.) Reposition Box #3 over the water portion of the QA#1 phantom image (Box #3 position in [Figure 10-8](#)).
- h.) Record the Mean and Standard Deviation values of the four images in the series for each of the three box positions in [Table 10-5](#).

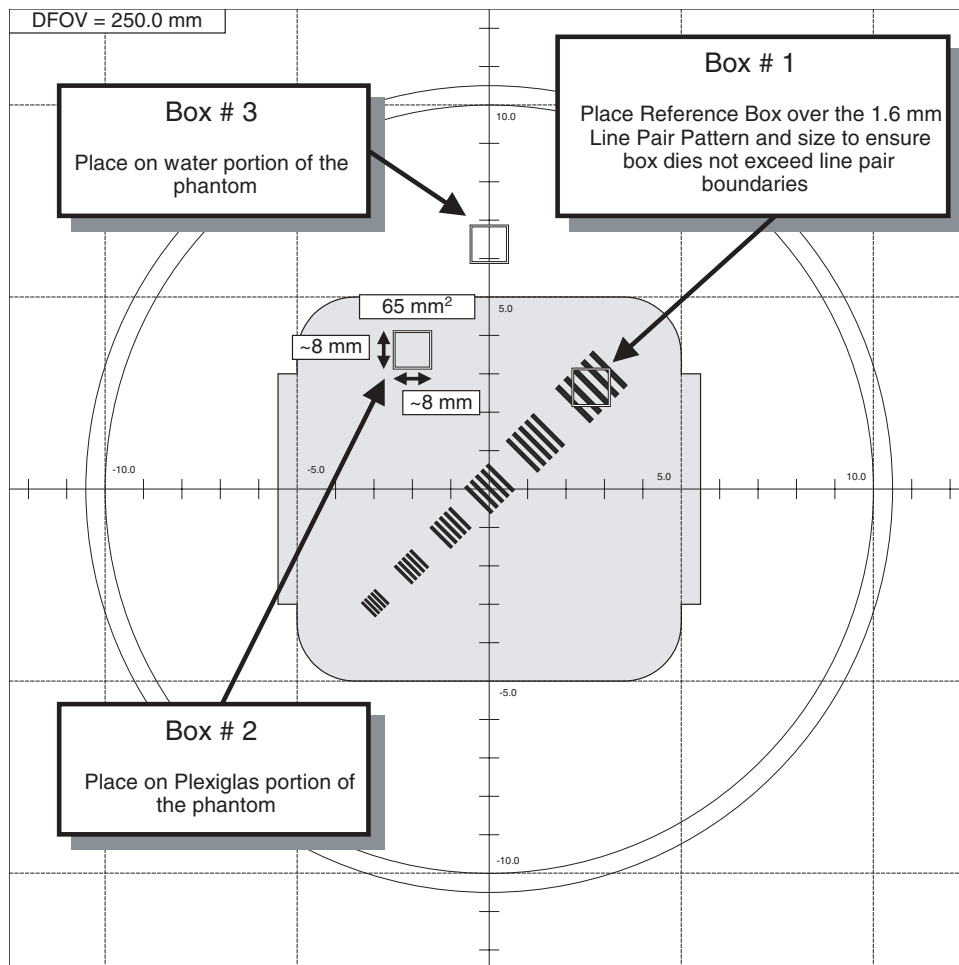


Figure 10-8 20cm QA#1 Phantom High Contrast Spatial Resolution Testing - Building And Placing Reference ROI Boxes

Image	Box 1		Box 2		Box 3		Contrast Scale (Box 2 Means – Box 3 Means)	Std Dev Average (Box 2 Std Dev – Box 3 Std Dev / 2)
	Means	Std Dev	Means	Std Dev	Means	Std Dev		
1								
2								
3								
4								

Table 10-5 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #1

Image	MTF	MTF 4-slice average	Contrast Scale	Comments
1		n/a		
2		n/a		
3		n/a		
4		n/a		
Specifications	n/a	0.65 to 1.0	110.0 to 130.0	n/a

Table 10-6 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #2

- 4.) Calculate the Contrast Scale value and four-image average MTF values for each image in the series; record the values in [Table 10-6](#), and compare them to the specifications.
 - a.) Calculate and record the Contrast Scale value (Means value of Box #2 - Means value of Box #3) for each image, and record them in [Table 10-5](#) and [Table 10-6](#).
 - b.) Calculate and record the MTF value for each image in [Table 10-6](#) using the following formula. Record Standard Deviation Average (Std Dev AVE) values in both [Table 10-5](#) and [Table 10-6](#).

$$\text{Std Dev AVG} = (\text{Std Dev}_{\text{Box 2}} + \text{Std Dev}_{\text{Box 3}}) / 2$$

$$\text{Modulation} = \sqrt{\text{StdDev}_{\text{Box 1}}^2 - \text{StdDev}_{\text{AVE}}^2}$$

$$\text{MTF} = 2.2 \times (\text{Modulation} / \text{Contrast Scale})$$
 - c.) Calculate the average MTF value for each image and record them in [Table 10-6](#).
 - d.) Verify the Contrast value for each image and the four-image MTF value meets specifications listed in [Table 10-6](#).
 - e.) Record the 20 cm QA Phantom High Contrast Spatial Resolution values (Contrast Scale for each image and four-image MTF value) in the HHS Record Tables.
- 5.) View each image of the Recon 2 Series, record all the visible Line Pair Patterns in [Table 10-7](#), and verify that each image meets specifications.
 - a.) From the Image Works desktop, select the IMAGE WORKS BROWSER window.
 - b.) Select the exam and the second series (Recon 2: Q) acquired in the previous section.
 - c.) Select the VIEWER button on the Image Works browser window. Set up the viewer window for four-image viewing.
 - d.) While viewing each image, indicate in [Table 10-7](#) all the Line Pair Pattern groups that can be visually distinguished (see [Figure 10-9](#)).
 - e.) Verify that the Line Pair Pattern visual check meets the specifications and record them in the HHS Record Tables.

Image	Line Patterns Visible	Comments
1		
2		
3		
4		
Specifications	B, C, D, E, F	n/a

Table 10-7 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #3

3.5.1.2 Failure Recovery

Specifications

Each image of the series must pass 20 cm QA#1 High Contrast Spatial Resolution parameter (Contrast Scale, Four-Image MTF Average and Visible Line Pair for the first and second series scan parameters) specifications:

- Contrast Scale: 110.0 to 130.0
- MTF Average: 0.65 to 1.0
- Visible Lines: B, C, D, E, F

Recommended Recovery

- A.) Check Phantom Alignment (leveling is critical) and repeat this scanning and High Contrast Spatial Resolution Performance Test.
- B.) Perform Alignment Procedures (POR Alignment, BOW Alignment, CBF/SAG Alignment, ISO Alignment, and Hot ISO Alignment), perform Full Calibration (Detailed Calibration and Auto CT# Adjust), and repeat this scanning and High Contrast Spatial Resolution Performance Test.

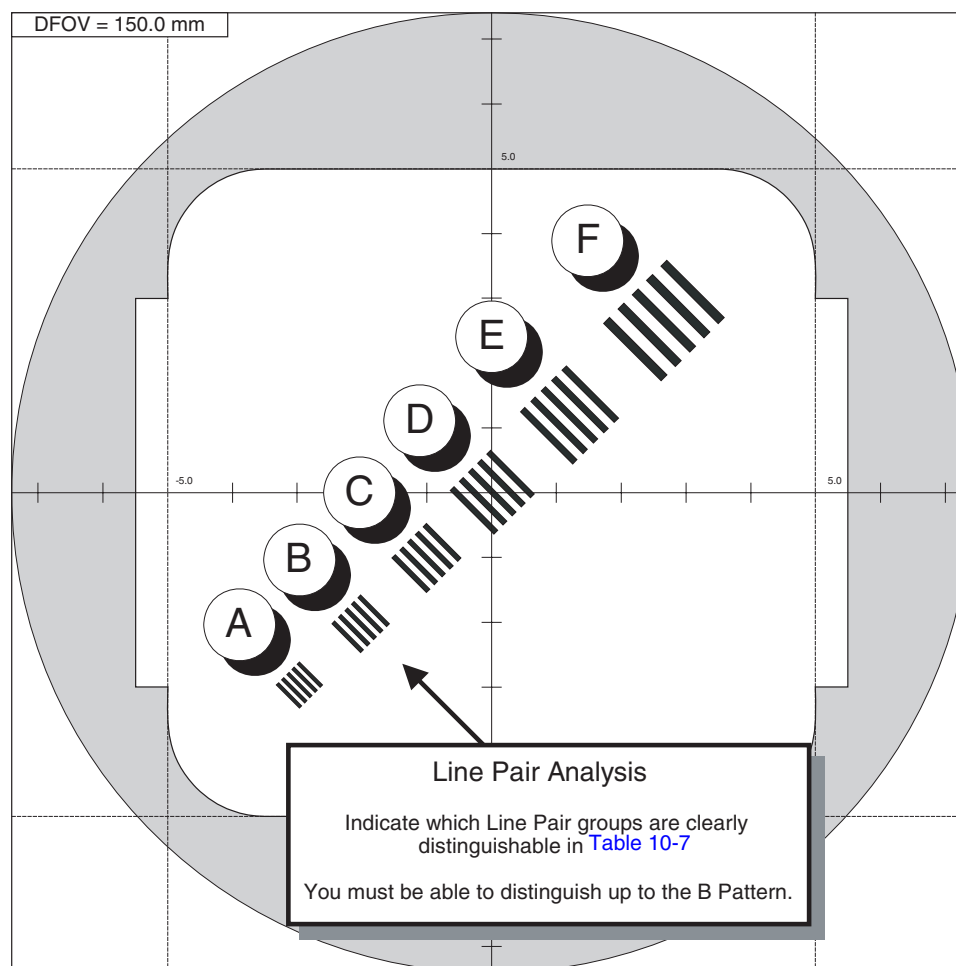


Figure 10-9 20cm QA#1 Phantom High Contrast Spatial Resolution (Visible Line Verification)

3.5.2 Low Contrast Detectability

3.5.2.1 Image Performance Verification

- 1.) Select the third 20 cm QA phantom image series (QA#2 Holes) from the exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS desktop.
 - b.) From the Image Works desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and the third series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 10 mm x 100 mm reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a rectangle of approximately 10 mm x 50 mm (500 mm²) box.

If required, magnify the image to adjust to proper dimensions. (See [Figure 10-10](#) for additional ROI size and placement information.)
- 3.) Collect Mean values for two reference ROI Box positions on four (1st, 3rd, 5th, and 7th) of the eight 20 cm QA#2 phantom images.

If required, magnify the image to adjust it to the proper dimensions. (See [Figure 10-10](#) for additional ROI placement information.)

 - a.) Adjust the Window setting of the image to 20 by simultaneously holding the center mouse button down while dragging the cursor to the left. Adjust the level for a viewable image(as shown in [Figure 10-10](#)).
 - b.) Position the reference ROI box built in step 2 on the plexiglas portion of the QA#2 phantom image without touching the water portion or the hole pattern (see [Figure 10-10](#)).
 - c.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box #1 ROI (size and position) on the remaining images in the series.
 - d.) Click on the MEASURE button and select the box ROI icon to display Box #2. The system places an ROI box labeled #2 at the center of the image with the exact same dimensions as Box #1.
 - e.) Reposition Box #2 over the water portion of the QA#1 phantom image and directly above Box #1 (Box # 2 position in [Figure 10-10](#)).
 - f.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box #2 ROI (size and position) on the remaining images.
 - g.) Record the Mean values of the 1st, 3rd, 5th, and 7th images in the series for each of the two box positions in Table 8.

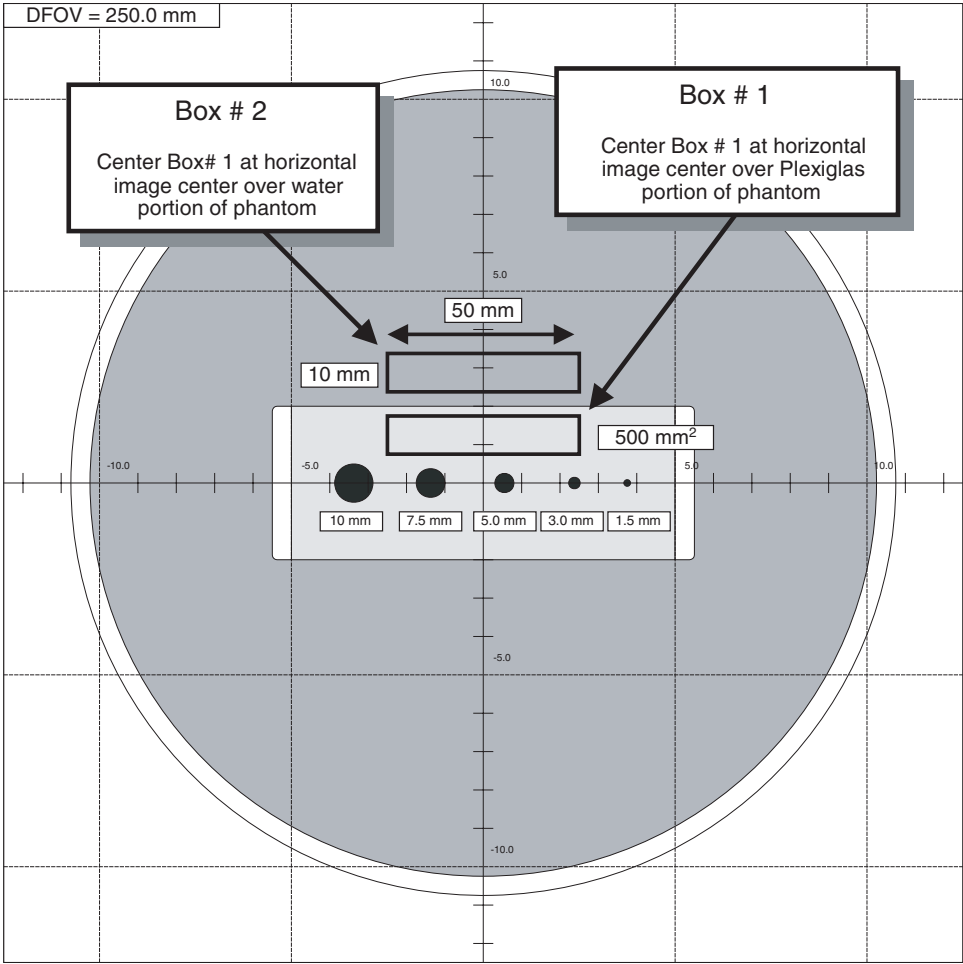


Figure 10-10 20cm QA#2 Phantom Low Contrast Detectability - Building and Placing Reference ROI Boxes

Image	Visible Holes Viewable at Window 20	Box 1 Means (Plexiglas)	Box 2 Means (Water)	Contrast Factor (Box 1 Means – Box 2 Means)	Comments
1					
3					
5					
7					
Specifications	See Table 10-9	n/a	n/a	2.0 to 12.0	n/a

Table 10-8 20cm QA#2 Phantom Low Contrast Detectability Image Performance Worksheet #1

- 4.) View images 1, 3, 5, and 7 of the QA#2 Holes Series; record the number of visible holes in Table 10-8, and verify each image meets specifications.
- a.) While viewing images 1, 3, 5, and 7, indicate in Table 10-8 the number of holes that are visually distinguishable (see Figure 10-10).
- b.) Verify that the Visible Hole visual check meets specifications listed in Table 10-9 for the calculated Contrast Factor and record them in the HHS Record Tables.

Contrast Factor Range (Box 1 Means – Box 2 Means)	Visible Number of Holes		Smallest Visible Hole Size
	Lower Limit*	Upper Limit*	
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm

*Required number of visible holes depends on the contrast factor.

Table 10-9 20cm QA#2 Phantom Visible Hole Specifications

3.5.2.2 Failure Recovery

Specifications

At least two out of the four images of the series must pass the 20 cm QA#2 Low Contrast Detectability parameter (Visible Hole Size for a calculated Contrast Factor) for the third series scan parameters specifications.

Recommended Recovery

- 1.) Check Phantom Alignment (leveling is critical) and repeat this scanning and High Contrast Spatial Resolution Performance Test.
- 2.) Perform Alignment Procedures (POR Alignment, BOW Alignment, CBF/SAG Alignment, ISO Alignment, and Hot ISO Alignment).
- 3.) Perform Full Calibration (Detailed Calibration and Auto CT# Adjust).
- 4.) Repeat this scanning and Low Contrast Detectability Performance Test.

3.5.3 QA#3 Phantom Brightness Uniformity and CT#

3.5.3.1 Performance Verification

- 1.) Select the fourth 20 cm QA#3 phantom exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS desktop.
 - b.) From the Image Works desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and fourth (QA#3) series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 31 x 31 pixel reference ROI box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square 15 mm x 15 mm (225 mm²) box. Tolerance: 15 mm ± 1 mm (196 mm² to 256 mm²).
If required, magnify the image to adjust it to proper dimensions. (See [Figure 10-11](#) for additional ROI size and placement information.)
- 3.) Collect Mean values for five reference ROI box positions on the eight 20 cm QA#3 phantom images.
If required, magnify the image to adjust it to proper dimensions. (See [Figure 10-11](#) for additional ROI placement information.)

- a.) Position the reference ROI box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
- b.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
- c.) Click on the MEASURE button and select the box ROI icon to display Box #2. The system places an ROI box labeled #2 at the center of the image with the exact same dimensions as Box #1.
- d.) Reposition Box #2 to the left center portion on the first image (Box #2 position in Figure 10-7).
- e.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box #2 ROI (size and position) on the remaining images.
- f.) Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in Figure 10-8 (Box #3 through #5).
- g.) Record the Mean values of the eight images in the series for each of the five box positions in Table 10-10.

Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

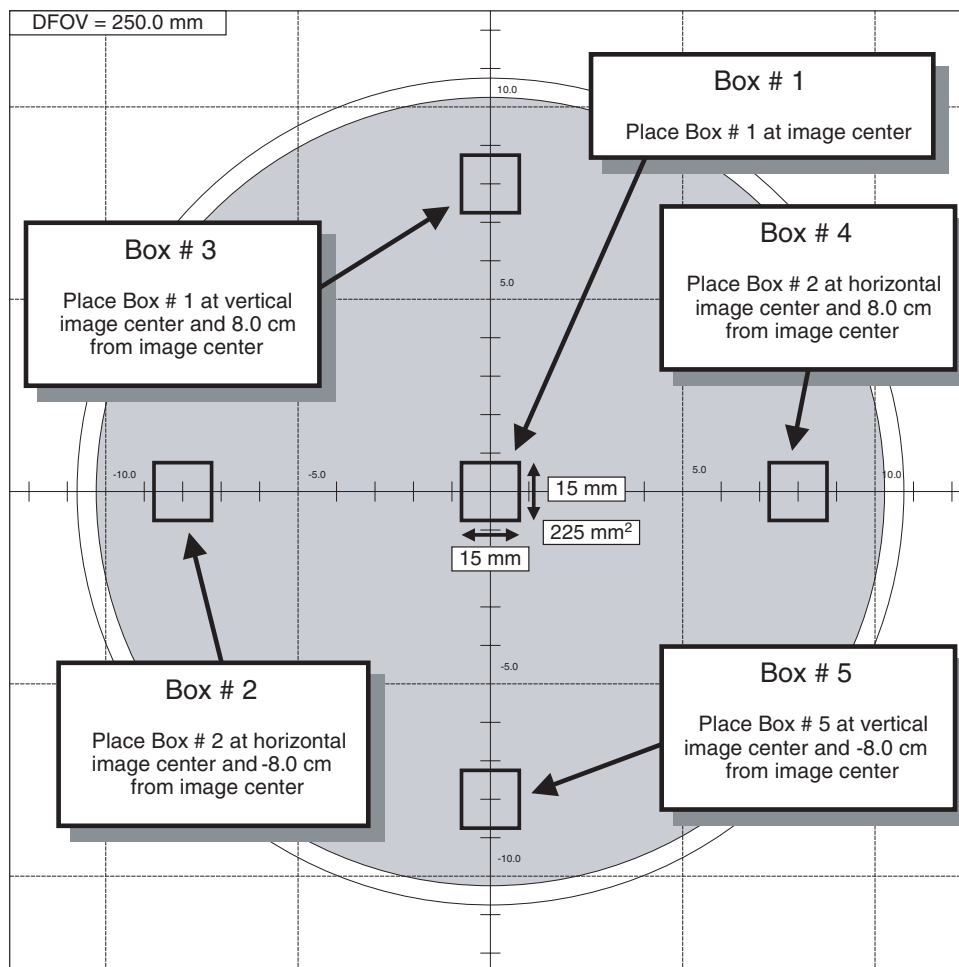


Figure 10-11 20cm QA#3 Phantom Brightness Uniformity & CT# Measurement - Building And Placing Reference ROI Boxes

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1			n/a		n/a		n/a		n/a		n/a
2			n/a		n/a		n/a		n/a		n/a
3			n/a		n/a		n/a		n/a		n/a
4			n/a		n/a		n/a		n/a		n/a
5			n/a		n/a		n/a		n/a		n/a
6			n/a		n/a		n/a		n/a		n/a
7			n/a		n/a		n/a		n/a		n/a
8			n/a		n/a		n/a		n/a		n/a

Table 10-10 20cm QA#3 Phantom CT# Brightness Uniformity & CT# Image Performance Worksheet

Box Size = 196 mm² to 256 mm²
 15 mm (± 1 mm) x 15 mm (± 1 mm)
 31 (± 2 pixels) x 31 (± 2 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -80 mm
 Box 3 = 80 mm x 0 mm
 Box 4 = 0 mm x 80 mm
 Box 5 = -80 mm x 0 mm

Row	Images	Center Box Means (AvXc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo - AvXc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
2A1A	1, 3, 5, 7				n/a	
1B2B	2, 4, 6, 8				n/a	
Specifications		0.0 ± 3.0	n/a	<± 3.0	n/a	

Table 10-11 48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet

- 4.) Calculate the Brightness Uniformity and CT# values for each image in the series; record the values in [Table 10-11](#), and compare them to the specifications.
 - a.) Calculate the average means values (AvXo) for the four outside boxes (boxes #2 through #5) for each of the images, and record them in [Table 10-10](#).
 - b.) Calculate the average center box (box #1) means values (AvXc) for each row (2A1A: Images 1, 3, 5, and 7; 1B2B: Images 2, 4, 6, 8), and record them in [Table 10-11](#).
 - c.) Calculate the average outside boxes (Box #2 through #5) means values (AvXo) for each row (2A1A: Images 1, 3, 5, and 7; 1B2B: Images 2, 4, 6, 8), and record them in [Table 10-11](#).
 - d.) Calculate the Brightness Uniformity (AvXo - AvXc) value for each row, and record it in [Table 10-11](#).
 - e.) Verify that the Brightness Uniformity (AvXo - AvXc) value and the average CT# value (AvXc) for each row meets the specifications listed in [Table 10-11](#).
 - f.) Record the 20 cm QA#3 phantom Brightness Uniformity (AvXo - AvXc) value and average CT# value (AvXc) for each row in the HHS Record Tables.

3.5.3.2 Failure Recovery

Specifications

Each Row (2A1A, 1B2B) of the series must pass the 20 cm QA#3 phantom Brightness Uniformity and average CT# specifications:

- AvXo - AvXc: $< \pm 3.0$
- AvXc: $< \pm 3.0$

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify the 20 cm QA#3 phantom Image Performance.

3.5.4 QA#3 Phantom Noise

3.5.4.1 Performance Verification

- 1.) Select the fourth 20 cm QA#3 phantom exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS desktop.
 - b.) From the Image Works desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and fourth (QA#3) series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 51 x 51 pixel reference ROI box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square, 25 mm x 25 mm (625 mm²) box. Tolerance: 25 mm $\pm$ 1 mm (576 mm² to 676 mm²).
If required, magnify the image to adjust it to proper dimensions. (See [Figure 10-12](#) for additional ROI size and placement information.)
- 3.) Collect Standard Deviation values for the single reference ROI box position on the eight 20 cm QA#3 phantom images.
If required, magnify the image to adjust it to proper dimensions. (See [Figure 10-12](#) for additional ROI placement information.)
 - a.) Position the reference ROI box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
 - b.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box #1 ROI (size and position) on the remaining images in the series.
 - c.) Record the Standard Deviation values of the eight images in the series for centered box position in [Table 10-12](#).
- 4.) Calculate the average Noise values for each image in the series, record the values in [Table 10-12](#), and compare them to the specifications.
 - a.) Calculate the average Noise values (AvSDc) for the inside Boxes for each of the two rows (2A1A and 1B2B), and record them in [Table 10-12](#).
 - b.) Verify that the Noise (AvSDc) values for each row meets the specifications listed in [Table 10-12](#).

c.) Record the 20 cm QA#3 Phantom Noise (AvSDc) for each row in the HHS Record Tables.

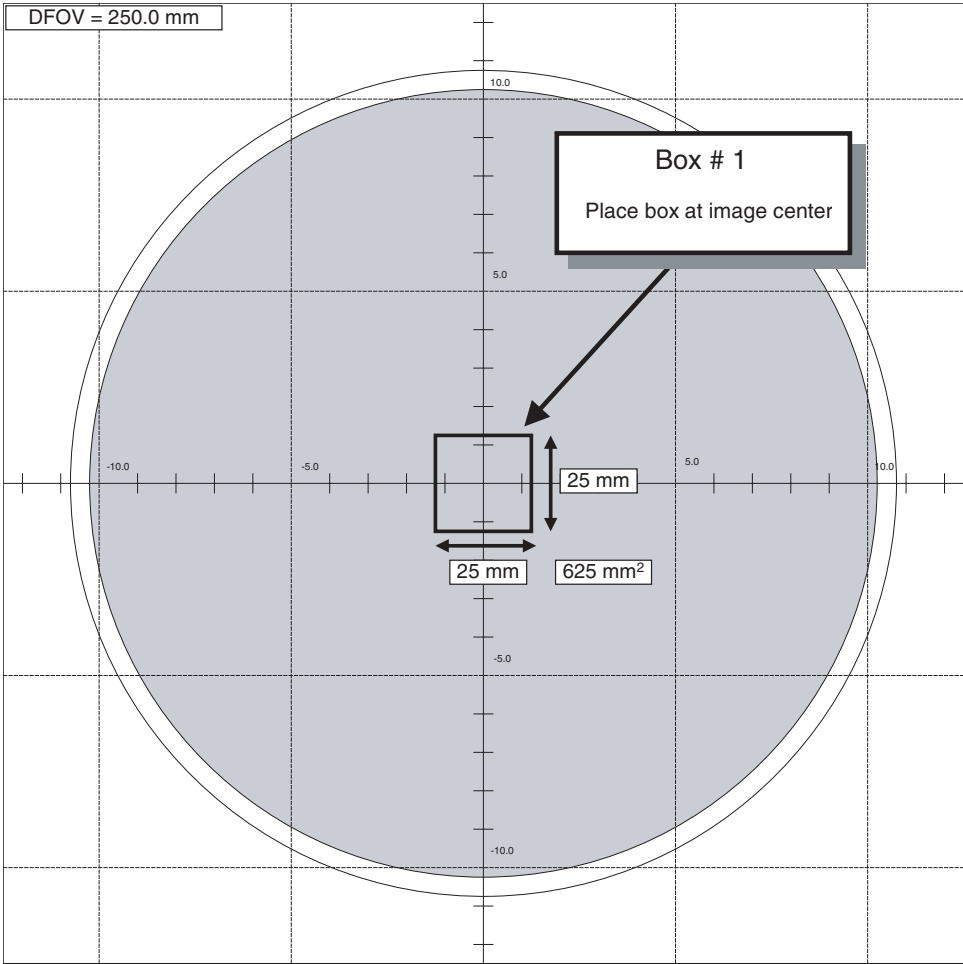


Figure 10-12 20cm QA#3 Noise Measurement - Building And Placing Reference ROI Box

Image	Row	Box 1 Std Dev (SDc)	Average Std Dev (AvSDc)	AvSDc Specifications		Comments			
				Systems with less than 5000 scans	Systems with more than 5000 scans				
1	2A1A			3.2 ± 0.3	3.2 ± 0.4				
3									
5									
7									
2	1B2B								
4									
6									
8									

Table 10-12 20cm QA#3 Phantom Noise Performance Worksheet

Box Size = 576 mm2 to 676 mm2
25 mm (± 1 mm) x 25 mm (± 1 mm)
51 (± 2 pixels) x 51 (± 2 pixels)

Box Position: Box 1 = 0 mm x 0 mm

3.5.4.2 Failure Recovery

Specifications

Each Row (2A1A, 1B2B) of the series must pass the 20 cm QA#3 phantom Noise specifications shown in [Table 10-12](#).

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify the 20 cm QA#3 phantom Image Performance.

Section 4.0

Save System State

Use the following commands to create the System State disk (DVD or MOD).

- 1.) Load the arechive medium (DVD or MOD) into the archive drive on the front of the console.

- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .

- 3.) Click on the PM icon, .

- 4.) Select SYSTEM STATE to open the System State Save/Restore menu.

- 5.) Select ALL

- 6.) Select SAVE

- 7.) When the save operation completes, select FILE and QUIT from the pull-down menu.

- 8.) Remove the archive medium (DVD or MOD) from the drive.

Chapter 11

System Alignments



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss or equipment damage, please do the following:

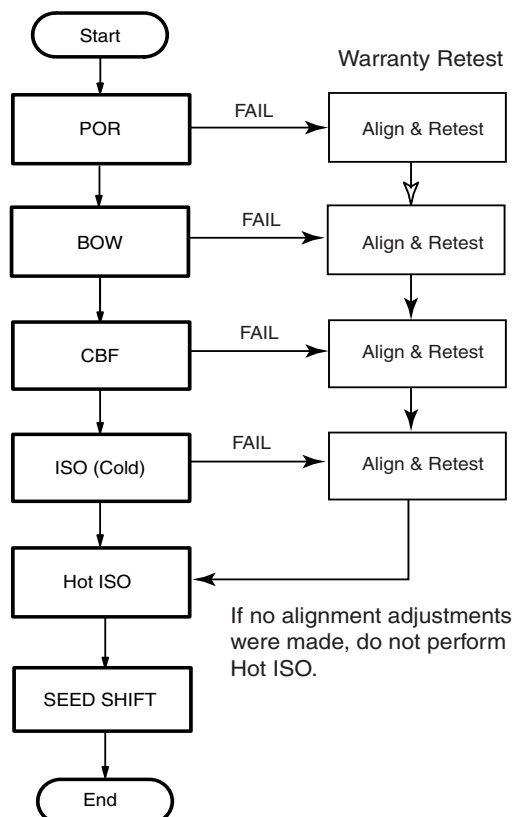
- When directed, record data collected from procedures in this chapter into GE Form 4879, located on the CD for this scanner's service manual.
- Locate the HHS data sent with your LightSpeed scanner. It can be found in the software collector carton. Retain this data with the HHS data collected at the site in a safe place.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Note: For each adjustment procedure, do not loosen anything until after you have run your first scan and calculated your first adjustment. Cool the tube at least two hours before starting the alignments in this chapter.

Section 1.0

Introduction

Use the procedures in this chapter to make sure the X-ray beam focuses directly upon the detector window. The flowchart in [Figure 11-1](#) shows an overview of this chapter.



IMPORTANT:

If alignment adjustments are made during this procedure, a FULL SET of ALL detail calibrations must be completed before proceeding with the system tests in [Chapter 7](#).

The procedure for completing detail calibrations is found in the *System Manual*, in the *Tube Change* section.

This should also be Code 10: Install Warranty.

Figure 11-1 System Alignment Overview

1.1 Required Tools

- Polaroid type 52 film
- Film developer
- Caliper Dial indicators
- Masking tape
- Socket set
- Torque wrench
- 48 cm Phantom
- Dial indicator mount
- Radial dial indicator

1.2 Gantry Geometry Definitions

Theta: (Also known as ISO) Along the tangent to gantry rotation, on the rotating structure. Move the X-ray tube to the 180° (six o'clock) position to equal 0° theta. You can adjust the tube unit and collimator in the theta direction.

Z-Axis: Parallel to the axis of gantry rotation, in the direction of cradle motion.

- You can adjust the tube unit and detector in the Z-Axis direction.
- Positive Z-Axis direction: Move the tube/detector toward the table.
- Negative Z-Axis direction: Move the tube/detector away from the table.

Radial: Along the radius, from the center of gantry rotation, in and out from gantry isocenter. You cannot adjust the LightSpeed detector in the radial direction.



Figure 11-2 Gantry Geometry

1.3 Tube Heat Definitions

Cold Tube:	No exposures for more than 6 hr. (overnight)
Cool Tube:	No exposures for more than 2 hr., but less than 6 hr.
Warm Tube:	No exposures for more than 1 hr., but less than 2 hr.
Hot Tube:	Exposures in less than 1 hr.

Section 2.0

Plane of Rotation (POR)

2.1 Overview

Plane-of-Rotation measurement/alignment puts the X-ray tube in the correct physical relationship to the detector and verifies it. This is normally only necessary when the X-ray tube is replaced or moved.

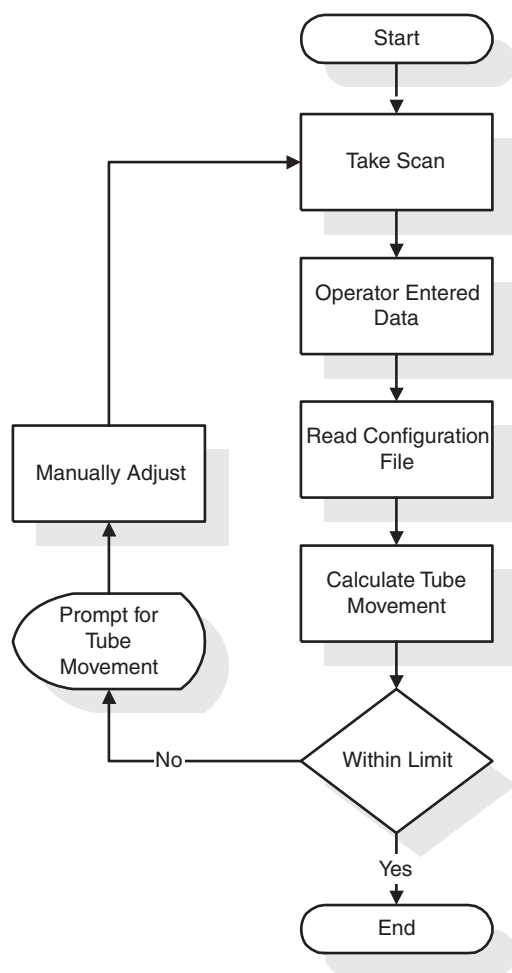


Figure 11-3 POR Procedural Flow

2.2 Tools Required



Figure 11-4 Vernier Caliper & Dial Gauge Indicator

- Calibrated Vernier Caliper (millimeters or inches).
- Dial Gauge Indicator (mm or in)
- Type 52 Polaroid Film (see [Figure 11-7](#))
- Type 52 Polaroid Film Developer

2.3 Procedure

Remember to always perform POR measurements when the X-ray tube unit is cool.

- If you have acquired any low technique scans prior to beginning the Plane-of-Rotation (POR) alignments, wait at least five minutes before starting the following procedure.
- If you recently acquired image scans and want to re-check the Plane-of-Rotation, wait at least 90 minutes to allow the tube unit to cool. (*Wait 90 minutes if the tube unit had more than 25 Kilo joules exposure, $[kV \times mA \times Sec \div 1000]$ within the last 30 minutes.*)

2.3.1 Start the POR Software

Note: Correct tube temperature is critical.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Click on INSTALL.
- 4.) Click on the SYSTEM ALIGNMENTS option.
- 5.) Click on the PLANE OF ROTATION option.



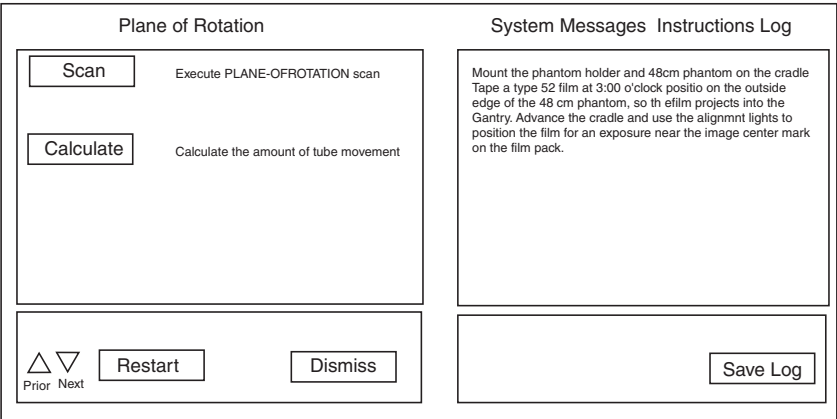


Figure 11-5 Plane Of Rotation Alignment Screen

2.3.2 Measure Tube Alignment

- 1.) Get the system’s phantom holder and attach the 48 cm phantom to it. (see [Figure 11-6](#)).

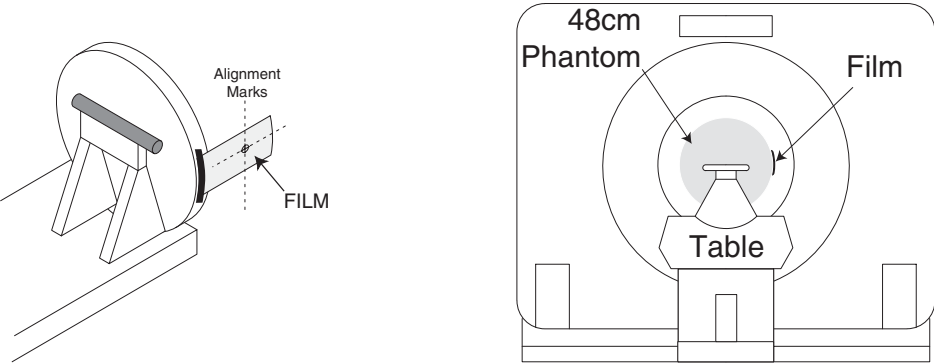


Figure 11-6 48 cm Phantom and Film on Table

- 2.) Attach one *Polaroid type 52* film on the outside edge of the 48 cm phantom, at the 3 o’clock position.

Note: Only the film should be projecting into the Gantry bore when complete. The phantom only positions and holds the film in the gantry bore (see [Figure 11-6](#)).

Orient the side of the film marked “**This side toward lens**” toward the iso center (see [Figure 11-7](#)). When exposed and developed later, the film shows the alignment of the X-ray beam with respect to the table, as viewed from the X-ray tube in the 3:00 o’clock position.

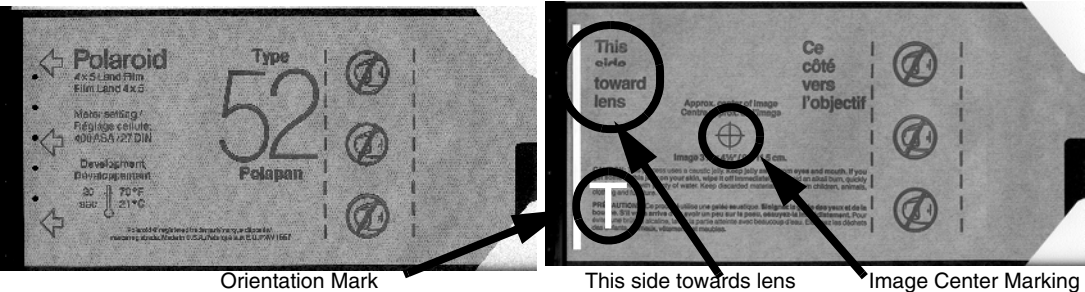


Figure 11-7 Polaroid Film, Type 52

- 3.) Advance the cradle, while using the alignment lights, to position the film’s center marking on the alignment light marks.

- 4.) Take a scan.
- 5.) Remove the exposed film and immediately mark the outside of the film nearest the table. For example, use a pen and print the letter **T** on the film nearest the table (Figure 11-8). Go and develop the exposed film.



Figure 11-8 Placement of “T” on Film

- 6.) After the film is developed, transfer the table marking to the film itself while keeping the orientation correct (Figure 11-9).

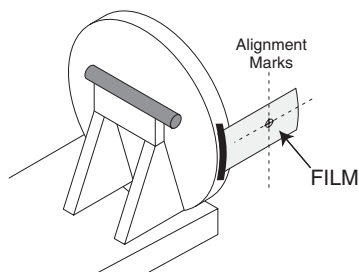


Figure 11-9 Alignment Marks on Film

- 7.) Verify the following (see Figure 11-10):
 - Film's narrow (white beam) slit lies within (between) the wider (gray) X-ray slit.
 - Film's edges in both Z direction are equally well defined by the exit slit of the collimator. The edges of the narrow beam should be much sharper than the wide beam.

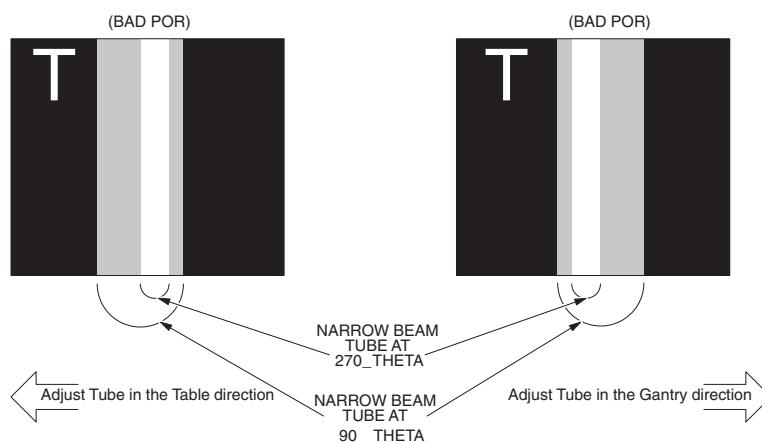


Figure 11-10 Plane of Rotation Film Interpretation

- 8.) Refer to [Figure 11-10](#) and [Figure 11-11](#) during the following instructions. Use a vernier calipers to measure the width of the two wider (dark gray) slits. They're the dark gray slits that extend past the edges of the narrower (off-white) slit and to the blackest part of the film.

- a.) Take three measurements to obtain an average value for X_F . X_f is the side of the film closest to the table. Using the same film measurement tool, take one X_f measurement at the top of the film, another near the middle and another near the bottom. Add these three X_f distances together and divide this sum by 3 or n . *Where,*

$$X_F = (X_{f1} + X_{f2} + X_{f3}) / (n)$$

Note:

It is important to take multiple measurements to reduce the likelihood of a measurement error, and consequently increase the accuracy of the alignment.

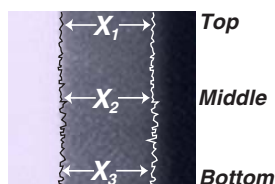


Figure 11-11 XF and XR measurement points

- b.) Now take three measurements to obtain an average value for X_R . Take one X_R measurement at the top of the film, another near the middle, and another near the bottom of the film. Add these three values together and divide the sum by 3 or n to obtain an average for X_R .

$$X_R = (X_{r1} + X_{r2} + X_{r3}) / (n)$$

- c.) Use the values obtained for X_F (front distance) and X_R (rear distance) in the calculation that follows.

- 9.) Click on the CALCULATE button (see [Figure 11-5](#)). Enter the values for X_F and X_R obtained in the steps above. The software does the distance calculation.

Note:

The specification limits are 0.059 - 0.082 in. or 1.50 - 2.083 mm.

- 10.) Check the results

- If the tube is out of specification, perform steps 1 and 2 ([page 263](#)), and move the tube. Refer to [Figure 11-12](#) for more details.
- If adjustment is necessary, a clockwise rotation (in) of the adjustment bolt moves the tube toward the table side. Repeat Steps 1 through 9, again, if you moved the tube, to check the accuracy of the adjustment. (Refer to [Figure 11-10](#) for more details.)

Note:

Mount the dial indicator onto the collimator/tube assembly as shown [Figure 11-12](#). *Make sure you zero the dial indicator when it's securely in place.*

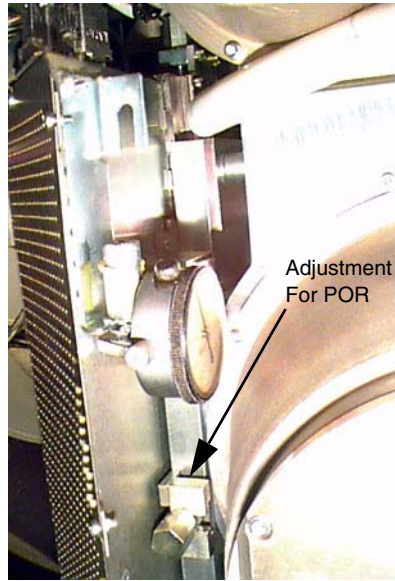


Figure 11-12 Dial Indicator Attached to Collimator/Tube Assembly

- When the tube is within specification, tighten the four M-12 Bolts on the tube and torque them to 50 ft-lbs (67.8 N-m) and you're done.

11.) Enter the results in the HHS table.

Section 3.0

Beam on Window Alignment (BOW)

3.1 Overview



NOTICE
Potential for
Image
Artifacts

Make sure the tube is cool before continuing. Wait 15-30 minutes after the completion of Plane-of-Rotation.

Purpose: Bring the detector into alignment with the X-ray beam of the correctly positioned tube. There are two specifications for BOW Alignment. The software expects the user to evaluate the raw vectors to determine if it meets the requirements.

- The first specification is for the initial check of BOW. The limits are -1.5 ± 0.25 mm.
- The second specification is for verification scans when the system is warmer. The limit for verification scans is -1.5 ± 0.5 mm.

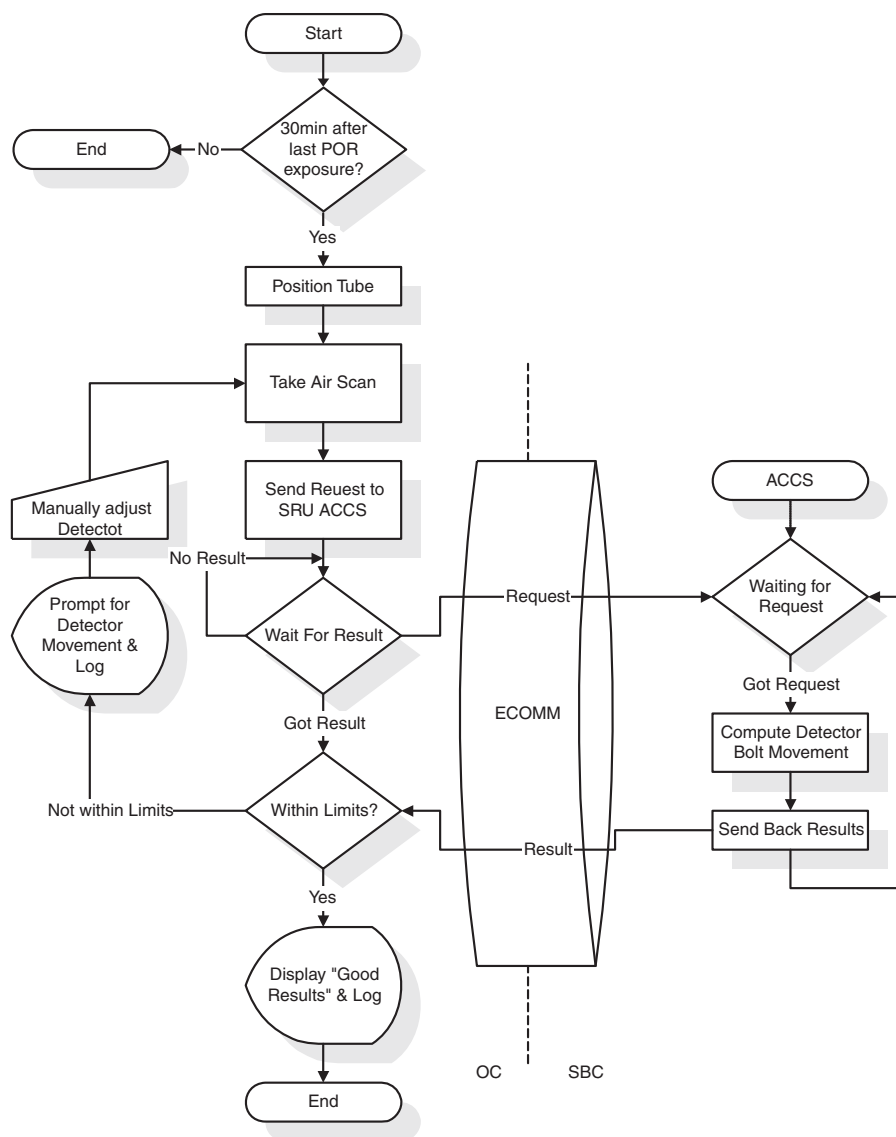


Figure 11-13 BOW Alignment

3.2 Cool Tube

Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of Plane-of-Rotation.

3.3 Access the Software

Never position the 1/8 in. metal shaft directly on the ISO center—always offset it. The software calculates tube movement for ISO center and looks for a sinusoidal waveform generated by the off center of the metal shaft. A metal shaft, positioned at ISO center, generates a straight line waveform.



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the UTILITIES icon.

3.) Select INSTALL.

4.) Select SYSTEM ALIGNMENTS.

5.) Select BEAM ON WINDOW.

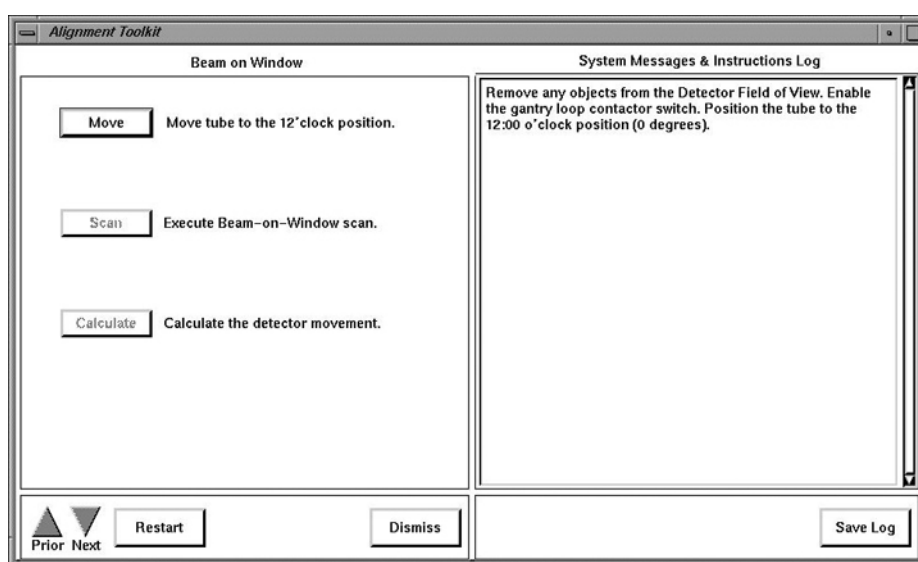


Figure 11-14 BOW screen

3.4 Adjustment Procedure

Tools Required: 21mm wrench
5 mm 6 in. long hex bit
5 mm thin wrench

Note: To make the adjustment, you must remove the front and rear covers (step 4).

Note: **Do not loosen anything until after you run your first scan and calculate your first adjustment.**

- 1.) Select MOVE to send the tube to the twelve o'clock position.
- 2.) Acquire a Beam on Window Scan.
- 3.) Select CALCULATE and make adjustments as indicated by the program.

You receive screen output similar to the following:

Upon intial inspection: spec = -1.5 +- 0.25mm

Upon verification scan: spec = -1.5 +- 0.50mm

Compare the raw alignment vectors to these specifications, make alignments suggested if specification is not met.

- 4.) Remove the front and rear covers.
- 5.) Loosen the M-6 cap head screws, located at each end of the detector (a total of two cap head screws). Refer to [Figure 11-15](#).
- 6.) Loosen the middle nut (jam nut) with a 21 mm thin profile wrench. Make adjustments using a 21 mm thin open end wrench, as requested by software. Tighten the jam nut and cap screws after adjustment.
- 7.) Acquire a Beam on Window Scan.
- 8.) Select CALCULATE.
- 9.) If the adjustments pass the calculation, proceed to step 10, otherwise return to step 4.
- 10.) Tighten all three M-6 cap head screws, using a 5 mm x 6 in. long hex bit, for 3/8 in. drive. (Torque to 10 ft-lb., or 14 N-m).

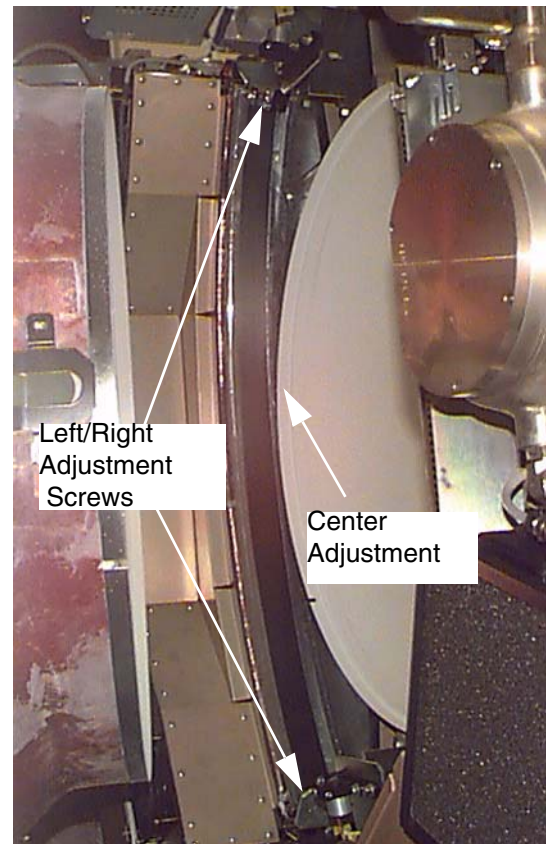
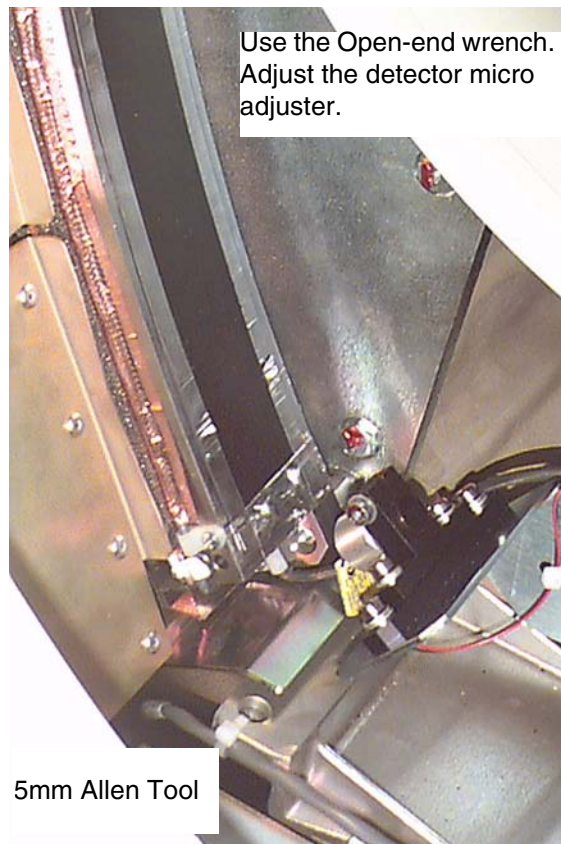


Figure 11-15 Detector Cap Head Screws

Note: **CW (Clockwise)** moves the detector toward the mounting plate. **CCW (Counter-clockwise)** moves the detector away from the mounting plate. **Right=Low Chnl, Center=Medium Chnl, Left=High Chnl**

- 11.) Enter the results in the HHS table.

Section 4.0

CBF/SAG Alignment Process

Purpose: CBF/SAG Alignment ensures the focal spot is accurate, the bowtie filter is centered, and center of rotation is in a straight line.

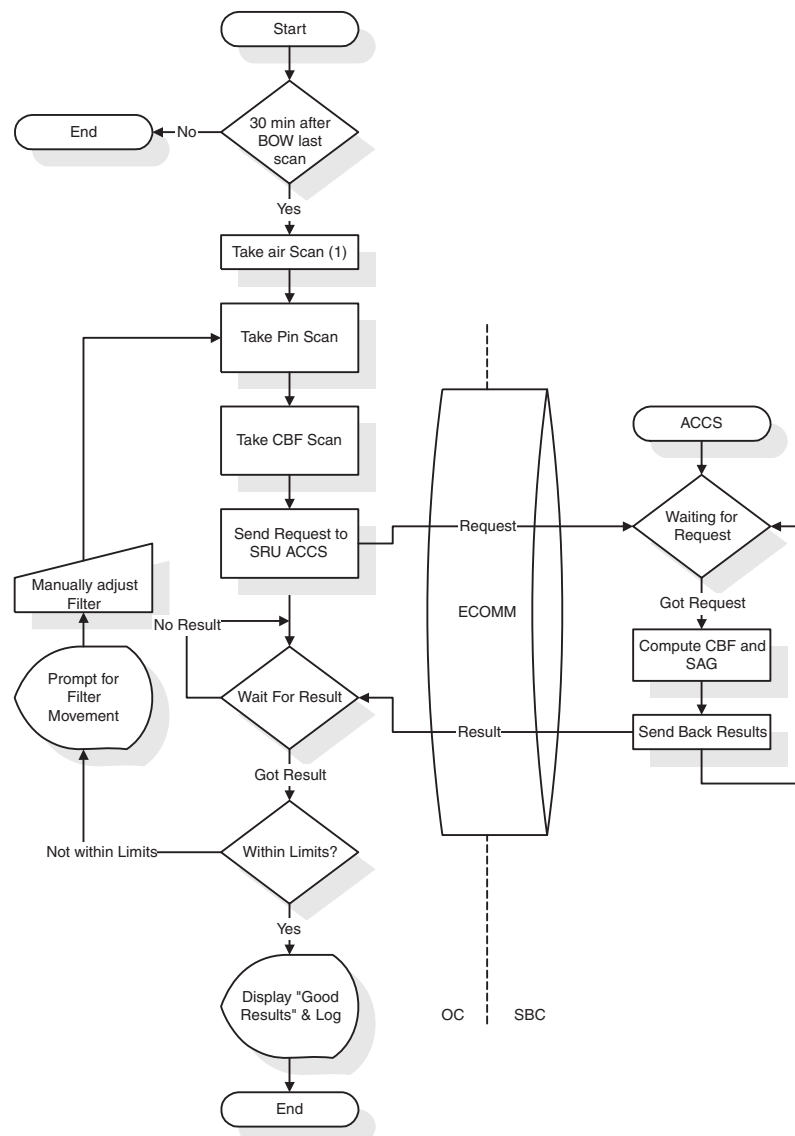


Figure 11-16 CBF Procedure

Never position the 1/8 in. metal shaft directly on the ISO center—always offset it. The software calculates tube movement for ISO center and looks for a sinusoidal waveform generated by the off-center of the metal shaft. A metal shaft, positioned at ISO center, generates a straight line waveform.

4.1 Cool Tube

Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of Beam On Window.

4.2 Access the Software

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENT.
- 5.) Select CBF/SAG ALIGNMENT.

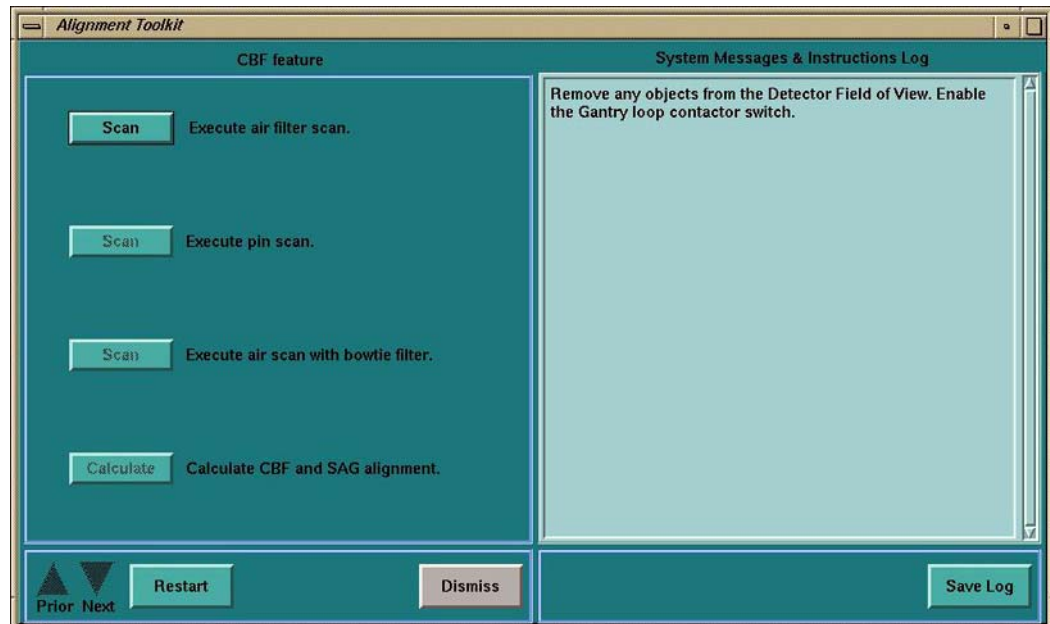


Figure 11-17 CBF/SAG Alignment Screen

4.3 Adjustment Procedure

Tools Required: 1/8 in. PIN (#0 Phillips) screw driver

Note: **Do not loosen anything until after running your first scan and calculating your first adjustment.**

- 1.) Press the flashing **START SCAN** button to acquire an air filter scan.
- 2.) Position the 1/8 in. Pin (#0 Phillips round) screw driver on the phantom holder, parallel to the Z-Axis to the end of the table nearest the gantry or on the phantom holder.
- 3.) Turn On the laser lights.
- 4.) Advance and adjust the table to position the metal shaft right of ISO center.
- 5.) Verify that the shaft remains level (perpendicular to the scan plane).
- 6.) Acquire a pin scan.
- 7.) Acquire an air scan with bowtie filter in place.
- 8.) Select **CALCULATE** to calculate the CBF and SAG alignment.
- 9.) Mount the dial indicator onto the collimator.
Remember to zero the dial indicator.
- 10.) Loosen the collimator bolts (4 in front, 2 in the rear). Use a swivel adapter on a ratchet wrench to loosen the bolt behind the cable connections.
- 11.) Adjust the collimator as indicated by the calculation results. Ignore the negative sign in front of the adjustments.
- 12.) Tighten the collimator bolts (4 in front, 2 in the rear). (Torque to 50 ft.-lb./ 68 N-m)
- 13.) Wait 15 minutes for the tube to cool.
- 14.) Rescan and calculate.
- 15.) Repeat steps 10 – 11 until the adjustment falls within the limit, then proceed to the next section.
- 16.) Enter results in the HHS table.

Section 5.0

ISO Alignment

5.1 Overview

Never position the 1/8 in. metal shaft directly on the ISO center—always offset it. The software calculates tube movement for ISO center and looks for a sinusoidal waveform generated by the off-center of the metal shaft. A metal shaft positioned at ISO center generates a straight line waveform.

5.2 Cool Tube

Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of CBF/SAG Alignment.

5.3 Access the Software

1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



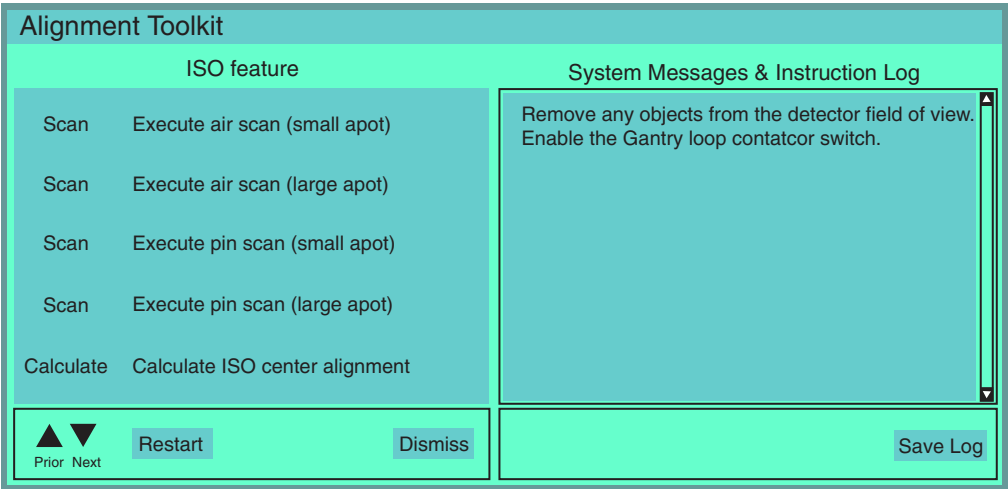
2.) Click on the UTILITIES icon.



3.) Select INSTALL.

4.) Select SYSTEM ALIGNMENTS.

5.) Select ISO.



↑ Use the Prior/Next Buttons to Access the Calculate Button

Figure 11-18 ISO Alignment Menu

5.4 ISO Adjustment Procedure

Tools Required: 1/8 in. PIN (#0 Phillips) screw driver

Note: Do not loosen anything until after you have run your first scan and calculated your first adjustment.

- 1.) Acquire an Air scan (small spot). (See [Figure 11-18.](#))
- 2.) Acquire an Air scan (large spot). (See to [Figure 11-18.](#))
- 3.) Position the 1/8 in. Pin (#0 Phillips round) screw driver on the phantom holder parallel with the Z-axis to the end of the table, nearest the gantry or on the phantom holder.
- 4.) Turn On the laser lights.
- 5.) Advance and adjust the table to position the metal shaft, right of ISO center.
- 6.) Verify that the shaft remains level (perpendicular to the scan plane).
- 7.) Acquire a small spot pin scan.
- 8.) Acquire a large spot pin scan.
- 9.) Select CALCULATE ISO. (If ISO falls outside 389.75 channels, no wait between tube adjustment and the test scan is required. If ISO falls between ± 0.02 and ± 0.1 channels, wait 15 minutes after you adjusted the tube before you scan.)
 - a.) If adjustments are required, proceed to step 10
 - b.) If adjustments are not required (pass), skip to step 15
- 10.) Zero the dial indicator and mount it on the tube assembly.
- 11.) Using a 10 mm hex, loosen the four (4) M-12 bolts on the tube assembly.
- 12.) Refer to [Figure 11-19](#) for the following steps:
 - a.) Locate the adjustment bolt on the top of the tube.
 - b.) Adjust the tube UP/DOWN as indicated by the calculation.
- 13.) Repeat steps 1 through 7.
 - If adjustments fall within the acceptable range, proceed to step 13.
 - If adjustments fall outside the limits, wait 15 minutes for the tube to cool and then return to step 9.
- 14.) Tighten the four M-12 bolts using a 10 mm hex (Torque to 50 ft.-lb./ 68 N-m).
- 15.) Enter results in the HHS table.

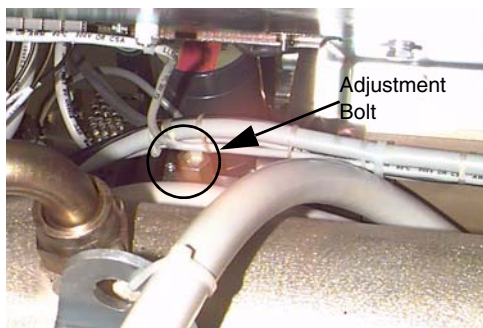


Figure 11-19 ISO Adjustment Bolt

- 16.) Wait 15 minutes, then recheck BOW to verify that the Beam-on-Window values are within the drift spec of $-1.5\text{mm} \pm 0.5\text{ mm}$.
 - If BOW is within spec, DONOT adjust anything
 - If BOW is out of spec, alignments must be done starting with POR and ending with BOW recheck.

Section 6.0

HOT ISO Alignment

COMPLETE THIS PROCEDURE ONLY IF ANY ADJUSTMENTS WERE MADE.

6.1 Accessing the Software

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENTS.
- 5.) Select HOT ISO ALIGNMENT.

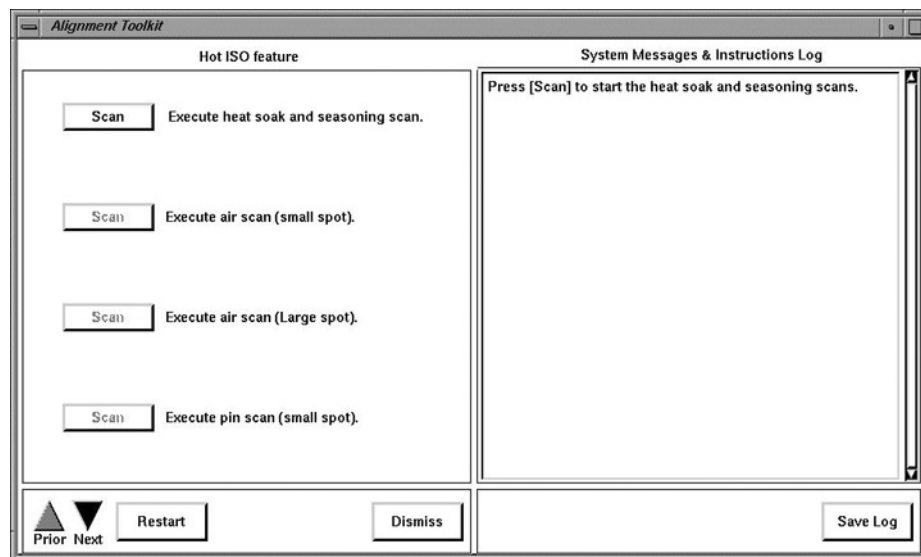


Figure 11-20 HOT ISO Screen

6.2 Adjustment Procedure

Note: **No movement of the tube is required. This correction is a software correction done automatically.**

- 1.) Execute the Heat Soak and Seasoning scan.
- 2.) Execute Air Scan (small spot).
- 3.) Execute Air Scan (large spot).
- 4.) Place the 1/8 in. screw driver on the phantom holder (should be pointing into the Z direction).
- 5.) Execute a Pin Scan (small spot).
- 6.) Execute Pin Scan (Large spot).
- 7.) Calculate correction factor with Calculate button.

Chapter 12

Generator Calibration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- When directed, record data collected from procedures in this chapter into GE Form 4879 located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

1.1 Process Overview

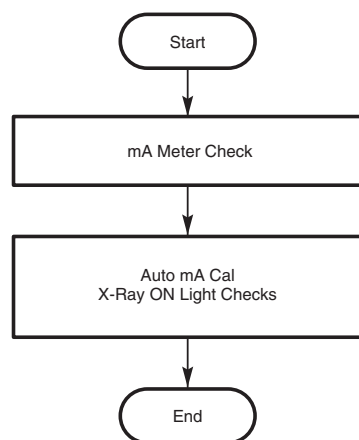


Figure 12-1 Chapter Flowchart

IMPORTANT:

- You must record the data collected in this chapter on GE Form 4879 (located in)
- Locate the HHS data, sent from the factory, in the software collector box.
- Retain this data with the HHS data collected at the site.
- A complete generator RECAL is not required.

Note: If any of the calibration procedures fail, the data collected is different, or you replace any component in the High Voltage section, YOU MUST RECALIBRATE THE SYSTEM. To re-calibrate the system, refer to procedures in the *LightSpeedSystem Service Manual for your scanner*.

1.2 Required Tool

DVM with meter leads

Section 2.0

mA Meter Check

- 1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.
- 2.) Click the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select GENERATOR CALIBRATION.
- 5.) Switch OFF the HVDC (550) ENABLE inside the gantry.
- 6.) Switch OFF the AXIAL DRIVE inside the gantry.
- 7.) Rotate the anode tank to the two o'clock position.
- 8.) Pin the gantry.
- 9.) Remove the OBC cover (if necessary).
- 10.) Use a DVM as a DC mA meter. Set the meter range to one decimal place.
- 11.) Connect the DVM black lead to TP8 (ACAL1) on the mA board.
- 12.) Connect the DVM red lead to TP11 (ACAL2) on the mA board.

Collimator
HEMRC
MA Control
KV Control
Gentry I/O
LSCOM
Heurikon (CPU)

Figure 12-2 OBC Board Slot Assignment

- 13.) Select MA METER CHECK.
- 14.) Select ACCEPT to begin the test after the time delay expires.
- 15.) Once the test begins, the software enables the meter circuit for four seconds.
- 16.) Record the displayed cathode and anode mA values on GE Form 4879 in the Circuit ON section of the kV Meter Verification table.
- 17.) The Circuit ON values of the cathode and anode sides should read approximately 15 mA to 25 mA 140 mA to 200 mA.

Note: The cathode side reads approximately 19 mA during Circuit ON. The anode should read 140 mA to 200 mA.

- 18.) Repeat the test for the cathode side:
 - a.) Use a DVM as a DC mA meter; set the meter range to one decimal place.
 - b.) Connect the DVM red lead to TP9 (CCAL1) on the mA board.
 - c.) Connect the DVM black lead to TP14 (CCAL2) on the mA board.

19.) Select ACCEPT.

20.) Record the displayed and measured cathode mA values in the Circuit OFF and Circuit ON table in space provided below. Acceptable Circuit ON values: 140 to 200 mA.

Note: **The anode side reads approximately 19 mA during Circuit ON. The cathode should read 140 to 200 mA, as stated above.**

- 21.) Disconnect the test equipment.
- 22.) Switch ON the HVDC (550) enable.
- 23.) Select DISMISS.

Circuit ON		
	A/D	Range
Anode mA		15 - 25
Cathode mA		15 - 25
DVM - A/D = Delta Compare Delta to Limits		

Table 12-1 mA Meter Verification (Table 1)

Section 3.0

Auto mA Cal/X-ray On Light Check

Use this procedure to verify and complete Auto mA Cal on the system. If you have just run a Heat Soak and Seasoning, wait 30 minutes before you run the Auto mA Cal.

Note: **The system automatically warms up the tube.**

- 1.) Select AUTO MA CAL.
- 2.) Press START SCAN when it flashes. The program automatically runs Ductility and Auto mA Cal.
- 3.) During this test, verify that the X-ray on indicators, located on the gantry display and the rear gantry shroud, illuminate during X-ray on.

If the system cannot complete the Auto mA Cal because HV aborts, run the "Tube Heat Soak and Seasoning" scan protocol (see the *LightSpeedSystem Service Manual* for your scanner for details). Place a 48 cm phantom in the X-ray beam to protect the detector from radiation damage during this procedure. Do not worry about the image quality of the scans reconstructed during the Tube Heat Soak and Seasoning.

- 4.) Record the data on Form 4879 (located in).

The MX200 tube uses a metal/ceramic frame, which means some electrons hit the target and actually bounce off, being captured by the frame. This causes a frame current, which results in an anode mA measurement of slightly less than the cathode mA, typically 5 % to 10% less. Because the cathode mA is the true indicator of radiation output, the CT QX/i system controls cathode mA rather than anode mA.

3.1 X-ray Lights

- 1.) Perform several scans (see below).
- 2.) Verify that the X-ray ON lights are on during the scans. Check the boxes in [Table12-2](#)
- 3.) Record data on **GE FORM 4879** (see).

PERFORMING A SCAN

- 1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.
- 2.) Select DIAGNOSTICS.
- 3.) Select DIAGNOSTIC DATA COLLECTION
- 4.) Set the scan time to 2.00
- 5.) Set the kV to 80
- 6.) Set the mA to 40.

Light On?	Light Locations
☐	SCIM
☐	Gantry Front
☐	Gantry Back (use a mirror to view)
☐	Room Light

Table 12-2 X-ray Light Chart

This Page Intentionally Left Blank

Chapter 13

Hospital/System Network Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- When directed, record data collected from procedures in this chapter into GE Form 4879 located on the CD for this scanner's service manual.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Overview

LightSpeed systems support two basic Networking Protocols:

- Advantage NET (IC, Signa 4.X, CT-HLA, CT/I ...)
- DICOM (CT/I, CT Synergy, Advantage Workstations, ...)

DICOM NETWORK INTRODUCTION

DICOM networks operate on the **tasks** or services that various devices on the network use or provide. These services are labeled as Application Entity Titles (AE Titles). The LightSpeedsystem is a **user** of six DICOM Network Services and is a **provider** of two DICOM Services:

AS A DICOM SERVICE USER:

- Send or **Push** images to another network device.
- Send or **Push** images to a DICOM Printer.
- Review the image database on another device and retrieve or **Pull** selected images from that device (Query/Retrieve User).
- Send or **Push** images to an image storage device and obtain confirmation that the images have been archived (Storage Commitment).
- Obtain Patient Worklist Information from the Hospital HIS/RIS System.
- Store images on MOD media.

AS A DICOM SERVICE PROVIDER:

- Receive **Pushed** images from another network device.
- Allow another network device to review the LightSpeed image database and to retrieve or **Pull** selected images (Query/Retrieve Provider)

For each DICOM Service that the LightSpeedsystem is a **User** (except for storing images on MOD media), you must **declare** this device on the LightSpeed system using three menu selections. For some devices, you must declare not only the device, but each service (AE Title) that the device provides. For example, you may be required to declare a PACS System twice on the LightSpeed system: once as a destination to **push** images and, second, as a destination that provides storage commitment capability after images have been **pushed**.

For each DICOM Service that the LightSpeed system is a **Provider**, you must **declare** the LightSpeed system on the network device that is using these services.

The hospital administrator provides information (hostnames, IP Addresses) required to complete configuring a hospital DICOM network. The DICOM Conformance Statement document (AE Titles, port numbers) is provided with each DCOM compatible network device on the network.

Section 2.0 Before You Start

2.1 Network Physical Requirements

Before setting up the LightSpeed system on the hospital network, verify that the following physical items are complete:

- LightSpeed console, monitor, keyboard, and mouse are installed and connected.
- LightSpeed system power is on
- Hospital Ethernet network RJ45 Class IV twisted pair cable is connected to the LightSpeed console network receptacle.
- Hospital network connection is operational and is running 10baseT or 100baseT.

2.2 LightSpeed Network Identity Information

To declare the LightSpeed system on the network, ensure that the following network identity information is available from the Hospital Network Administrator:

- Hostname (No more than 16 Characters)
- Internet Protocol (IP) Address
- Subnet Net Mask IP Address (if applicable)
- Broadcast Address (if applicable)

2.3 LightSpeed to DICOM Remote Hosts Network Information

To declare DICOM remote hosts (PACS systems, archival devices, review workstations) on the LightSpeed system, ensure that the following information is available for each remote host:

- From the Hospital Network Administrator:
 - Hostname
 - Internet Protocol (IP) Address
 - Network Protocol (DICOM for LightSpeed Systems)
- From the Remote Host Device DICOM Conformance Statement Document:
 - DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
 - DICOM Listening Port Number

2.4 LightSpeed to DICOM HIS/RIS Interface Network Information

To declare DICOM Hospital HIS/RIS Interface devices (Mitra and others) on the LightSpeed system, ensure that the following information is available:

- From the Hospital Network Administrator: *Internet Protocol (IP) Address*
- From the HIS/RIS Interface Device DICOM Conformance Statement Document:
 - DICOM Application Entity Title or AE Title (DICOM Service that the HIS/RIS interface provides)
 - DICOM Listening Port Number

2.5 LightSpeed to DICOM Printers Network Information

To declare DICOM on the LightSpeed system, ensure that the following information is available for each printer:

- From the Hospital Network Administrator:
 - Hostname
 - Internet Protocol (IP) Address
- From the Printer DICOM Conformance Statement Document:
 - DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
 - DICOM Listening Port Number

Section 3.0

Declaring the System on the Hospital Network

3.1 Enter Configuration Routine

- 1.) On the OC, open a Unix shell window.
- 2.) Enter root as a superuser:
Type: `su -` ENTER at the prompt.
Type the password and press ENTER at the password prompt.
- 3.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the root prompt.
- 4.) Launch the Install Utility:
Type: `reconfig` ENTER at the prompt.
The OC displays the Install Utility Window as shown in [Figure 13-1](#).



Figure 13-1 Install Utility Window

- 5.) Enter the Configuration Routine:
Using the mouse, click the CONFIG button.
The OC displays the System Configuration - System Settings screen, as shown in [Figure 13-2](#).

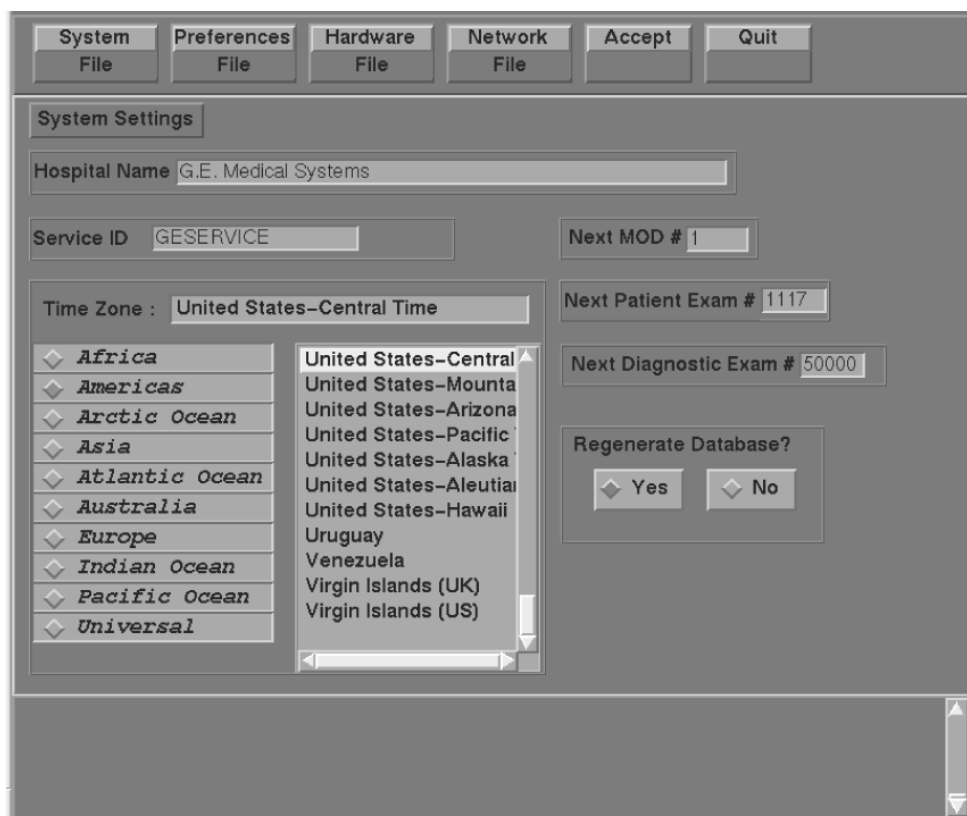


Figure 13-2 System Settings Screen

3.2 Configure Network Settings

This screen provides the ability to declare the system on a hospital network. Key information such as Host Name, IP Address, and Net Mask (for systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 13-3](#).
- 2.) Enter the Suite Name.

The Suite Name is a means of identifying this particular LightSpeedsystem as a part of a group of CT Systems in a suite configuration. This Suite Name appears on all image headers.

The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface is <Suite Name>_oc and the SBC interface is <Suite Name>_sbc.

Figure 13-3 Networks Settings Screen

- 3.) Enter the hospital-provided Host Name.
The Host Name identifies the network hostname and AE Title of the LightSpeedsystem.
The Host Name:
 - **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_OC.
 - **MUST NOT** exceed 16 Characters.
 - **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _
- 4.) Enter the hospital-provided IP Address.
- 5.) Enter the hospital-provided Net Mask (if the system is on a subnet).
- 6.) Enter the Broadcast Address:
The Broadcast Address should be the same as the IP Address. The bits of the host ID portion (last digit group) may be set to 1 or 0, depending on the configuration of the network. The standard default is 1, but older SunOS machines use 0.
For example:
If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255, if the network is configured to use 1's to specify the broadcast address.
If the network contains genesis-based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.
- 7.) Enter the hospital-provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- 8.) Select NIS (Yellow Pages database) Advanced Option, only if requested by the hospital network administrator, as follows:
 - a.) Select the ADVANCED OPTIONS button on the Network Settings screen.
 - b.) Select the Use NIS? button.
 - c.) Enter the hospital-provided Domain Name.
- 9.) Record all the Network parameters in the *LightSpeedSoftware Installation Procedures* Document, or on the worksheet in [Chapter 6](#).

3.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the System Configuration Screen.
The system loads the application software, OS patches, and kernel changes, and configures the system on both the OC and the SBC.
This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window, which closes when the loading process is complete. All the window output is logged to a file named:
`/var/adm/install.log.YYYYMMDDWWHHMMSS.`
(Where `YYYYMMDDWWHHMMSS` is the Date/Time that the loading process was started.)
- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click YES.
- 3.) The system automatically logs in as ctuser after the reboot. Select OK on the Autostart Disabled pop-up message.
- 4.) To start-up Application: in the console shell window, type **startup** ENTER.

Section 4.0

Declaring Remote Hosts on the System

4.1 Enter Remote Host Configuration Screen

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.



4.2 Declaring Advantage NET Remote Hosts on the LightSpeed

Use Advantage NET Protocol networks to communicate with older CT or MR Systems:

- MR Signa 4.x
- CT-HLA
- CT/I Systems
- Workstations that support the Advantage NET protocol

Advantage NET Protocol does not offer full compatibility with LightSpeed DICOM formats.

Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to communicate with this system.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter Screen as shown in [Figure 13-4](#).

Figure 13-4 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital-provided Host name.
- 3.) Enter the hospital-provided Network Address (IP Address).
- 4.) Select ADVANTAGE NET as the Network Protocol.
The system automatically de-highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.
- 5.) Select SAVE to store the parameter settings of the remote host.

4.3 Declaring DICOM Remote Hosts on the LightSpeed

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third-party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to communicate with this system.

- 1.) Select REMOTE HOSTS from the pull-down menu. The system displays the Remote Host Parameter screen as shown in [Figure 13-5](#).

Figure 13-5 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital-provided Host name.
- 3.) Enter the hospital-provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.
The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.
- 6.) Enter the AE Title from the DICOM Conformance Statement provided with the device.
Application Entity Titles (also known as ACR-Nema or DICOM Name) refer to the DICOM Network Services that a device provides to the LightSpeed system. For most devices, the AE Title is the same as the hostname (LightSpeed systems are equipped with this feature).
Some devices, such as PACS systems, may have separate AE Titles and port numbers for each of the services that the PACS system provides.
In these cases, enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services. the host provides. There is one host as an image **push-to** destination, another host as a **query/retrieve** provider, and a third host as a **storage/commitment** provider.
Review the DICOM Conformance Statement for each device that provides a remote host network service for the LightSpeed system. This includes image **push-to** or store destination, Query/Retrieve, and Storage Commitment. This ensures that each service is correctly configured.

- 7.) Select the correct Archive Node choice for the device. The Archive Node selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider. It indicates to the operator that a study/series/image was archived. Select one of the following:
 - AUTO: the LightSpeedsystem automatically checks if the designated remote host is a DICOM Storage/Commitment Provider.
 - YES: the device is the hospital-designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen indicates Archive Status = Y to the operator.
 - NO: the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct Access to local host: settings. These two selections allow you to block the remote host from using the LightSpeed DICOM services as a provider (image **push-to** destination, and a **Query/Retrieve** provider).
 - Send Images? Set to YES if the customer wants the LightSpeedsystem to allow images **pushed** to the system from the applicable remote host. Set to NO if the customer wants to block an image **push** from the applicable remote host.
 - Query/retrieve images? Set to YES if the customer wants the remote host to allow review of the LightSpeedimage database (query) and **pull** selected images from the database. Set to NO if the customer does not want the remote host to have this ability.
- 9.) Select the correct Custom search? setting. This selection allows the LightSpeedsystem to search through the remote host image database when the operator is using the remote browser screen to **query** the remote host. The search parameters that the LightSpeedsystem allows the customer to use are:
 - last name contains
 - patient ID
 - exam number
 - accession number
 - exam dateSelect ON if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
Select OFF if the device does not support custom searches.
- 10.) Record all the remote host network parameters for each remote host in the *LightSpeedSoftware Installation Procedures* Document.
- 11.) Select SAVE to store the parameter settings of the remote host.

Section 5.0

Declaring the System on Remote Hosts

5.1 Declaring LightSpeed on Advantage NET Protocol Devices/Systems

Instructions for declaring the system an Advantage NET remote host are found in:

- the Advantage NET Protocol device service manual
- the system service manual.

5.2 Declaring LightSpeed on DICOM Protocol Devices/Systems

Instructions for declaring the system a DICOM remote host are found in:

- the DICOM protocol device service manual
- the system service manual

The LightSpeedsystem provides two DICOM services as a provider to remote hosts:

- A remote host can **push** images to the LightSpeed image database.
- A remote host can review the LightSpeed image database (query) and **pull** selected images (retrieve).

Use the following LightSpeed parameter information to configure the DICOM device/system to either **push** images to the LightSpeed system and/or perform a **Query/Retrieve** operation:

- Hostname: Provided by the Hospital Network Administrator. (The same LightSpeed -assigned hostname entered in the Network Configuration Screen.)
- Application Entity Title: the same entry as the Hostname.
- Network Address: Provided by the Hospital Network Administrator. (The same LightSpeed-assigned IP Address entered in the Network Configuration Screen.)
- Network Protocol: DICOM 3.0.
- Port Number: For all DICOM service that the LightSpeed system provides, use 4006.
- Provider Type: This field concerns the LightSpeed DICOM **Query/Retrieve** provider capability. All LightSpeed systems are study root systems, which allow queries at the exam, series, and image level.
- Support Worklist: This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All LightSpeed systems support a filtered search of the image database as part of the LightSpeed DICOM **Query/Retrieve** provider capability.

Section 6.0

Declaring DICOM HIS/RIS Interfaces on the System

6.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the LightSpeed system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer.

In addition, the LightSpeed system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

6.2 Loading ConnectPRO Software Option on the System

- 1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.



- 2.) Click the UTILITIES icon.



- 3.) Click INSTALL.

- 4.) Select INSTALL OPTIONS and click START. The console displays the Software Options window as shown in [Figure 13-6](#). (Refer to the appropriate *Software Installation Procedures* manual for screen details.)

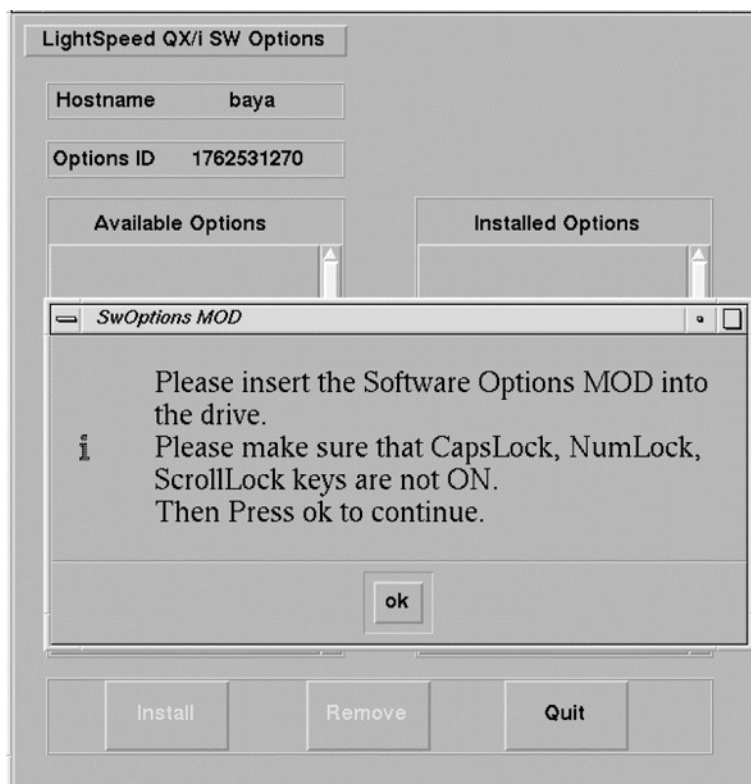


Figure 13-6 Options Window when First Selected

- 5.) Insert the ConnectPRO Options MOD into the MOD drive .
- 6.) Click OK to continue.
- 7.) On the Software Option window, select the CONNECTPRO option in the Available Options window.
- 8.) Click the INSTALL button.

The console may display a message box during the software loading operation. When the system has completed loading the software:

- the Installed Options window displays the ConnectPRO option.
- the console displays the ConnectPro Setup window as shown in [Figure 13-7](#).

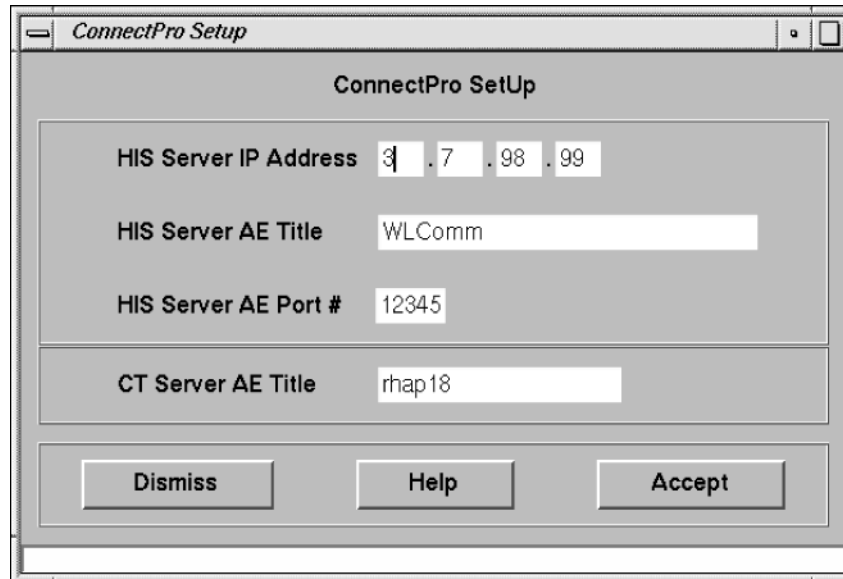


Figure 13-7 ConnectPro Setup Screen

- 9.) Enter the hospital-provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the LightSpeed ConnectPRO HIS/RIS Network parameters in the appropriate *LightSpeedSoftware Installation Procedures* document.
The system automatically loads the LightSpeed CT Server AE Title (Hostname of the system).
- 13.) Click the ACCEPT button. The system performs the following routines:
 - a.) Checks for valid setup entries.
 - b.) Executes a perl script that changes the configuration files for HIS/RIS.
 - c.) Returns to the Software Option window shown in [Figure 13-6](#).
- 14.) Click the QUIT button, and the subsequent OK button, in the message window.
- 15.) Remove the MOD from the MOD drive, and write-protect the side containing the ConnectPRO option. Remove the options

Section 7.0

Declaring DICOM Filming Devices on the System

7.1 Prerequisites

Before configuring DICOM filming devices (cameras, printers) on the LightSpeed system, ensure that the following are complete:

- Filming Device Service Representative to assist in camera/printer setup for best image quality film presentation.
- Hospital DICOM network is operational.
- Filming device is connected to the DICOM network with the correct filmer DICOM interface.
- Filming device is DICOM protocol compatible.
- Filming device DICOM Conformance Statement document is available.

Note: **Filmer DICOM Application Entity Titles may be site specific. Check with the Filming Device Service Representative and the hospital network administrator to ensure that you are using the correct AE Title for the destination filming device.**

7.2 Declaring DICOM Filming Devices on the System



NOTICE Potential For Data Loss

Empty all filming queues before modifying camera parameters.

This section contains procedures for recording important Camera setup information. Use the tables at the end of this section to record information from the setup screens.

- 1.) Click the SERVICE DESKTOP button.
- 2.) On the Desktop Toolbar, select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears with a pop-up box reminding you that all filming queues must be empty before updating or deleting a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed (see [Figure 13-9](#)). First, you must click OK in the warning message box (see [Figure 13-8](#)).

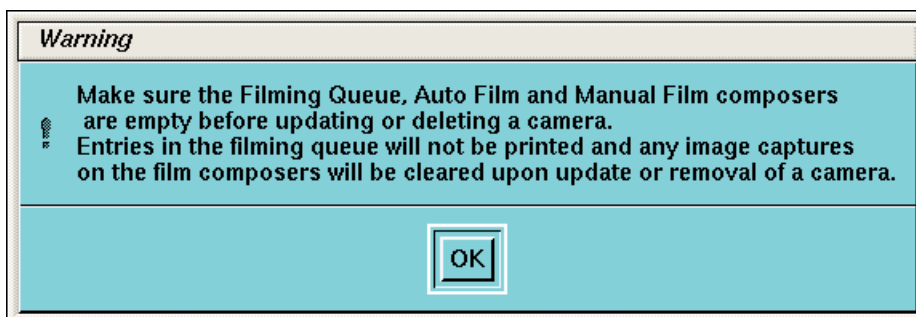


Figure 13-8 Warning Screen

- 4.) Now you are asked a series of questions.
 - a.) To add a new camera, click the ADD button (see [Figure 13-9](#)).
 - b.) A dialog window for the camera type (DASM/DICOM) is displayed. If no DASM is detected during the OC boot, the DASM button is disabled (see [Figure 13-10](#)). If a DASM is present and is not detected, reboot the OC and run the camera configuration tool again.

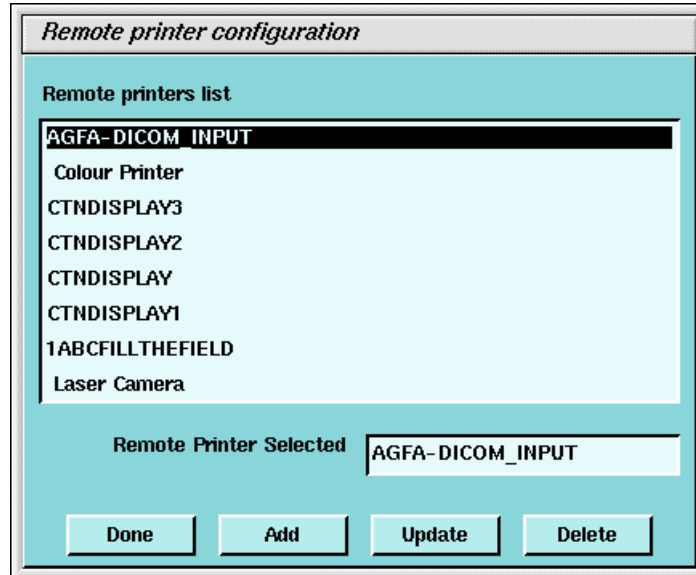


Figure 13-9 Printer Configuration GUI

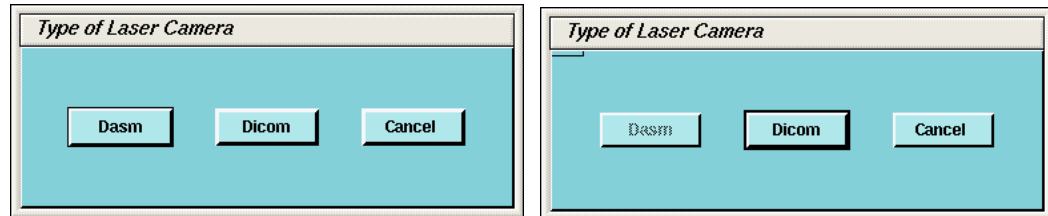


Figure 13-10 Dialog Box for camera Type (with and without DASM)

- 5.) To add a new laser camera, click DASM in the *Type of Laser Camera* dialog box. This displays a list of available camera models. Select the appropriate model from the list and click SELECT (see [Figure 13-11](#)). Now you must configure it.

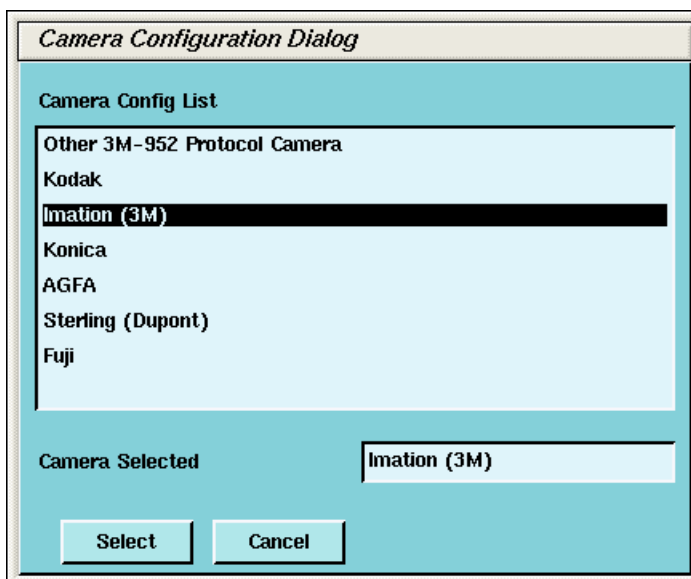


Figure 13-11 Camera Model Dialog (with DASM)

- a.) DASM Interface is automatically detected as either Analogue or Digital
- b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.
- c.) Camera manufacturers provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak:	<u>SMOOTH</u>
Dupont/Sterling:	<u>SMOOTH</u>
3M/Imation (Laser Camera):	<u>SHARP</u>
3M/Imation (Dry View):	<u>SMOOTH</u>
Agfa:	<u>SMOOTH</u>

- d.) Select the appropriate File Format. Select ON from the drop-down list boxes on the menu. The camera manufacturer determines valid film formats. For example, IMATION does not support 4x4, 2x4 or 1x2. AGFA does not support 2x4. The DICOM print convention designates film formats by column and row (e.g. 12 on n1 film is 3x4). When finished setting parameters, click DONE and proceed to step 8.

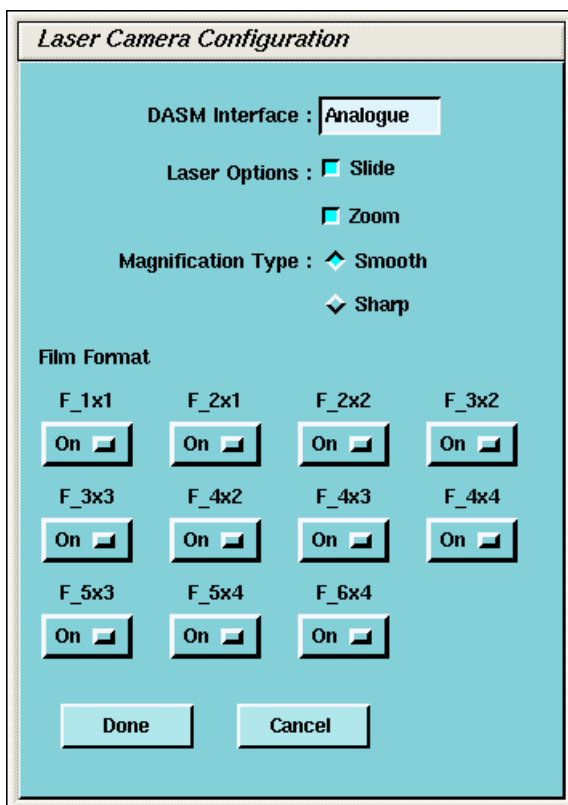


Figure 13-12 Laser Camera Configuration

- 6.) To add a new DICOM camera, click ADD and then DICOM in the dialog box that is displayed.
 - a.) A list of camera models appears (see Figure 13-13). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that model, except the Network parameters.

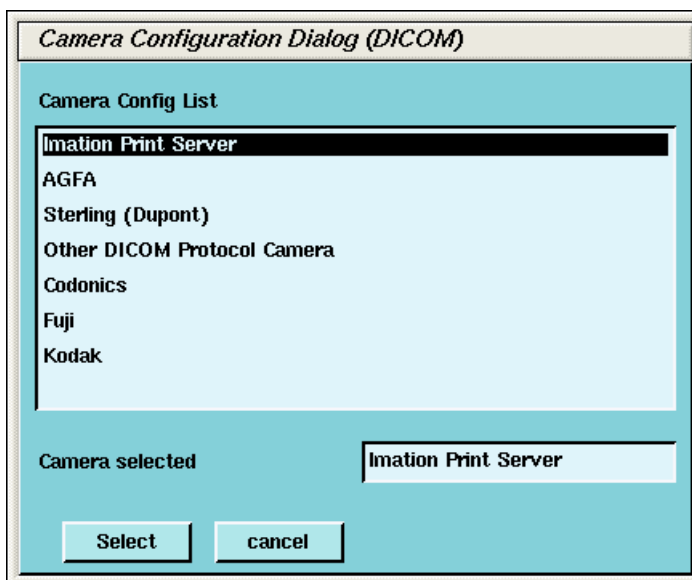


Figure 13-13 Camera Model Dialog (DICOM)

Comment: Recheck the preset information with the camera manufacturer's representative.

Selection of a different camera model clears the Image Quality parameters, because these are camera manufacturer dependent.

b.) Enter the Network Parameters (see [Figure 13-14](#))

- > Device Name A unique name used to identify the camera.
- > Host Name DICOM print server host name, as defined by the hospital.
- > IP Address DICOM print server IP address, as defined by the hospital.
- > AE Title DICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site-specific. Check with the Camera Manufacturer's Representative and the hospital network administrator to ensure using the correct AE Title for the destination DICOM Print Camera.

- > TCP/IP Listen Port DICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturer's DICOM Conformance Statement.*
- > Comments Optional comments used by the DICOM print server.

Figure 13-14 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and CLEAR FILM are supported.

- d.) Set Destination to the final location for film output: either MAGAZINE or PROCESSOR.
- e.) Orientation selects the film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacturer to ensure optimal image quality. Choices are described below:
 - > None No interpolation. Not all cameras support this option.
 - > Replicate Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normally preferred.*
 - > Bilinear A first order interpolation of pixels. This results in images usually described as blurred. *This algorithm is not normally preferred.*
 - > Cubic A third order interpolation. This is used with a large number of possible formulations. Camera manufacturers define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing coefficients" is camera dependent.

Comment: For most Camera Manufacturers, the preferred selection is CUBIC

- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-down menu box located below each selection (see [Figure 13-14](#)). The camera manufacturers sets the valid film formats. For example, IMATION does not support 4x4, 2x4 or 1x2. AGFA does not support 2x4. The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).
- h.) After the camera is configured, click the ADVANCED button. This creates the camera device file automatically, and displays the Advanced Parameters screen (see [Figure 13-15](#)).

The screenshot shows a dialog box titled "Camera Attribute Dialog". Inside, there is a section titled "Image Quality Parameters". The parameters are arranged in a grid-like fashion:

- Configuration:** A text box containing "PERCEPTION_LUT=KANAMC".
- Maximum Density:** A text box containing "320".
- Minimum Density:** A text box containing "32".
- Smoothing Type:** A text box containing "140".
- Empty Density:** A text box containing "Black".
- Border Density:** A text box containing "Black".
- Film Size:** A text box containing "<empty>".
- Trim:** A text box containing "Yes".
- Priority:** A text box containing "High".

Each text box has a small square icon to its right. At the bottom of the dialog, there are two buttons: "Done" and "cancel".

Figure 13-15 Advanced Parameters (Camera Attribute Dialog)

- 7.) Advanced camera parameters control the image quality of films.
- Note: For more information on the proper settings for these parameters, please see the Camera DICOM Print Device Conformance Statement or the Camera Manufacturer's Representative.

Refer to a copy of the Conformance Statement when working with the Camera Manufacturer's Representative, to set up the DICOM Print Camera I/Q and Time-out Settings correctly.

- a.) Configuration - This parameter is camera manufacturer dependent, as it is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field scrolls automatically as text is entered. To review your entry:

- 1.) Click and hold the middle mouse button while the cursor is in the field.
- 2.) Drag the mouse toward the right (or left), as needed.

Note:

Recommended Configuration Setting Values:

Agfa Drystar (MG3000)	PERCEPTION_LUT=KANAMORI (100)
Imation Dryview (8700)	LUT=0,7
Kodak Laser Printer 190	CS434\CN0\PD1.20

- b.) Smoothing Type - Set Smoothing Type to ON, and input the selected value. Use this parameter when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note:

Recommended Smoothing Type Starting Values and Ranges:

Agfa DryStar (MG3000)	Start Value: 140	Range: 137-150
Imation Dryview (8700)	Start Value: 3	Range: 3-13
Kodak Laser Printer 190	Start Value: Enhanced	Range: Normal

- c.) Minimum and Maximum Density - Sets the brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacturer dependent. For Maximum Density, input the correct value into the text box. For Minimum Density, set it to ON and input the correct value in the text box.

Note:

Recommended Minimum and Maximum Density Starting Values:

Agfa Drystar (MG3000)	Min.: 20 or 23	Max: 300
Imation Dryview (8700)	Min.: (Blank)	Max: 300
Kodak Laser Printer 190	Min.: 20	Max: 300

- d.) Empty Image Density - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- e.) Border Density - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- f.) Film Size - Allows the system to specify a particular film size.
- g.) Trim - YES produces a white (clear) box surrounding each image.
- h.) Priority - This sets the print priority.
- i.) When you have completed entering the advanced parameters, click DONE.

- 8.) **Camera Data Tables:** To locate Install Camera Information: Click the SERVICE DESKTOP button. On the Desktop Toolbar, select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window is displayed. Select each of the cameras that are installed from the list of installed cameras, and click UPDATE to view the camera settings. Record the values used to set up each camera in the tables that follow. Extra tables are provided for multiple cameras.

Note: You can determine this information by looking at the contents of the following files:

- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
- For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg` where, `name.cfg` is the camera device name from the printer configuration GUI.

Example: `more <filename from above> ENTER`

DASM CAMERA #1

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 13-1 DASM Camera #1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click ADVANCED.

Table 13-2 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	
NOTES		

*To view Advanced DICOM Camera Settings, you must click ADVANCED.

Table 13-3 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click ADVANCED.

Table 13-4 DICOM Camera #3

7.3 Troubleshooting Tips

PROBLEM	SYMPTOM	SOLUTION
Error bringing up the camera installation/filming application	After creating/modifying the DICOM print camera, the install camera interface does not come up and the filming application indicates it cannot bring up the film composer	The configuration field contains invalid information such as a backslash (\) or brackets ({}) Manually edit the camera.dev file in <code>~ctuser/app-defaults/devices</code> to remove the offending characters in the set configuration line.
Configuration information field	Cannot view the entire configuration field (>25 characters)	Hold down the middle mouse button and move the field contents
Set DICOM print attributes not supported by software	User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera. DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for the LightSpeed (14x17). (Otherwise the camera queues the films or returns an error causing the queue to pause, based upon the DICOM print camera specifications.)	After the DICOM print device is configured, use an editor to add the following line to the camera.dev file located in <code>~ctuser/app-defaults/devices</code> For Trim Off - set Trim NO For Trim On - set Trim YES After the DICOM print device is configured, use an editor to add the following line to the camera.dev file located in <code>~ctuser/app-defaults/devices</code> . To force 14x17 - set filmSize 14INX17IN
Prevent DICOM print attributes from being sent to the DICOM print camera	Some DICOM print attributes are optional and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.	After the DICOM print device is configured, use an editor to add the following line(s) to the camera.dev file located in <code>~ctuser/app-defaults/devices</code> To prevent sending the: - Smoothing Parameter - set FB_Smooth FALSE - Border Density - set FB_Border FALSE - Empty Image Density - set FB_EID FALSE - Minimum Density - set FB_MinD FALSE - Trim Parameter - set FB_Trim FALSE
Error trying to connect to the DICOM print camera	DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface", The help message refers to running the install.dasm (Association Error).	The system is unable to complete the association. Check the AE Title and the Port number of the DICOM print server and correct them through the Install Camera procedure.
Film composer error not usable	Film composer error says "unrecognized status - code ()"	Review the log file, and the attention and status windows. These areas have the correct filming status (for example, film jam and supply empty).
Debugging connection issues difficult	The time-outs for the DICOM print are long, which means waiting a long time before knowing the application is not working.	The time-outs for the DICOM print were set up to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print time-outs for the association and DIMSE classes (for example, N_GET, N_DELETE) can be modified within the DICOM print camera installation. They can be reduced to 90 seconds.

Table 13-5 Troubleshooting Tips

PROBLEM	SYMPTOM	SOLUTION
DICOM print error on N-GET time-out confusing	When the N-GET time-out goes off, the error message in the prslog file is "Could not get printer status, invalid command sequence for N-GET".	This error indicates that the issue may be an inactivity timer on the N-GET DIMSE service.
DICOM print camera slide support	Current implementation of DICOM print does not allow selection of slide format	Feature not currently supported; possible in future release.
Confusion on film format notation	GEHC Laser Camera and DICOM Print film format notations are opposite.	GEHC Laser Camera film format notation is row x col (for example 12 on 1 = 4x3) Dicom Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Table 13-5 Troubleshooting Tips

Section 8.0

Teleradiology (Framegrabber Type) Systems and Auxiliary Monitors

8.1 Introduction

Some Genesis-based systems have teleradiology (TR) systems that framegrab the Genesis GFB video (512 x 512 50/60Hz). LightSpeedDOES NOT directly support this type of TR. The LightSpeed RGB color display video is a much larger format at a much higher pixel frequency. GE does not promise any direct compatibility with framegrabbing TR systems. (DICOM 3.0 TR systems may work depending on the DICOM implementation, but GEHC does not, and cannot, validate all the various TRs.)

In the framegrabber case, a high quality (300Mhz bandwidth) video splitter/amplifier intercepts and re-drives the display CRT RGB video. Composite greyscale would be available on Green #2 (1280x1024 pixels at 72 Hz). Any framegrabber hardware attempting to capture this signal must be capable of a 140 Mhz pixel rate. This also involves TR system configuration parameters. The TR capture software may also need upgrading to deal with 1280x1024 and/or **crop** the signal. The TR remote display software may need upgrading to view the larger format. The image transmission times to the remote TR may be up to four times as long. GEHC supplies all technical information necessary to assist TR suppliers in making their systems work with LightSpeed, but GEHC cannot be responsible for this third-party TR equipment, software, or compatibility with LightSpeed.

The following common parts are available from GE Healthcare:

PART NUMBER	DESCRIPTION
2237018-2	Four-way Video Splitter (BNC Converter)
2154425	BNC to BNC 7' Splitter Cable
2142221	BNC to Octane 7' video Cable
2256482	DB15 to Octane 7' Video Cable
2256485	DB15 to BNC 7' Adapter Cable

Table 13-6 Common Parts Available From GEHC

8.2 Auxiliary Monitors

Figure 13-16 illustrates the **only approved GE CT configuration** for remote monitors, including the part numbers for the only approved components. The use of other configurations and/or components should be strictly avoided.

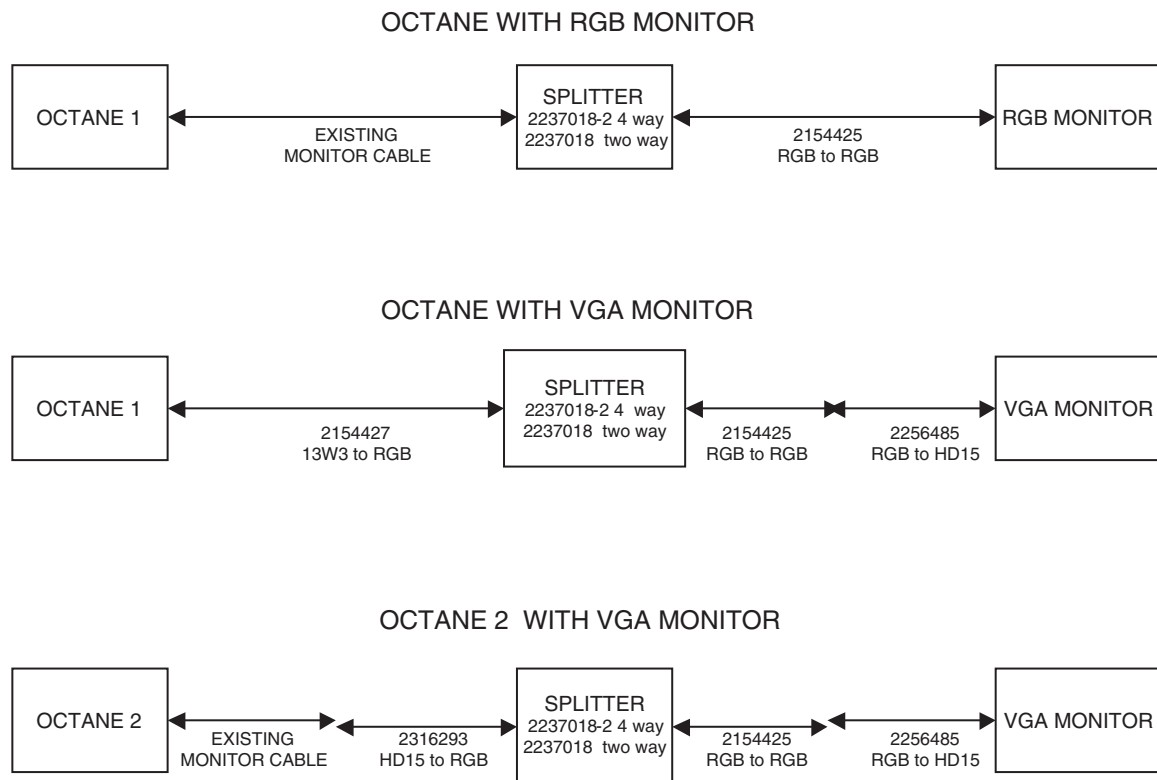


Figure 13-16 Video Splitter/Monitor Cabling

8.3 GEHC Analog Filming Interface Specs (Video & Serial)

Note: The analog (VDB) video output specs and serial interface were specified from, and are the same as, the Genesis interface output specs. This does NOT apply to the digital (LCAM) filming interface.

OUTPUT	VALUE
amplitude	1 volt peak-to-peak
video	0.643V $\pm$ 10%
setup	0.071V $\pm$ 10%
sync	0.286V $\pm$ 10%
DAC resolution	8 bits
diff. linearity	$\pm$ 1 LSB max
glitch area	80 picovolt seconds max, for any step size
rise/fall times	> 10 nsec, $\pm$ 10-90%
FS settling time	7.5 nsec typical to 1 LSB
transfer function	guaranteed monotonic
noise level	> 5.0 millivolt peak-t~peak, combined sync/async noise
DC offset	$\pm$ 1 VDC referenced to ground

Table 13-7 Video Output (Measured Into 75Q at BNC Output)

OUTPUT	VALUE
logic family	F series TTL
output low level	0.8VDC max
output high level	2.0VDC min
output period	41.336 nsecs $\pm$ 10%
transition times	10 nsec max, $\pm$ 10-90%

Table 13-8 Pixel Clock Output

VIDEO TIMING CHARACTERISTIC	60 HZ	50 HZ
pixel frequency	24.192 Mhz	24.192 Mhz
pixel period	41.336 nsec	41.336 nsec
horizontal line frequency	33.6 Khz	33.6 khz
horizontal line width	720 pixels	720 pixels
horizontal active	544 pixels	544 pixels
horizontal blanking	176 pixels	176 pixels
horizontal front porch	26 pixels	26 pixels
horizontal sync	76 pixels	76 pixels
horizontal back porch	74 pixels	74 pixels
vertical frame frequency	60 Hz	50 Hz
vertical frame time	560 lines	672 lines
vertical active	524 lines	524 lines
vertical blanking	36 lines	148 lines
vertical sync	3 lines	3 lines
vertical back porch	30 lines	86 lines
vertical front porch	3 lines	59 lines
scanning format	non-interlaced	non-interlaced

Table 13-9 Video Timing Characteristics

FORMAT	VALUE
visible field	544 pixels by 524 lines
image field	512 pixels by 512 lines
greyscale field	32 pixels by 16 level gray bar on left side of image
greyscale	software selectable on/off
greyscale off value	0 (black)
initial greyscale	255 (white) at upper left corner
border field	12 lines at bottom of visible field
border field value	any 8-bit value, software programmable

Table 13-10 Video Display Format

ATTRIBUTE	VALUE
interface	R5422
25D conn pinout	pin 8 (RX+), pin 21 (RX-), pin 9 (TX+), pin 22 (TX-), pin 7 (GND)
baud rate	1200 baud
word length	8 bit, 1 start bit, 1 stop bit
parity	even
type	asynchronous

Table 13-11 Host Communications/Control Serial Port (analog interface only)

Note: The digital DASM/LCAM serial control has standard R5232 on pins 2, 3, and 7.

This Page Intentionally Left Blank

Chapter 14

System Level Safety Tests

Section 1.0

Patient Touch Leakage Test

1.1 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
1	none	1 hr.	none

1.2 Overview

Complete these procedures after all options are installed. This test covers three safety and leakage current checks:

- Patient Touch current test (completed after installation)
- System Ground resistance measurement (completed during installation)
- Ground current typical (completed after installation - optional)

1.3 Preliminary Requirements

1.3.1 Tools and Test Equipment

ITEM	QTY	PART#
Standard service tool kit	1	
Fluke 87	1	
AMP Clamp-on AC Probe	1	2338920
Dale 600 Leakage Current Meter	1	46-328406G1
Dale 600 Leakage Current Meter Ultra Sound Kit	1	46-285652G1
Dale 600 Leakage Current Meter European 220 VA	1	46-328406G2

1.3.2 Consumables

No consumables required.

1.3.3 Replacement Parts

No replacement parts required.

1.4 Safety

 **WARNING** **POTENTIAL FOR SHOCK**
 **GROUND WIRES HAVE GROUND CURRENT PRESENT WITH POWER “ON.”**
FOLLOW APPROPRIATE SAFETY PROCEDURES FOR WORKING WITH AN ENERGIZED SYSTEM.

1.5 Required Conditions

No required conditions.

1.6 Patient Touch Leakage Current Test Procedure

- 1.) To measure leakage current using the Dale 600 meter, see [Figure 14-1](#).

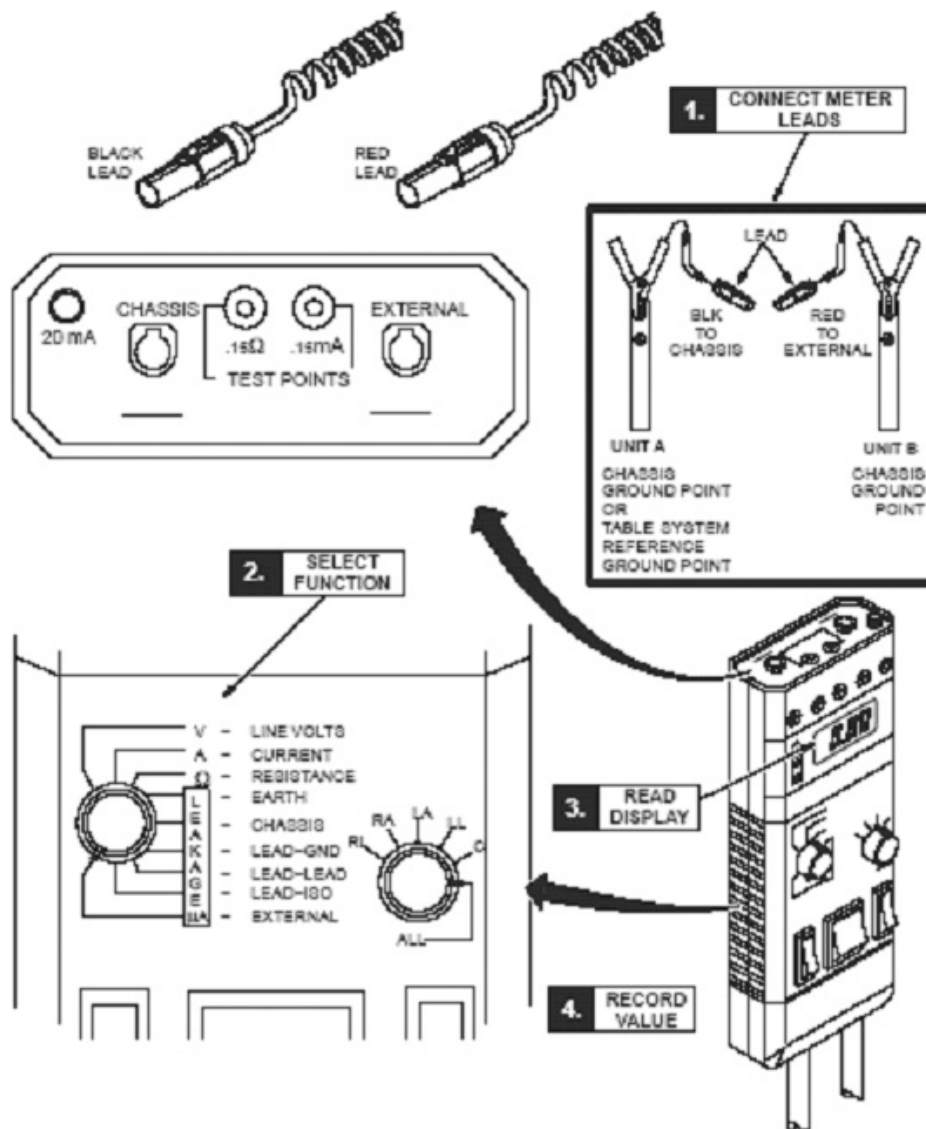


Figure 14-1 Using Dale 600 to Measure Leakage Current

- 2.) Plug the Dale 600 Meter into the outlet nearest the table. (Table base covers must be removed to use table outlet.) Confirm the outlet is correctly wired per the three LED indicators on the meter.
- 3.) Connect the black lead to the chassis plug, and the red lead to the external plug on the top of the Dale 600 meter. Set the function switch to external and display readings are in micro-amps. Connect the black lead to the table base ground buss, and the red lead to the devices (components) under test.
- 4.) Test with power on. The leakage current cannot exceed the following specifications:
 - a.) **Critical care areas (invasive) - 10 μ a**
 - b.) General care areas - 20 μ a
 - c.) Not intended for patient area - 50 μ a

- 5.) Complete the testing between the system reference ground point (table base) and the unit reference ground points (i.e. gantry and table, see illustration 1).
- 6.) Test all conductive surfaces and components within patient reach or within 2 m (6 ft) around the table and 2.3 m (7.5 ft) above the table. Measure at table maximum travels. On some sites, wall outlet cover plates and sinks may become an issue.
- 7.) Test all Optional components, such as in room monitors, injector, overhead monitor suspensions, and table options.

Record these results in the data sheet. These tests must be completed at installation and during PM inspections.

Gnd Bus to	Install <10 μ A	6-month PM <10 μ A	6-month PM <10 μ A
Table gnd points			
Gantry gnd points			
Injector gnd points			
Boom in Room			
Monitors			
Sinks			

- 8.) Test conditions: Table max elevation Min - Max travel; Patient face up and face down, and head first/feet first. Table min elevation 2.2 m (7.2 ft.) patient access to all conductive surfaces including: IV poles, any smart step monitors and stands, table bearing rails, and all other conductive surfaces. Be aware of static discharge: from the scan window, keypads, display, and touch pads or other plastic surfaces.
- 9.) Procedure Hints:
 - a.) Look for items that have abnormal measurements—high or low.
 - b.) These could indicate mis-wiring, loose connections, or poor connections due to corrosion, painted surfaces, etc.
 - c.) High leakage could indicate a wiring error, such as a neutral connected to the ground.
 - d.) Fluctuating ground currents could indicate a short, a poor connection, or a facilities ground problem. This can cause leakage currents from other areas of the facility to flow through our system grounds.

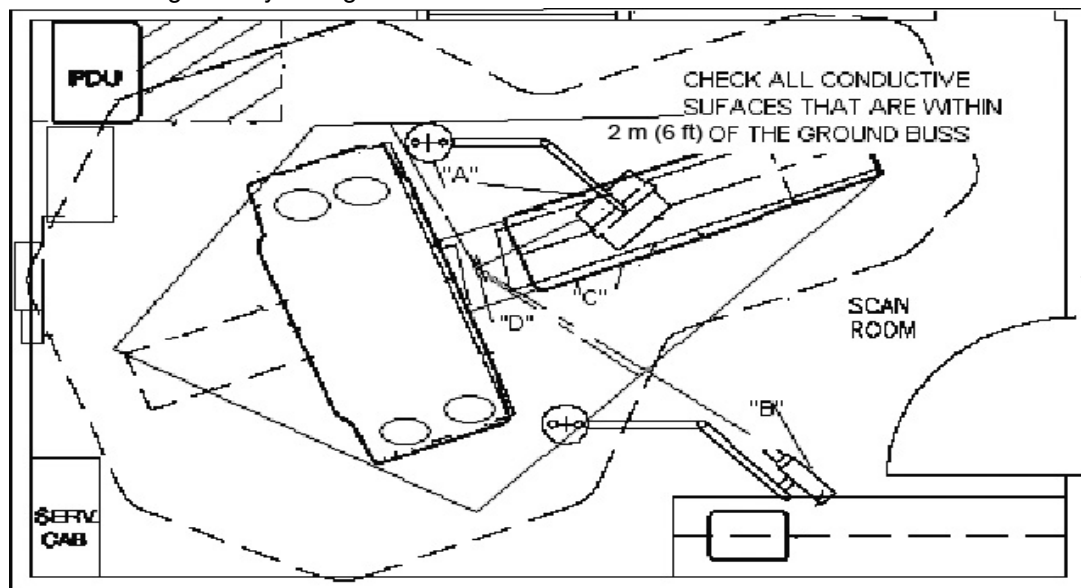


Figure 14-2 CT Leakage Current Map

1.7 Ground Resistance Checks Procedure (done during mechanical install)

Tools Required: Use a Fluke 87 to measure the ground resistance.

- 1.) Use a Fluke 87 or equivalent RMS Meter to verify the presence of less than 0.5 ohm of resistance between each of the following points:

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

Table 14-1 Resistance Verification - Site Ground

- 2.) Connect the meter leads to the Fluke 87 meter.
- 3.) Set the function switch to resistance ohms.
- 4.) Connect the black and red leads per Table 14-1.
- 5.) Do all ground resistance testing with the power off. The ground current causes errors in the resistance readings.
- 6.) Use lockout/tag out procedures.
- 7.) Maximum ground resistance is 0.5 ohms impedance.
- 8.) Use the table footswitch ground bar as a reference to check the ground resistance to all five system grounds.

Record the measurements in the data sheet:

SYSTEM GROUND BUS	TYPICAL READING	SITE DATA
Gantry Ground Stud	0.5 Ohms	
Console Ground Stud	0.5 Ohms	
PDU Ground Stud	0.5 Ohms	
Table Ground Stud	0.5 Ohms	

Hints:

- Look for undersized ground wires and repair.
- Ground resistance readings should not fluctuate. If so, this indicates an intermittent

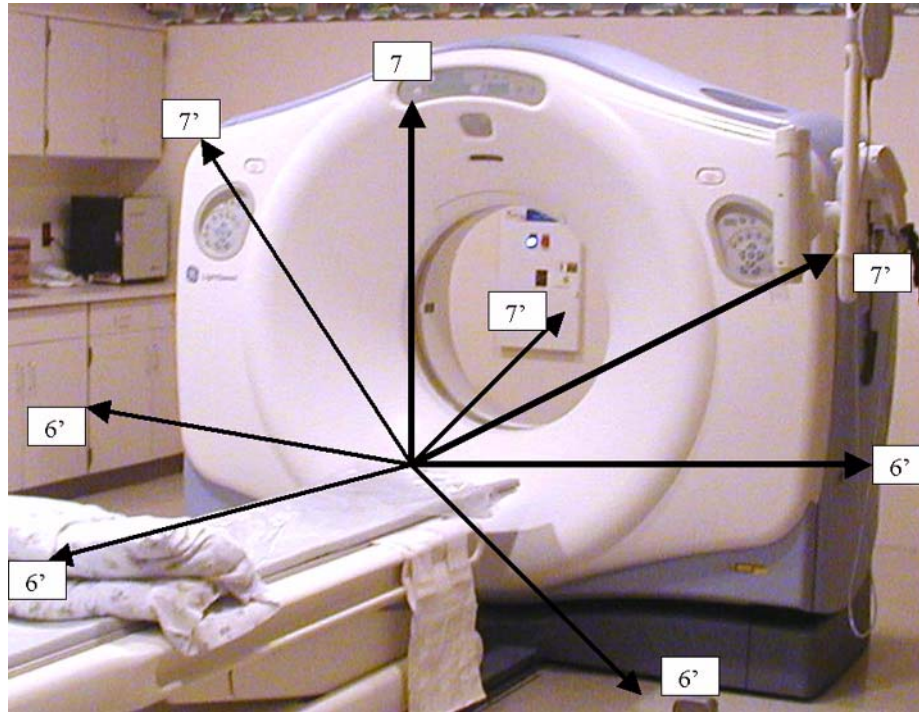


Figure 14-3 Measurement Points

1.8 Finalization

No finalization required.

Section 2.0

CT System Chassis Leakage Test

The CT system Chassis Leakage Test is an optional procedure, unless required by your state.

2.1 Preliminary Requirements

2.1.1 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
1 Engineer	N/A	120 min	N/A
1 Customer Electrician	N/A	120 min	N/A

TIME

Two hour labor on-site, includes:

- Removal of covers
- Leakage test
- Reinstallation of covers
- Recording of site data on the CT System Chassis Leakage Test Completion Form, if this test is required by your state. Forward the completed form to your Project Manager of Installation.

2.1.2 Tools and Test Equipment

ITEM	QTY
Standard FE Service Tool Kit	1
GEH LOTO Kit	1
Dale 600 or 601 leakage Current meter*	1
*GE Healthcare performed validation of this procedure using the Dale 600/601 meter ONLY. due to the unique nature of this meter, GE cannot guarantee the accuracy of this procedure if you use another meter.	

2.1.3 Safety



DANGER POTENTIAL FOR ELECTRIC SHOCK

Ground wires have ground current present with power “ON.” This test is performed with Power “ON”; system is in standby!



NOTICE Follow ALL required safety, PPE, and arc-flash procedures customary for your organization when working on this product.

2.1.4 Required Documentation

Refer to Equipment Service section - LOTO PPE for detailed procedure.

2.2 Procedure



NOTICE Follow LOTO and other safety procedures found in this manual before starting this procedure.

- 1.) Remove the gantry top and side covers. Guide rails on each side of the top cover aid removal. Refer to [Appendix A, Section 1.0](#) and [Section 2.0](#) for details.

Note: Fans located on the top cover make that cover heavier.

- 2.) Power down the console and follow the appropriate GE Healthcare Lockout/Tagout procedures.

- 3.) Remove the gantry front cover following the procedure in [Appendix A, Section 3.0](#).

- 4.) Remove the table footswitch top cover to gain access to the ground cables and ground bar (see [Appendix A, Section 8.0](#)).

Note: Do NOT disconnect ANY grounds at this time.

- 5.) Confirm that all system grounds are securely attached to the system ground bus and NOT the table base.

- 6.) Direct the electrician to remove all external electrical connections made during installation, including:

- a.) Main system ground at PDU
- b.) Power feeder flex connection at PDU
- c.) Room door interlocks and room warning light connections
- d.) Any and all other external ground connections to the system

- 7.) Confirm that all external gantry, table, console, and PDU connections, have been removed.

Note: Some wires, such as the room warning light, may have external power and wire nuts, which should be installed to protect from arcing.

- 8.) Follow the Lockout/Tagout procedure for re-energizing power and boot to application level. If not already done, remove the footswitch assembly top cover while the system boots.

- 9.) Plug the Dale 600/601 Leakage Current Meter into one of the outlets on the gantry.

Note: Some meters include a black lead and a red test lead, while other meters include two black leads. If your meter includes two black leads, the longer black lead is the test lead.

- 10.) Connect the meter leads to the meter as follows:

- a.) Connect one end of the shorter black lead to the chassis plug and the other to the table ground bus.
- b.) Connect the red or longer black test lead to the external plug on top of the Dale 600/601 meter.

- 11.) Set the function switch on the Dale 600/601 meter to EXTERNAL. Using the red or longer black external lead, touch the meter test terminal to confirm that the meter is operational.

Note: For more information, refer to the Dale 600/601 Operator's Manual or see [Figure 14-1](#).

- 12.) With the system at application-level and all components functional, test the system ground wires as follows:

- a.) Remove a system ground wire.
- b.) Test that system ground wire.
- c.) Replace that system ground wire.
- d.) Repeat, testing all system ground wires ONE-AT-A-TIME. [Table 14-2](#) lists each system ground wire.

Note: The measured leakage current must not exceed 5 mA in any ground wire.

Components	Results	Date	Tester
PDU Leakage 1/0			
Gantry leakage 1/0			
Console Leakage #2			
Table Leakage #10			
Option Grounds, if present			

Table 14-2 System Ground Measurements

- 13.) After completing all tests, follow the Lockout/Tagout procedure to power down the system.
- 14.) Direct the electrician to re-install all electrical connections, conduits, cables, and wires, and to secure all connections per the NEC code.
- 15.) Check to confirm that all connections are securely tightened.
- 16.) Re-install all removed system covers, except for the gantry right-side cover, located near the service switch panel.

2.2.1 System Power-Up and Test

Note: Complete this section of the installation procedure on-site.

- 1.) Follow the Lockout/Tagout procedure for re-energizing power.
- 2.) Turn on the gantry service switches and power up the console.
- 3.) Check that no cables remain in the gantry rotating path.
- 4.) Return cover dollies to storage areas.
- 5.) Check that the table controls and footswitch function properly.
- 6.) Re-test the system by completing a system functional scan. If installed, test the room warning light and the door interlock at this time.
- 7.) If this test is required by your state, complete the CT System Chassis Ground Leakage Completion form located on the CD for this system's Service Manual, and return it to your Project Manager of Installation.

2.3 Finalization

No finalization required.

Section 3.0

System Chassis Leakage Test Completion Form

CUSTOMER SITE DATA

Where Installed

City

State

Customer's Telephone Number

Room No.

System ID #

GON #

DVM Serial #

Cal Due Date

Leakage Meter Manufacturer

Model #

Serial #

Cal Due Date

Installer's Name

Company

This system was tested in accordance with NFPA 99: Standard for Health Care Facilities.

The measured leakage current must not exceed 5 mA in any ground wire.

Table Ground Bus	Results	Test Date	Tester
Gantry ground wire			
Table ground wire			
PDU ground wire			
Console ground wire			
Options—if present			
Boom in room ground wire			
(other)			
(other)			
(other)			

This system was tested in accordance with NFPA 99 Sections 8.4.1.3.4.1, and 8.4.1.3.4.2: Standard for Health Care Facilities.

All tests as listed above: ☐ Passed ☐ Failed

Return this form to: **Project Manager of Installation**

This Page Intentionally Left Blank

Chapter 15

Installation Completion Checklist

Complete the following checklist:

CT SYSTEM

- ☐ Functional tests completed.
- ☐ Power and ground audit completed.
- ☐ Seismic kit installed (if needed in your area).

CT OPTIONS

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
- ☐ Teleradiology Connections (Refer to [Chapter 13, Section 8.0](#))

SYSTEM CLEANUP

- ☐ Clean system with supplies shipped with your system.
- ☐ Touch-up nicks and scratches (as needed).
- ☐ Install pads and stickers (as needed).
- ☐ Store customer materials appropriately.
- ☐ Store service documents, tools, and materials appropriately.

DOLLIES RETURN

- ☐ Confirm that dollies have been picked up. If not, complete the necessary documentation.

INSTALLATION COMPLETION MATERIAL CHECKS

- ☐ Complete all HHS documentation. Complete Install CQAs as needed.
- ☐ Complete all dispatching activities (03-04-10 codes).
- ☐ Complete customer acceptance checks/test.
- ☐ Complete the system transfer, and notify the appropriate GE personnel.
- ☐ Communicate with the customer and address installation issues.

This Page Intentionally Left Blank

(c) 2000, General Electric Company.

GE Medical Systems, a General Electric Company, going to market as GE Healthcare.

3200 N. Grandview Boulevard

Waukesha, Wisconsin 53149

USA

www.gehealthcare.com

